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APPLICATION IMPLEMENTATION

PROCESS AND TASK REFERENCE

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In memory of a great friend and colleague, Josee, who would not let us forget the silent
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Preface

The Application Implementation Method *Process and Task Reference* provides detailed descriptions of the tasks and deliverables of the Application Implementation Method (AIM). AIM defines a set of organized and flexible process steps that guide a project team through the Application Implementation process.

The material in this reference includes a description of AIM tasks and deliverables. In addition, guidelines on prerequisites; approach and techniques; roles and responsibilities; deliverable components; component descriptions; audience, distribution, and usage; completion criteria; and deliverable template information are provided. This reference is provided to assist in expediting and standardizing all of the tasks executed and deliverables produced in an Application Implementation project.

This reference, and the Application Implementation Method itself, are part of Oracle MethodSM—Oracle’s integrated approach to technical and functional delivery.

Audience

The Application Implementation Method *Process and Task Reference* is written for implementers, developers, team leaders, project team members and project leaders. Implementers, project team members, and team leaders use this reference in detail to execute tasks and create deliverables in AIM. Team leaders and project leaders use this reference as an overview to better understand the nature of tasks and deliverables in order to better manage the execution of tasks and creation and completion of deliverables.

How the Reference Is Organized

This reference consists of an introduction followed by a chapter on each of the processes of the Application Implementation Method. Due to the size of this reference, it is divided into three volumes. Volume 1 contains chapters 1 through 4; volume 2 contains chapters 5 through 8; and volume 3 contains chapters 9 through 11 and the appendices.

Introduction: The introduction presents a brief overview of processes and phases. It also provides conventions used within the reference.

Process Chapters: Each process chapter consists of two main sections: an overview, and detailed task guidelines. The overview section provides the following for each process:

- **Process Flow** — depicts the flow of tasks for this process
- **Approach** — describes the approach for this process
- **Tasks and Deliverables** — lists the tasks executed and the deliverables produced. This table also provides the responsible role and the type of task. Task type definitions can be found in the Glossary at the end of the Application Implementation Method Handbook. Task types are:
 - SI — singly instantiated task (standard task)
 - MI — multiply instantiated task
 - MO — multiply occurring task

- IT — iterated task
- O — ongoing task
- **Objectives** — describes the objectives of the process
- **Deliverables** — lists and defines the deliverables for the process
- **Key Responsibilities** — lists and defines the key roles for the process
- **Critical Success Factors** — lists the success factors for the process
- **References and Publications** — lists any references and publications for the process

The task detail section provides the following for each task:

- **Optional Criteria** — where applicable, criteria specifying under what conditions the task, or some of its task steps, should be executed
- **Prerequisites** — lists the prerequisites for the task and their source
- **Task Steps** — lists the steps of the task
- **Approach and Techniques** — describes the approach and suggested techniques for the task
- **Linkage to Other Tasks** — identifies which downstream tasks use the deliverable produced as input
- **Role Contribution** — provides the role contributions for the task
- **Deliverable Guidelines** — lists the guidelines and deliverable components for the task and describes each section of the deliverable document
- **Tools** — lists the template and tools to use when creating the deliverable for each task

Appendix A: This appendix provides an index which cross-references the AIM tasks in version 2.0 and version 3.0.

Appendix B: Appendix B provides a description of roles used in the Application Implementation Method.

How to Use this Reference

The *Application Implementation Method Process and Task Reference* provides the detailed task and deliverable information that makes up the Application Implementation Method. Each task produces a deliverable and these deliverables are described in this reference. Each task has an identifier (task ID) which makes it easy to locate. The task ID corresponds directly to the deliverable ID of the task that creates the deliverable.

All users of this reference should read the Introduction to better understand tasks, deliverables, and processes.

Oracle recommends that users of all of the AIM handbooks, and the Application Implementation Method itself, take advantage of AIM training courses provided by Oracle University. In addition to the Application Implementation Method handbooks and training, Oracle Consulting or one of Oracle's Implementation Services Partners can also provide experienced AIM consultants, automated work management tools customized for AIM, and Application Implementation Method deliverable templates.

Conventions Used in this Manual

We use several notational conventions to make this reference easy to read.

Capitalization

Names of tasks, deliverables, process and phases are always give title capitals. For example, Design Audit Facilities task, System Data Model deliverable, Technical Architecture process and Build phase.

Abbreviations and Acronyms

Occasionally, it is necessary to use abbreviations and acronyms when adequate space is not available. The Glossary lists meanings of all acronyms and abbreviations.

UPPERCASE TEXT

Uppercase text is used to call attention to command keywords, object names, filenames, and so on.

Italicized Text

Italicized text indicates the definition of a term or the title of a manual.

Bold Text

Bold text is designed to attract special attention to important information.

Attention

We sometimes highlight especially important information or considerations to save you time or simplify the task at hand. We mark such information with an attention graphic, as follows:



Attention: Since project team training occurs simultaneously with this task, some recommendations (or decisions) from training may be implemented in the mapping environment. In this case, these training inputs become predecessors to this task.

For More Information

Throughout the reference we alert you to additional information you may want to review. This may be a task section, appendix, manual reference, or web site. We highlight these references with an easy-to-notice graphic. Here is an example:



Reference: For more information about content for the Solution Design presentation, review the Critical Success Factors, page 3-7.



Web Site: You can find further information on Oracle's Home Web Page <http://www.oracle.com/>

Suggestions

We provide you with helpful suggestions throughout the reference to help you get the most out of the method. We highlight these suggestions with an illuminated light bulb. Here is an example of a suggestion:



Suggestion: Verify your backup and recovery plan with your hardware and software vendors.

Warning

Considerations that can have a serious impact on your project are highlighted by a warning graphic. Read each warning message and determine if it applies to your project. Here is an example:



Warning: Any time you insert data directly into Oracle Application tables, you run the risk of corrupting the database. Oracle strongly discourages inserting data directly into Oracle tables that are not designed as an Open Interface.

Optional Criteria

Where applicable, optional criteria specifying under what conditions the task, or some of its task steps should be executed is highlighted by a delta symbol graphic. Here is an example:



If your project includes *process change*, you should perform this task.

Related Publications

Books in the Application Implementation Method suite include:

- *AIM Method Handbook*
- *AIM Process and Task Reference*, volume 1
- *AIM Process and Task Reference*, volume 2
- *AIM Process and Task Reference*, volume 3 (this book)
- *AIM FastForward Add-In Handbook*

You may also refer to the following Project Management Method (PJM) suite of reference books:

- *PJM Method Handbook*
- *PJM Process and Task Reference*

Obtaining Additional AIM Advantage Licenses

Each key member of your project team should have a licensed copy of AIM Advantage installed on their workstation. Key members are those individuals who will be leading areas of the implementation project and generating key deliverables.

Oracle provides AIM Advantage licenses to select Oracle Applications Implementation Partners and Oracle Consulting for their project staff. Customers should also equip their key project team personnel with licensed copies of AIM Advantage. This facilitates improved understanding of the implementation process by all team members, as well as improved overall cross-project team productivity.

Additional copies of AIM Advantage may be obtained from the Oracle Direct Marketing or telesales group in your country, or you can contact your local Oracle sales representative.

EMM Advantage and Oracle Application Upgrades

EasiPath Migration Method (EMM Advantage), Oracle's packaged methodology for application upgrades, is complementary to the Applications life-cycle of installation and subsequent upgrades. Produced by Oracle Corporation, EMM Advantage is available to help you structure and manage your Oracle Applications upgrade project.

EMM Advantage includes:

- EasiPath Migration Method (EMM) — a proven, structured approach used successfully worldwide by Oracle consultants
- Project Management Method (PJM) — a standardized Oracle approach to project management

The EMM Advantage toolkit, in combination with your skills, experience, and business knowledge, helps promote a higher-quality migration and lead you to business results faster. It is available from the Oracle Direct Marketing or telesales group in your country, or you can contact your local Oracle sales representative.

Your Comments are Welcome

Oracle Corporation values and appreciates your comments as an Oracle AIM user and reader of the reference. As we write, revise, and evaluate our documentation, your comments are the most valuable input we receive. If you would like to contact us regarding comments on this or other Oracle Method manuals, please use the following address:

email: aiminfo.us@us.oracle.com

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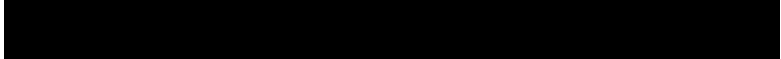
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Introduction

Processes, tasks, and deliverables are the basis of Oracle's Application Implementation Method. They are the building blocks from which project managers construct Application Implementation projects. The *AIM Process and Task Reference* provides the details of every process that plays a part in the Application Implementation Method and every task included in each process. As an introduction, this section provides a brief overview of the concepts of *process*, *task*, and *deliverable*, and describes the information in this reference that is given for each.

What is a Task?

A **task** is a unit of work that results in the output of a single deliverable. Tasks are the most elementary unit of work that one would put into a project plan — they provide the basis of the work breakdown structure. A task produces a single, measurable deliverable and is usually assigned to be the responsibility of a single team member (although many others may play contribution, review, and approval roles). Project progress is usually measured by the successful completion of tasks.

Task IDs and Names

Each task in AIM has a unique ID. That ID is composed of the alphabetical ID of the process in which the task plays a part (more on this later), followed by a three digit sequence number, that usually increments by 10. The sequence number generally reflects the order in which tasks are to be completed. For example, you can be reasonably sure that task BP.020 will be completed well in advance of BP.090 within the Business Process Architecture (BP) process.

Note that the task ID does not reflect the strict order or phase in which the task occurs within a project. A project manager may combine tasks and processes in different ways to meet the needs of different development approaches. Therefore tasks may have different sequences, relative to each other, in different types of Application Implementation projects.

Each task also has a unique name. A task name always consists of a verb (such as *create...*, *determine...*, *design...*), followed by an object. In most cases the object is the name of the deliverable that the task produces. You will find that the text always refers to task names with title case letters, for example, Develop Current Process Model (BP.040).

Task Information

Each task in AIM has a task guideline. If you know a task's ID, it is easy to find the guideline for that task in the *AIM Process and Task Reference*. Locate the process chapter by the first part of the ID, and then locate the

task within the chapter by using the numerical sequence number part of the task ID.

If you only know the name of a task, you can use Appendix A to find the ID. Appendix A contains an alphabetical listing of tasks by task name. It also contains an alphabetical listing of tasks by deliverable name.

Optional Criteria

Tasks are divided into two types — Core and Optional. A core task must occur on every implementation. An example of a core task is Setup Applications (PM.050). An optional task is performed only if the project requirements mandate its use. An example of an optional task is Design Database Extensions (MD.060), which only needs to occur on projects where application extensions that drive database changes will be developed.

Many of the tasks in the *AIM Process and Task Reference* have criteria that define when the task or some of the task steps should be executed. The optional criteria, where applicable, is located just below the task description. In the case of optional task steps, the delta symbol (Δ) will appear to the left of the task step ID.

Prerequisites

Each task assumes that certain things (such as information, programs, and hardware platforms) have been previously produced or compiled, and are available for use. In most cases, these **prerequisites** of the task are specific deliverables of previous tasks. In some cases, they are expected from the client business organization.

Each task guideline lists that task's prerequisites. Under each prerequisite you will find an indication of which components or specific information within the prerequisite is used in the task. The text also indicates how you will use that component or information when carrying out the task.

Task Steps

Many tasks may be broken down into smaller units of work called **task steps**. In some cases, a task guideline may indicate a suggested set of task steps to follow. Many times, the team member responsible for the task (the “owner” of the task) will want to specify the task steps. The task owner will want to base those steps upon techniques that are appropriate to the overall development approach and the tools and resources that are available to the project. Any set of task steps that reaches the deliverable is acceptable as long as it includes adequate quality assurance steps.

From time to time, the reader will see a task step that has a delta symbol to the right of the task step ID. This indicates that the task step may be optional. The reader should consult the task optional criteria located just below the task description for advice regarding when a particular task step may be optional. If no delta symbol is present, then the task is assumed to be recommended as mandatory.

Role Contribution

In addition to the task owner, many other project team members may spend time on a task. Each of these team members will be fulfilling a particular **role**. Their responsibilities may be to contribute information, analyze, document, set up, review, administer, or control. The effort they spend on the task may be significant or nominal.

Each task guideline provides a suggested list of roles that may play a part in completing the task. Next to each role is an indication of the percentage of the total task effort that that role may contribute. These are suggestions that can be used in planning — actual role contributions will depend on the task steps that the task owner assigns.

What is a Deliverable?

AIM is a deliverable-based method. This means that each task produces one or more deliverables, or output, whose quality you can measure. Deliverables can have many formats, such as documents, schedules, program code, or test results.

Each deliverable in AIM is **recognizable** (it has a specific name and ID) and **measurable**. Each deliverable has the same ID as its corresponding task. Each deliverable also has a unique name, which the text always refers to using title case letters. An example is the Current Process Model (BP.040). If you know the name of a deliverable, you can find its ID, and the name of its corresponding task, by using Appendix A.

An AIM deliverable can be further broken down into smaller units called **deliverable components**. For example, the deliverable Current Process Model (BP.040) contains the following deliverable components:

- Enterprise Process Model
- Core Process Models
- Process Flow Diagrams
- Process Step Catalog
- Event Catalog
- Performance Measures

A **deliverable definition** is a specification of the content, organization, and usage of the product from an AIM task. This reference attempts to specify a standard set of deliverable definitions for the Application Implementation Method.

Dependencies between Application Implementation Method tasks are based on AIM deliverables. Each task requires specific deliverables from prior tasks before it can commence. A single task may impact many subsequent tasks in the current and future phases, based on the deliverable it produces.

The *AIM Process and Task Reference* lists the prerequisite deliverables for each task in the task guidelines, and also indicates this information on the process flow diagram at the start of each chapter. The *AIM Method Handbook* does the same in the diagram at the start of each phase chapter. This makes it easy for a project team member or manager to

verify which deliverables must be collected as input to a particular task, before that task can be started.

Project Deliverables

You identify your deliverables using the project workplan. Each task in the project workplan should produce a single unique deliverable. As you tailor the tasks in your workplan, you need to tailor your deliverables as well.

When you begin preparing the project workplan using AIM routes (work breakdown structures), each Application Implementation Method task initially refers to the name of its corresponding deliverable. As you create or revise the tasks in your workplan, make sure that your deliverable names are unique and meaningful. For example, if you create a separate instance of an AIM task for multiple project teams, you would append a qualifier, such as the team name, to the deliverable name for each new task as well.

Project tasks and dependencies can also be tailored based on the prior availability of project deliverables. If a deliverable is already available prior to a project or phase, the task that normally produces it can be reduced to a “review” task. In some cases, it can be eliminated.

Deliverable Review

The production of deliverables is a way to measure the progress of a project. You verify successful completion of a deliverable by performing a quality review. The quality review should be based on the quality criteria specified for the AIM deliverable definition in this reference. You can also establish alternate or additional criteria, such as those required by the business management. Whatever the case, make sure that the completion criteria for each deliverable are clearly understood by the entire project staff.

Deliverable Documentation

Project deliverables can take many formats. Paper and electronic formats are the most common, but other formats include computer hardware and software (for example, System Test Environment), and

even human beings (for example, Skilled Users). In many cases, you will want to produce not only the project deliverable itself, but also a record or representation of that deliverable that may be easier to review, record, and signoff. For example, in addition to producing the actual Skilled Users deliverable, document the learning events that were actually attended by each student, including an indication of which users were not prepared as expected.

You should keep in mind that in many cases, the document only **represents** the project deliverable, or only documents the parts or aspects of the deliverable that are most relevant to communicate. Much more information can often be required to actually meet the quality criteria of the deliverable. In some cases you may not need to produce a document at all. The production of the document alone should not be the goal.

Deliverable Control

You should determine the level of control of each project deliverable as either **controlled** or **uncontrolled**. You control a deliverable and its corresponding documentation in order to protect its integrity and manage its approval and revision.

As a rule, every key deliverable of the project should be controlled. You control the content of the deliverable using configuration control procedures, to restrict access and changes to the deliverable to only those authorized. You also track each version of the deliverable over time and reconstruct any previous version as needed.

You control documentation for a deliverable using document control procedures, to define how documents are prepared, approved and distributed. A controlled document is assigned a version number and its date and distribution list is clearly indicated. You may also want to number each copy of the document with a copy number. As authorized changes are made to the contents of the document, new versions are periodically created and sent out to the original distribution list (at a minimum). You also include a log of changes within each version. If you had numbered copies, you may also want to request that superseded copies be returned.

A deliverable may be uncontrolled because it will not be updated or will be shortly replaced by a controlled deliverable. Changes to an uncontrolled deliverable would not go through the project's configuration control process. However, you should still review the

deliverable for quality and retain a copy of the deliverable in the project's repository.

In the case of uncontrolled documents, such as meeting minutes or memos, document control requirements can be reduced. You will not need to associate a version number with an uncontrolled document. You may not even need to formally distribute it, although a copy should always be retained in the project library.



For more information on configuration control and document control, see the *Project Management Method Process and Task Reference*.

What is a Process?

Major project objectives are achieved by processes. A **process** is a series of **tasks** that results in one or more critical project deliverables. The tasks in a process are usually highly dependent upon one another, and often span much of the project. For example, the Data Conversion process begins early in the development life cycle by defining the scope of the conversion project. This is followed by designing and building the programs and tools for conversion. After testing, the data is converted and available for production.

Figure I-1 shows the processes that are a part of AIM and their relative durations.

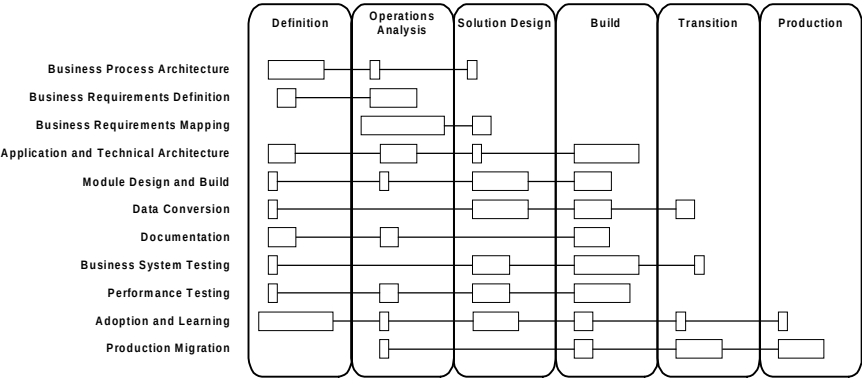


Figure I-1 AIM Context

Process Guidelines

Each chapter of the AIM Process and Task Reference is devoted to a single process. The first part of each chapter gives guidelines on the process as a whole. It shows the relationships of tasks within the process, lists the critical deliverables of the process, and provides guidance on the skills needed to execute the process.

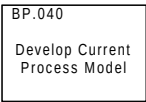
The process guidelines do not indicate exactly where each task falls in the project task plan, since this may vary by the development approach chosen. For more information on choosing and structuring a development approach using Application Implementation Method processes, see the AIM *Method Handbook*.

Process Flow Diagrams

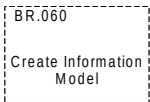
Each process is represented by a process flow diagram. This diagram shows the tasks that are part of the process, and the dependencies between those tasks. It also shows the key deliverables of the process, and the roles that are responsible for each task. The “Approach” section that follows explains the reason behind the organization of the tasks in the process flow diagram.

One thing to keep in mind is that a process flow diagram may indicate that two tasks are strictly dependent upon one another (task “B” may not begin until task “A” has completed) when, in fact, the two tasks will most likely overlap in real life. An example is in the Documentation process. The task Produce Documentation Prototypes and Templates (DO.50) is a predecessor task to Publish User Reference Manual (DO.060). Ideally, all problems would be analyzed before making any changes to the application so that changes to the modules could be made efficiently. However, this is rarely the case in the course of a project, where schedule demands usually require that application changes be made as soon as possible.

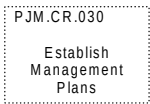
The following describes the symbols that are used in the process flow diagrams.



Core Task. This shows a core task that is contained in AIM. The text provides the AIM ID and the task name.



Optional Task. This shows an optional task that is contained in AIM. The text provides the AIM ID and the task name.



External Task. This symbol depicts a task that should be performed, but is not contained in the same process.



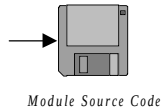
Decision. A decision indicates that there are two or more possible branches to the process flow, depending upon the outcome of the stated question. A decision symbol does not indicate another task -- all work that is done in order to indicate a particular outcome is included in previous tasks.



Process Flow. An arrow between two tasks signifies that the task at the end of the arrow generally should not start until the previous task has been completed. Example: you should not start the task *Map Business Requirements (BR.030)* until you have finished the task, *Gather Business Requirements (RD.050)*. In some cases, it may be desirable to overlap such tasks in an actual project plan.

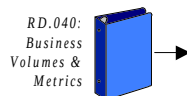


Key Document Deliverable. This represents an important textual output of the process. It includes the name of the deliverable.



Key Software Deliverable. This represents an important software product of the process.

Other symbols are used for key deliverables, as appropriate.



Required Prerequisite. This represents a key input deliverable for a task. The name of the deliverable and the ID of the task that produces the deliverable are given. Different symbols may be used to represent the medium of the deliverable.



Agent Role. Each diagram is divided horizontally into sections or *agent channels*. Each agent channel is labeled with a role. All tasks within that section are the responsibility of that project role.



Attention: Required Prerequisites and Deliverables do not correspond to the agent channel in which they are drawn.

Performance Testing (PT)

This chapter describes the Performance Testing process.

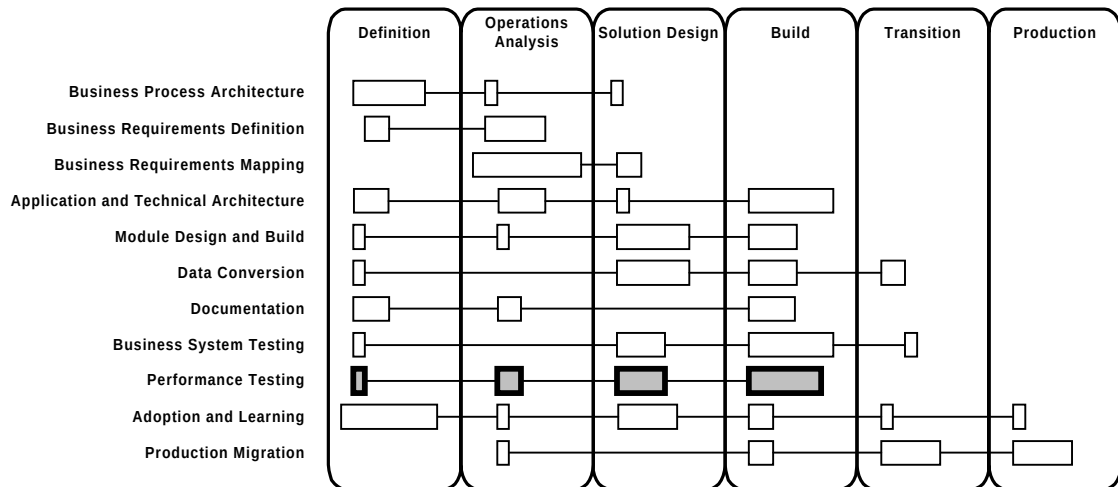


Figure 9-1 Performance Testing Context

Process Flow

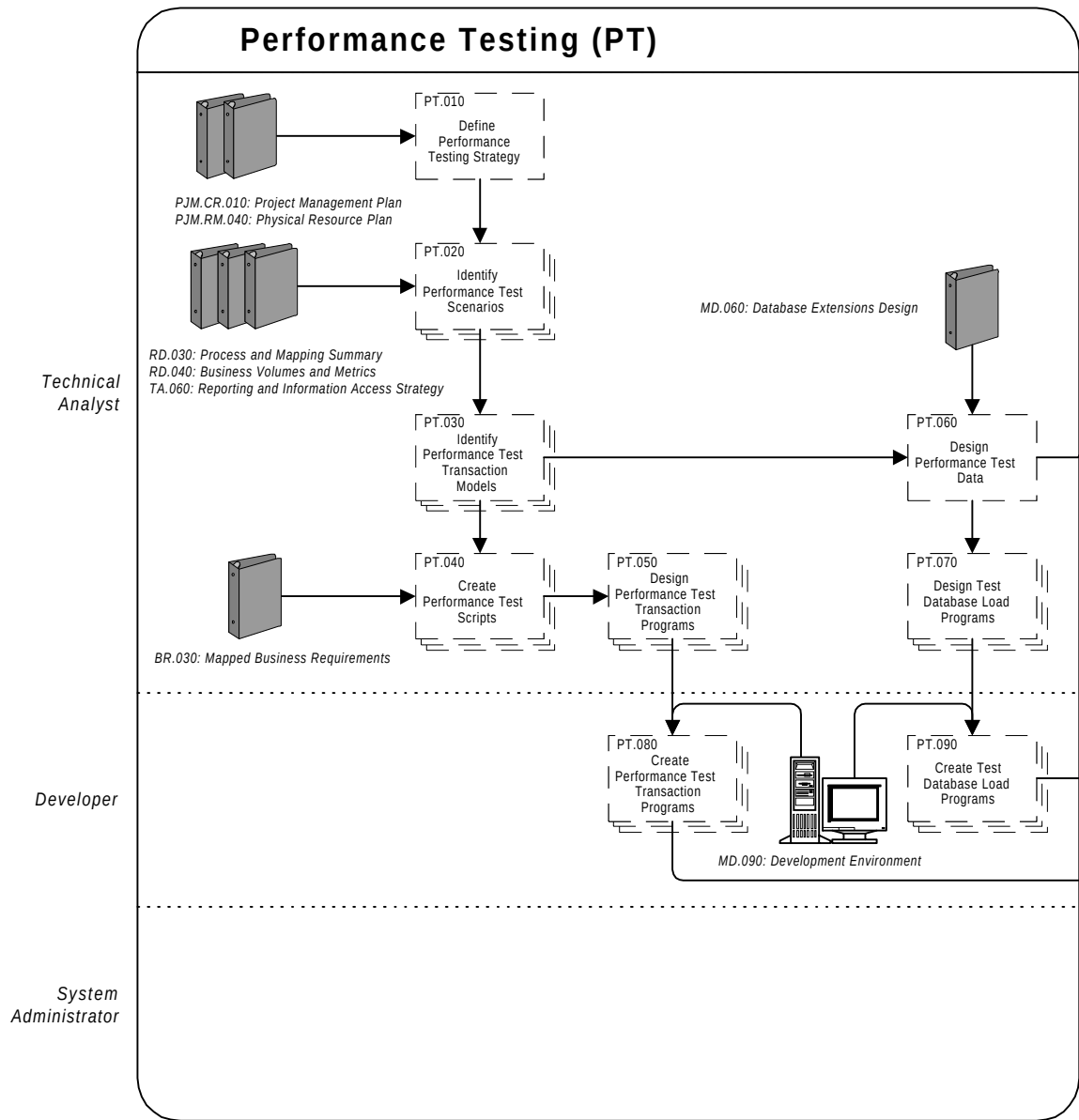


Figure 9-2 Performance Testing Process Flow Diagram

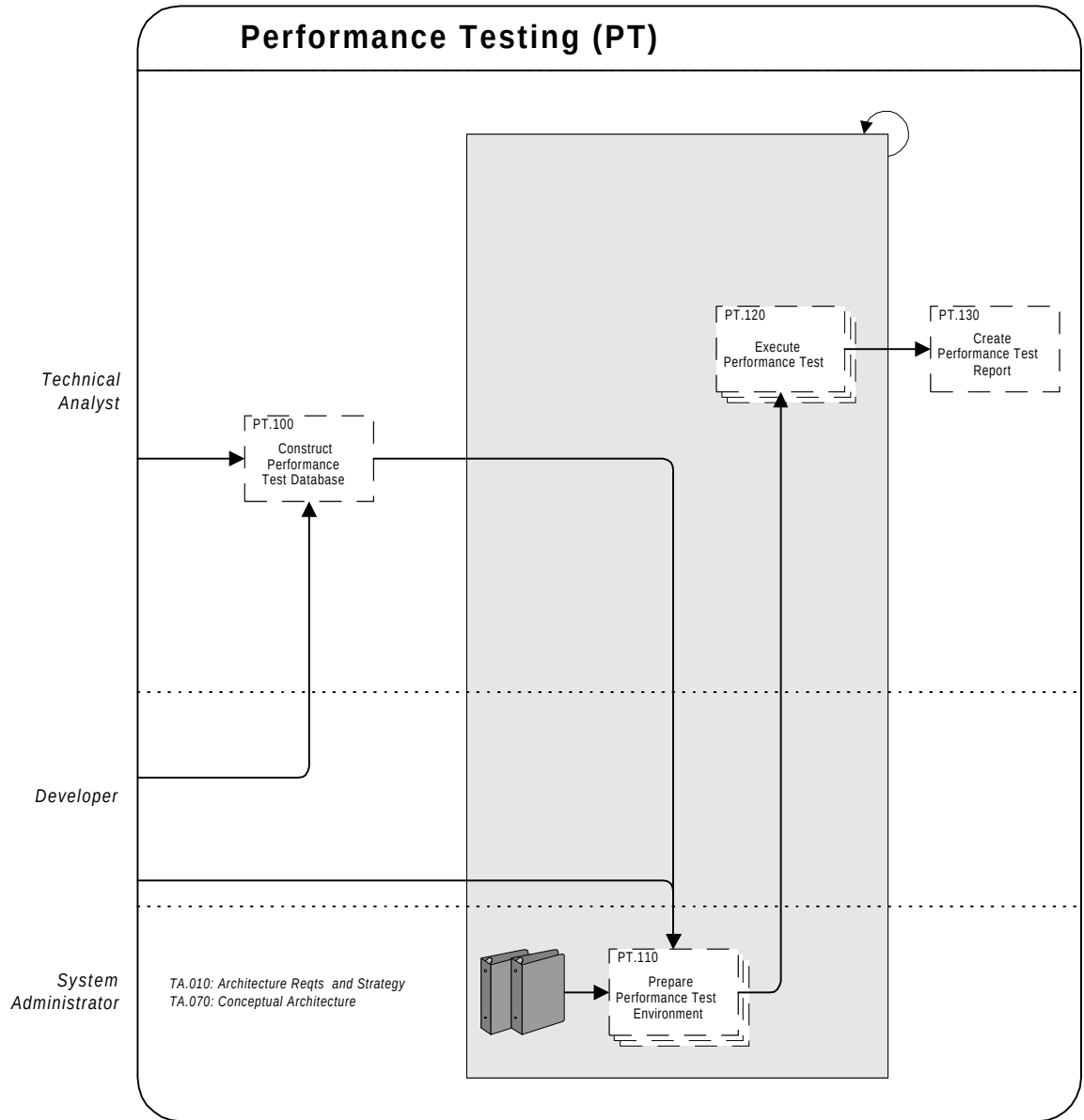


Figure 9-2 Business System Testing Process Flow Diagram (cont.)

Approach

The objective of Performance Testing is to define, construct, and execute a performance test on an entire system or on a particular component of a system. By implementing the process, you can establish the performance of the system or component under test and use the results to make decisions on whether the performance is acceptable for the business. If the performance characteristics you measure in the test prove to be unacceptable, you can implement tuning efforts to improve the performance quality, or more drastically, propose a change in the architecture of the system to provide the dramatic improvement you desire. Performance Testing is closely related to Application and Technical Architecture (TA) and both are mutually interdependent.

You can manage the performance quality of the system you are implementing through various project practices and methods, but the only means of getting direct information about the likely performance characteristics of your new system is to implement a performance test.

Performance Testing Extended into Production System

In some situations, you may choose to continue to use the Performance Test Environment throughout an implementation project and even after the system has gone into live production. The test environment can be a powerful tool for maintaining the performance quality of the live system if you need to make changes to it. For example, you may need to reconfigure the database after you add more disk space or apply patches or upgrades. If you choose to perpetuate the process as a continuing performance quality management technique, you will repeat the same performance test on the live system. Then you can compare the performance metrics gathered from the test environment to the metrics from the actual live system. This assessment of the actual system performance enables you to validate the original test model and to make changes, if necessary.

Early Comparison of Competing Architectures or Configurations

In general, a performance test is conducted to test and monitor the performance system components that you are in the process of implementing; however, you may also consider instigating a performance test to simulate a business system in lieu of having to fully implement the proposed business environment. This might be relevant

if you wish to compare the performance of alternative system architectures as the basis for selecting the preferred one.

Performance Testing Approach

This process uses a top-down approach to the analysis, design, construction, and execution of a performance test. The top-down approach comes from emphasizing the relationship of the test system to real production processing situations at the outset, and only then exposing the details of the test transaction and database models. The test transaction and database models may need to be designed with imposed approximations, but these are always applied to the system in a measured way. The benefit of the top-down approach is that the test results are put in a proper context by always relating them back to a real system. In the more traditional bottom-up approach, performance test models are built without the same rigorous tieback to the production system, which makes interpreting the test results more difficult and prone to risk.

Performance Testing uses the concept of a performance test scenario to make this tieback to the real system. Performance test scenarios are point-in-time snapshots of the processing occurring (or projected to occur) in the real system. Once the scenarios have been identified, they can be characterized and used to fix the relative mix of different transactions, inquiries, or reports that should execute during the performance test of that particular processing situation. You can scale the scenario mix to measure the performance of the system under different total loads (transaction models).

Business System Testing and Performance Testing

Performance Testing is closely related to standard Business System Testing (TE), but with some key differences. Both processes validate some aspect of the system, but standard Business System Testing (TE) confirms whether the system functions as designed. It validates that the transactions, modules, and interfaces produce the correct results individually and when combined into the integrated system. In contrast, Performance Testing is not so concerned with the results of a business transaction, module, or process, but with the speed and efficiency that a system component, or the system as a whole, processes the transactions.

Performance Testing versus Benchmarking

A system performance test is often referred to as a benchmark; however that term is something of a misnomer. A benchmark test contains one or more of the following elements:

- performed for pure performance comparison of multiple hardware and software configurations
- the test transaction model is usually built upon a few highly simplified transactions
- overly simplistic model of user processing
- usually uses a very sparsely populated test database
- may be prescribed or audited by an external body, such as the Transaction Processing Council (TPC)
- very highly (and artificially) tuned and optimized environment with performance being the only consideration
- has limited predictive or diagnostic use for real user environments

Contrast those elements with a performance test conducted within the present context:

- often only a single hardware or software configuration is of interest (although performance testing may be performed for comparison purposes)
- should have many complex transactions proceeding concurrently
- attempts to model real user work patterns
- test conducted against a production-sized database environment
- attempts to simulate, to a determined degree of approximation, a real user environment

In cases where the performance test has a very limited scope and a highly simplified transaction model, it could be argued that a performance test is little different from a benchmark. Conceptually, the two are still very different. In a performance test, you use the particular hardware and software configuration and test transaction model that you have arrived at by making judicious compromises on how much of a complex system you can realistically model, given time and cost

constraints. In contrast, a benchmark tests raw performance of a system with an artificial transaction mix and database, and makes no attempt to model a complex system environment. Changes to the test system are made almost entirely on the basis of performance gains for the small transaction set and will not consider impact on the system as a whole.

System Components

The figure below illustrates the general layers in any system architecture, each of which can have an impact on the overall performance of the system. In general terms, a problem in a lower layer affects the performance of all the layers above it and restricts the ability to tune those layers. For example, an inadequate capacity in hardware and networks causes performance problems in the server and applications that no amount of sophisticated performance analysis can completely remedy.

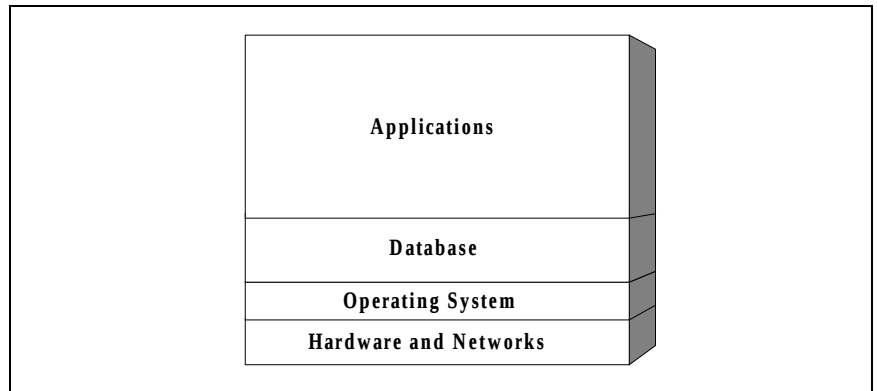


Figure 9-3 System Components

Of the top three software layers, the tuning of the applications may have the largest impact on overall system performance. A rough rule of thumb is that 70 percent of initial performance improvement can come just from tuning badly coded applications. However, even a well-tuned application performs badly in a poorly tuned server, operating system, and hardware environment.

Performance Testing is a key performance quality management technique that can be applied to any of these layers individually or, in its most powerful and comprehensive form, used to validate all the layers of the system acting together in the combined business system.

Scope of Performance Testing

The scope of Performance Testing is potentially very wide. At one extreme you can instigate a system-level test where you assess the complete business system for performance quality. At the other extreme, you can performance test the individual pieces of code or modules that a developer creates.

You can use performance testing to accomplish any of the following performance quality goals:

- Test critical end-to-end business process throughput in the complete system.
- Test key activities or functions within a business process to establish the performance of individual process steps.
- Elucidate dynamic system performance effects that may be too complex to model or predict.
- Study system tuning across the application, database, and operating system.
- Establish performance viability of an architectural configuration.
- Accurately plan system capacity or size a particular server.
- Accurately plan network capacity or size a particular network segment.

Deciding what you need to performance test in order to manage the performance quality of your implemented systems is a major part of the performance testing scope.

Performance Testing Strategy — Automated versus Manual

The strategy you use to conduct performance testing depends on the size and complexity of the implementation, as well as the resources available. Because it is a quality assurance technique, the strategy should take into account the perceived risk associated with poor performance in the final system and the probability of performance problems occurring. The strategy you choose to mitigate these risks directly affects the complexity and cost of performance testing in the project.

Decide first whether to use an automated form of testing or to execute the tests manually. If the scope of the test transactions is limited to purely batch processing, you probably do not need to consider extra automated tools, and the testing process primarily relies on the batch scheduling mechanism. If the scope includes multiple simultaneous online processes, consider an automated testing tool to help manage the processing. These tools can require extra learning events or expertise and add an extra layer of complexity to the management of performance testing.

Manual testing helps reduce the complexity of the test environment itself but puts greater demand on the time of individuals and limits the scope of what you can achieve.

Types of Performance Testing Projects

This section discusses several types of performance testing that you may choose to initiate during the implementation of your system.

System Performance Testing

System performance testing is a technique for testing the performance of the overall business system, including applications, servers, and technical infrastructure. The design and execution of a full system performance test with realistic business volumes and multiple simultaneous business processes is costly and not always practical. When constructing the test, you need to keep in mind the following factors that may affect the accuracy and interpretation of the results:

- simulation of peak period transaction rates in the processes tested
- tests conducted using a server containing a volume of data that is a significant portion of the data that would be present in the production server at the end of the first quarter of operations
- inclusion of non-Oracle applications that will share machine resources with Oracle Applications processing concurrently
- inclusion of mission-critical business processes tested in their entirety or the most important subprocesses thereof
- inclusion of a simulated background reporting load in the test

A more sophisticated test might include the following elements:

- transaction rates to include projections for business growth
- mixtures of realistic volumes of online and batch processing
- the number of separate application sessions each online user would typically have open on their desktop
- structured test cycles to simulate different system scenarios of the business peak processing (for example, a heavy batch processing nighttime scenario plus a scenario of heavy online quarter-end daytime processing)

Module or Code Performance Testing

You may choose to performance test individual modules or programs that constitute activities or functions within a particular business process. Often the individual business process or module owners (if custom developed) may be testing performance as part of their individual roles and responsibilities on the project. Centrally controlled testing with a controlled test area can help provide rigor in performance testing. A centrally controlled test can also help make sure that all business process modules or programs are tested against an *industrial-sized* server, and that test results are interpreted correctly.

Capacity Planning of Hardware and Networks

The need for careful system capacity planning is often overlooked in a project; however, an overloaded server or network connection can be as devastating on performance as badly tuned code or databases. You may have performed theoretical or semi-theoretical sizing of the hardware and networks as part of Application and Technical Architecture (TA), which helps manage the performance of the servers and networks. However, this is no substitute for, and is not as accurate as, actual testing of the system. Performance Testing can identify the selection of transaction flows to use for planning capacity, and the individual flows themselves can be monitored in a controlled Performance Test Environment to accumulate capacity metrics.

Wide Area Network (WAN) connections are limited in bandwidth compared to Local Area Network (LAN) connections and generally have slower latency time (round trip time) because of the longer data transport distances and different transport mechanisms; therefore, they are the most prone to inadequate performance. WAN connections to smaller remote sites with relatively few users can be simply tested by arranging for the users to log in and perform transactions online or

print reports in a typical system performance scenario. The network connections that transfer interface data should also be performance tested at the peak projected rates of data transfer, if possible.

Common Elements of Performance Tests

Whatever the scope or goals of your performance test you need to consider certain common elements. Deciding how you achieve or implement these common elements is what constitutes performance testing. Common elements to consider in developing your Performance Testing strategy include:

- aspects of the system you test
- performance scenarios you model in the test
- special tools or packages you need to run and monitor the test
- transaction programs you use — do you need to design and build any new programs to automate the test or is the system be used as is and be manually tested?
- how much data you need in the test database and how you construct the test database
- whether you intend to extend the testing into production operations and use the method and test components as a performance regression testing tool

Automated Performance Testing Tools

Automated performance testing tools enable you to automate the management and results collection for performance tests in some way. The most sophisticated tools enable you to construct, manage, and record results for a system performance test that requires the simulation of multiple simultaneous online user transaction flows. Such sophisticated tools also enable you to simulate the effects of multiple online users working on a system without the need for users to perform the transactions manually during the test. This type of tool is sometimes referred to as an automated load or stress testing tool. In a controlled performance test environment, the automated tool usually runs on a separate server so that it does not consume the machine resources of the system under test.

The automated testing tool typically has a means to capture the keystrokes and build in transaction delays and user *think* times. You should also be able to specify one or more transaction performance

metrics that you wish to capture during the test. Various third-party vendor products on the market perform this in different ways and some may require specialized programming knowledge. At least one vendor of such tools requires the technical analyst to be conversant with C programming to be able to modify the test programs.

Performance Measurements and Tools

In order to obtain the results you need to meet the performance test objectives, make sure that you have appropriate performance measurement tools installed in the performance test environment. Make measurements on the usage of system resources such as CPU, memory, and I/O. Establish a way to collect this information for each performance test cycle, and display the information graphically so that you can quickly visualize the state of the system and any resource conflicts or bottlenecks. Your hardware vendor may have tools that are appropriate for this task, and many third-party systems management tool vendors may have appropriate tools as well. Vendors that participate in Oracle's Systems Management Tools Initiative (SMTI) program are likely to have tools that interface to Oracle servers for system interrogation and metrics gathering.

Performance Tests Definition and Interpretation

For any system that has more than a few different types of users or possible transaction flows, you need to make approximations when you design a performance test for the system.

Any implementation of a complex Enterprise Resource Planning (ERP) application suite (such as Oracle Applications) generally has, in its scope, multiple business processes, different user groups accessing the system, many complex transaction flows, and a dynamic mix of active sessions and transactions that fluctuate with time. For practical purposes you have to make approximations when you define a performance test for an implementation of this type.

When you interpret the results from performance testing, bear in mind the approximations you made when you defined the performance test strategy, test scenarios, transaction model, and test database. Be careful not to over-interpret results or draw conclusions that may not be totally trustworthy, given the uncertainty about the areas or aspects of the system that you could not include or account for in your test. The performance risk is reduced compared to not performing the test at all,

but beware of overstating the reduction in risk, and have a risk mitigation or contingency plan for those areas that you did not include.

System Performance Testing

This type of performance test is the most ambitious and complex to construct. Because you are attempting to simulate the entire business system, you need to make approximations in deciding which business processes or transaction flows you are going to include or exclude in the transaction model. If you choose business processes on the basis of their importance to the business, be aware that they may not account for a majority of the system resource usage. One technique is to artificially inflate the rate and number of transactions in the transaction model you adopt to account for those that are missing. Care needs to be taken when interpreting results from a test where this technique is employed.

Module or Code Performance Testing

When you define this type of performance test, you are usually trying to break down the system into its logical code modules or programs and then test the performance of each module or piece of code individually. This is the type of test that an individual developer might do in a prepared Performance Test Environment. The testing is fundamentally single-threaded in nature (you are simulating only one user or batch process performing transactions through the module or program). This type of performance testing helps assess the performance quality of individual modules or pieces of code, but does not allow for dynamic effects in the system as a whole, for the tuning of the whole system, and for contention and competition for resources that occurs in the complex multiprocess system.

Performance Test Database Strategy

The test database is a key part of any Performance Test Environment and creating a well-populated *industrial-sized* database is a critical success factor for a performance test that uncovers performance bottlenecks or system tuning problems. Unfortunately, the design and construction of the test database can be one of the more time consuming and onerous tasks of a performance test project, and it is a task that carries a high degree of risk. Depending on when you instigate performance testing, one or more of the following situations may be true:

- Definition of setup data is incomplete.
- Not all setup data has been entered into a database.
- Data conversion programs are not available or fully tested.
- Data conversion volumes are not sufficient for the performance testing.
- Some types of business object data are not converted into the new system.

The approach you take to populating the database with the data you need depends on what is available to you from the implementation project. The simplest approach is to make use of as much data as possible from the master setup data instance (if one exists in your project), and converted data or programs. If you need to pad the existing volumes of transaction data from data conversion, you must load manufactured data into the test database. The following figure illustrates the different approaches you can take in populating a performance test database in an implementation project.

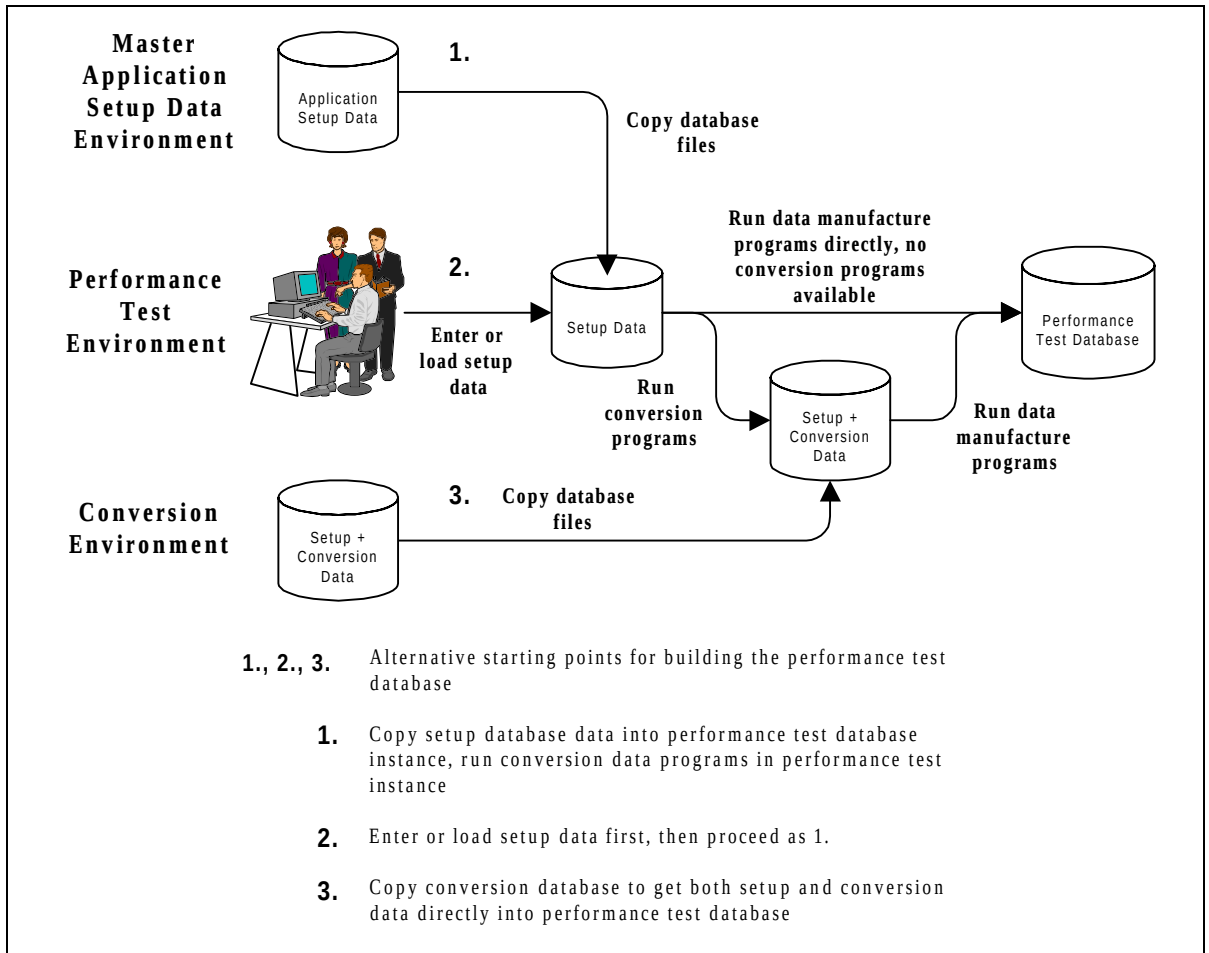


Figure 9-4 Pathways for Performance Test Database Population

Tasks and Deliverables

The tasks and deliverables of this process are as follows:

ID	Task Name	Deliverable Name	Required When	Type*
PT.010	Define Performance Testing Strategy	Performance Testing Strategy	Project includes <i>performance testing</i>	SI
PT.020	Identify Performance Test	Performance Test Scenarios	Project includes	MI

ID	Task Name	Deliverable Name	Required When	Type*
	Scenarios		<i>performance testing</i>	
PT.030	Identify Performance Test Transaction Models	Performance Test Transaction Models	Project includes <i>performance testing</i>	MI
PT.040	Create Performance Test Scripts	Performance Test Scripts	Project includes <i>performance testing</i>	MI
PT.050	Design Performance Test Transaction Programs	Performance Test Transaction Program Designs	Project includes <i>performance testing</i>	MI
PT.060	Design Performance Test Data	Performance Test Data Design	Project includes <i>performance testing</i>	SI
PT.070	Design Test Database Load Programs	Performance Test Database Load Program Designs	Project includes <i>performance testing</i>	MI
PT.080	Create Performance Test Transaction Programs	Performance Test Transaction Programs	Project includes <i>performance testing</i>	MI
PT.090	Create Test Database Load Programs	Performance Test Database Load Programs	Project includes <i>performance testing</i>	MI
PT.100	Construct Performance Test Database	Performance Test Database	Project includes <i>performance testing</i>	SI
PT.110	Prepare Performance Test Environment	Performance Test Environment	Project includes <i>performance testing</i>	MI, IT
PT.120	Execute Performance Test	Performance Test Results	Project includes <i>performance testing</i>	MI, IT
PT.130	Create Performance Test Report	Performance Test Report	Project includes <i>performance testing</i>	SI

*Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Table 9-1 Performance Testing Tasks and Deliverables

Objectives

The objectives of Performance Testing are:

- Define, construct, and execute one or more performance test on the entire implemented system or particular components of the system.
- Provide a direct means for assessing the performance characteristics of some aspect of the new system.
- Complement the functional testing performed in Business System Testing (TE) with a method for managing the performance quality in the implementation.
- Encourage the use of a structured performance testing approach as a means to mitigate the performance risks in a project.
- Establish a performance test model and environment that can be used for continuing performance regression testing throughout the implementation project and into production operation.

Deliverables

The deliverables of this process are as follows:

Deliverable	Description
Performance Testing Strategy	The strategy you will use for performance testing. It includes a discussion of the scope, methods, tools, and techniques you employ.
Performance Test Scenarios	The point-in-time snapshots of the total production system processing environment you model in your performance test. There may be more than one performance test scenario to model and each may have a different set of transactions.

Deliverable	Description
Performance Test Transaction Models	The transaction models you use to simulate each performance test scenario. In general, these contain a subset of the total transactions and processing that would occur in a real system and are approximations to the real processing environment in a live system. This deliverable also discusses the transaction rates you use for the testing and the measurements you make.
Performance Test Scripts	The details of the individual steps you execute for each transaction test user profile.
Performance Test Transaction Program Designs	The detailed designs for any transaction programs you need to execute during the test. This may include programs that execute transactions against the server, or automated programs that manage the execution of multiple simultaneous transaction flows.
Performance Test Data Design	The detailed design for the construction of the test database. It specifies the business objects and volumes needed in the test database and how the test database will be populated with the prerequisite data.
Performance Test Database Load Program Designs	The designs for any programs you need to create specifically for the purposes of loading data into the test database.
Performance Test Transaction Programs	The coded and tested transaction programs.

Deliverable	Description
Performance Test Database Load Programs	The coded and tested data load programs.
Performance Test Database	The databases you use for executing the performance tests. It should have a significantly greater volume of data than is usual for business system test databases. In general, it includes both application setup data and transaction data. Depending on the exact circumstances and timing of the project, the database may be sparsely populated, but it should have all the data necessary for the Performance Test Transaction Programs (PT.080) to execute flawlessly.
Performance Test Environment	The fully configured environment in which you execute the performance test. The environment includes hardware and networks, test databases, automated test execution tools, and monitoring tools. All steps should have been taken to allow the performance test team to connect to the system and start the execution of the performance tests.
Performance Test Results	Your raw measurements and findings from the execution of one or more performance tests.
Performance Test Report	Summarizes the work done in defining the performance test and presents the results from performance testing. It includes the testing approach, the test models, test hardware and software configuration, test results, and conclusions.

Table 9-2 Performance Testing Deliverables

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Business Analyst	Provide input about the key processes and functionality that will be used in the new system, gather and communicate current and future business volumes, and review proposed performance test model and test scripts.
Client Staff Member	Provide documents and information about the existing technical infrastructure covering applications, servers, interfaces, hardware and networks, system capacity, and performance.
Database Administrator	Provide input on database performance risks, support performance test database development, configure servers in development and performance test environments, tune databases, identify database performance measurements, recommend database monitoring tools and techniques, and monitor server performance during the test.
Developer	Create the performance test transaction and data population programs that will be used to execute the transactions, reports, and inquiries against the performance test database.

Role	Responsibility
IS Manager	Provide input to performance testing about current and future systems, and articulate the concerns of senior management regarding technical and performance aspects of the new system.
Network Administrator	Provide input on network performance risks, support performance test network environment, identify network performance measurements, recommend network monitoring tools and techniques, and monitor network performance during test.
Project Manager	Conduct detailed process planning and assign tasks, establish roles, brief users and staff, manage team, manage changes, manage issues, and maintain quality.
Project Sponsor	Provide format, conclusion, recommendation, and overall presentation input for the performance test report that will be delivered to the executive sponsors, project managers, and project team.

Role	Responsibility
System Administrator	Provide input on hardware and infrastructure performance risks, identify hardware and infrastructure needed for performance test environment, order new hardware for test environment (if needed), coordinate preparation of the performance test environment, tune performance test environment, recommend system monitoring tools and techniques, monitor hardware and operating system performance during test, and manage test execution.
Technical Analyst	Identify test data needed for the Performance Test Database (PT.100); design, build, and test database load programs; construct performance test database; and design, build, and test transaction programs.
Tester	Identify the scope and strategy for performance testing, coordinate with project and business managers, develop test models, provide advice and guidance during technical development, manage dependencies to other processes, create and present the final Performance Test Report (PT.130), and secure acceptance for performance testing work.
User	Participate in interviews and provide input about detailed work patterns and transaction flows. Enter transactions manually during performance test execution, if performance testing is non-automated.

Table 9-3 Performance Testing Key Responsibilities

Critical Success Factors

The critical success factors of Performance Testing are as follows:

- clear scope definition for Performance Testing as a means to manage performance quality or establish and plan system capacity
- recognition of the inherent limitations of performance testing as a performance quality tool and realistic assessment of risk
- availability of expertise to define and interpret performance tests
- availability of sufficient quality information about current and future business volumes to enable the definition of test simulation rates and volumes
- test definition input from the business analysts and technical analysts who are defining, mapping, and configuring the new system
- availability of automated testing tools and expertise to use them, or availability of users to provide the manual test transactions
- availability of a controlled environment where a performance test can be set up and run

References and Publications

For information about Oracle products and services relating to this process, consult Oracle's web page.



Web Site: Oracle's Home Page on the World Wide Web.
<http://www.oracle.com/>

Automated Testing Tool Vendors

Several vendors provide automated testing tools to support Performance Testing (PT).



Web Site: For a list of vendors with testing tools visit the AIM Advantage web page.
<http://www.oracle.com/methods/aim/>

For examples of performance tests on Oracle Applications using automated testing tools:



Reference: Fox, Mark and Roets, Anna. Pre-Installation Performance Benchmarking and Tuning of ORACLE Manufacturing at AT&T GIS. OAUG Proceedings, Fall 95.



Reference: Lim, Cindy, Schuettinger, Steven and Packer, Allan. Capacity Planning and Tuning of Oracle Financials on Sun. IOUW Proceedings, Fall 95.

Many sources for discussions of systems management tools and techniques may be relevant to your performance test. The International Oracle User Group (IOUG) holds an annual International Oracle User Week (IOUW) conference. Many papers presented in past conferences deal with the issues and techniques for managing and monitoring Oracle-based systems. For a sample of papers, you can refer to the following:



Reference: International Oracle User Week Conference Proceedings, Fall 94, 95. International Oracle User Group.

PT.010 - Define Performance Testing Strategy (Optional)

In this task, you define the scope and objectives of performance testing in your project and outline the strategy that you will use to conduct the performance test.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Testing Strategy**. It documents the scope and objectives of performance testing for the project. It outlines the Performance Testing approach that is to be followed, and the strategy for achieving the stated objectives.

Prerequisites

You need the following input for this task:



Project Management Plan [initial complete] (PJM.CR.010)

The Project Management Plan provides a high-level discussion of the scope of the testing, what performance concerns, risks or questions it should address, and how the project should be organized and run.



Physical Resource Plan (PJM.RM.040)

The Physical Resource Plan provides information about the environment strategy for the overall project and may include information about the environment resources for performance testing.



Oriented Project Team (AP.020)

The Oriented Project Team has been introduced to the project and is able to contribute to the development of the Performance Testing Strategy.

Optional

You may need the following input for this task:

 Preliminary Conceptual Architecture (TA.030)

The Performance Testing Strategy you define depends on the architecture options outlined in the Preliminary Conceptual Architecture. For example, the Performance Testing process may need to evaluate three possible configurations that have been defined in the Preliminary Conceptual Architecture.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Discuss the project structure with the project manager and establish the specific scope and objectives for performance testing.	
2.	Review the Project Management Plan (PJM.CR.010).	
3.	Document the context for the Performance Testing.	Introduction
4.	Gather relevant materials from the project library.	
5.	Identify Performance Testing scope, milestones, constraints, and risks.	Scope
6.	Identify Performance Testing objectives and critical success factors.	Objectives

No.	Task Step	Deliverable Component
7.	Define policies and procedures unique to Performance Testing.	Approach
8.	Identify Performance Testing dependencies to the main project and other subprojects.	Approach
9.	Establish logistics and environment requirements.	Approach
10.	Establish the method for Performance Testing.	Approach
11.	Identify the mechanism for executing the performance tests.	Technical Strategy
12.	Define the strategy for obtaining a test database.	Technical Strategy
13.	Establish high-level test execution procedures.	Technical Strategy
14.	Establish the procedures for refreshing the test environment after a test cycle.	Technical Strategy
15.	Document the test technical architecture.	Technical Architecture
16.	Describe the resource requirements needed to support the testing process.	Resource Requirements
17.	Document the technical test infrastructure requirements.	Resource Requirements
18.	Document the staffing requirements for the tests.	Resource Requirements

No.	Task Step	Deliverable Component
19.	Document specific tool requirements for the testing process.	Tool Requirements
20.	Identify risks in the strategy.	Risks
21.	Review the deliverable with project and IS managers.	
22.	Secure acceptance for the Performance Testing Strategy.	

Table 9-4 Task Steps for Define Performance Testing Strategy

Approach and Techniques

This task requires the identification of the strategies, standards, and policies for Performance Testing. The information needed to prepare the strategy and document the results generally comes from either the existing business information systems strategy or documents, or discussions with senior project and IS organization management. Once you define and document the Performance Testing Strategy, it should be reviewed and accepted by management before progressing with the rest of performance testing.

Project Organization

In a large and complex implementation project, performance testing may need to be semi-autonomous because of its size and complexity, and a separate Performance Testing subproject may be needed to provide effective control. In these types of implementation projects, the main Project Management Plan (PJM.CR.010) defines the high-level scope and policies but does not provide enough detail about individual subprojects.

Process Scope and Objectives

When setting scope for performance testing, it is important to understand that a full, complex performance test can be expensive to

define, set up, and execute. Proper scope setting and management of expectations are essential. The project managers should specify their high-level goals for the performance test and how the project will use the results. It is important to communicate that the test is not a benchmark and draw a distinction. In addition, the test is not usually designed to provide functional recommendations, nor does it indicate processing gaps. A useful by-product may be information about the processing speed of certain business processes, which may be useful in suggesting process improvement options and the relative timing of interdependent tasks. However, unless categorically stated as an objective, the processing speed of specific business processes is not usually the central goal.



Attention: Performance testing may be one aspect of an overall performance quality management program in a project. If this is the case, the scope should indicate how the testing fits into that program.

Hardware Vendor Coordination

Hardware vendors should have up-to-date metrics for the capacity and performance of their servers. The vendor may also offer technical consulting services with consultants that are familiar with the applications you are attempting to implement and performance testing methods. Discuss your performance testing and capacity planning requirements with your hardware vendor representative and establish the levels of assistance they can provide for this task.

Automated Testing Risks

A key decision in formulating the Performance Testing Strategy is whether to use an automated testing tool or execute the tests manually. If the scope of the test transactions is limited to purely batch processing, you may not need to consider extra automated tools — the testing process primarily relies on the batch scheduling mechanism. If the scope includes multiple simultaneous online processes, you may need to consider employing an automated testing tool to help manage the processing. Use of these tools can require extra events or expertise and add an extra layer of complexity to all phases of a performance test. Another disadvantage is that it is easy to drastically overestimate the productivity of the system users and the speed of transactions resulting in an unintended inflated demand for system resources. This error in designing the test programs can make the results more conservative than they might otherwise be.

Manual Testing Risks

The alternative to using automated tools for the online processes is to conduct the tests manually. This approach helps reduce the complexity of the test environment itself, but puts greater demand on the time of individuals and limits the scope of what you can achieve. In smaller projects with limited resources and time, you can merge the Performance Testing of the system with the Business System Testing and employ the users performing the system test to execute the performance test also. In such circumstances it is very easy for the Performance Testing goals and objectives to merge with those of the Business System Testing, with the result that the performance testing that is done is less than rigorous. It may not benefit from special performance testing techniques. This includes the use of a volume database or test transaction rates and volumes related to specific periods of peak demand in the real system. It also has a lower probability of uncovering performance risks.

A performance test conducted manually using newly trained system users may lead to lower than expected performance demand because of unfamiliarity with the system processes and unstreamlined data entry. When you organize a manual project, keep in mind the competence and familiarity of your performance testers with the system under test.

Performance Test Approach

The method that AIM employs for performance testing is robust and proven in prior implementations. There are other techniques for developing the performance test transactions and database; however, they are all very similar to the process presented here. The advantage of the AIM approach is that it is top-down in nature and attempts to tie the transaction model and database that you use for your performance testing back to actual or predicted situations (point-in-time processing snapshots) in the production system. Other approaches to the problem are often more bottom-up in nature, with the result that relating the results back to the actual system you are modeling is difficult and unclear. The frequent overriding need for approximations in defining performance tests to simulate complex systems means that correct interpretation of the results is critical.

Staffing Resources

The staff resources needed for performance testing are directly dependent upon whether the test execution method is automated or manual. If the testing is automated, you need technical analysts who

are experienced in the automated testing tool chosen, or you need to arrange learning events for the identified technical analysts on the project. If the testing is manual, you need to arrange for enough coordinated user resources to manually enter the transactions at test execution time.

Required Environments

The Performance Testing team should have access to an environment for development of any special transaction programs and a separate controlled environment to conduct the performance test in what will become (or is) the real production environment. The team needs access to a fully functioning installation of Oracle and third-party applications, which should have all or most of the required setup data loaded. This is not always practical in a project, but even if compromises are necessary, you should still attempt to perform the testing in a controlled environment, insulated from the spurious transactions and processing of other project users on the system. In extreme cases where a server has to be shared with the wider project team, you may have to schedule the performance testing for times outside normal working hours.

Linkage to Other Tasks

The Performance Testing Strategy is an input to the following tasks:

- TA.010 - Define Architecture Requirements and Strategy
- PT.020 - Identify Performance Test Scenarios
- PT.030 - Identify Performance Test Transaction Models
- PT.110 - Prepare Performance Test Environment

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	70
Project Manager	15

Role	%
Database Administrator	5
Network Administrator	5
System Administrator	5
IS Manager	*

Table 9-5 Role Contribution for Define Performance Testing Strategy

* Participation level not estimated.

Deliverable Guidelines

Use the Performance Testing Strategy to establish and document the strategy for the performance testing part of the implementation project.

This deliverable should address the following:

- goals of the performance test
- milestones
- critical success factors
- project policies
- risks and assumptions
- acceptance criteria
- change management
- potential impacts of Performance Testing on the overall project

Deliverable Components

The Performance Testing Strategy consists of the following components:

- Introduction
- Scope
- Objectives

- Approach
- Technical Strategy
- Technical Architecture
- Resource Requirements
- Tool Requirements
- Risks

Introduction

This component details the purpose of the deliverable, provides background information on the project, and includes references to related documents.

Scope

This component describes the scope of performance testing for the project in as much detail as possible. Scope statements can be made in terms of whether solution elements are in or out of scope for the project. Examples of elements that can define the scope include:

- business processes or functions within scope
- applications to be tested
- technical architecture configurations
- interfaces, data conversion programs, and databases
- automated or manual performance testing
- capacity planning

In addition to discussion of the scope, this component should also include a description of the key process milestones, the constraints that this process will be subject to and any assumptions made, the risks inherent in the application of performance testing within the project, and the relationship of the process to other systems projects and initiatives already underway.

Objectives

This component lists the high-level objectives communicated by the business and project managers. Since this is a high-level document, the stated objectives should not be too detailed, but they should be specific

and measurable. It should have a description of the critical success factors for Performance Testing.

Approach

This component describes the process, policies and procedures, project dependencies, and the technical background for Performance Testing.

The description of the process should include a high-level discussion of the tasks and deliverables, including any variations from the standard Performance Testing process that has been adopted for your project. If there is specific terminology associated with the process or selected tools, a glossary of terms related to Performance Testing (PT) should also be included.

The subproject policies and procedures should be related to the corresponding policies and procedures adopted for the main project. If the subproject is to use different policies, procedures or standards from the main project in any of the key control and reporting areas, the deliverable should document the differences in detail, explaining why they differ. Examples of areas where the subproject will typically inherit standards and procedures from the main project are listed below:

- project management plan
- issue management and resolution
- change management
- reporting format
- reporting relationship to the main project
- acceptance
- project policies and procedures
- subproject team meetings
- logistics and administrative support

The project dependencies section of the component should describe the dependencies between Performance Testing and other processes or subprojects taking place within the overall implementation project.

The technical background should describe the technical circumstances affecting the approach to the project. Examples of the points that may be included are:

- implementation sites
- technical architecture direction
- computing platforms and technical infrastructure
- major system or application requirements
- innovative or unusual technical requirements

In relation to the statements about the technical circumstances for the project, the deliverable should indicate the environments that will be necessary to support the subproject. Typically, at least one separate environment will be needed for controlled performance test measurements.

Technical Strategy

This component describes the technical strategy for Performance Testing, including the means for executing the tests, the database preparation, and the tools that are to be used. Specific areas that you should highlight include:

- **Transaction Models** — specific scenarios or models of system processing that should be included in the tests
- **Transaction Execution** — how transactions are to be executed in the tests; a key decision is whether an automated load testing tool is to be used
- **Database Construction** — the strategy to be used for constructing the test database, reusing other project data or programs, if possible
- **Performance Monitoring** — how monitoring of performance is to be done during the tests and the specific measurements to be taken
- **Test Execution and Refresh** — specific techniques for execution of the tests and refresh of the test environment after completion of a test cycle

Technical Architecture

This component describes the technical architecture of the test environment. It should include both the technical architecture of the system under test, as well as additional automated test systems. For example, it may need to include any special servers that act as test driver machines processing the automated test tool software. You should include a diagram of the technical architecture of the test environment.

Resource Requirements

This component describes the specific resources needed for Performance Testing which may include resources in the areas of:

- software
- hardware and networks
- hardware and software delivery schedule
- staff

Tool Requirements

This component describes the specific tool requirements identified for building, executing, and monitoring the performance tests. Include the following areas:

- application development
- automated performance testing
- performance monitoring and other system management tools
- configuration management

Risks

This component describes the risks in the strategy to be used for Performance Testing. Examples of risks areas are:

- insufficient funding to meet the required Performance Testing goals
- lack of availability of expert resources
- lack of an automated testing tool needed for performance measurements to meet the scope and objectives of the project

- inability to construct sufficiently detailed system models due to time or cost constraints

Audience, Distribution, and Usage

Distribute the Performance Testing Strategy to the following:

- project manager who should sign off the performance testing requirements and resources needed
- IS manager who should sign off the Performance Testing Strategy
- Performance Testing team members
- system, database, and network administrators who are responsible for configuring and managing the Performance Test Environment (PT.110)

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Are the project scope and objectives clearly identified?
- Are specific critical success factors and risks documented?
- Has the impact of dependent tasks from other processes been taken into account?
- Are the performance testing requirements clearly defined?
- Is the strategy understood by those on the distribution list for this deliverable?
- Are all resource and tool requirements that could affect performance testing stated and understood?

Tools

Deliverable Template

Use the Performance Testing Strategy template to create the deliverable for this task.

PT.020 - Identify Performance Test Scenarios (Optional)

In this task, you define the Performance Test Scenarios that you wish to include in performance testing. These are either scenarios that business management has dictated for inclusion or that you identify from an analysis of the performance quality objectives for the project, coupled with detailed input of business solutions and processing business volumes. Performance test scenarios are point-in-time snapshots of the processing that occurs in a real system.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Scenarios**. These scenarios describe the processing situations (point-in-time processing snapshots) in the future production system where there will be critical performance requirements. The Performance Test Scenarios form the basis for models of the system transactions in future tasks.

Prerequisites

You need the following input for this task:

 Project Management Plan [initial complete] (PJM.CR.010)

The Project Management Plan provides a high-level discussion of the scope of testing, performance concerns, risks or questions it should address, and how the project should be organized and run.

 Process and Mapping Summary (RD.030)

Consult the Process and Mapping Summary to get an initial overview of how the business functions using the new system. If Establish Process and Mapping Summary was not performed, this deliverable will not exist. (See the task description for RD.030 for more information on when this task should be performed.)

Business Volumes and Metrics (RD.040)

The Business Volumes and Metrics provides the business metrics that you use to create the scenarios or relative mixes of the different active business functions and processes. This deliverable should enable you to identify peak processing periods for particular business functions and transactions and the user counts that perform the functions or transactions.

Reporting and Information Access Strategy (TA.060)

The Reporting and Information Access Strategy describes standard and ad hoc reporting and inquiry requirements and the strategy for satisfying these requirements for all users. Reporting and inquiry activity is an important consideration for defining test scenarios. If Define Reporting and Information Access Strategy was not performed, this deliverable will not exist. (See the task description for TA.060 for more information on when this task should be performed.)

Performance Testing Strategy (PT.010)

The Performance Testing Strategy provides information about the detailed strategy to be used for Performance Testing, including an analysis of how the performance test objectives are addressed and the tools and techniques to be used. If Define Performance Testing Strategy was not performed, this deliverable will not exist. (See the task description for PT.010 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the project scope and objectives for performance-related concerns.	
2.	Review business volumes documentation and deliverables.	

No.	Task Step	Deliverable Component
3.	Gather performance requirements and concerns from business analysts and users.	
4.	Prepare an introduction for the document giving the context for the deliverable.	Introduction
5.	Create an initial list of Performance Test Scenarios.	Test Scenario Descriptions
6.	Review the Processing and Mapping Summary (RD.030) to assist in business function decomposition of scenarios.	Test Scenario Descriptions
7.	Review the list of Performance Test Scenarios with user management and functional teams.	Test Scenario Descriptions
8.	Identify the performance test metrics for each Performance Test Scenario.	Business Metrics
9.	Document the assumptions made in defining the Performance Test Scenarios and their decomposition.	Assumptions
10.	Document the risks in the Performance Test Scenario analysis.	Risks

Table 9-6 Task Steps for Identify Performance Test Scenarios

Approach and Techniques

Definition of a Performance Test Scenario

A performance test scenario is a point-in-time snapshot of the total production system processing environment that you model in your performance test. There may be more than one performance test scenario to model in the testing, and each may have a different set of active business functions and transactions. There can be one or more Performance Test Scenarios included in a particular performance test.

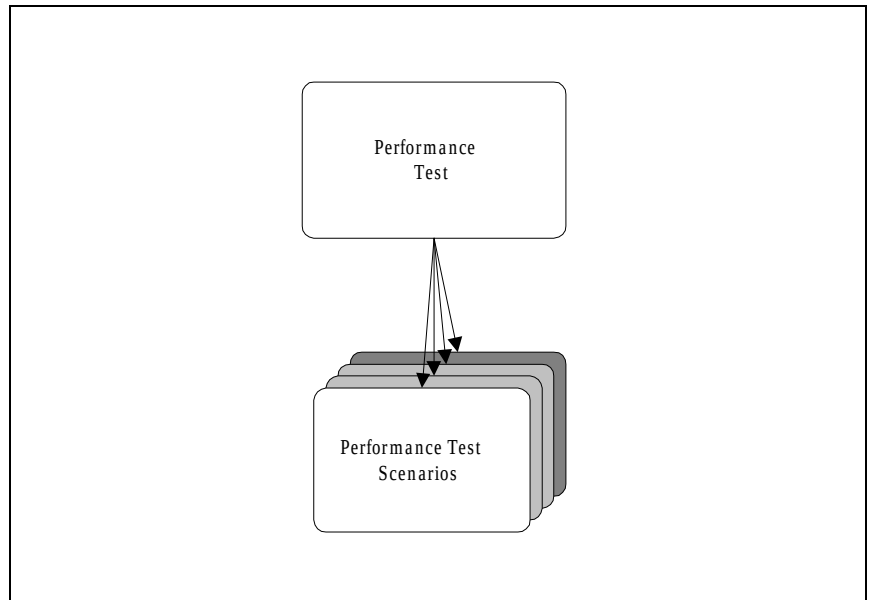


Figure 9-5 Performance Test Scenario Relationship

Types of Scenarios

Performance test scenarios are generally divided into two types: those that have significant online activity and those that are primarily or totally batch processing in nature. Each scenario generally includes users working in multiple business processes, performing various business function steps of the processes, and accessing different applications through a mixture of online and batch processing. The list of scenarios that you create provides the context for the test transactions and scripts that you identify and document subsequently.

Distinguish Performance Test Scenarios and Business Scenarios

Make the distinction between the Business Requirements Scenarios (RD.050) in Business Requirements Definition (RD) and Performance Test Scenarios. The former refers to a specific instantiation of a single business process, with a single input and set of decisions. A business scenario does not refer to a point in time. Performance Test Scenarios are point-in-time snapshots of system processing, consisting of multiple separate business processes and functions that are all concurrently active. In general, a performance test scenario has many business process scenarios active at that point-in-time.

Set of Scenarios Creation

You create the list of Performance Test Scenarios by understanding the performance concerns of the user management and functional teams, analyzing the performance test scope and objectives, and then factoring in the projected business volumes for the new system. Usually the scenario is a list of processing situations that constitute the greatest performance risk in the implemented system. The definition of the set of Performance Test Scenarios is also directly affected by the scope of Performance Testing (PT) as detailed in the Performance Testing Strategy (PT.010).

Review Management Performance Concerns

You may have identified definite concerns on the part of business management regarding the performance of the implemented system during the early performance testing scope and strategy work. Take care to review the prior specification of the project at this point, because the scenario analysis sets the context for the tests you perform and directly affects the effort needed to design and execute the performance test.

Identify Processing Bottlenecks

Using the Business Volumes and Metrics (RD.040) deliverable and having identified the application features employed in the business solution, attempt to identify those critical periods in the business processing cycle when system load is likely to be heaviest. In the process of identifying system load, include transaction load and reporting load because running inquiries and reports also consumes system resources. Typically, these periods of heaviest load occur at financial period close (either month end or quarter end).

Include a Comparative Scenario

As a comparison for the periods when the system is under extraordinary load, you may optionally wish to include a regular weekday scenario that baselines how the system performs under ordinary conditions. You can then compare the performance of the system under regular processing conditions with the periods of peak system usage.

Factor in Cost and Complexity

In determining the list of scenarios that you must test, bear in mind the extra cost in resources and time for each scenario you add. If the scenarios contain essentially the same business functions, but only vary in the relative mix of the functions, the design and build work in the test are much less than if scenarios include different sets of business functions. Even if the business function decomposition is the same for two scenarios, there is still significant work to be done to re-execute the tests and take measurements for the different relative mixes and detailed transaction models.



Warning: Scope creep can happen during the definition of the Performance Test Scenarios (PT.020) and Performance Test Transaction Models (PT.030). It is easy to add extra scenarios and transaction models without fully considering the impact of each extra scenario on the amount of time and resources needed to define the extra transaction programs and the new data that the technical analysts may need to load in the performance test database. Additional scenarios and transaction models also require additional test execution time. Be sure to assess the downstream effect of the scenarios you choose and select scenarios carefully.

Performance Test Scenarios Characterization

Although you often need to make approximations in designing performance tests because of the complex nature of the processing environment of a production system, Performance Test Scenarios are defined to be exact in nature. For example, it is possible to specify a particular scenario as being the morning of the last day of external sales order processing in a business financial quarter. It is only when you attempt to represent the scenario in terms of the business processing and transaction flows that you usually need to make approximations and consider the most important transactions that occur in the system. By defining the Performance Test Scenarios as exact snapshots of the

total processing on a system, you contain the approximations within the design of the performance tests — you can discuss the analysis of the business volumes and the test processing environment in exact terms.

Scenario Decomposition Reflects Scope

As mentioned above, the performance testing scope in the project affects the list of Performance Test Scenarios you can consider including in the test but the decomposition of the scenarios into constituent business functions will also reflect the scope. For example, you may wish to performance test only the batch processing in the system, but at business volumes and rates that are appropriate for a daytime scenario that also has online activity in the real production system. The scenario still reflects a daytime processing situation, but you are artificially ignoring the online processing and thereby introducing an approximation as a result. Therefore, Performance Test Scenarios enable you to assess the approximations you are making in defining the performance test in relation to the situation in the real production system.

Artificial Performance Test Scenarios

If you plan to define and execute a performance test early in an implementation project you may not wish, or be able, to attempt a reasonable simulation of the future system processing. You might still want to run a performance test to gain information about some basic system parameters (for example, those that are relevant to system capacity planning such as memory or CPU usage and I/O metrics). Under these circumstances, you probably need to select a few sample transactions for which you measure the processing statistics, and then define your single test scenario as being the sum total of the business functions corresponding to these transactions.

This type of performance test scenario is not a realistic simulation of the actual future system and is artificial in nature, although its results can be useful if interpreted and used with the understanding of the limitations of the test they were derived from. Such a scenario is similar to a benchmark test, which usually has a highly simplified and artificial scenario model and accompanying transaction set.

Examples of Performance Test Scenarios

Example 1: Daytime, Last Day of a Financial Quarter

This scenario is one that is often important to manufacturing companies that experience heightened demand and volumes of sales orders to ship at the very end of a quarter. It is often specified for performance testing by management, because of concerns about the ability of the system to meet the extra performance demands of elevated sales order volumes, and concern that the mission-critical order fulfillment process can provide the end-to-end throughput needed to recognize all possible shipment revenue. Often, other business functions that would normally be active are curtailed at this time to give the system over to revenue generation, but there may still need to be other important business functions active during the processing that should be represented in the scenario in some way.

Example 2: End of Financial Period GL Close

Depending on when this occurs and its duration, it may be just a batch scenario that occurs overnight, but could also include a portion of a day when online activity occurs as well.

Example 3: Night Run of MRP or Payroll

These two large batch processes usually have critical performance criteria and the period when they run may often be a scenario within its own right. The simplest representation of such a scenario would be to ignore other processing that occurs on the system at the same time and only consider the processing volumes of the single batch program, however there may be other significant batch reports and programs that the business would like to run concurrently with the main program. The performance testing of such a scenario may be an opportunity to help tune the processing of the batch programs and determine what can and cannot process concurrently with them. If the batch program execution occurs in normal online working hours, then you will have to consider the effects of the online activity competing for system hardware resources.

Example 4: Overnight Batch Processing

If suites of batch programs are regularly executed overnight, you may wish to consider a scenario to measure their performance when executing together. The processing that might occur overnight includes sales order statement generation, invoice printing, other financial reporting, planning and forecasting programs, and so on.

Optional Processing Identification

IS managers may grant latitude about when certain batch processes are run. Indeed, the way the user performed the processing in the legacy system and the time required to do it may be completely different from the way it is done in the new system, and at the very least, some limited re-engineering of processing is to be expected. You need to clearly establish with the users and the implementation team which processes are immutable and need to be performed at a certain time and which may not be so critical. In this way, the scenario analysis becomes a tool to identify optimal scheduling of the processes from a performance standpoint.

A common example of a process that may be rescheduled in this way is a Material Requirements Planning (MRP) run, which is often performed weekly by companies to plan their material requirements. During the weekend of month end, there is a heavier than normal load on the system and the business analysts may be amenable to rescheduling the MRP run to take place early during the following week, after the books are closed. Rescheduling in this way frees machine resources for the more critical processing associated with the added volume of business during month or quarter end.

Linkage to Other Tasks

The Performance Test Scenarios are an input to the following task:

- PT.030 - Identify Performance Test Transaction Models

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	65
Business Analyst	35
IS Manager	*
User	*

Table 9-7 Role Contribution for Identify Performance Test Scenarios

* Participation level not estimated.

Deliverable Guidelines

Use the Performance Test Scenarios to analyze Performance Testing requirements for your project and define a set of situations that you plan to simulate.

This deliverable should address the following:

- all scenarios that represent performance risks
- online processing scenarios
- batch processing scenarios
- current number of users of each transaction
- future projected user volume for each transaction
- transaction rates
- target business metrics
- risks and assumptions

Deliverable Components

The Performance Test Scenarios consists of the following components:

- Introduction
- Test Scenario Descriptions
- Business Metrics
- Assumptions
- Risks

Introduction

This component details the purpose of the deliverable, provides background information on the project, and includes references to related documents.

Test Scenario Descriptions

This component describes the system processing scenarios that you separately model in the performance tests. For each test scenario identified, the business functions considered to be active on the system at that time, the roles of users that perform those business functions, and the number of users performing that role are identified. If you are attempting to model an end-to-end business process, you need to decompose it into constituent elementary business functions and include them separately. Each step in the process may be performed by a different role.

Business Metrics

This component describes the business metrics for each performance test scenario that are used to specify the transaction rates and user volumes for the performance testing of each scenario. The key metrics for each performance test scenario are:

- transaction rates for the constituent business functions (either current or projected for a certain date)
- number of users executing each business function (either current or projected for a certain date)

Assumptions

This component describes the assumptions made in detailing the test scenarios and their corresponding metrics. It is critical that the

assumptions be detailed and that they be reviewed with the project and IS managers prior to proceeding with test definition. Examples of assumptions include:

- assumptions about projected volumes
- assumptions about archive or purge
- assumptions about the future business operations model

Risks

This component describes the risks in the test scenarios. Examples of risks include:

- limited number of business functions included for a scenario
- ignoring online processing effects for simplicity of models
- ignoring batch processing load for simplicity of models
- some processing scenarios in the real system may not be simulated due to insufficient time or resources

Audience, Distribution, and Usage

Distribute the Performance Test Scenarios to the following:

- Performance Testing team members
- project manager
- IS manager

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Are the test scenarios fully described and decomposed into their active business functions?
- Are the metrics included for each test scenario?
- Does the deliverable include the assumptions and risks for the selected test scenarios?

Tools

Deliverable Template

Use the Performance Test Scenarios template to create the deliverable for this task.

PT.030 - Identify Performance Test Transaction Models (Optional)

In this task, you define in more detail the sets of transactions that are to be executed during the performance test and the detailed transaction rate metrics for each transaction. A transaction model for a scenario is the precise mix of transactions included in the performance test for that particular scenario.

The other part of this task is to identify the specific application configuration you need to support the test transaction models. This is less of a concern if you are performance testing a system that is already in production operation. However, if the applications are to be reconfigured in some way or the application configuration of the system has not been fully defined, you need to define how you will set up the applications to support the test models.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Transaction Models**. These models define the transaction sets that are to be used to simulate test scenarios during performance testing execution.

Prerequisites

You need the following input for this task:

 Process and Mapping Summary (RD.030)

The Process and Mapping Summary provides an overview of how the business functions use the new system. You may also be able to identify decisions made about key application configuration parameters that are important for your test environment, depending on the progress of Performance Testing relative to Business Requirements Mapping (BR). If Establish Process and Mapping Summary was not performed, this deliverable will not exist. (See the task description for RD.030 for more information on when this task should be performed.)

❑ Performance Testing Strategy (PT.010)

The Performance Testing Strategy includes the scope and objectives of the performance test, which help you choose the appropriate number and complexity of transaction models. If Define Performance Testing Strategy was not performed, this deliverable will not exist. (See the task description for PT.010 for more information on when this task should be performed.)

❑ Performance Test Scenarios (PT.020)

The Performance Test Scenarios describe the scenarios for which you create one or more detailed transaction models. If Identify Performance Test Scenarios was not performed, this deliverable will not exist. (See the task description for PT.020 for more information on when this task should be performed.)

Optional

You may need the following input for this task:

❑ Application Reference Manuals and Documentation

The reference documentation that accompanies the applications being tested is a valuable source of information about the transactions that a particular type of system user should execute in order to perform their work.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Identify how many transaction models you need for each Performance Test Scenario (PT.020), taking into account the scope and objectives for the performance test.	

No.	Task Step	Deliverable Component
2.	Identify the transactions that constitute each test business function in each test scenario	
3.	Review the high-level business operations model pertinent to each test business function.	
4.	Identify the test mechanism that will be used to execute each particular transaction in each scenario.	
5.	Prepare an introduction for the document giving the context for the deliverable.	Introduction
6.	Calculate transaction rates for each transaction model of each scenario, using the business metrics gathered previously.	Performance Test Scenario
7.	Identify the key application configuration parameters that need to be defined for the test environment.	Application Configuration
8.	Review the Process and Mapping Summary (RD.030) for key application parameter decisions.	Application Configuration
9.	Identify suitable values for application configuration parameters not currently defined.	Application Configuration

Table 9-8 Task Steps for Identify Performance Test Transaction Models

Approach and Techniques

Performance Test Transaction Models

Performance Test Transaction Models are the sets of transactions, including transaction rates and user metrics, that you define to simulate the Performance Test Scenarios (PT.020) during the actual performance test executions. Each Performance Test Transaction Model is defined for a particular scenario and there must be at least one transaction model for every test scenario you wish to include in the overall performance test. However, there may be more than one transaction model for a scenario.

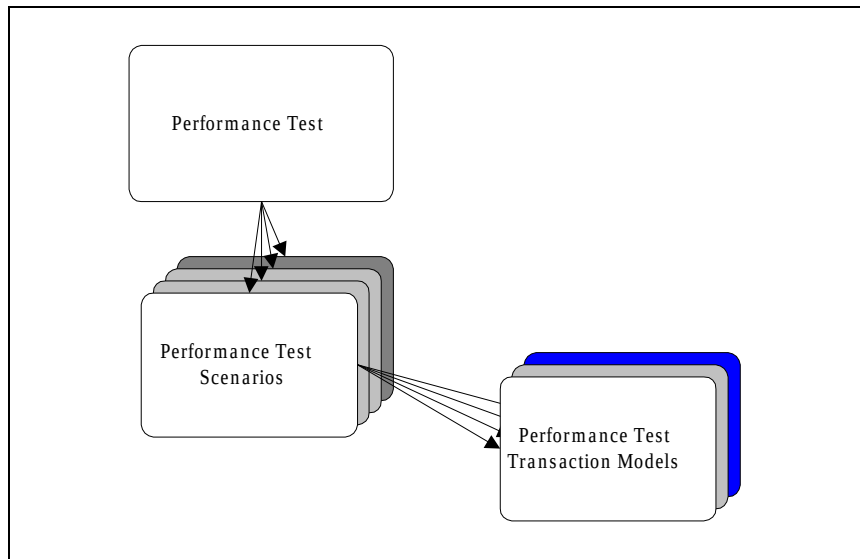


Figure 9-6 Relationship between Performance Test Scenarios and Transaction Models

During this task you also identify the specific application configuration you need to support the models. This is less of a concern if you are Performance Testing a system that is already in production operations, but if the applications are to be reconfigured in some way or the application configuration of the system has not been fully defined, you need define how you will set up the applications to support the test models.

Multiple Transaction Models for a Performance Test Scenario

You may need to define multiple Performance Test Transaction Models for a particular Performance Test Scenario (PT.020). Examples of where this technique is appropriate are listed below:

Example 1: Compare Performance at Initial Live Cutover and After Twelve Months Business Growth

A common performance test objective is to test and tune the system as it would be at initial live cutover and in addition, compare the performance to the performance it would have after twelve months of operation with projected future growth of business volumes and user headcount. This is particularly important for capacity planning predictions for the future. In this case you would create two transaction models for the test scenario, one being the initial cutover model and the other being the twelve-month model. The scenario is the same in both cases as are the transactions within the scenario, but what varies are the transaction rates and the user counts accessing the system (the relative mix of transactions remains constant, but the absolute rates vary).

Example 2: Compare Performance With 250, 500, and 750 Users on the System

Another common performance test objective is to compare the system performance with different total numbers of users working in a scenario with the same transaction rates. This example would simulate three stages of business growth, but with each user performing work at the same rate each time. This would require three transaction models for the test scenario.

Actual User Roles versus Test User Roles

When defining your transaction models, you need to clearly distinguish between actual user roles and test user roles. A user role is a role in the business that performs the processing and work within a particular business function. It represents a business role that is fulfilled by real business users working on the actual system. A test user role may or may not directly and uniquely map to a single real user role. Test user roles are artificial constructs that are devised with the technical execution of the test in mind. You define them to help simplify the design, development, and management of the transaction programs within the test. For example, it may be simpler to break the processing of a single user role into multiple test user roles.

Example 1: Sales Order Administrator

A sales order administrator may have to deal with orders for different types of products. Technically, it may be simpler to treat the processing of the separate product orders as if they were processed by two different types of system users, even though in reality, the work is done by the single real user role of sales order administrator. In this case, one real system user role gives rise to two test user roles:

- Test User Role (product A)
- Test User Role (product B)

You would calculate the total number of order administrator users performing transactions on the system by summing these two test user counts in the performance test.

Example 2: Multiple Open Sessions per User

Another example of an artificial test user role is a role that is needed for multiple session connections. A real system user may have multiple application sessions open simultaneously, but typically only performs work in one session at a time. In order to verify these extra sessions are reflected in the test transaction model, you could define a single test user role whose effect on the performance test is merely to open an extra application session by logging into the system and then going into a dormant state. The technical reason for including this information and simulating these multiple connections in the test is that although the real system user will generally only be able to perform work in one session at a time, the other open sessions may still consume system resources (such as server memory).

When you define test user roles that do not map directly and uniquely to actual system user roles, you should be careful in your computation of test user sessions and transaction rates. The sum total of the test transaction processing and test user sessions should equal the overall target rates and session counts for your particular scenario.

Automated and Manual Performance Testing

When you are conducting a manual performance test and are using manual entry of transactions for the online processing, the distinction between real user roles and test user roles is less important — you will probably be employing real system users for the test and can have them work through portions of their expected system workflow.



Warning: Scope creep can happen during the definition of the Performance Test Scenarios (PT.020) and the Performance Test Transaction Models. It is easy to add extra scenarios and transaction models without fully considering the impact of each extra scenario on the amount of time and resources needed to define the extra transaction programs, and the new data that the technical analyst may need to load in the performance test database. Additional scenarios and transaction models also require additional test execution time. Be sure to assess the downstream effect of the transaction models you choose for each scenario.

Transaction Models Design

Be careful not to oversimplify the design of the Performance Test Transaction Models so as to render the results of the performance test invalid or unrepresentative of production. Using the sales order administrator example from above, if a particular scenario has 50 sales order administrators entering sales orders, you may unwittingly design the transaction model and the test scripts for the transaction flows to each use the same customer for the sales orders being entered. This is unrealistic if the sales order administrators in reality enter orders for a wide variety of different customers who phone the sales office to place their orders. Technically, this restriction of the transaction models, scripts, and programs to a single customer means that the customer table is not fully accessed during the test. The database block with the customer record is less likely to be flushed from database memory, and the time to retrieve the customer from the database might be much less than the real access time for random customer records, especially if the customer table has a large number of rows.

This need to spread the range of setup data referenced on transactions has the effect of complicating the transaction models, scripts, and programs. The effect is particularly marked for automated test tool programs where some mechanism for reading a set of allowed setup data values for each setup data field may be needed. This range of setup data values is often stored in a data file and each setup field in the transaction model scripts may need a corresponding data file of allowed values. The need to manage these technical issues can have an important effect on the way you structure your transaction model, transaction scripts, and transaction programs. For more information, see Create Performance Test Scripts (PT.040) or Design Performance Test Transaction Programs (PT.050).

Linkage to Other Tasks

The Performance Test Transaction Models are an input to the following tasks:

- PT.040 - Create Performance Test Scripts
- PT.050 - Design Performance Test Transaction Programs
- PT.060 - Design Performance Test Data
- PT.110 - Prepare Performance Test Environment

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	90
Business Analyst	10
IS Manager	*

Table 9-9 Role Contribution for Identify Performance Test Transaction Models

* Participation level not estimated.

Deliverable Guidelines

Use the Performance Test Transaction Models to identify the transaction models (mixes) corresponding to your Performance Test Scenarios (PT.020).

This deliverable should address the following:

- the full mix of users, user sessions, and transaction rates for each scenario
- likely maximum transaction models
- worst-case transaction models

- key setup parameters
- assumptions

Deliverable Components

The Performance Test Transaction Models consist of the following components:

- Introduction
- Performance Test Scenario
- Application Configuration

Introduction

This component details the purpose of the deliverable, provides background information on the project, and includes references to related documents.

Performance Test Scenario (repeating)

This component describes the decomposition of the individual test scenarios into test transaction models. There should be one component for each Performance Test Scenario (PT.020) that documents all the test transaction models for that scenario. The individual Performance Test Transaction Models often have the same *relative* mix of active transactions as the Performance Test Scenario and only differ in the total number of users or transactions per minute per transaction. This simulates increasing business growth for each basic Performance Test Scenario.

Application Configuration

This component describes the key setup parameters and application functionality that is relevant to the scope of the performance test and the transaction models. In addition, describe any assumptions that you need to make about the configuration. For example, if the process and mapping teams have not finalized the use of a particular piece of functionality in an application, you should discuss the issue with business analysts and make a decision about its inclusion in the performance test.

Audience, Distribution, and Usage

Distribute the Performance Test Transaction Models to the following:

- project manager
- Performance Testing team members

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Are the Performance Test Scenarios (PT.020) fully decomposed into their Performance Test Transaction Models?
- Have the transaction rates been calculated for each transaction model?
- Does the deliverable include the assumptions and risks for the selected scenarios?

Tools

Deliverable Template

Use the Performance Test Transaction Models template to create the deliverable for this task.

PT.040 - Create Performance Test Scripts (Optional)

In this task, you define the detailed Performance Test Scripts for entering the data to perform online and batch transactions, inquiries, and reports that model the processing occurring during a test scenario.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Scripts**. These scripts define the way in which each scenario transaction model executes during the performance test and includes the response measurement you make as part of each scripted transaction flow.

Prerequisites

You need the following input for this task:



Mapped Business Requirements (BR.030)

The Mapped Business Requirements provide detailed information about the business processes and process scenarios that users of the system follows when performing their work. It also includes the specific system modules accessed during each process step.



Performance Test Transaction Models (PT.030)

The Performance Test Transaction Models provide information about the sets of transactions that model the system in the test. You need to script out the individual transactions that are mentioned in each model. If Identify Performance Test Transaction Models was not performed, this deliverable will not exist. (See the task description for PT.030 for more information on when this task should be performed.)

Optional

You may need the following input for this task:

 Application Reference Manuals and Documentation

The reference documentation that accompanies the applications being tested is a valuable source of information about the detailed system steps each type of user should go through in order to perform their work.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Prepare an introduction for the document giving the context for the deliverable.	Introduction
2.	Identify transaction scripts for each test business function in each test scenario.	Test Scripts
3.	Review the Performance Test Scripts with business analysts.	
4.	Revise the Performance Test Scripts.	Test Scripts
5.	Identify the test response measurements to make during the execution of the Performance Test Scripts.	Test Script Response Measurements
6.	Document assumptions applied to create the Performance Test Scripts.	Assumptions

Table 9-10 Task Steps for Create Performance Test Scripts

Approach and Techniques

This task defines the detailed system steps and events that a user would go through in order to perform the transactions detailed in the Performance Test Transactions Models (PT.030). Whereas the transaction model is defined at the business object level (for example, *create a sales order*), this task describes how the business object is transacted in the application in detail.

User Workflow Patterns

When designing the Performance Test Scripts, keep in mind the workflows that occur in the real production system. Work with business analysts in the project team that are experts in the business functions to understand these workflows. If you can work directly with the system users, then they may be able to give you insight into their detailed work patterns. However, if the system is being newly implemented, information about their work patterns in the legacy system may be of questionable value for the new system.

Performance Test Scripts Review

To establish the degree to which the Performance Test Scripts constitute a reasonable model of the real system transactions, have them reviewed by business analysts on the project team

Test Script Response Measurements

Test Script Response Measurements are timing measurements that you make during the execution of the test scripts within each test execution cycle. Depending on the exact scope and objectives of performance testing for your project, it may not be worthwhile to make such measurements. For example, if the Performance Test Transaction Models (PT.030) are highly simplified, the test response measurements made in the test environment may not be useful as metrics for the real system. In more sophisticated and complex simulations, however, you may wish to take timings of particular transactions or system event points. You may want to sample the response times for the following types of system events:

- time to commit a transaction
- time to display a list of values

- time to open a form or a graphical window

The design and construction of a performance test can be a lengthy process and the desire to capture as many timings or measurements as possible is understandable. But beware of information overload. If you have too many timing event points in the scripted transaction flows, you will have a great deal of information to process and interpret after the testing has finished. Rather than capturing the response for every single event point in the system, you should attempt to capture sample event points in the transaction flows and have the other event points monitored visually during the test execution. If an especially slow system response becomes evident from visual inspection during the actual test execution, you may be able to add it to the test script response points already defined in the automated tool you are using, or in the list of event points that users are recording timings for.

Linkage to Other Tasks

The Performance Test Scripts are an input to the following task:

- PT.050 - Design Performance Test Transaction Programs

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	70
Business Analyst	30

Table 9-11 Role Contribution for Create Performance Test Scripts

Deliverable Guidelines

Use the Performance Test Scripts to document the workflows that each test user should follow to execute the transactions for each scenario. The Performance Test Scripts also enable you to record the detailed response measurements that are taken as a particular test user executes the script.

This deliverable should address the following:

- all required test steps to support each scenario and transaction model
- application referenced by each step
- module accessed by each step
- required measurements

Deliverable Components

The Performance Test Scripts consist of the following components:

- Introduction
- Test Scripts
- Test Script Response Measurements
- Assumptions

Introduction

This component details the purpose of the deliverable, provides background information on the project, and includes references to related documents.

Test Scripts

This component describes the scripts for the test transactions of each test scenario. For each test scenario, describe the scripts for the test transactions that are included in the transaction models for that scenario.

A useful technique is to diagram the script flow in the same way that Business Process Architecture (BP) creates process flows for the business processes. You can use the Future Process Model (BP.080) as a template.

Test Script Response Measurements

This component describes the measurements for the test transactions flows. The measurements need to be detailed for each test scenario.

Assumptions

This component documents the assumptions made in the course of identifying the scripts and test script measurements.

Audience, Distribution, and Usage

Distribute the Performance Test Scripts to the following:

- business analysts for review
- Performance Testing team members

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Are the Test Scripts fully described and decomposed into their individual business events and actions?
- Are the Test Script Response Measurements included for each scenario test script?
- Does the deliverable include the assumptions for the scenario test scripts?

Tools

Deliverable Template

Use the Performance Test Scripts template to create the deliverable for this task.

PT.050 - Design Performance Test Transaction Programs (Optional)

In this task, you design the performance test transaction programs that will execute the transactions, reports, and inquiries against the performance test environment.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Transaction Program Designs**. These designs are a set of design documents for the special test programs needed to execute transactions during the performance tests.

Prerequisites

You need the following input for this task:



Performance Test Transaction Models (PT.030)

The Performance Test Transaction Models provide information about the sets of transactions that model the system in the test. You need to script the individual transactions that are mentioned in each model. If Identify Performance Test Transaction Models was not performed, this deliverable will not exist. (See the task description for PT.030 for more information on when this task should be performed.)



Performance Test Scripts (PT.040)

The Performance Test Scripts provide information about the detailed transaction flows for the transactions in the test models as well as the test script response measurements that you need to make during the test. If Create Performance Test Scripts was not performed, this deliverable will not exist. (See the task description for PT.040 for more information on when this task should be performed.)

Optional

You may need the following input for this task:

 Design Standards (MD.030)

The Design Standards provide information about the standards for new modules and programs for the system. The standards may not be relevant for the design of automated multi-user testing tool programs. If you are designing programs to simulate user processing (such as a script that simulates online inquiries), you should try to conform to these standards. If Define Build Standards was not performed, this deliverable will not exist. (See the task description for MD.030 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review Performance Test Transaction Models (PT.030) and identify new programs that are necessary to run the performance test.	
2.	Prepare an introduction for the document giving the context for the deliverable.	Introduction
3.	Identify the high-level approach for each new program that is needed for the performance test.	Summary of Transaction Program Designs
4.	Create detailed designs for each program you have identified.	Detailed Transaction Program Designs

Table 9-12 Task Steps for Design Performance Test Transaction Programs

Approach and Techniques

You must design performance test transaction programs if you need special programs to automate the simulation of multiple users or execute transactions. You can bypass this task if you do not need to build any special programs to execute transactions within the performance test. This would be the case if you are conducting a manual system performance test that uses standard batch scheduling and manual online transaction entry, or if you are just testing a batch processing scenario.

Two Types of Transaction Programs

There are two types of transaction programs that you might need to design within this task:

- automated multi-user testing tools
- general transaction simulation programs

Automated Multi-User Testing Tools

The exact nature of this task depends on the testing tool employed. A test of a character interface system requires the generation of keystroke files that may then be embedded into other programs written in languages such as C. The testing tool, which you switch on before this exercise, captures keystrokes or events that mimic the associated back-end communication with the database server. The testing tool technical analyst then modifies the generated script or program to scale the number of system users to reflect the number of users in the transaction model.

In addition to recording the transaction flows in the tool corresponding to the test scripts, the tool technical analyst needs to code in transaction delay times, user “think” times, and so on.

General Transaction Simulation Programs

You may also need to design programs that simulate transactions in a real system that you cannot reproduce easily in the test. A few examples follow:

Example 1: Ad hoc query users run a special application to query the database but you cannot simulate the effects of the queries on the

system using an automated multi-user testing tool because of cost or resource constraints. You could design SQL query programs that simulate the effects of the real ad hoc query application and schedule them from a separate operating system program.

Example 2: You want to simulate the effects of online user transactions on a database server but do not have access to an automated multi-user testing tool because of project cost constraints. For simple transactions it may be feasible to simulate the execution of the transactions from a user machine by running the scripts from a separate server machine. You would run the scripts in separate flows to simulate online user transactions submitted from desktop machines.

Example 3: Perhaps you are not easily able to include the execution of the interfaces to legacy applications in your test. You can adopt an approach similar to *Example 1* and simulate its effect.

Spurious Program Events Avoidance

When designing the flows through the transaction programs be careful not to introduce spurious test events that would not occur for a user performing a workflow in the real system.

Example 1: Batch processes often auto reschedule. Do not start a batch process multiple times if it will reschedule itself after its initiation.

Example 2: Do not introduce spurious login/logout events during the course of a workflow that a user would not normally perform within a single system session.

Application Development Installation Access

To help with Design Performance Test Transaction Programs, arrange for access to an application installation where you can access all the application products within your performance testing scope. Access to a functioning application environment may be a necessity if you need to record the keystrokes or graphical events of scripted transaction flows.

Linkage to Other Tasks

The Performance Test Transaction Program Designs are an input to the following task:

- PT.080 - Create Performance Test Transaction Programs

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	100

Table 9-13 Role Contribution for Design Performance Test Transaction Programs

Deliverable Guidelines

Use the Performance Test Transaction Program Designs to document the design of the programs you need to build to conduct the performance test.

This deliverable should address the following:

- program requirements
- design approach
- technical requirements
- process logic

Deliverable Components

The Performance Test Transaction Program Designs consist of the following components:

- Introduction
- Summary of Transaction Program Designs
- Detailed Transaction Program Designs

Introduction

This component details the purpose of the deliverable, provides background information on the project, and includes references to related documents.

Summary of Transaction Program Designs

This component summarizes the designs for the individual test transaction programs needed for the performance test. The summary should take the form of a statement about how the individual programs are to be implemented or created. In a project where you are using an automated testing tool, this would require summarizing the structure of the testing tool programs.

Detailed Transaction Program Designs

This component describes the detailed designs for the individual test transaction programs.

Tools

Deliverable Template

Use the Performance Test Transaction Program Designs template to create the deliverable for this task.

Oracle PL/SQL

If you need to build general transaction simulation programs, you may find that Oracle PL/SQL is a powerful and portable language for creating 3GL programs that can also access Oracle Server. You can also execute PL/SQL programs directly from the Oracle Applications Concurrent Manager if, for example, you wanted to simulate interfaces to external systems that periodically load data into the Application Open Interfaces.



Reference: *Oracle Applications Object Library Reference Manual*, Vol. 2.

PT.060 - Design Performance Test Data (Optional)

In this task, you perform the detailed data design work for performance testing on the target environment. Specify the data and volumes for the performance test database, the source of the data, and the proposed mechanism for getting the data into the test database.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Data Design**. It documents the database requirements for setup data and transaction data in the performance testing environment.

Prerequisites

You need the following input for this task:



Business Volumes and Metrics (RD.040)

The Business Volumes and Metrics provides the data and transaction volumes that you use to define the volume of data you need in the test database to meet the test objectives in accordance with the test strategy.



Database Extensions Design (MD.060)

The Database Extensions Design provides information about *customizations* that require changes to the underlying application database. If the transactions that constitute the test models do not impinge on functional areas where analysts have identified extensions, you do not need this deliverable. However, if these transactions do impinge on such areas, this deliverable is a required prerequisite. If Design Database Extensions was not performed, this deliverable will not exist. (See the task description for MD.060 for more information on when this task should be performed.)

❑ Performance Test Transaction Models (PT.030)

The Performance Test Transaction Models provide information about the sets of transactions that model the system in the test. Make sure that the test database setup data can enable the error-free execution of these transactions and that the database has the full volumes of setup and transaction data dictated by the performance test objectives and strategy. If Identify Performance Test Transaction Models was not performed, this deliverable will not exist. (See the task description for PT.030 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Identify the data needed in the test database for the transaction models.	
2.	Calculate volumes of test data objects needed to satisfy the performance test scope and strategy.	
3.	Review progress in the specification of application setup data by implementation team (if applicable).	
4.	Review progress by the Data Conversion team (if applicable).	
5.	Identify existing application setup or transaction data in other databases.	

No.	Task Step	Deliverable Component
6.	Prepare an introduction for the document giving the context for the deliverable.	Introduction
7.	Identify the high-level strategy for the construction of the test database.	Test Data Strategy
8.	Identify gaps in the existing application setup data.	
9.	Establish mechanisms for filling gaps in the setup data.	
10.	Establish values of keys for missing setup data.	
11.	Document setup data analysis.	Application Setup Data Analysis
12.	Identify gaps in the existing transaction data.	
13.	Establish mechanisms for filling gaps in the transaction data.	
14.	Establish values of keys for missing transaction data.	
15.	Document transaction data analysis.	Application Transaction Data Analysis
16.	Identify risks in the test database strategy.	

Table 9-14 Task Steps for Design Performance Test Data

Approach and Techniques

General Considerations

The test database is a key part of any performance test environment and creating a well-populated database is a critical success factor for a performance test that helps uncover performance bottlenecks or system tuning problems. Much of the regular Business System Testing in an implementation project is performed against databases that are populated with relatively small and unrealistic volumes of data, with the result that the Business System Testing does not exercise the technical configuration and processing to a degree that would uncover major performance issues (nor is it intended to). Under these circumstances, it is even more important that the performance test is executed against a database that is *industrial-sized*, and has a volume of data in it that is at least equal to, or a significant fraction of, the volume of data expected in the real production database after a financial quarter of operations processing.

The reason why Business System Testing is not generally performed against volume databases is that there is rarely a volume database available in an implementation project. Data Conversion programs may not be available or fully tested early or midway in an implementation project, or the data conversion volumes may not be sufficient for the Performance Testing. The work involved in artificially populating a database to the required volumes is time consuming.



Warning: The design and construction of the test database can be one of the more time consuming and onerous tasks of a performance test project, and it is also a task that carries a high degree of risk.

Network Performance Testing

If your Performance Testing is focused more on the network components of your system, rather than the overall system performance including applications and databases, you may not need to invest as much time and effort in the design and construction of the test database. You still need to identify a test database to execute transactions and make sure that you have access to it within the Performance Test Environment. You may not need to load large volumes of data, since you are not focusing on the performance of the system processes or database.

Production Database Copy

If you are already in production with your new system and are using Performance Testing to assess the performance impact of a change or reconfiguration of the production system, the effort involved in this task diminishes dramatically. You should be able to use the real production database as a source for your test database (assuming you have sufficient disk space available).

Test Database Volumes Calculation

To calculate the data volumes you need in the database, review the scope and strategy for the performance testing, and in particular, the test transaction models that may simulate different situations after the new system has gone into live production. Use the business volumes information and the transaction models to define the type and volume of data that needs to be in the test database. If the information is available in Business Volumes and Metrics (RD.040), you can factor in decisions about the archive and purge cycles for different business objects. The volume of data in the database is a critical factor affecting the test performance results, so make sure that the database properly represents the processing environment.

Test Database Strategy

If a direct copy of the live processing database is not available for transference into the test environment, you need to construct the test database from scratch. Get the setup data loaded and then load the volume transaction data.

Reuse Implementation Team Data and Programs

If a parallel implementation project is proceeding in conjunction with the performance testing, you can reduce risk by reusing as much of the data and programs created within that project as possible. The following sources of data and programs are useful to this task:

- data in Business Requirements Mapping (BR) or Business System Testing (TE) databases
- data in a dedicated application setups database
- programs from Data Conversion (CV)

Even if the data does not yet exist or programs have not been written, you may be able to use module designs from Data Conversion (CV) for this purpose.

Remember that the data in the test database may not require the same integrity that the production system must have. If the data in a table is not accessed during a transaction and is not part of a report or inquiry data set, the data can be present as “junk” data that fills out the table volume for the presence of table scans and index searches.

When filling the gaps in the data, the database designer needs to decide whether to load data that accurately represents the data in the live processing environment or whether to manufacture artificial data to fill out the database. The latter option is easier, but care needs to be taken to verify that a realistic distribution is chosen for the data of the table keys, so that indexes perform as they would in the live environment.

Populate Only What Is Needed

It may not be necessary to populate all the tables in the test database to full production volume. If the performance test transactions, inquiries or reports do not access a particular table, it may not need to be populated for the test. However, if the user wants to use the performance tests to test systems operations as well (for example, backup and recovery) it may be necessary to populate all tables to their limiting volume. The tables that are not affected by the test transactions can be populated wholly with “junk” data, just to get the correct total database size.

Loading Data to Support a Business Process

If you plan to performance test many or all business function steps within an overall business process, you must consider loading data in tables that are accessed during all the individual business functions embedded in the process. In such a situation, you may have to run transactions through the business process to be able to populate the tables corresponding to the business objects transacted within the middle steps of the process. In the order fulfillment business process, for example, if you want to test picking and shipping of products, you will have to get sales orders to a state where they can be picked or shipped.

Executing the linked transactions of a business process to populate a database can be very time consuming, repetitive, and difficult to

automate, but it may be the only reasonable way to populate the database to support this type of complex business process. You should keep this in mind when designing the approach for the various business objects you need to populate in the test database.



Suggestion: In addition to the straightforward execution of business process flows to populate mid-process transaction tables, there may be other creative ways to use application functionality to populate tables. The tight integration between ERP/Supply Chain applications can be problematic for database population; however, you may be able to use it to your advantage. For example, if you have forecast or MPS data available, you can run MRP to create volumes of purchase orders for testing — even if you are not going to performance test MRP functionality.

Data Multiplication

A very simple approach to populating a table to full volume is to multiply the number of rows already in the table with a script. This approach does not provide for complete integrity of the data, but is a fast way to pad out a table to its full production volume. Develop a method of continuously changing the primary keys of the rows so that the primary key integrity is not violated — this is important for index searches to work correctly.

One means of changing the primary key is to add a changing prefix to a primary key field, for example:

Customer Name: Global Computers Inc.
 AAA-Global Computers Inc.
 AAB-Global Computers Inc.

The prefix is also a means of tracking where a particular record lies in the database blocks storing the table data and can be used to help make sure that you access a wide distribution of records during a particular transaction or inquiry (if the business data processing model requires it).

Application Development Installation Access

To help with the data design task, you should arrange for access to an application installation where you can access all the application products within your performance testing scope. The database should

be a development or scratch database of some type where you can enter and prototype data creation.

Linkage to Other Tasks

The Performance Test Data Design is an input to the following tasks:

- PT.070 - Design Test Database Load Programs
- PT.100 - Construct Performance Test Database

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	70
Business Analyst	30

Table 9-15 Role Contribution for Design Performance Test Data

Deliverable Guidelines

Use the Performance Test Data Design to define the data that you need in your test database and the volumes of data you require. For the data that you need to populate artificially, you should designate the approach that is to be used to populate the test database.

This deliverable should address the following:

- approach for populating the test database
- source of setup and reference data
- specific data volumes
- database padding strategy
- primary key distribution / randomization strategy

Deliverable Components

The Performance Test Data Design consists of the following components:

- Introduction
- Test Data Strategy
- Application Setup Data Analysis
- Application Transaction Data Analysis
- Risks

Introduction

This component details the purpose of the deliverable, provides background information on the project, and includes references to related documents.

Test Data Strategy

This component describes the strategy to be used to assemble a Performance Test Database from the data and programs available in the project environments, generating any missing data that you might need. The Performance Testing Strategy (PT.010) may already contain a discussion of the Test Database Strategy, in which case this component can either reference the component in the Performance Testing Strategy (PT.010) or reproduce that text here.

Application Setup Data Analysis

This component describes the setup data required for the Performance Test Database. For each business object that needs to be present in the test database, indicate the volume of data needed, the source of the data, and the specifications of any new data you need to create artificially.

Application Transaction Data Analysis

This component describes the transaction data required for the Performance Test Database. For each business object that needs to be present in the test database, indicate the volume of data needed, the source of the data, and the specifications of any new data you need to create artificially.

Risks

This component describes the risks in the strategy and analysis of the test data presented in the Performance Test Data Design.

Audience, Distribution, and Usage

Distribute the Performance Test Data Designs to the following:

- project manager
- Performance Testing team members
- technical analysts participating in Data Conversion (CV)

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Are the setup and transaction data fully specified for the test database?
- Is the new data (which may need to be created) specified?
- Are the data design risks documented?

Tools

Deliverable Template

Use the Performance Test Data Design template to create the deliverable for this task.

SQL Trace Utility

If you are not precisely sure which tables a transaction accesses, you can determine the tables that are accessed during a transaction by turning on SQL trace for the duration of the transaction. You can then infer the business objects from the tables that the transaction accesses, using the technical reference manuals.



Suggestion: Turn SQL trace on and then rerun the performance test transactions either manually, or by replaying the keystroke files against another suitable database. By reviewing the trace log file you can easily determine those tables that were accessed during a transaction, report, or inquiry, as well as the nature of the data access.

Technical Reference Manuals

Oracle Applications Technical Reference Manuals have a complete list of forms, batch processes, and tables documented for each application, together with a module-table usage section that details the data manipulation functions a particular module performs and which tables are accessed by the module. These can be useful in specifying which tables need to be populated for the transactions in the test model.

Import and Export

Oracle Server has special utilities for importing and exporting data out of tables in a database. These utilities can be useful for copying data between databases, rather than having to write PL/SQL or SQL scripts specifically for this action, or manually re-entering data. When using the import and export utilities, you must be careful to export and import all the tables that have foreign key relationships in order to preserve the integrity of the data you wish to use in the Performance Test Database.

PT.070 - Design Test Database Load Programs (Optional)

In this task, you design the programs that load test data into the Performance Test Database.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Database Load Program Designs**. These designs are a set of documents for special programs needed to load data into the performance test database.

Prerequisites

You need the following input for this task:

Performance Test Data Design (PT.060)

The Performance Test Data Design provides process information regarding population of the test database and the types and volume of data to be used. If Design Performance Test Data was not performed, this deliverable will not exist. (See the task description for PT.060 for more information on when this task should be performed.)

Optional

You may need the following input for this task:

Design Standards (MD.030)

Design Standards provide information about the standards needed for designing new modules or programs. If Define Design Standards was not performed, this deliverable will not exist. (See the task description for MD.030 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Identify the tables you need to populate for each type of business object that needs to be loaded into the performance test database.	
2.	Prepare an introduction for the document giving the context for this deliverable.	Introduction
3.	Create a detailed approach for each type of setup business object that you need to load into the test database.	Application Setup Data Load Programs
4.	Design the programs you need for each setup business object type.	Application Setup Data Load Programs
5.	Create a detailed approach for each type of transaction business object that you need to load into the test database.	Application Transaction Data Load Programs
6.	Design the programs you need for each transaction business object type.	Application Transaction Data Load Programs

Table 9-16 Task Steps for Design Performance Test Database Load Programs

Approach and Techniques

You can bypass this task if you do not need to build any extra programs specifically to load data into the performance test database. This would be the case if, for example, you are conducting a performance test against a copy of an existing production database that has the approximate volumes of data you need, or, if you can use existing Data Conversion programs to load all transaction and setup data.

Existing Programs Reuse

Try to reuse specific programs and techniques that are developed elsewhere in the project for loading data into the performance test database. For example, if programs have been designed as part of the Data Conversion process to migrate legacy data to the new system, you may be able to use these programs in the construction of the performance test database.

Data Conversion

Conversion from legacy systems is usually implemented using flat-file extracts on the source system. Load the flat-file data using SQL loader to Oracle temporary tables, where it can be manipulated before being transferred to the actual application tables. In some cases the data can be moved directly from database to database when both Oracle protocol converters and utilities are available for real-time communication between the databases.

Data Multiplication

One useful technique for manufacturing data is to multiply the data already present in tables. If you do not have any data in the tables to start with, you can perform row counts on the tables you believe will be affected, create new transactions through the regular application forms, and then perform the row counts again to find precisely which tables you have populated. You can then select and reinsert the rows that you just created using data multiplication scripts. You need to be careful with primary and foreign key relationships when multiplying the data using this technique.

Design Documentation Time Minimization

The specific programs used for populating the performance test database are often “throwaway” in the sense that they are designed specifically for loading performance test data and therefore do not need to be documented with the same degree of rigor as other custom code developed in the project. Some design documentation is necessary to record how the database was actually populated with data, and this may be mentioned in the final Performance Test Report (PT.130), but do not spend undue amounts of time documenting these designs.

Linkage to Other Tasks

The Performance Test Database Load Programs are an input to the following task:

- PT.090 - Create Test Database Load Programs

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	90
Business Analyst	10

Table 9-17 Role Contribution for Design Test Database Load Programs

Deliverable Guidelines

Use the Performance Test Database Load Program Designs to document the designs for the programs you use to populate the Performance Test Database.

This deliverable should address the following:

- specific tables to load for each business object
- foreign key relationships
- required processing logic

Deliverable Components

The Performance Test Transaction Program Designs consist of the following components:

- Introduction
- Application Setup Data Load Programs
- Application Transaction Data Load Programs

Introduction

This component details the purpose of the deliverable, provides background information on the project, and includes references to related documents.

Application Setup Data Load Programs

This component describes the designs for the individual application setup data load programs. For each business object required, the approach you will take for loading the data, the database objects you need to load, and the structure of the program you will use should be defined.

Application Transaction Data Load Programs

This component describes the designs for the individual application transaction data load programs. For each business object required, the approach you will take for loading the data, the database objects you need to load, and the structure of the program you will use should be defined.

Tools

Deliverable Template

Use the Performance Test Database Load Program Designs template to create the deliverable for this task.

Application Open Interfaces

Oracle Applications have built-in open interfaces to many of the most commonly converted application business objects — for both setup and

transaction objects. These interfaces should be used whenever possible for loading data into the test database.



Warning: Any time you insert data directly into Oracle Application tables, you run the risk of corrupting the database. Oracle strongly discourages inserting data directly into Oracle tables that are not designed as an Open Interface.



Reference: *Oracle Financials and Government Financials Open Interfaces Manual.*



Reference: *Oracle Manufacturing Open Interfaces Manual.*

PT.080 - Create Performance Test Transaction Programs (Optional)

In this task, you create the special Performance Test Transaction Programs that execute the transactions, reports, and inquiries against the Performance Test Database.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Transaction Programs**. These programs are the actual program code for the programs defined in Design Performance Test Transaction Programs (PT.050)

Prerequisites

You need the following input for this task:



Development Environment (MD.090)

You need the Development Environment where you can build the new programs. This Development Environment may be shared with the environment where the developers create new system modules or may be an environment that is solely used for the performance test development work. If Prepare Development Environment was not performed, this deliverable will not exist. (See the task description for MD.090 for more information on when this task should be performed.)



Performance Test Transaction Program Designs (PT.050)

The Performance Test Transaction Program Designs provide the design information for the new programs you build in this task. If Design Performance Test Transaction Programs was not performed, this deliverable will not exist. (See the task description for PT.050 for more information on when this task should be performed.)

Optional

You may need the following input for this task:

 Build Standards (MD.040)

The Build Standards provide information about the standards that you conform to when you build new modules or programs for the system. They may not be relevant for the build of automated multi-user testing tool programs; however, if you are building programs to simulate the effect of user processing (such as a script that simulates online inquiries) you should conform to these standards whenever possible. If Define Build Standards was not performed, this deliverable will not exist. (See the task description for MD.040 for more information on when this task should be performed.)



Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the detailed Performance Test Transaction Program Designs (PT.050).	
2.	Code programs.	
3.	Unit test programs.	Tested Source Code for Programs
4.	Update the design documents as needed.	

Table 9-18 Task Steps for Create Performance Test Transaction Programs

Approach and Techniques

Non-Technical Resources Utilization

If you are using an automated testing tool, you may not need to utilize technical resources when building the test programs. The generation of performance test character keystroke or graphical event files does not require technical resources and the entry of the data according to the documented test plans can be done by business or functional specialists, or even users. If an implementation project is proceeding in tandem with the performance test, the data entry required for the test programs can be a useful opportunity for users to gain experience on the new system.

Testing Tool Programming Expertise

Some automated testing tools require C programs to manage the test transactions. Although the tools may generate some of the programming automatically, they may still need some manual changes by skilled C developers. The types of changes that need to be made may include:

- changing record primary keys to prevent clashes
- setting transaction “think” times
- setting delay times for the commencement of different test users

You should already know how your testing tool functions and what technical skills you need to build and make changes to the programs.

Transaction Programs Testing

Test the new programs by executing them against a database. If possible, test the programs against the performance test database as it is being created for the project. If not, at least try to test against a database with similar setup data loaded in it. Errors and bugs discovered when building the programs mean less time spent coding corrections when testing the execution of the programs in the Performance Test Environment.

Linkage to Other Tasks

The Performance Test Transaction Programs are an input to the following tasks:

- PT.110 - Prepare Performance Test Environment
- PM.090 - Measure System Performance

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Developer	85
Technical Analyst	10
Business Analyst	5

Table 9-19 Role Contribution for Create Performance Test Transaction Programs

Deliverable Guidelines

The Performance Test Transaction Programs are the tested program code. If Performance Testing is part of a larger implementation project, you may share the development environment with other technical work on the project and need to define procedures to separate your code and testing from the other development work.

Tools

Deliverable Template

A deliverable template is not provided for this task.

Automated Multi-User Testing Tool

If you are using automated multi-user testing software, it may have built-in tools to assist in the generation of the test programs. Make sure you are familiar with the facilities and tools that your testing tool provides to expedite the creation and maintenance of the test programs.

PT.090 - Create Test Database Load Programs (Optional)

In this task, you develop and test the programs that you use to populate the Performance Test Database with data.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Database Load Programs**. These are the actual program code for the programs defined in Design Test Database Load Programs (PT.070).

Prerequisites

You need the following input for this task:



Development Environment (MD.090)

You need the Development Environment where you can build the new programs. The Development Environment may be shared with an environment where new system module development also takes place or may be an environment that is solely for the performance test development work. If Prepare Development Environment was not performed, this deliverable will not exist. (See the task description for MD.090 for more information on when this task should be performed.)



Performance Test Database Load Program Designs (PT.070)

The Performance Test Database Load Program Designs provide the design information for the new programs you build in this task. If Design Test Database Load Programs was not performed, this deliverable will not exist. (See the task description for PT.070 for more information on when this task should be performed.)

Optional

You may need the following input for this task:

 Build Standards (MD.040)

The Build Standards provide information about the standards you conform to when you build new modules or programs for the system. You should conform to these standards where possible and appropriate. If Define Build Standards was not performed, this deliverable will not exist. (See the task description for MD.040 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the detailed Performance Test Database Load Program Designs (PT.070).	
2.	Code programs.	
3.	Unit test programs.	Tested Source Code for Programs
4.	Update the design documents as needed.	

Table 9-20 Task Steps for Create Test Database Load Programs

Approach and Techniques

Load Programs Testing

Test the programs by executing them against a database. You can either test them directly against the database that you are constructing for the performance test, or test and debug them against a scratch database

before finally running them against the actual performance test database. The latter option is preferred, but may not be possible if disk space is limited and other development teams on the project will not allow you to create scratch data in their database. If you are testing the creation of data in what will be the Performance Test Database, be careful about creating volumes of data that cannot be transacted properly because of bugs in your data load programs. Take a pragmatic approach and only create large volumes of data during testing when you are fairly confident the bugs have been fixed.

Linkage to Other Tasks

The Performance Test Database Load Programs are an input to the following task:

- PT.100 - Construct Performance Test Database

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Developer	85
Technical Analyst	10
Business Analyst	5

Table 9-21 Role Contribution for Create Test Database Load Programs

Deliverable Guidelines

The Performance Test Database Load Programs represent the coded and tested programs in the performance test development environment. If Performance Testing is part of a larger implementation project, you may share the development environment with other technical work on the project and need to put procedures in place to separate your code and testing from the other development work.

Tools

Deliverable Template

A deliverable template is not provided for this task.

Data Manipulation Tools

Oracle supplies various standard development tools that you can use to directly manipulate data in an Oracle database:

- SQL*Loader, (especially for loading flat file data)
- SQL*Plus and PL/SQL
- Pro*C
- Developer/2000 tools — Oracle Forms, Reports, and so on
- J-Developer

Pro*C

If your conversion or data manufacture logic is complex, you can implement the programs in Pro*C. However, PL/SQL can perform many of the 3GL style programming functions that Pro*C can and there is no need to compile and link PL/SQL. If you choose to use Pro*C, keep in mind that Pro*C programs are more difficult to debug and test than SQL*Plus and PL/SQL and are not as portable.

Oracle Forms

Oracle Forms is the main interface for inserting and manipulating data in an Oracle database. You can create simple forms interfaces to manually enter data sets directly into Oracle tables.

Oracle Reports

Oracle Reports is a powerful report development tool, but is also capable of extensive data manipulation and validation. Its GUI interface makes it an attractive development tool for general purpose batch programs.

Other Development Tools

There are many other third-party vendor tools available that you can use to load and manipulate data in Oracle databases. In a Microsoft Windows client-server environment, you could, for example, use a Visual Basic client program to connect, via ODBC, to the Oracle database.

Automated Testing Capture Tool

If you intend to use an automated tool for your performance testing, the tool probably has some form of capture utility that records keystrokes or graphical interface events. By replaying these scripts repeatedly, you can load multiple transactions into the database. When loading the data this way, you benefit from the fact that you are entering transactions through the actual forms used for entering transactions in the package application product, and you do not have to create special data load programs. The forms also perform all necessary validation on the data for you. The disadvantage is that if you intend to run the captured scripts multiple times, you may get primary key violations if you attempt to replay the script without considering the effects of data duplication.

You need to be able to define the parameters for the captured script and designate a list of primary key values that you select for each iterative execution. This may require some recoding of the original captured script. Check the documentation that the vendor supplies with the tool in order to assess the programming skills needed to make these changes and whether the time and effort required is more cost effective than performing a batch data load through applications open interfaces (for example, this recoding may require C programming expertise).

PT.100 - Construct Performance Test Database (Optional)

In this task, you populate the Performance Test Database up to the full desired volume in preparation for subsequent migration of the database to the testing environment.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Database**. It is an Oracle database fully populated with data for Performance Testing purposes.

Prerequisites

You need the following deliverables for this task:



Performance Test Data Design (PT.060)

The Performance Test Data Design provides information about the strategy for constructing the database and including manually entered data, data copied or imported from other databases, or data loaded programmatically. If Design Performance Test Data was not performed, this deliverable will not exist. (See the task description for PT.060 for more information on when this task should be performed.)



Performance Test Database Load Programs (PT.090)

You need the Performance Test Database Load Programs if the performance test database strategy requires the loading or manufacture of data. If Create Test Database Load Programs was not performed, this deliverable will not exist. (See the task description for PT.090 for more information on when this task should be performed.)

Optional

You may need the following input:

☐ **Custom Database Objects (MD.100)**

You need the Custom Database Objects only if the Performance Test Transaction Models impinge on areas of the database that have custom extensions. If Create Database Extensions was not performed, this deliverable will not exist. (See the task description for MD.100 for more information on when this task should be performed.)

☐ **Conversion Programs (CV.080)**

You need the Conversion Programs only if the Test Database Strategy component of PT.060 requires the loading of conversion data into the test database using the programs developed by the Data Conversion team. If Develop Conversion Programs was not performed, this deliverable will not exist. (See the task description for CV.080 for more information on when this task should be performed.)

☐ **Conversion Database Data Files**

You need these data files only if the Test Database Strategy component of PT.060 requires the import of converted data from a separate conversion database in the project.

☐ **Application Setup Database Data Files**

You need these data files only if the Test Database Strategy component of PT.60 requires the import of application setup data from a separate application setup database in the project.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Create a new database instance.	Initial Performance Test Database Instance

No.	Task Step	Deliverable Component
2.	Copy the database files from another project database into the Performance Test Database (if applicable).	
3.	Enter or load application setup data into the Performance Test Database.	Test Database Setup Data
4.	Verify the application setup data.	Verified Setup Data
5.	Load the converted data into the test volume database.	Test Database Converted Transaction Data
6.	Verify the converted data.	Verified Converted Transaction Data
7.	Load the new generated test data into the test database.	Test Database Generated Transaction Data
8.	Verify the new data.	Verified Generated Data
9.	Verify the final test environment and application operation.	Verified Test Database Operation Against Application
10.	Backup the database for migration to the Performance Test Environment.	Backup Copy Of Database Operating System Files

Table 9-22 Task Steps for Construct Performance Test Database

Approach and Techniques

Database Creation Environment

You may have chosen to construct the test database directly in the Performance Test Environment rather than construct it fully populated in the Development Environment and then migrate it (which might be

the case if disk space is limited, for example). You do not need to migrate it after it is built, but you do need to verify that the appropriate components of the Performance Test Environment are in place prior to the construction task and that you can connect to the application and database correctly and enter data or run the data load programs.

Database Build

You either create the database from an empty database instance and enter or load all of your data directly into the empty database, or you take a copy of another database in the project that has good quality and quantity of setup data, and use that when you start up your new database instance.

Database Copy or Backup

After you finish loading all the volume data into the test database, you need to take a copy or backup of the database. If you are constructing the database in the development environment, the copy of the database is needed to migrate the database to the actual test environment. Another reason for taking a copy of the database is so that you can refresh the test database after you complete a particular performance test cycle.

The simplest means of taking a backup of a database that you can fully shut down is to copy the database files. Shut the database down in normal mode and then copy the appropriate database, log, and control files into the project source control area. You can then migrate these files to the test environment for a simple reconstruction of the database.

Linkage to Other Tasks

The Performance Test Database is an input to the following task:

- PT.110 - Prepare Performance Test Environment

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	75
Database Administrator	25

Table 9-23 Role Contribution for Construct Performance Test Database

Deliverable Guidelines

Load the Performance Test Database to the full planned volume. If you are unable to do so, update the data design to reflect the designed data volumes versus the actual data loaded. A shortfall in the amount of data you can load commonly results from technical difficulties and complexity in building the database that could not be predicted when the estimates for resources and time were produced.

Tools

Deliverable Template

A deliverable template is not provided for this task.

Data Loading Tools

The data loading tools that you can use for constructing the test database are the same tools that you have used for building the test data load programs. You may be able to create operating system jobs (for example, UNIX scripts) that can be executed in the background and do not need direct manual execution or monitoring.

PT.110 - Prepare Performance Test Environment (Optional)

In this task, you prepare and document the separate environment where performance testing will take place. Preparing the environment involves configuring the hardware and networks, installing the Performance Test Database (PT.100), verifying that the Performance Test Transaction Programs (PT.080) execute correctly against the database, initial tuning of the environment, installing any monitoring tools needed, and documenting the test execution procedures. After this task is complete, the performance test environment should be ready to support execution of performance test cycles in which measurements are made.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Environment**. It consists of the physical environment plus a document of the same name that describes the initial parameter settings for the environment and the test execution procedures.

Prerequisites

You need the following input for this task:

☐ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy includes requirements for all project environments and thus provides information necessary for preparing the Performance Test Environment. It also describes the strategy for reusing environments for different purposes throughout the project, which can impact the timing of this task.

☐ **Conceptual Architecture (TA.070)**

The Conceptual Architecture outlines the technical architecture configuration that is subject to testing. The Performance Test Environment should match the proposed architecture and use the actual

hardware and software defined therein. If Revise Conceptual Architecture was not performed, this deliverable will not exist. (See the task description for TA.070 for more information on when this task should be performed.)

☐ **Performance Testing Strategy (PT.010)**

The Performance Testing Strategy provides information about the detailed test environment, procedures, and tools to be used. If Define Performance Testing Strategy was not performed, this deliverable will not exist. (See the task description for PT.010 for more information on when this task should be performed.)

☐ **Performance Test Transaction Models (PT.030)**

The Performance Test Transaction Models provide information about the application configuration needed to allow the Performance Test Transaction Programs (PT.080) to function properly. If Identify Performance Test Transaction Models was not performed, this deliverable will not exist. (See the task description for PT.030 for more information on when this task should be performed.)

☐ **Performance Test Transaction Programs (PT.080)**

If special Performance Test Transaction Programs have been designed and built for the test, then they are a required prerequisite for a fully configured test environment. If Create Performance Test Transaction Programs was not performed, this deliverable will not exist. (See the task description for PT.080 for more information on when this task should be performed.)

☐ **Performance Test Database (PT.100)**

The fully populated Performance Test Database is needed to execute the test transactions against. If Construct Performance Test Database was not performed, this deliverable will not exist. (See the task description for PT.100 for more information on when this task should be performed.)

☐ **Hardware and Software for the Performance Test Environment**

The hardware and software are needed so that the technical infrastructure of the Performance Test Environment can be configured prior to migrating the test data and the transaction programs.

Optional

You may need the following input for this task:

Validation-Tested Conversion Programs (CV.110)

If the scope of Performance Testing on your project includes performance testing of Conversion Programs (CV.080), then the Validation-Tested Conversion Programs must be loaded into the Performance Testing Environment in preparation for testing. If Perform Conversion Validation Test was not performed, this deliverable will not exist. (See the task description for CV.110 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Prepare an introduction for the Performance Test Environment document giving the context for this deliverable.	Introduction
2.	Install the performance test servers.	Performance Test Environment Checklist
3.	Install the operating system on the servers.	Performance Test Environment Checklist
4.	Connect the servers to the network.	Performance Test Environment Checklist
5.	Create user accounts on the servers.	Performance Test Environment Checklist
6.	Configure the file system on the servers.	Performance Test Environment Checklist

No.	Task Step	Deliverable Component
7.	Install and connect to the network user machines in the Performance Test Environment (if applicable).	Performance Test Environment Checklist
8.	Install the Oracle database on the database servers.	Performance Test Environment Checklist
9.	Install the Oracle Applications on the organization's servers.	Performance Test Environment Checklist
10.	Migrate the database files and data from the performance test database development environment.	Performance Test Environment Checklist
11.	Startup the database and verify operation.	Performance Test Environment Checklist
12.	Startup Application Concurrent Managers.	Performance Test Environment Checklist
13.	Verify functioning of the Oracle Applications and database.	Performance Test Environment Checklist
14.	Install the automated test software (if applicable).	Performance Test Environment Checklist
15.	Verify functioning of the automated test software (if applicable).	Performance Test Environment Checklist
16.	Verify the execution of the transaction programs against the test database.	Performance Test Environment Checklist
17.	Develop any custom tools or scripts that are used for monitoring the tests.	Performance Test Environment Checklist

No.	Task Step	Deliverable Component
18.	Install the performance monitoring tools.	Performance Test Environment Checklist
19.	Tune the Performance Test Environment.	Performance Test Environment Checklist
20.	Document the test cycle execution steps.	Test Cycle Execution
21.	Document the baseline Performance Test Environment configuration parameters.	Baseline Test Configuration Parameters
22.	Configure the concurrent manager program execution for programs that execute test transactions.	Baseline Configuration Parameters
23.	Update the Installation Plan and Record for the new environment configuration.	Updated Installation Plan and Record (PJM.RM.050)

Table 9-24 Task Steps for Prepare Performance Test Environment

Approach and Techniques

Performance Test Database Migration

If you have constructed the fully populated test database in a separate development environment, you need to migrate the backup or copy of the database to the Performance Test Environment. If you have chosen to construct the test database directly in the Performance Test Environment, you do not need to migrate it, but instead need to prepare some components of the Performance Test Environment prior to constructing the test database itself.

Multiple Test Architectures Reconfiguration

If you are conducting comparative tests on multiple hardware and network architectures, you may need to reconfigure the Performance Test Environment between the individual architecture performance tests. If the tests involve comparison of multiple application or technical architectures, you may need to reconfigure the test database and the test transactions as well.

After receiving an affirmative decision from management to continue with the original scope of the test and test the performance of other hardware and network configurations outlined in the project scope, you need to repeat some of the tasks necessary to perform the first configuration test. The tasks you need to revisit are:

- Prepare Performance Test Environment (PT.110)
- Execute Performance Test (PT.120)

You may also need to perform Construct Performance Test Database (PT.100) if the reconfiguration requires changes to the logical or physical architecture of the test database. The physical architecture would change if, for example, you were comparing the performance of UNIX raw devices versus file systems or you were comparing a distributed data architecture for the test database in a decentralized versus a centralized database configuration. You also need to revisit Create Performance Test Transaction Programs (PT.080) if you need to make changes to the programs to reflect the reconfiguration of the test environment.

Performance Test Transaction Programs and Database Verification

Prior to executing your performance test, you need to test the execution of the test transaction programs against the test database. If you developed both the test transaction programs and test database within the same environment (either a separate project development environment or within the Performance Test Environment itself), you may already have been able to verify that the test programs and test database work correctly together. If, because of segregation, timing, or some other reason, you have not been able to perform this verification task step, you should do so before attempting to begin a results-gathering performance test cycle.

Even if you did test the operation of the transaction programs and the test database within the Development Environment, you should take the time to retest within the actual test environment. Retesting saves

the effort of getting system and people resources ready for a performance test cycle, only to find that you have to postpone the run because of errors resulting from the change of environment (such as different server or file system names). A delay in declaring the test transaction programs and database ready for a full performance test cycle execution can save frustration later.

Linkage to Other Tasks

- The Performance Test Environment is an input to the following task:
- PT.120 - Execute Performance Test

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Administrator	60
Technical Analyst	20
Database Administrator	10
Network Administrator	10

Table 9-25 Role Contribution for Prepare Performance Test Environment

Deliverable Guidelines

The Performance Test Environment consists of a physical hardware and software project testing environment. Also prepare a document with the same name that describes the initial parameter settings for the environment and test execution procedures. Add as much detail as possible so that this document acts as the source of all the information needed for the performance team to execute a performance test cycle. Include any special system parameters and logins required for starting scripts or batch programs.

If you are testing multiple hardware architectures or configurations during the test, you can either create a separate deliverable for each configuration or subdivide the deliverable into sections for the different hardware configurations. If the parameters and test cycle execution sequences are unchanged, you do not need to duplicate this level of detail for the multiple configurations.

This deliverable should address the following:

- steps required to complete the configuration of the environment
- hardware and network parameters
- database parameters
- application parameters
- instructions for executing a performance test cycle

Deliverable Components

The Performance Test Environment document consists of the following components:

- Introduction
- Performance Test Environment Checklist
- Baseline Test Configuration Parameters
- Test Cycle Execution

Introduction

This component details the purpose of the deliverable, provides background information on the project, and includes references to related documents.

Performance Test Environment Checklist

This component lists the steps required to configure the Performance Test Environment. As you complete each step, enter the current date, your initials, and any relevant comments.

Baseline Test Configuration Parameters

This component documents the initial values of the main test configuration parameters prior to performing the test. The test configuration parameters include both parameters that are associated

with the system (such as database tuning parameters, operating system parameters, and so on), and parameters set for the test itself, such as number of test users for each role or timing delays between transactions. Some of these parameters may change during the test for tuning purposes, and this is the reason for the designation of the parameters as baseline.

Test Cycle Execution

This component describes the process for executing a performance test cycle (one iteration of a performance test run). There should be a sufficient level of detail to enable this document to be used by the Performance Testing team as a checklist for starting up and managing the performance test.

Audience, Distribution, and Usage

Distribute the Performance Test Environment document to the following:

- Performance Testing team members

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Is the Performance Test Environment fully configured and able to support the commencement of performance test execution for the purpose of taking measurements?
- Are the initial (baseline) test parameters documented in detail?
- Is the test execution sequence sufficiently detailed to enable test execution from this document alone?

Tools

Deliverable Template

Use the Performance Test Environment template to create the supporting document for this task.

Performance Measurement and Monitoring Tools

You may have already documented many of the major performance test measurement tools as part of Define Performance Testing Strategy (PT.010), but there is an opportunity in this task to define and add further tools that may be helpful in gathering the required data.

System Resource Monitoring Tools

Typically, you include tools that measure the system resources (such as CPU, memory, and I/O) as a function of time and system process in the test. If possible you should use tools with a graphical capability that allows you to easily visualize system resource usage variation and bottlenecks as the tests proceed.

There are a number of system resource and monitoring tools available. The following examples apply to a UNIX environment:

- **sar—System Activity Reporter**

This is a generic UNIX utility that collects low-level performance statistics over a period of time. This information can be used to identify disk and CPU bottlenecks and balance system load. For more information, consult your UNIX documentation. Examples are:

- **sar -u 5 5** reports CPU utilization, sampling every 5 seconds
- **sar -d 5 5** reports activity for disk and tape drives, sampling every 5 seconds

- **vmstat—Virtual Memory Statistics**

This is also an operating utility supplied on UNIX platforms that you use for reporting CPU, process, virtual memory, and disk activity. An example is:

- **vmstat 5** summarizes activity over the last 5 seconds

Oracle Data Dictionary Monitoring Scripts

You can develop scripts that query key data dictionary tables for database-level statistics gathering. You can use scheduling programs built into your operating system to query the database periodically during the test and gather the information for direct analysis or graphical display.

Oracle has a number of data dictionary dynamic performance views that you can use to check the tuning of the database. Some examples of the use of these V\$ views are :

- V\$LIBRARYCACHE and V\$ROWCACHE to assess cache hits ratios
- V\$SESSTAT for memory usage
- V\$FILESTAT to assess disk access and I/O to database files
- V\$LOCK_ACTIVITY and V\$PING for assessing “pinging” in a parallel server cluster

The following tools are also available:

- BSTAT
- ESTAT
- MON

BSTAT and ESTAT are standard Oracle utilities. BSTAT begins the performance monitoring process and ESTAT stops performance monitoring and stores the results in system tables. You can then run reports against the system tables that hold the performance data. MON allows you to view ongoing use of the database.

Table Row Counting Scripts

You can use the variation of row counts in selected key tables in the database as a means of monitoring the progress of transactions in a performance test. When coupled with a means of graphically displaying the row counts as a function of time, this can be a very powerful way of following the test progress on a transaction-by-transaction basis. In Oracle Applications, you can monitor open interface tables to see whether the validation and import of interfaced data are keeping pace with the rate of loading data into the interface. This is also a valuable technique if you are simulating the business function steps that make up part or all of a business process. By graphically monitoring the intermediate stages of a business process by counting rows in tables for each step of the process, you can determine the detailed transaction flow rates for each step and establish process bottlenecks.

In order to create a tool of this type, you need a way of polling the rows in selected tables and then displaying them in a graph. You can use SQL*Plus scripts that are periodically executed by the operating system

to write the row counts to operating system files. You can use several standard UNIX programs to display the row counts in the files graphically such as plot, xplot, or gnuplot. Another way to accomplish this is to count the rows directly from the program that also displays the counts graphically (for example, you could use Oracle Graphics, Oracle Web DB, or Excel with ODBC).

Batch Performance Measurements

You can record the performance of your batch processing by monitoring and collecting the log files produced by the batch programs during the test. You can write a script that looks for specific batch log files and copies them from the test environment into the results consolidation directory. You can also query the FND_CONCURRENT_REQUESTS table periodically to get summary information about the time taken to process batch programs, but also to monitor the backlog of concurrent requests in a particular concurrent manager job queue.



Attention: If the applications concurrent manager is a major focus area for your performance testing efforts, you should monitor it carefully during the test. Performance problems often result from poorly tuned and configured concurrent manager job streams. You should monitor both the time taken to execute a particular batch job and also the backlog of jobs. If the backlog grows during your test execution, the concurrent manager is unable to cope with job submission rate and you will know that there is a bottleneck in your concurrent manager job stream.

PT.120 - Execute Performance Test (Optional)

In this task, you execute the Performance Test Transaction Programs against the test database and measure various performance metrics for individual transactions and system components. Execute the test programs for each transaction model within each test scenario and take the prescribed measurements of system performance. Repeat for each hardware configuration.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Results**. These results record the test date and time, specific test or configuration parameters, as well as the individual measurements made.

Prerequisites

You need the following input for this task:



Performance Test Environment (PT.110)

The Performance Test Environment is the only prerequisite for this task. Once the environment is fully configured, all of the hardware, software, monitoring tools, automated testing tools, transaction programs, and test execution procedures should be in place for the formal performance test execution to begin. If Prepare Performance Test Environment was not performed, this deliverable will not exist. (See the task description for PT.110 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Prepare an introduction for the document giving the context for this deliverable.	Introduction
2.	Set configuration parameters for the test cycle.	
3.	Modify the test programs for the test cycle.	
4.	Tune the system for the test cycle.	
5.	Run the test cycle and take the prescribed measurements.	Test Cycle
6.	Analyze the test measurements and compare them to agreed upon metrics.	
7.	Establish and schedule the next test cycle based on the results analysis.	
8.	Review the results for the test transaction models and scenarios against the scope and objectives of the test.	

Table 9-26 Task Steps for Execute Performance Test

Approach and Techniques

Test Cycle Iterations

There is a good deal of tuning and re-tuning necessary during the execution of a large-scale, multiple transaction performance test. Collecting system measurements for a particular set of test parameters or a test configuration may require multiple cycles, before the system and the test programs are properly tuned, to provide realistic or useful results.

Automated Load Testing Tool Program Design

It is at this point that flexibility and good design in the grouping of the test scripts into test programs pays dividends. Some automated testing tools require technical programming to develop test scripts, and if the suite of test programs has been badly designed, you may need to reprogram to tune parameters such as transaction rates, delay times, and “think” times.

Effects of Automated Test Driver Machine

If you are executing a large-scale test using an automated testing tool that simulates multiple online user sessions, you need to verify that the server that is acting as the driver for the automated tool is not a bottleneck to your test. The software that is automating the test should be running on a different server than your system under test and the server that it runs on should have sufficient processing power so it does not limit the test transaction rates. If you have planned the capacity of the test driver machine and the network that connects it carefully, you can ignore the effects of the test driver altogether. If, however, you have doubts about the effect of the test driver, you may need to monitor it during the test execution to make sure it is not affecting the results.

There are techniques you can employ to satisfy yourself that the effects of the test driver machine can be ignored. For example, if you increment the user sessions, you should see little or no degradation of transaction response time. Executing such tests on the driver machine is another reason why you may need to iterate through multiple test execution cycles.

System Stabilization

The more complex a performance test, the more time will be needed for all the transactions to begin processing and to reach a stable state where you can gather meaningful performance metrics. Depending on how you have structured your test, there may be initial spurious variations in the system and transaction measurements as particular transaction flows begin. Allow the various overall system performance and load metrics to stabilize before collecting detailed data.

Results and Review Scope Analysis

Analysis of the results is intended as a checkpoint for management to decide whether they wish to proceed with testing further configurations. The first configuration tested may be the preferred configuration and once they have reviewed the preliminary results of the test, they may choose not to proceed with the other configurations. You can present the results in the form of an Executive Summary by using a scaled down version of the full Performance Test Report (PT.130).

Test Execution Team

You need to assemble a test execution team. The precise makeup of the team depends on how the test is to be conducted; however, you need technical resources that can address issues in any aspect of the infrastructure and technical analysts that understand how the transaction model is being implemented in the test environment. In a testing project that is using an automated testing tool, the need for technical analysts that understand the tool is even more important. During the testing, the technical analysts who created the test transaction may be required to tune the testing scripts and make slight program modifications. Database, network, and system administrators need to be available to monitor the technical infrastructure and make recommendations. This is the case whether the test scenarios and transactions models are batch only, or contain significant online processing.



Attention: The most successful performance tests are the ones in which detailed hands-on monitoring of the test environment is done while the test execution is in process. The reason is that while it is possible to automate the collection of almost all of the measurements in the test, you inevitably miss dynamic effects or subtle problems if you attempt to completely automate the testing. It is also easier to analyze performance problems as the test proceeds, rather than trying to recreate the test conditions subsequently. Unfortunately this also means that resources need to be freed up to participate in the tests, either directly or by being *on call* to the test execution team.

Performance Test Log

Although not a formal deliverable for the process, establishing a performance test log is a good practice to provide structure around the test results gathering. Record the details of each test cycle, the measurements taken, and any other unusual events or occurrences during the test.

Linkage to Other Tasks

The Performance Test Results are an input to the following tasks:

- PT.130 - Create Performance Test Report
- PM.040 - Prepare Production Environment
- PM.130 - Propose Future Business Direction

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	40
System Administrator	30
Database Administrator	20

Role	%
Network Administrator	10
User	*

Table 9-27 Role Contribution for Execute Performance Test

* Participation level not estimated.



Attention: Users have a significant role in a manual test that relies on them to enter the test transactions during the test execution task. At the other extreme, they may not have a role in a sophisticated performance test that uses automated multi-user load testing tools. Remember this dichotomy when scheduling and planning resources for this task.

Deliverable Guidelines

Use the Performance Test Results to record and catalog the results you obtain during your test execution.

This deliverable should address the following:

- test cycle start and end times
- primary objective of each test cycle
- changes to the baseline parameters for each cycle
- general observations
- specific measurements

Deliverable Components

The Performance Test Results consist of the following components:

- Introduction
- Test Cycle

Introduction

This component details the purpose of the deliverable, provides background information on the project, and includes references to related documents.

Test Cycle (repeating)

This component Includes the following key items:

- **Test Cycle Objective** — describes the objective of the test cycle
- **Test Cycle Parameters** — describes the changes in parameter settings for the test cycle
- **General Observations** — describes the general observations made during the test cycle execution
- **Specific Results and Measurements** — describes the results and measurements during the performance test execution of this cycle

Tools

Deliverable Template

Use the Performance Test Report template to create the deliverable for this task.

PT.130 - Create Performance Test Report (Optional)

In this task, you document the results and the description of Performance Testing in a format suitable for presentation to project managers, executive sponsors, and the project team.



If your project includes *Performance Testing*, you should perform this task.

Deliverable

The deliverable for this task is the **Performance Test Report**. It details the approach used in the project, tests performed, results, conclusions, and recommendations.

Prerequisites

You need the following deliverable for this task:



Performance Test Results (PT.120)

Analyze the results of the performance test and document your findings. If Execute Performance Test was not performed, this deliverable will not exist. (See the task description for PT.120 for more information on when this task should be performed.)

You may also need to use deliverables from other Performance Testing tasks to create the Performance Test Report.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Gather and review deliverables from the Performance Test.	
2.	Summarize the original scope, objectives, and approach of the performance testing subproject.	Project Scope, Objectives, and Approach
3.	Summarize the strategy to Performance Testing.	Performance Testing Strategy
4.	Summarize the test scenarios, transaction models, and test scripts.	Test Models
5.	Describe how the Performance Test Database was constructed and populated.	Test Database
6.	Summarize the test system configuration including hardware, network, database, and applications.	System Architecture and Configuration
7.	Document the test assumptions in detail.	Assumptions
8.	Document the test risks in detail.	Risks
9.	Analyze the performance test measurements and results.	

No.	Task Step	Deliverable Component
10.	Document the performance test measurements and results in detail.	Test Results
11.	Formulate conclusions and recommendations from the test results.	Conclusions and Recommendations
12.	Create the report executive summary.	Executive Summary
13.	Review the Performance Test Report with management, business analysts, and technical analysts.	
14.	Secure acceptance for the Performance Test Report and the test project.	

Table 9-28 Task Steps for Create Performance Test Report

Approach and Techniques

Task Effort Depends on the Test Scope and Objectives

The effort that you need to put into this task depends on the nature of the performance test you have conducted and whether it is of strategic importance to the overall implementation project and the business.

If Performance Testing has been primarily focused on final validation and tuning of the business system, then the project manager may not require you to create a separate and formal Performance Test Report. The results may be documented within other project Business System Testing (TE) documents that summarize and discuss all of the testing efforts on the project. At the other extreme, Performance Testing may have been organized as a separate subproject providing critical capacity planning or architecture information that senior management will use to make business or project strategic decisions. In this situation, you may need to create a formal report that is presentable to senior management.

Presentation of Results to Project Management

Distribute the final test report to the project sponsor and relevant senior project managers. Organize a meeting to present the results, conclusions, and recommendations arising from the results of the test. Present answers to the specific performance concerns posed by the users and identify any project shortcomings. The presentation of the results is an important element of the meeting, but also try to indicate how the results are of direct use to the business. Present suggestions for incorporating the test methodology and reusable test programs into future work that the information systems administrators may perform. Encourage the future use of the results for comparison against the production measured performance.

Linkage to Other Tasks

The Performance Test Report is not a direct input to other tasks; however, if you have identified serious performance issues, you need to revisit some Application and Technical Architecture (TA) tasks and modify the design of your architecture.

If you have serious performance problems, repeat the following task:

- Revise Conceptual Architecture (TA.070)

If you have minor performance issues, you can address them when you execute the following task:

- Refine Production System (PM.110)

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	65
Database Administrator	10
Network Administrator	10
System Administrator	10

Role	%
Business Analyst	5
IS Manager	*
Project Sponsor	*

Table 9-29 Role Contribution for Create Performance Test Report

* Participation level not estimated.



Attention: The roles IS manager and project sponsor are needed to review the Performance Test Report; however, they do not contribute to the creation of the report.

Deliverable Guidelines

Use the Performance Test Report to document the work you have performed in Performance Testing and, if appropriate, to enable you to communicate the results and conclusions to project managers and sponsors. Tailor the report to the intended audience and reflect the scope of the Performance Testing for your project in the style of deliverable you create.

This deliverable should address the following:

- a summary of previous deliverable produced for Performance Testing
- a summary of the analysis and design deliverables
- detailed test results
- how the test results answers posed in the original Performance Testing Strategy (PT.010)
- specific recommendations

Deliverable Components

The Performance Test Report consists of the following components:

- Executive Summary
- Project Scope, Objectives, and Approach
- Performance Testing Strategy
- Test Models
- Test Database
- System Architecture and Configuration
- Assumptions
- Risks
- Test Results
- Conclusions and Recommendations

Executive Summary

This component summarizes Performance Testing for executive consumption. It should introduce the performance testing effort and then describe its scope, objectives, results, and conclusions.

Project Scope, Objectives, and Approach

This component describes how the scope of Performance Testing was defined. Scope statements can be made in terms of whether solution components are in or out of scope for the project. Examples of elements that define the scope include:

- business processes or functions within scope
- applications to be tested
- technical architecture configurations
- interfaces and databases
- automated or manual performance testing
- capacity planning

The objectives should list the high-level objectives for the project. The extent to which the report is able to fulfill the original objectives for the project determines the relative success of the project.

The approach should describe the methods and approaches used to accomplish the project. Included in this should be a description of the dependencies between Performance Testing and other processes or subprojects taking place within the overall implementation project.

Performance Testing Strategy

This component describes the detailed approach used for Performance Testing and documents its benefits. The description of the approach should include a high-level discussion of the work breakdown structure, the tasks and deliverables, reasons for selection, and the benefits of the particular testing method adopted. It should describe the unique concepts used in Performance Testing (in particular, Performance Test Scenarios - PT.020 and Performance Test Transaction Models - PT.030).

The Performance Testing Strategy component should also describe the strategy used for executing the tests. Specific areas that you should highlight include how transactions were executed in the tests (including whether an automated load testing tool was used), and specific techniques for execution of the tests and refresh of the test environment after completion of a test cycle.

Test Models

This component describes the models of the system constructed for the performance tests. Generally these models are approximations of the complex transaction environment of a real production system, so it is important to document how the real production system was represented in the tests. It should include:

- test scenarios
- transaction models
- test scripts

Test Database

This component describes the method for creating the performance test database. It should describe the strategy used for the construction of the database with reference to reuse of existing project data and data load programs, and the volumes of data loaded for the key business objects accessed by the performance test transactions.

System Architecture and Configuration

This component describes the system architecture and configurations implemented for the performance test. It includes detailed specifications for the technical infrastructure of the test such as hardware, operating system and network components, as well as the application configuration.

You should include a section on the technical architecture of the test environment. It should include both the technical architecture of the system under test, as well as additional automated test systems. For example, it may need to include any special servers that act as test driver machines processing the automated test tool software. In addition, it should include a diagram of the performance test environment technical architecture.

Assumptions

This component lists all of the key assumptions made during the project into a single integrated list. The assumptions will pertain to different aspects of the test — business, functional, and technical — so you may wish to create a separate section for each category of assumptions. An example of the categories you could create is:

- business definition
- application functionality and configuration
- technical architecture
- test execution

Risks

This component summarizes the major risks of the project arising from the assumptions made in the project, or from some inherent source of risk in the implementation.

Test Results

This component documents the results obtained from the performance test execution. If you have collected a great volume of detailed results from numerous performance test cycles, you need to find the best method of forcefully presenting the results to the project team and project and business managers, without losing the impact of your results in volumes of complex detailed data. If you wish to include the detailed data, you could include it in an appendix to the report. One

way to structure the component is to create separate subsections for the following categories:

- summary
- performance measurements
- issues identified and resolution
- patches applied

Conclusions and Recommendations

This component describes the conclusions you have drawn from the results of the tests and the recommendations based on the results and the conclusions. It is convenient to present these discussions in separate sections:

- conclusions
- recommendations

This is the most important component of the deliverable and should address the original performance issues or concerns that the performance test was intended to answer. The recommendations you document may include the following:

- specific recommendations on technical infrastructure and application configuration
- identification of areas that may constitute future risk and techniques that could be used to mitigate the risk
- follow up on testing or performance monitoring projects or activities
- escalation of internal project issues
- escalation of issues that are relevant for vendor support, such as performance bugs

Audience, Distribution, and Usage

Distribute the Performance Test Report to the following:

- project manager of the main project
- IS manager and other senior business managers
- Performance Testing team members

- IS operations staff and administrators
- other process or subproject leaders who have an interest in the results of the performance tests

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Is the Performance Test Report comprehensive?
- Does the report discuss how the performance test was designed?
- Does the Performance Test Report describe *and* interpret the test results?
- Are the assumptions and risks clearly documented to qualify the results and conclusions?
- Are the original performance concerns and project objectives addressed in the Performance Test Report?

Tools

Deliverable Template

Use the Performance Test Report template to create the deliverable for this task.

If you are writing a formal report for consumption by senior management and project sponsors, you may want to include all the template component sections. If however, you have performed a limited scale project, you can omit some of the components (such as the Executive Summary).

Adoption and Learning (AP)

This chapter describes the Adoption and Learning process.

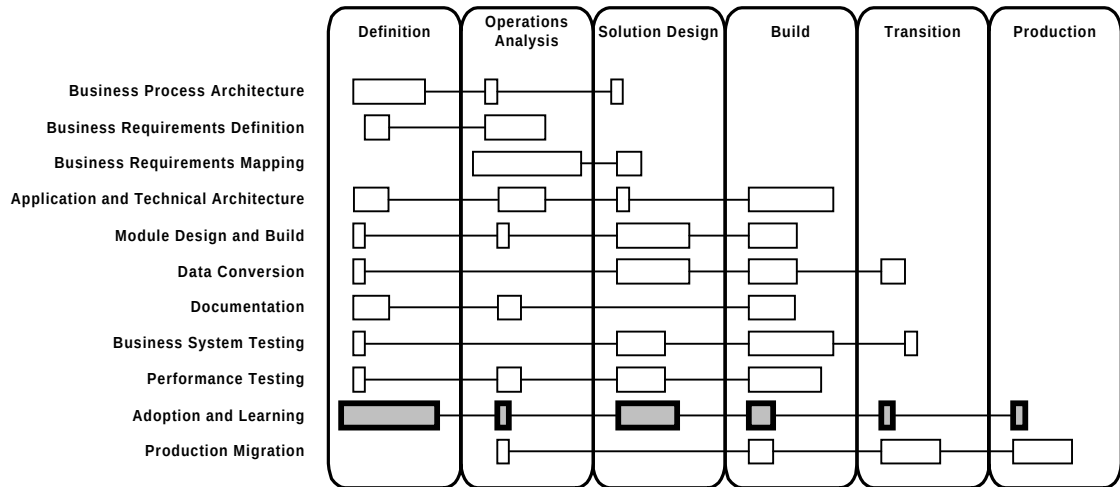


Figure 10-1 Adoption and Learning Context

Process Flow

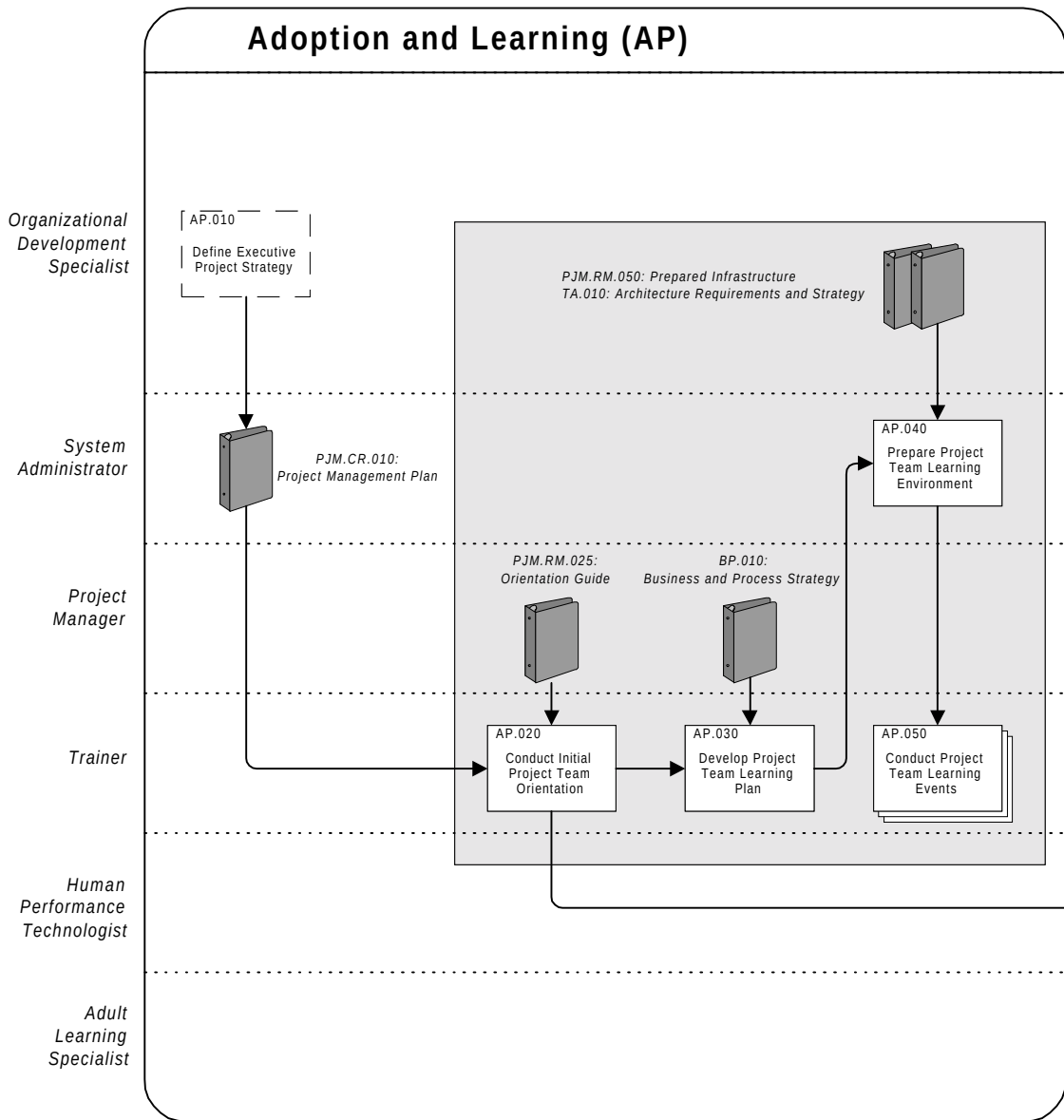


Figure 10-2 Adoption and Learning Process Flow Diagram

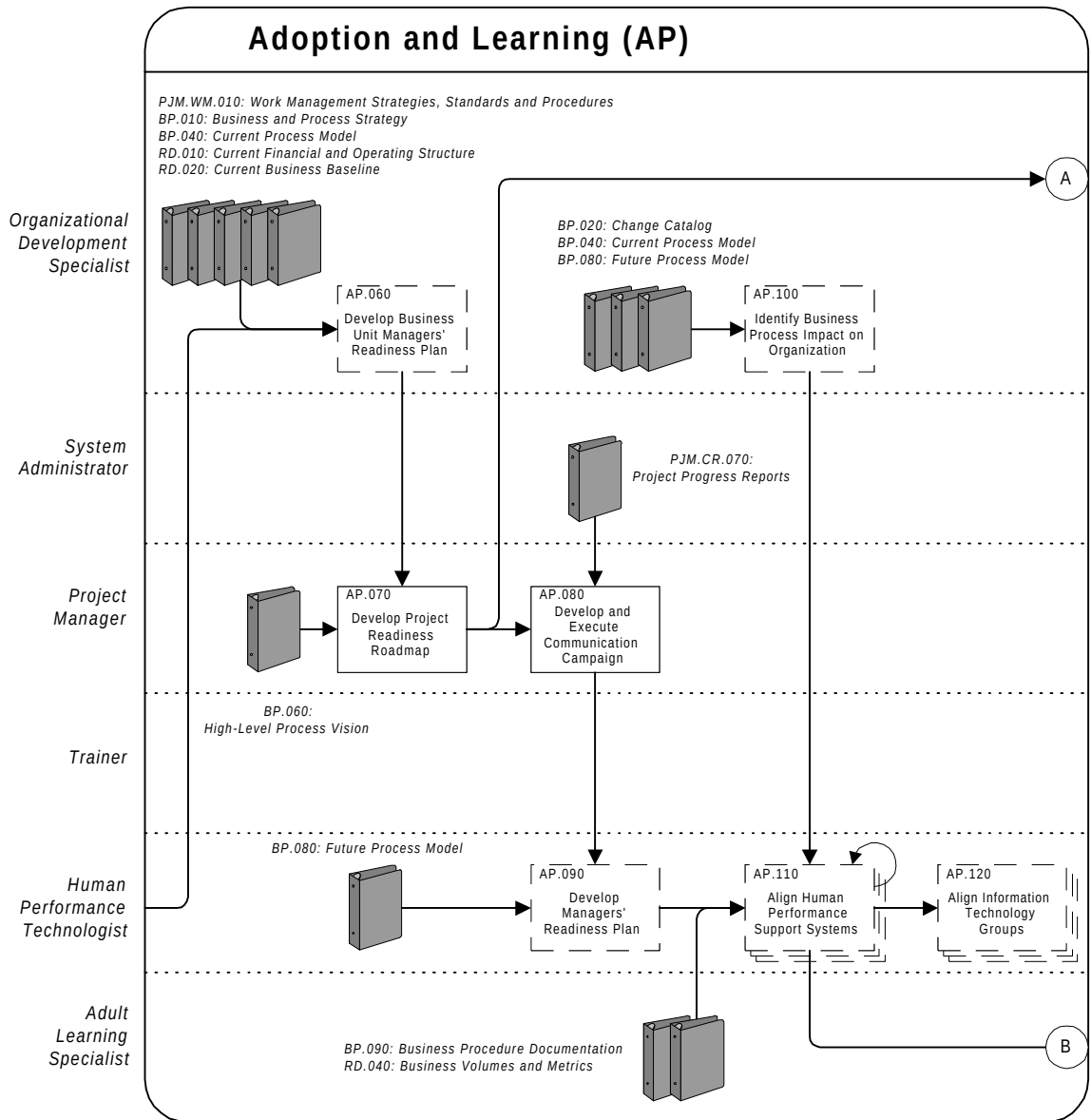


Figure 10-2 Adoption and Learning Process Flow Diagram (cont.)

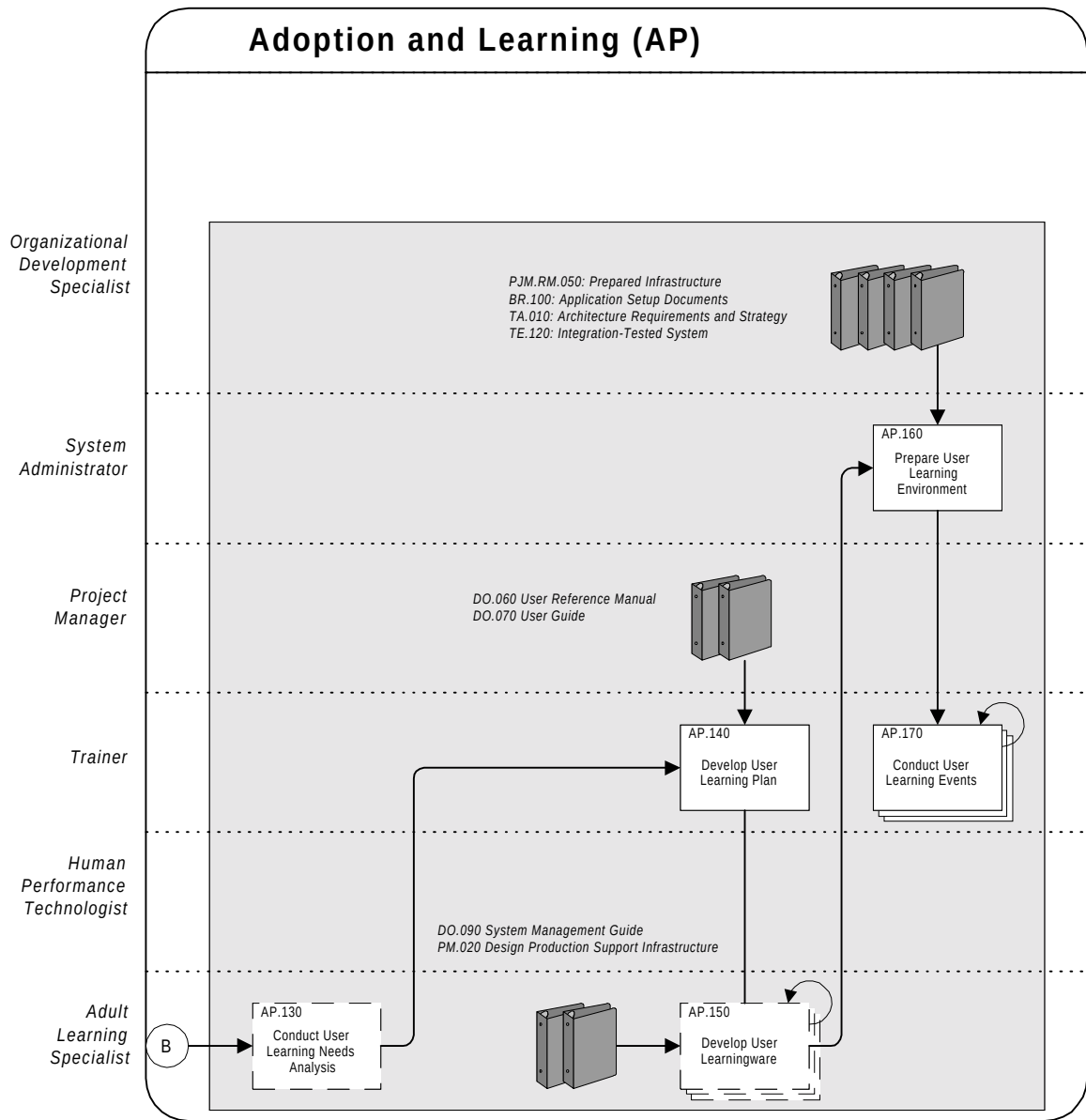


Figure 10-2 Adoption and Learning Process Flow Diagram (cont.)

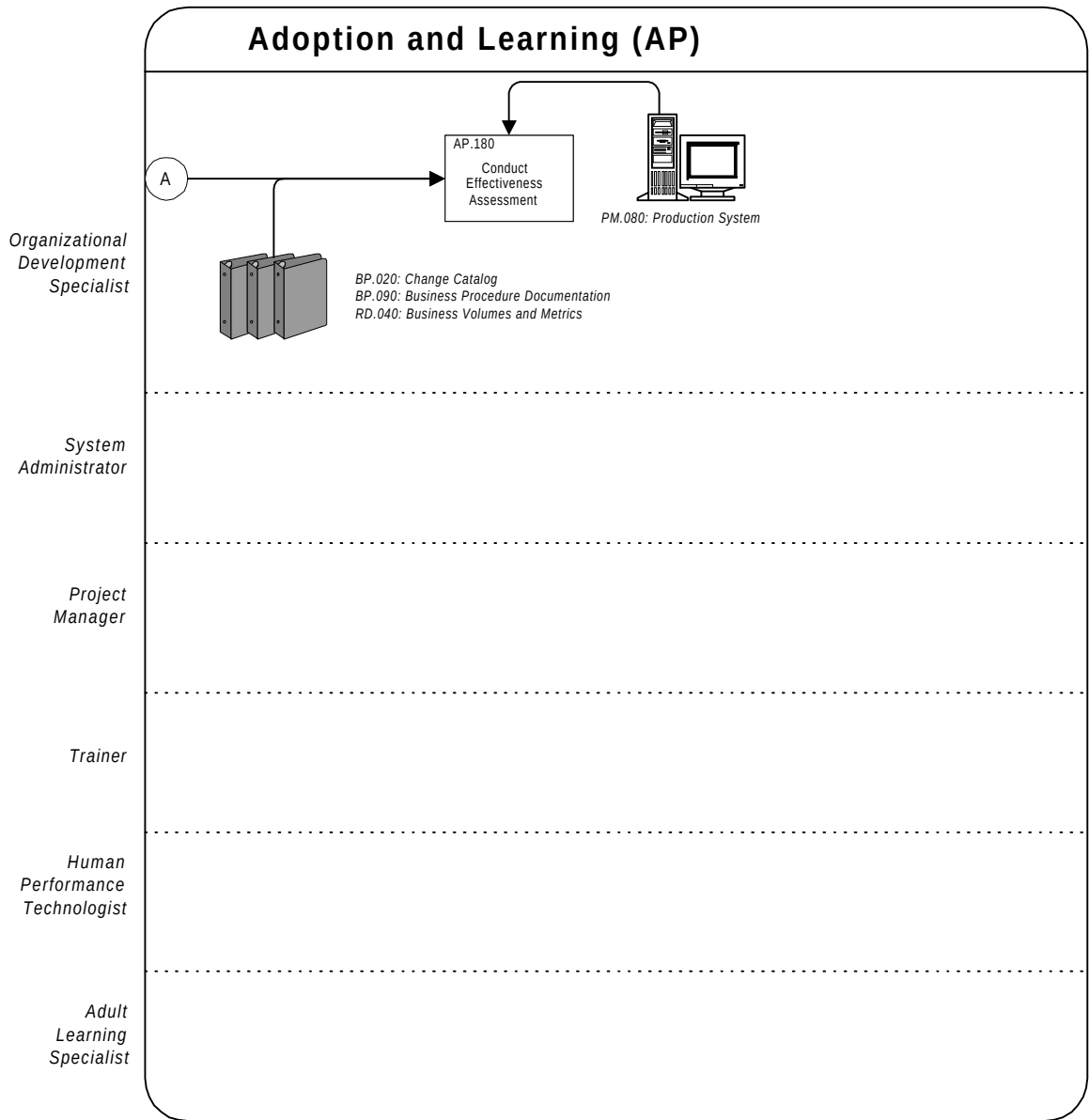


Figure 10-2 Adoption and Learning Process Flow Diagram (cont.)

Approach

Adoption and Learning focuses on the use and acceptance of new applications by all users and the optimization of organizational performance.

Inherent in every technology change is the opportunity to become a stronger, more cohesive organization; a more efficient performer; a more agile competitor. The benefits are great, but the challenges are many. Organizations who implement new technology cannot afford to neglect key areas that might ultimately compromise the value of the technology investment. The goal is for full adoption of the new business system.

The human and organizational aspects of an implementation are often the decisive factors in its success. Adoption and Learning provides a structured approach that addresses the human and organizational acceptance and use of new applications. Adoption and Learning increases the organization's return on investment by reducing the risks of implementing applications, increasing the productivity of all user groups, and assisting the organization to reach peak performance with the new business system.

The more quickly and effectively the business can adopt new technology, the more productive and competitive is the organization.

Adoption and Learning impacts the following five major audiences:

- executives
- implementation project teams
- functional managers
- users
- information technology groups

Each of these audiences has a stake in the expected performance of the new system and can impact the success of the implementation. However, the audiences are not mutually exclusive. For example, functional managers or members of the information technology groups may also be users or on the implementation project team.

Benefits for the Organization

Adoption and Learning focuses on addressing the human and organizational factors and aligning them to minimize the implementation risk and optimize system performance from a human perspective. Adoption and Learning increases stakeholder commitment to the new technology and resulting changes and builds support for the implementation by informing, involving, and including stakeholders throughout the process.

Adoption and Learning includes the following benefits for the organization:

- identifies organizational issues that will either lengthen the implementation or harm it and overcomes those barriers
- improves organizational knowledge and skills for the new environment, as well as organizational effectiveness during implementation
- realizes projected benefits by focusing on post-implementation issues like user acceptance, productivity, and human performance support

Impacts on Various Audiences

Executives — Advocate Successful Change

The tasks targeted for executives consist of facilitated work sessions. These assist executives to effectively develop strategies for the successful execution of the organization-wide implementation and the management of implementation risks. The tasks include a sponsorship program and the formulation of realistic performance expectations for the implementation.

By performing the tasks recommended in Adoption and Learning, executives:

- understand the role they are to play in the implementation, as well as the time commitment and energy required for that role
- understand the benefits the implementation provides to their business, as well as the costs and time frames
- comprehend the barriers to success, based on experience with a number of similar implementations

- make timely and informed project startup decisions
- initiate a sponsorship network to sustain the momentum of the project

Implementation Project Team — Give the Team a Jump Start

The tasks targeted for the project team include a series of activities, processes, tools, and learning events to quickly orient the implementation team to the project. Project team members acquire the skills they need to implement the applications, to function as a team, and to represent the project to the rest of the organization.

By performing the tasks recommended in Adoption and Learning, the implementation team:

- becomes a high-performance team from the onset, reducing startup time and costs
- establishes appropriate rules of engagement
- provides just-in-time skills so members fulfill their role on the team
- specifies clear performance criteria against which the project team members are measured
- establishes project team orientation, knowledge transfer requirements, and transition / roll-off guidelines
- functions as a more cohesive working group throughout the life of the implementation

Functional Managers — Create a Stepladder for Performance

Functional managers learn how the technology impacts their work groups' processes and procedures. They are provided with tools to assume their role as change leaders, and with learning events to acquire necessary application skills. They define newly required roles, competencies, and performance support and contribute to the improvement of the workflows to take advantage of the new technology and to manage to new performance expectations.

By performing the tasks recommended in Adoption and Learning, the functional managers:

- develop a reasonable expectation of the benefits the implementation provides to their business units
- increase performance through more effective change leadership
- design optimal job roles to meet business objectives
- develop new organizational structures to help maximize productivity
- define recognition and rewards to encourage user performance in their new roles and responsibilities

Users — Achieve Business Performance

Through a series of learning and communication events, users acquire the skill and the motivation to perform their new role using the full potential of the technology. This includes the procedural, functional, and technical proficiencies users need to achieve their new performance expectations.

By performing the tasks recommended in Adoption and Learning, users:

- get involved in the definition of application and business requirements
- participate in just-in-time learning and receive performance reinforcement
- increase their acceptance and effective use of the system
- create a post-implementation environment to facilitate productivity

Information Technology Groups — Sustain Technical Gains

The tasks targeted to the information technology community provide the right skills base to enable ongoing technical support to users after the implementation and leverage business and technical system capabilities. It includes certification paths and retention strategies for the information technology population.

By performing the tasks recommended in Adoption and Learning, the information technology groups:

- smoothly transitions from implementation to on-going technical performance
- develops proficiency for ongoing system improvement, furthering technical performance gains

Training versus Learning

The Adoption and Learning process includes a number of tasks that were identified in AIM 2.0 as Training tasks and are now labeled “learning.” The field of andragogy (adult learning) makes a clear distinction between training and learning. Whereas training is often a one time occurrence, driven by the trainer rather than the learner, it stops short of providing the actual performance results expected from a skilled performer. Learning expands the concept of training so it applies to an entire process where a learner participates in learning events organized in a structured path that favors gradual learning from simple to more complex. Learning here is better mapped to the ways adults learn and is generally self-driven by the learner. It includes repetition activities, reinforcement and measurement to ensure that the learning has actually occurred and is now translated into the expected behavioral performance. It is also delivered in a variety of formats to cater to the variety of adult learning styles.

Tasks and Deliverables

The tasks and deliverables for this process are as follows:

ID	Task Name	Deliverable Name	Required When	Type*
AP.010	Define Executive Project Strategy	Executive Project Strategy	Project includes an <i>organizational change management focus</i> or <i>impacts a large user population</i>	SI
AP.020	Conduct Initial Project Team Orientation	Oriented Project Team	Always	SI
AP.030	Develop Project Team Learning Plan	Project Team Learning Plan	Always	SI

ID	Task Name	Deliverable Name	Required When	Type*
AP.040	Prepare Project Team Learning Environment	Project Team Learning Environment	Always	SI
AP.050	Conduct Project Team Learning Events	Skilled Project Team	Always	MI
AP.060	Develop Business Unit Managers' Readiness Plan	Business Unit Managers' Readiness Plan	Project includes an <i>organizational change management focus</i> <u>and</u> impacts a large user population	SI
AP.070	Develop Project Readiness Roadmap	Project Readiness Roadmap	Always	SI
AP.080	Develop and Execute Communication Campaign	Communication Campaign	Always	SI
AP.090	Develop Managers' Readiness Plan	Managers' Readiness Plan	Project includes an <i>organizational change management focus</i> or impacts a large user population	SI
AP.100	Identify Business Process Impact on Organization	Business Process Organizational Impact	Project includes <i>process change</i>	SI
AP.110	Align Human Performance Support Systems	Human Performance Support Systems	Project includes an <i>organizational change management focus</i> or <i>process change</i>	MI, IT
AP.120	Align Information Technology Groups	Aligned Information Technology Groups	Project impacts a large user population <u>and</u> includes an <i>organizational change management focus</i>	MI
AP.130	Conduct User Learning Needs Analysis	User Learning Needs Analysis	Project includes an <i>organizational change management focus</i> <u>and</u> impacts a large user population	SI
AP.140	Develop User Learning Plan	User Learning Plan	Always	SI

ID	Task Name	Deliverable Name	Required When	Type*
AP.150	Develop User Learningware	User Learningware	Project includes an <i>organizational change management focus, customizations, interfaces or process change</i>	MI, IT
AP.160	Prepare User Learning Environment	User Learning Environment	Always	SI
AP.170	Conduct User Learning Events	Skilled Users	Always	MI, IT
AP.180	Conduct Effectiveness Assessment	Effectiveness Assessment	Always	SI

*Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Table 10-1 Adoption and Learning Tasks and Deliverables

Objectives

The objectives of Adoption and Learning are:

- Align business objectives and technology capabilities throughout the organization.
- Establish a measurement system that provides an evaluation of organizational performance to determine whether expectations were met during implementation and after production cutover.
- Establish executive sponsorship and management advocacy.
- Promote open communication among levels of the organization for involvement and buy in, and minimize resistance to technological change.
- Increase stakeholder commitment to the new technology and resulting changes; build support for the implementation by informing, involving, and including stakeholders throughout the process.
- Accelerate the implementation project team's ability to work together through team building and organization-specific application learning.

- Determine human performance support implications so that the organizational structures and job roles align to meet new performance expectations resulting from the technology change.
- Create a user-friendly environment for learning about the new technology by developing learning and performance strategies and plans that promote the optimum performance of users on the new system.
- Optimize the information technology groups’ infrastructure to help enable the ongoing support of the applications (including ongoing learning and certification plans so that information technology employees can continuously optimize system functionality to meet business needs).

Deliverables

The deliverables of this process are as follows:

Deliverable	Description
Executive Project Strategy	Captures the executives’ strategic vision, project direction, approach for mitigating implementation risk and how to manage the impact of the project on their business operations.
Oriented Project Team	The current team members who have a thorough knowledge of the project vision, direction, and strategies, and their individual mandates.
Project Team Learning Plan	The learning paths for all project team members to guide the acquisition of skills required to fulfill the project roles, such as technical, tool, product, functional, and project management.
Project Team Learning Environment	Application servers, configuration, and setups containing a controlled set of data used for the project team’s learning events.

Deliverable	Description
Skilled Project Team	All members of the team who have participated in the learning events intended to give them the knowledge and skills they need to perform their role(s) on the team.
Business Unit Managers' Readiness Plan	A series of work sessions that prepare leaders at this level to do their part to make the implementation successful.
Project Readiness Roadmap	A plan for addressing the human and organizational factors that impact the success of the implementation. It includes a readiness strategy, implementation decisions, a communication strategy, and a learning strategy for users.
Communication Campaign	A description and planning of two-way communication events, organized by audiences, to promote shared understanding and feedback for acceptance and support from targeted stakeholders.
Managers' Readiness Plan	A description of a series of events which prepare the middle and first-line managers, who are not on the project team, to understand the goals for the project, the project strategy, and their roles in making the project successful.
Business Process Organizational Impact	A description of the impact to the work, people, organizational structure, organizational culture, and physical layout caused by business process change.

Deliverable	Description
Human Performance Support Systems	Approach to facilitate the changes on human resource and work management tools and processes. Definition of new roles, performance metrics, competencies, human performance support systems processes, and protocols. Includes learning materials for managers to learn how to implement the new human performance support systems.
Aligned Information Technology Groups	Members of the Information Technology Groups with their updated mission and skills to perform their new roles.
User Learning Needs Analysis	Insights and knowledge, skills, and aptitudes (SKA) gap analysis findings for all audiences of users, including overall readiness state for the project.
User Learning Plan	Learning paths to provide procedural, system literacy, business knowledge, and skills to users. Includes plans for learning administration.
User Learningware	Learning materials for users covering procedural, system literacy, business knowledge, and skills. Includes materials for preparing learning agents; for assisting user learning, reinforcement, measurement; and for administering learning events.
User Learning Environment	Physical and technical setup, including application servers, configuration, and containing a controlled set of data that is synchronized with the learningware.

Deliverable	Description
Skilled Users	Users from all audiences who have acquired system literacy, procedural, and business skills as per their learning paths.
Effectiveness Assessment	Recommendations for continuous improvement in all organizational, business and technical systems (for example, system, business and organizational performance, and so on).

Table 10-2 Adoption and Learning Deliverables

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Adult Learning Specialist	Assist in developing learner group-specific curriculum plans and instructional materials for the project team and users.
Application Specialist	Provide knowledge and guidance regarding application implementation strategies and change impact. Also support and provide interpretation for application functionality, tools, templates, and method.
Assessment Specialist	Assist in conducting the assessment and generating the Project Readiness Roadmap (AP.070).

Role	Responsibility
Business Analyst	Conduct interviews and facilitate working sessions with executives and project team to define organizational strategies and change impact.
Business Line Manager	Participate in interviews and working sessions. Provide information about organizational performance measures, and ways in which the units operate and respond to events in order to achieve those objectives. Describe requirements for organizational acceptance. Participate in readiness events and development of new human performance support systems.
Client Executive	Define executive project strategy and prepare project delegation to the steering committee and project team. Participate in communication and measurement events.
Client Project Manager	Participate in the determination of how well the production system and the business and organizational performance meet the business objectives identified at the beginning of the project.
Client Staff Member	Support the organization's change process. Provide information about the organizational culture, values and practices. May participate in readiness and communication events.

Role	Responsibility
Communication Specialist	Assist in developing the communication model and standards for the project team, and the communication strategy to involve, inform and generate acceptance and support from the stakeholders throughout the organization.
Database Administrator	Assist in installing and configuring the Project Team and User Learning Environments and maintaining database access controls.
Human Performance Technologist	Assist in defining the new human performance requirements, to include: roles that are derived from the new business processes and procedures, and new competency profiles. Assist in developing the performance support systems and human resource management protocol to meet the expected organizational performance.
IS Manager	Participate in defining support requirements for users.
Key User	Represent constituencies. Participate in readiness, communication, and acceptance events.
Organizational Development Specialist	Provide expertise in the human and organizational facets of change. Develop the readiness strategies and facilitate an efficient organizational transition to the technology-induced change.

Role	Responsibility
Project Manager	Participate in definition and maintenance of an optimal project environment. Support events to help the project team function as a high-performance team.
Project Sponsor	Lead and facilitate alignment behind project vision and momentum.
Steering Committee Member	Guide overall project strategy and review, adopt recommendations, and provide sponsorship for implementing changes. Responsible for project approach acceptance and support, funding, issue resolution, and signoff.
Subject Matter Specialist	Help define organizational requirements (for example, performance) and provide valuable input for readiness events. Provide user input and support learning and user acceptance activities.
System Administrator	Assist in administering the project team and user learning environments, including ensuring hardware is correctly configured; installing, configuring, maintaining operating and development software, and controlling access to the applications.
Tool Specialist	Provide knowledge and guidance regarding specific tool functionality, for example, facilitation software, such as <i>The Memory Jogger Plus+</i> or procedural documentation software, such as Oracle Tutor.

Role	Responsibility
Trainer	Define learning requirements, prepare learning plan, produce learning material, and deliver courses.
User	Participate in readiness, communication, and acceptance events.

Table 10-3 Adoption and Learning Key Responsibilities

Critical Success Factors

The critical success factors of Adoption and Learning are as follows:

- definition of clear and realistic business expectations and organizational performance measures for the project
- availability of the organization's stakeholders to participate fully in readiness tasks
- provisions made for workload while project team members participate in readiness tasks
- inclusion of key stakeholder constituencies in the change process and clear description of project impact and benefits
- visible support and participation of key leaders and sponsors throughout the impacted business units
- mechanism to listen and respond to top concerns about the new systems
- early establishment of ongoing communication and feedback/evaluation mechanisms that fit the organizational environment
- management activities that help employees understand their new performance objectives and expectations, as well as the importance of their job to the change effort
- clear description of new job evaluation and grades

- clear explanation of the mechanisms of the rewards and recognition program around the new roles
- just-in-time learning events that are tailored to the learning and performance needs of all impacted roles
- commitment of management to post-performance support deployment measurement

References and Publications

Oracle Tutor

Oracle Tutor provides users with relevant process documentation and learning materials. These materials are process- or role-based, easy to update, cost-effective and developed in conjunction with the implementation. Oracle Tutor helps apply functionality specific to Oracle Applications within the context of the organization's unique set of policies and procedures. The product is comprised of three components: an authoring tool, a publishing tool, and a repository of Oracle Application courseware and procedure document objects.

Memory Jogger Plus+

The Memory Jogger Plus+ describes facilitation techniques are to be used in work sessions for this process.

ERP Pioneers

The cover story in *Computerworld*, January 1999, titled "ERP Pioneers" describes the major challenges related to implementing Enterprise Resource Planning applications.

AP.010 - Define Executive Project Strategy (Optional)

In this task, you develop the organizational strategy for the project.



If your project includes an *organizational change management focus* or impacts a *large user population*, you should perform this task.

Deliverable

The deliverable for this task is the **Executive Project Strategy**. It identifies and captures the executives' strategic vision, project direction, approach for mitigating implementation risk and how to manage the impact of the project on their business operations.

Prerequisites

You need the following input for this task:

☐ **Organizational Chart**

The Organizational Chart assists in the stakeholder analysis.

☐ **Contractual and Business Agreement Documentation**

This documentation may include collective and ancillary documents such as vendor agreements, contract / labor agreements, system requirements documentation, board level directives and communications, and requests for proposal. These documents assist in the identification of the business drivers for the project and provides details on the scope of the implementation, the modules purchased, the agreements, and so on.

☐ **Return on Investment (ROI) Analysis**

A Return on Investment (ROI) analysis, such as Oracle Solution Value Assessment (SVA), can document the business case and the return that the executive management team can realistically expect from its investment in technology. If an ROI, such as Oracle Solution Value Assessment (SVA) has been performed, you need it as an input.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Gather executive insights.	Appendix - Interview Guide
2.	Prepare for work sessions.	Introduction
3.	Review implementation guidelines.	Implementation Model
4.	Facilitate work session on project vision and business objectives.	Project Vision and Business Objectives
5.	Facilitate work session on project direction.	Project Direction
6.	Facilitate work session on roles.	Roles
7.	Define steering committee group operating norms.	Roles
8.	Develop executive action plan.	Roles
9.	Secure signoff of participants.	Acceptance Certificate (PJM.CR.080)

Table 10-4 Task Steps for Define Executive Project Strategy

Approach and Techniques

Interactive Work Sessions

Key executives are assembled to identify the scope and organizational impact of the project on the business, and to plan for a successful delegation to the project team. The interactive work sessions look at key considerations when introducing new enabling technology, in order to maximize the business benefit to the organization and reduce the implementation risks. Consider using tools that help in developing, exchanging, and capturing ideas to assist you with this facilitation task, such as The Memory Jogger Plus+ software. In order to support the discussion on the business objectives for the project, you may also consider using a balanced scorecard approach.



Reference: Brassard, Michael, P., *The Memory Jogger Plus+, Featuring the Seven Management and Planning Tools*, GOAL/QPC 1996, MA ISBN: 1-879364-83-2.

If the steering committee members are not on the board of directors or are not available at this time, consider conducting separate sessions with the executives, followed by work sessions with the members of the steering committee when available. If the steering committee members have been selected and represent the key members of the board of directors, consider merging the work sessions.

Sessions with Executives

First, schedule the sessions with executives. In these sessions, executives explore the impact of the new applications on their organization, its process, its technology, and its people. The discussions with the executives in this task are high-level and are detailed in further deliverables, such as the Define Business and Process Strategy (BP.010), Define Architecture Requirements and Strategy (TA.010), and Develop Project Readiness Roadmap (AP.070). From these initial discussions, executives identify barriers to a successful implementation early in the project and create acceptance from the executive stakeholders in the organization. Executives prepare for the delegation to the steering committee and the project team.

Sessions with Steering Committee

Assemble the members of the steering committee to help them further define the Executive Project Strategy and to prepare to launch the project with the project team. These sessions look at project issues relevant to the launch of the project team for a successful delegation. As a result of the session, the steering committee builds on the initial project decisions and is ready to assemble and launch the core project team.



Attention: If the organization does not have a clear vision for the project, expressed in terms of business benefits, consider holding a work session to define the vision for the project before conducting the sessions.

When the executive team is not aligned behind the project vision, consider holding a vision alignment work session before conducting the sessions.

Also make sure the authority and empowerment levels of the steering committee and project team members are clearly defined and aligned with the expected accountabilities



Suggestion: The establishment of a robust and capable internal and vendor external support infrastructure is critical to the success of your project. Oracle Support Services provides a unique service to assess your current support capabilities and assist with planning, updating and aligning these capabilities to Oracle's support infrastructure. Contact your local Oracle Support Services representative to schedule an Oracle Support Assessment (OSA). This should coincide with the activities planned for this AIM task.

Linkage to Other Tasks

The Executive Project Strategy is an input to the following tasks:

- BP.010 - Define Business and Process Strategy
- BP.070 - Develop High-Level Process Designs
- RD.010 - Identify Current Financial and Operating Structure
- AP.020 - Conduct Initial Project Team Orientation
- AP.030 - Develop Project Team Learning Plan

- AP.060 - Develop Business Unit Managers’ Readiness Plan
- AP.070 - Develop Project Readiness Roadmap
- AP.080 - Develop and Execute Communication Campaign
- AP.090 - Develop Managers’ Readiness Plan
- AP.110 - Align Human Performance Support Systems
- AP.120 - Align Information Technology Groups

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Organizational Development Specialist	50
Business Analyst	25
Project Manager	20
Application Specialist	5
Client Executive	*
Project Sponsor	*
Steering Committee Member	*

Table 10-5 Role Contribution for Define Executive Project Strategy

* Participation level not estimated.

Deliverable Guidelines

Use the Executive Project Strategy to assist the executives and steering committee to create a successful launch and charter for the project.

This deliverable should address the following:

- insights from the executives
- implementation guidelines
- business objectives and project vision confirmation
- risk and impact assessments inherent to the implementation
- direction for the project and guidelines, such as budget management
- links between the project organization and the business
- project performance expectations
- steering committee's group norms and project environment
- project team charter clarification
- action plans for moving forward

Deliverable Components

The Executive Project Strategy consists of the following components:

- Introduction
- Implementation Model
- Project Vision and Business Objectives
- Project Direction
- Roles
- Appendix - Interview Guide

Introduction

This component documents the outline of the activities for the sessions.

Implementation Model

This component is used to position a model and the guidelines for a successful technology implementation, to minimize risks. The model serves as a proactive framework to position the criteria for success by addressing the technical, business, and people requirements.

Project Vision and Business Objectives

This component captures the project vision and goals, and how they relate to the corporate strategy. It helps to confirm shared understanding of how the goals for the project create business benefits and how success is measured. It also helps to generate executive acceptance and alignment behind the project vision.

Project Direction

This component asks the executives to visualize exactly what they hope to achieve and to determine how their technology decision impacts the people, processes and technology, including their technical support infrastructure. It facilitates a discussion with executives to identify the likely risks / barriers to overcome. It also captures strategies to manage impact and mitigate risks and reflect them in the project direction.

It assists the steering committee members in defining requirements for key elements of the project environment, such as guidelines for budget management within their forum and for the project team. It uses these requirements to complete the delegation grid and clarify how the roles of the steering committee and the project team complement each other.

Roles

This component helps executives define their role in the project (for example, sponsorship, communication, and so on) and their expectations for the steering committee's and project team's involvement in the project. It assists the steering committee members to help them define the norms that govern the way they interact with each other and operate as a group for the duration of the project. It also helps to refine the mandate of the project team, including the performance expectations and the relationship of the project team to the steering committee and the rest of the organization.

It captures the initial action items to be carried out by the executive and steering committee members, including sponsorship and communication tasks.

Appendix - Interview Guide

This component provides an interview guide to gather insights from the executives.

Tools

Deliverable Template

Use the Executive Project Strategy template to create the deliverable for this task.

Use the Session Planning Checklist template to prepare for the Executive Project Strategy Sessions.

Use the Invitation Memorandum template to invite the participants to the sessions.

Use the Acceptance Certificate (PJM.CR.080) template when securing signoff from executives.

AP.020 - Conduct Initial Project Team Orientation (Core)

In this task, you orient the current project team members by communicating the project vision and charter, the rationale for key decisions, and presenting a high-level application overview.



If your project includes an *organizational change management focus*, you should perform the additional task steps indicated below.

Deliverable

The deliverable for this task is the **Oriented Project Team**. This team represents the current team members who have a thorough knowledge of the project vision, direction, and strategies, and their individual mandates.

Prerequisites

You need the following input for this task:

☐ **Project Management Plan [initial complete] (PJM.CR.010)**

The Project Management Plan defines the scope, objectives, and approach for the project and refers to the detailed standards and procedures employed during the execution of the project.

☐ **Project Orientation Guide (PJM.RM.025)**

The Project Orientation Guide contains a collection of key project information for team members. This information includes, but is not limited to, standards, policies, and procedures.

☐ **Executive Project Strategy (AP.010)**

The Executive Project Strategy guides the delegation of the project mandate to the project team and details the delegation of the project mandate and responsibilities to the project team. If Define Executive Project Strategy was not performed, this deliverable will not exist. (See

the task description for AP.010 for more information on when this task should be performed.)



Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Gather project team insights.	Appendix A - Implementation Team Interview Guide
2.	Prepare for work sessions.	Introduction
3.	Conduct Project Team Induction.	Project Team Induction
4.	Conduct a high-level overview of the application functions and features for the project team.	Application Overview
Δ 5.	Facilitate cross-functional decision making and rules of engagement discussions.	Cross-Functional Decision Making Setup; Rules of Engagement
Δ 6.	Facilitate change leadership work session.	Change Leadership Guiding Principles and Roles
7.	Identify the steps required to induct a new team member.	Induction Checklist
8.	Update Project Orientation Guide (PJM.RM.025).	

Table 10-6 Task Steps for Conduct Initial Project Team Orientation

Approach and Techniques

During this task, the project team learns about the vision and rationale for the project to which they are assigned, their role as sponsors and change leaders. They start identifying some of their team practices, such as working in a cross-functional environment and determining their decision making approaches.

The work sessions orient the project team members to their mandate and give them an overview of the applications. The sessions are highly interactive and focus on decision making around key questions to help make sure the team engages in a successful charter and addresses the following:

- How can they best fulfill their role in the project?
- What practices will the team adopt for their rules of engagement?
- How do the project team members fulfill their role as change leaders for the project?

The work sessions look at ways of answering these questions and examine the issues relevant to the team members' understanding and acceptance of their mandate. Consider using available software packages to assist with the facilitation.



Reference: Brassard, Michael, P., *The Memory Jogger Plus+, Featuring the Seven Management and Planning Tools*, GOAL/QPC 1996, MA ISBN: 1-879364-83-2.

Linkage to Other Tasks

The Oriented Project Team is an input to the following tasks:

- BP.010 - Define Business and Process Strategy
- RD.010 - Identify Current Financial and Operating Structure
- TA.010 - Define Architecture Requirements and Strategy
- MD.010 - Define Application Extension Strategy
- CV.010 - Define Data Conversion Requirements and Strategy
- DO.010 - Define Documentation Requirements and Strategy

- TE.010 - Define Testing Requirements and Strategy
- AP.030 - Develop Project Team Learning Plan
- AP.060 - Develop Business Unit Managers’ Readiness Plan
- AP.070 - Develop Project Readiness Roadmap
- PM.010 - Define Transition Strategy

Role Contribution

The percentage of total task time required for each role follows:

Core Task Steps

Role	%
Trainer	35
Project Manager	30
Application Specialist	20
Business Analyst	15
Client Staff Member	*
Project Sponsor	*

Table 10-7 Role Contribution for Core Task Steps of Conduct Initial Project Team Orientation

* Participation level not estimated.

Optional Task Steps

If your project is addressing the optional task steps for this task, then you need to allocate additional time and the following resources:

Role	%
Organizational Development Specialist	100
Client Staff Member	*

Table 10-8 Role Contribution for Optional Task Steps of Conduct Initial Project Team Orientation

* Participation level not estimated.

Deliverable Guidelines

The Oriented Project Team represents the project team members after they participate in the induction activities, application overview, and other work sessions. During these sessions the participants define the decision making approach to be used and refine the practices that will be put in place for successful management of their charter as a team and their role as change leaders.

To facilitate the orientation of the project team, you can augment the Project Orientation Guide (RM.025) by capturing the information relayed in these sessions.

The Oriented Project Team should understand the following:

- project direction, strategies, and charter
- basic functionality of the applications being implemented
- cross-functional team dynamics
- project team rules of engagement
- change leadership guiding principles and roles

Deliverable Components

To support the orientation of the project team, create a document with the following components:

- Introduction
- Project Team Induction
- Application Overview
- Cross-Functional Decision Making Setup
- Rules of Engagement
- Change Leadership Guiding Principles and Roles
- Induction Checklist
- Appendices

Introduction

This component outlines the activities for the sessions.

Project Team Induction

This component introduces the current project team members to the project and assists them in understanding, and if appropriate, refining and furthering the project direction. The deliverable also helps them clarify their respective roles and performance measures. It uses the Executive Project Strategy (AP.010) and /or the Project Management Plan (PJM.CR.010) as the source of key messages.

Application Overview

This component positions the Application Overview within the context of the key project decisions around implementation. Tailor the overview to the project specifications.

Cross-Functional Decision Making Set Up

This component helps the project team to define decision-making approaches to be used throughout the project.

Rules of Engagement

This component helps the project team to define their group operating norms that will guide their rules of engagement throughout the project.

Change Leadership Guiding Principles and Roles

This component facilitates a consensus among the project team on how to best lead change in the organization and the project team members' actual role as change agents for the project. It starts from the guiding principles for effective change leadership and the typical reactions to change.

Induction Checklist

This component contains a checklist that can be tailored to the needs of the new team members, one-on-one and in groups. It leverages the opportunity to quickly orient the new member to the project, his/her role, the expectations, and practices for the project.

Appendices

This component contains Appendix A and B. Appendix A, Implementation Team Interview Guide, provides an interview guide to gather insights from project team members. Appendix B, Project Team Performance measures, contains project team performance measures.

Tools

Deliverable Template

Use the Project Team Orientation template to create the supporting document for this task.

Use the Session Planning Checklist template to prepare for the project team orientation.

Use the Invitation Memorandum template to invite current project team members to the orientation session.

Use the Application Overview Customization template to guide the preparation of the on-site application overview.

AP.030 - Develop Project Team Learning Plan (Core)

In this task, you assess the learning requirements for the entire project team based on the project vision, charter, and scope in order to develop the learning paths aligned to project roles.



If your project includes an *organizational change management focus* or *process change*, you should perform the additional task steps indicated below.

Deliverable

The deliverable for this task is the **Project Team Learning Plan**. It contains the schedule of learning events for all the project team members to prepare them to fulfill their project roles. The learning focuses on technical, product, functional, project management, team management, and any other skills required of the project team members.

Prerequisites

You need the following input for this task:

☐ **Project Management Plan [initial complete] (PJM.CR.010)**

The Project Management Plan defines the scope, objectives, and approach for the project and refers to the detailed standards and procedures employed during the execution of the project.

☐ **Business and Process Strategy (BP.010)**

The Business and Process Strategy gives a clear idea of the scope of the required changes and organization priorities. If Define Business and Process Strategy was not performed, this deliverable will not exist. (See the task description for BP.010 for more information on when this task should be performed.)

❑ Executive Project Strategy (AP.010)

The Executive Project Strategy guides the delegation of the project mandate to the project team and details the delegation of the project mandate and responsibilities to the project team. If Define Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for more information on when this task should be performed.)

❑ Oriented Project Team (AP.020)

The Oriented Project Team provides the right framework to understand and identify the skills required of each project team member.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
Δ 1.	Review and assess the Business and Process Strategy and/or the Executive Project Strategy deliverables for the impacts associated with business processes or organizational changes	
2.	Determine the learning scope by identifying the required skills per role. Include technical, functional, project management, tools, product content as required.	Project Team Learning Scope
3.	Review learning options and determine learning paths per individual/role.	Project Team Learning Scope

No.	Task Step	Deliverable Component
Δ 4.	Select high-performance team development modules and add to content lists.	Project Team Learning Scope
5.	Put in place the measurement mechanisms to verify the effectiveness of the learning events.	Project Team Learning Scope
6.	Create inventory of learning materials/events and need for customization.	Administration of Project Team Learning Materials/Events
7.	Identify the logistics for delivery of the learning events.	Administration of Project Team Learning Materials/Events
8.	Secure approval for plans and obtain confirmation from Oracle Education and others (contract resources, organization's training group).	Administration of Project Team Learning Materials/Events
9.	Schedule learning events, instructors and attendees.	Administration of Project Team Learning Materials/Events
10.	Publish the learning plan to the entire team.	
11.	Coordinate technical site visit by technical operations.	

Table 10-9 Task Steps for Develop Project Team Learning Plan

Approach and Techniques

During this task, the project team identifies the learning requirements of the team members and puts in place learning paths by project team role. They determine who needs what learning, when, in what form, what sequence, and deployed in what ways.

The focus of the learning plan is multifold and can apply to only a subset of roles, or to all roles. The learning plan focuses on product and tool knowledge as applicable; on technical, functional, and project management skills; and on the people skills required by the project team to function smoothly as a cohesive work group, throughout the phases of the project.

The audience for the learning plan includes all members currently identified, as well as all members who may join the team in the future (who may not be identified yet by name but are identified by planned role).

Project Team Learning Scope

Based on the skills required from the project team, the attendees determine the skills requirements by comparing required and existing skills for all project team members, by role. They then develop learning paths per individual or role, taking into consideration all learning options, and the need to develop or customize learning materials. Finally, they identify the requirements to administer the learning events for all team members.



Web Site: For a list and description of the latest Oracle Education offerings, such as the curriculum paths, course details (Reusable Content Object Matrix), and interactive seminars, refer to Oracle Education's web site (<http://education.oracle.com/>).



Attention: Each team member is responsible for reading the strategy documents generated from other processes. By doing this they are ready to provide input to the Project Team Learning Plan and make sure it is exhaustive in that it reflects all the requirements derived from the various process strategies.

Linkage to Other Tasks

The Project Team Learning Plan is an input to the following tasks:

- AP.040 - Prepare Project Team Learning Environment
- AP.050 - Conduct Project Team Learning Events

Role Contribution

The percentage of total task time required for each role follows:

Core Task Steps

Role	%
Trainer	70
Project Manager	30
Client Staff Member	*
Project Sponsor	*

Table 10-10 Role Contribution for Core Task Steps of Develop Project Team Learning Plan

* Participation level not estimated.

Optional Task Steps

If your project is addressing the optional task steps for this task, then you need to allocate additional time and the following resources:

Role	%
Organizational Development Specialist	100
Client Staff Member	*

Table 10-11 Role Contribution for Optional Task Steps of Develop Project Team Learning Plan

* Participation level not estimated.

Deliverable Guidelines

Use the Project Team Learning Plan to guide the project team members on the development of the learning plan that allows all members of the team to acquire the skills and knowledge (for example, applications, tools, and project management) they require to fulfill their respective roles on the project and to work efficiently and perform as a team. The deliverable also helps the project team set up the best administrative infrastructure for the learning; for example, preparing participation in onsite and outside learning events.

This deliverable should address the following:

- gaps between current and required skills per role (technical, functional, project management, tools, products, people skills)
- learning option selections
- learning paths per role
- high-performance team development module selections
- measurement mechanisms
- inventory of learning materials / events and need for customization
- delivery of learning event logistics
- confirmation of education providers (Oracle Education, contract resources, organization's training group)
- schedule of learning events

Deliverable Components

The Project Team Learning Plan consists of the following components:

- Introduction
- Project Team Learning Scope
- Administration of Project Team Learning Materials / Events
- Appendix - Data Gathering Tool for Project Team Skills Inventory

Introduction

This component gives an overview of the Project Team Learning Plan.

Project Team Learning Scope

This component gives an overview of the process and input required from each participant. Using the list of all known project team members and their role on the project, compare their existing skills to the required skills and determine the gap. Include technical skills, functional skills, project management skills, use of tools, and team building.



Web Site: For a list and description of the latest Oracle Education offerings, such as the curriculum paths, course details, and interactive seminars refer to Oracle Education's web site (<http://education.oracle.com/>).

Learning paths per individual/role include a relevant variety of learning options, such as instructor-led onsite training learning, computer-based learning, coaching, tutorial, and video. Also include measurement activities to determine the effectiveness of the learning.

Administration of Project Team Learning Materials/Events

This component inventories the existing learning materials and events by content. The inventory includes materials/events from Oracle Education and other parties; for example, partners, third-parties (education providers), and the organization's own training group. Based on this inventory, we can derive the need for customization and development of project specific materials. This component also includes the schedule of learning events and administration of logistics.

Appendix - Data Gathering Tool for Project Team Skills Inventory

This component contains a data gathering tool for project team skill inventory.

Tools

Deliverable Template

Use the Project Team Learning Plan template to create the deliverable for this task.

AP.040 - Prepare Project Team Learning Environment (Core)

In this task, you establish the technical and physical infrastructure required for the actual project team learning, including either installing a new application environment or preparing an existing application environment.

Deliverable

The deliverable for this task is the **Project Team Learning Environment**. It includes the setup of learning centers and production requirements for learning materials, and specifies the application servers, configuration, and set ups containing a controlled set of data for the project team's learning events.

Prerequisites

You need the following input for this task:

☐ **Prepared Infrastructure (PJM.RM.050)**

The Prepared Infrastructure includes hardware systems that are the foundation for all project support environments including the learning environment.

☐ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy helps define the requirements for project team learning environment.

☐ **Project Team Learning Plan (AP.030)**

The Project Team Learning Plan defines the learning requirements to deliver knowledge to the project team.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review and update checklists.	Preparation Checklists
2.	Install Project Team Learning Environment (if necessary).	Systems Environment
3.	Set up applications.	
4.	Review baseline (current system) user procedures.	
5.	Obtain actual baseline business scenarios (if necessary).	
6.	Gather necessary sample data.	

Table 10-12 Task Steps for Prepare Project Team Learning Environment

Approach and Techniques

This task provides a system for the project team to practice what they have learned. This includes hardware, software, sample data, and documentation. This environment is separate from the testing environments in order to preserve the integrity of learning administration, such as starting points. In addition, the learning facilities have protection from interruptions and the distractions of ongoing business. You can install a new environment or modify the existing environment for the learning activities. It is useful to construct learning environments that reflect some real production data and system configuration.

Preparation of Learning Environment

Project team learning is critical to the success of the project. Without proper understanding of supported application functions, business requirements may be incomplete, resulting in reduced traceability from business solutions to requirements.

Engage key project team members who have subject matter expertise in gathering relevant information to drive the learning events; they have credibility with users who possess such information, and by asking them, you involve them in the learning effort.

Modification of the Demonstration Environment for Learning

Use the standard applications demonstration database for learning. As part of learning preparation, gather information from analysis sessions or briefly interview key users to collect data such as commonly used part numbers, terms and conditions, and priority codes. Use organization-specific data as examples during learning. Make sure to adapt the discussion to the organization's particular business needs.



Attention: Examples in the demonstration database may conflict with the new data that is set up for the organization. Practice demonstrations and labs that use new data before the learning events. Make sure that you allow some time for making modifications to the demonstration environment as required.

Linkage to Other Tasks

The Project Team Learning Environment is an input to the following task:

- TA.110 - Define System Capacity Plan
- AP.050 - Conduct Project Team Learning Events

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Administrator	35
Database Administrator	25
Application Specialist	20
Trainer	20
Client Staff Member	*
Key User	*

Table 10-13 Role Contribution for Prepare Project Team Learning Environment

* Participation level not estimated.

Deliverable Guidelines

The Project Team Learning Environment includes newly installed standard application modules, plus all setups and data required to support project team learning events.

This deliverable should address the following:

- separate or existing environment selection
- application setups
- demonstration data
- learning facilities reservation and preparation
- backup and contingency plans

Deliverable Components

The Project Team Learning Environment consists of the following components:

- Preparation Checklists
- Systems Environment

Preparation Checklists

This component provides checklists to guide the preparation of the physical and technical learning environments, such as setting up the desktop clients, printers, and other facilities.

Systems Environment

This component describes the application installations, application server and database server instances needed to support the learning environment.

Tools

Deliverable Template

Use the Project Team Learning Environment template to create the deliverable for this task.

AP.050 - Conduct Project Team Learning Events (Core)

In this task, you facilitate the learning events for the project team. These learning events cover tools, applications, and all knowledge and skills areas deemed necessary for the effective functioning of the team as described in the Project Team Learning Plan (AP.030).



If your project includes an *organizational change management focus*, you should perform the additional task step indicated below.

Deliverable

The deliverable for this task is the **Skilled Project Team**. This team represents all members of the team who have participated in the learning events intended to give them the knowledge and skills they need to perform their roles on the team.

Prerequisites

You need the following input for this task:

☐ **Project Orientation Guide (PJM.RM.025)**

The Project Orientation Guide contains a collection of key project information for team members. This information includes, but is not limited to, standards, policies, and procedures.

☐ **Project Team Learning Plan (AP.030)**

The Project Team Learning Plan includes the plan which is executed in this task for all project team members.

☐ **Project Team Learning Environment (AP.040)**

The Project Team Learning Environment provides a system on which the project team can learn and practice. This includes hardware, software, sample data, and documentation.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Prepare for on-site learning.	Administration of Learning Events for Project Team
2.	Provide functional and /or technical learning to current project members.	Administration of Learning Events for Project Team
Δ 3.	Facilitate high-performance team development sessions	Appendices
4.	Conduct orientation sessions using the Project Orientation Guide (PJM.RM.025) for new project team members as they join.	
5.	Guide new team members through their learning path per role as described in the Project Team Learning Plan (AP.030).	

Table 10-14 Task Steps for Conduct Project Team Learning Events

Approach and Techniques

Functional, Technical, and General Learning

Align the project team’s functional, technical, and general learning content with the project objectives by covering only the standard applications functionality. In addition, include content to help the project team members to complete the required tasks, such as process mapping and modeling. If you are using business process models, such as Oracle Business Models, include them in the learning content. If you are using a procedural documentation software tool, such as Oracle

Tutor, include the software workshops for the process owners to learn the tools. These skills are leveraged during Develop Future Process Model (BP.080) and Document Business Procedures (BP.090). Throughout the life of the project, you may also need to consider providing addition learning as applicable.

High-Performance Team Learning

Each project team has its own collective personality, its own areas of strength, and its own areas needing improvement. As the need arises, select the team development activity best suited to help the team accelerate its growth toward becoming a high-performing team. These modules are intended to be facilitated as “work sessions,” for example, using actual issues experienced in the project team. Some of the topics covered in the modules include: communication, meeting management, leadership, supportive team climates, goal alignment, and managing conflict.



Reference: Scholtes, Joiner and Streibel, *The Team Handbook*, Second Edition, Oriel. ISBN: 1-884-73111-2.

New Team Members

As new members are added to the project team, it is important to orient them to the project. The project manager or project lead (if applicable) should personally invite each new member to join with a one-on-one interview. If the new team members are added one at a time, the orientation can simply be an extension of the one-on-one interview which covers an induction checklist. If several members join at once, you might consider conducting the process in a group setting. Whether individually or in a group setting, facilitate the process in an interactive way. You might consider assigning a project team member specifically to the new member (“buddy”) to help the new member quickly become productive on the project.

Following the induction, the new member should then acquire the required skills identified in the Project Team Learning Plan (AP.030) by following the learning path associated by roles.

Linkage to Other Tasks

The Skilled Project Team is an input to the following tasks:

- BP.040 - Develop Current Process Model
- BP.080 - Develop Future Process Model
- BR.030 - Map Business Requirements

Role Contribution

The percentage of total task time required for each role follows:

Core Task Steps

Role	%
Trainer	100
Client Staff Member	*

Table 10-15 Role Contribution for Core Task Steps of Conduct Project Team Learning Events

* Participation level not estimated.

Optional Task Steps

If your project is addressing the optional task steps for this task, then you need to allocate additional time and the following resources:

Role	%
Organizational Development Specialist	100
Client Staff Member	*

Table 10-16 Role Contribution for Optional Task Steps of Conduct Project Team Learning Events

* Participation level not estimated.

Deliverable Guidelines

The Skilled Project Team represents the team members after they participate in the learning events. These learning events address functional, technical, general, and high-performance learning requirements.

The Skilled Project Team should be able to perform in the following areas:

- cross-functional project team
- team contribution as per their roles

Deliverable Components

To support the Project Team Learning Events, create a document with the following components:

- Introduction
- Administration of Learning Events for Project Team
- Appendices

Introduction

This component provides an overview of the project team learning events administration.

Administration of Learning Events for Project Team

This component is used to prepare for and carry out the learning events planned for the project team. It helps to track, reinforce, and measure the effectiveness of the learning.

Appendices

This component includes appendices A, B, and C to support the team's ability to cohesively work together. Appendix A, Ice Breakers, suggests icebreakers. Appendix B, Team Building Interludes, describes team building interludes. Appendix C, High Performance Team Development Modules, lists facilitation guidelines for high-performance team development modules.

Tools

Deliverable Template

Use the Project Team Learning Events Administration template to create the supporting document for this task.

AP.060 - Develop Business Unit Managers' Readiness Plan (Optional)

In this task, you conduct work sessions to assist business unit managers in understanding their roles towards making the project successful.



If your project includes an *organizational change management focus* and it impacts a *large user population*, you should perform this task.

Deliverable

The deliverable for this task is the **Business Unit Managers' Readiness Plan**. It describes a series of work sessions that prepare leaders at this level to do their part to make the implementation successful.

Prerequisites

You need the following input for this task:

☐ **Project Management Plan [initial complete] (PJM.CR.010)**

The Project Management Plan defines the scope, objectives, and approach for the project and refers to the detailed standards and procedures employed during the execution of the project.

☐ **Work Management Strategies, Standards, and Procedures (PJM.WM.010)**

The Work Management Strategies, Standards, and Procedures define how the PJM Work Management Process supports and implements the Project Management Plan (PJM.CR.010).

☐ **Business and Process Strategy (BP.010)**

The Business and Process Strategy includes an initial assessment of what will be changed, both in the organization and in the applications. If Define Business and Process Strategy was not performed, this deliverable will not exist. (See the task description for BP.010 for more information on when this task should be performed.)

☐ **Current Process Model (BP.040)**

The Current Process Model provides an understanding of the processes and systems that support the current business, with indications of the opportunities for process improvement in cycle time, cost, and quality. If Develop Current Process Model was not performed, this deliverable will not exist. (See the task description for BP.040 for more information on when this task should be performed.)

☐ **Current Financial and Operating Structure (RD.010)**

The Current Financial and Operating Structure provides a complete picture of the organization’s financial and operating structure.

☐ **Current Business Baseline (RD.020)**

The Current Business Baseline identifies the existing business systems and how they meet the business requirements.

☐ **Executive Project Strategy (AP.010)**

The Executive Project Strategy defines the vision for the project and the business results expected, that should guide the roles and performance improvement strategy discussions. If Define Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for more information on when this task should be performed.)

☐ **Oriented Project Team (AP.020)**

The Oriented Project Team provides a refinement of the project decisions, plans, and insights to the direction of the project.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Gather information for business unit managers’ insights.	Appendix A - Business Unit Managers’ Interview Guide

No.	Task Step	Deliverable Component
2.	Prepare for work session.	Introduction
3.	Facilitate business unit managers' induction work session.	Business Unit Managers' Induction
4.	Conduct high-level systems overview of impacts to their business unit.	High-Level Systems Overview
5.	Facilitate the organizational performance improvement strategy work session.	Organizational Performance Improvement Strategy
6.	Facilitate work session change leadership guiding principles and roles.	Change Leadership Guiding Principles and Roles

Table 10-17 Task Steps for Develop Business Unit Managers' Readiness Plan

Approach and Techniques

In this session, the business unit managers gain understanding of the project vision, how it will impact their business unit, and how they will get involved to achieve the targeted business benefits. The business unit managers define how their business unit can best improve their organizational performance with the benefit of the technology. As a result of the session, the business unit managers will lead the organizational improvement strategy that will cascade down through the business units. They also develop a communication plan related to the project and the organizational improvement strategy.

Business unit managers have the opportunity to understand the importance of the project for the organization, and how their business unit can benefit from the features and functions of the applications to improve their organizational performance. This session includes an on-site high-level systems overview tailored to the specific needs of the business unit managers.

Organizational Performance Improvement Strategy

Business unit managers start developing the organizational strategy by translating the corporate business objectives, enabled by the new technology, into organizational performance expectations at the business unit level. They decide on the performance improvement areas where they want to focus their efforts, identify the priorities, devise how they will delegate the completion of the performance measures to their own management team, and decide upon a Roll-Out strategy for the new organizational performance measures. Finally, they begin a proactive discussion on how to best transition the organization from the current system to the new, with minimum hardship. Discussions around the organizational performance improvement strategies are aligned with the business models, such as Oracle Business Models (OBMs), and used to facilitate discussions around the business rationale for consistency.



Reference: Kaplan, Robert S. and Norton, David P., *The Balanced Scorecard*, Harvard Business School Press, Boston, 1996. ISBN: 0-87584-651-3.

Change Leadership Guiding Principles and Roles

In this session, the business unit managers are introduced to the fundamental leadership principles associated with managing change. The session begins with a discussion of changes in the organization's business and industry and moves to a discussion of the characteristics of technology-induced change. Since resistance is a natural reaction to change, the discussion focuses on the kinds of resistance leaders are likely to encounter in a systems technology implementation, and presents a strategy for moving people past resistance and toward acceptance.

Linkage to Other Tasks

The Business Unit Managers' Readiness Plan is an input to the following tasks:

- AP.070 - Develop Project Readiness Roadmap
- AP.090 - Develop Managers' Readiness Plan
- AP.120 - Align Information Technology Groups
- PM.010 - Define Transition Strategy

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Organizational Development Specialist	40
Project Manager	30
Business Analyst	20
Trainer	10
Business Line Manager	*
Project Sponsor	*

Table 10-18 Role Contribution for Develop Business Unit Managers' Readiness Plan

* Participation level not estimated.

Deliverable Guidelines

Use the Business Unit Managers' Readiness Plan to facilitate the work session where business unit managers determine how their business unit can best take advantage of the new technology, and how they can help in the success of the implementation.

This deliverable should address the following:

- business unit manager insights
- business managers' orientation
- consensus on the guiding principles for effective change leadership and the role of change agents
- high-level systems overview tailored to the needs of the organization

- organizational performance improvement strategy priorities
- initial organizational performance transition strategy (finalized in the Develop Transition and Contingency Plan - PM.030)

Deliverable Components

The Business Unit Managers' Readiness Plan consists of the following components:

- Introduction
- Business Unit Managers' Induction
- High-Level Systems Overview
- Organizational Performance Improvement Strategy
- Change Leadership Guiding Principles and Roles
- Appendices

Introduction

This component documents the activities in order to develop the organizational performance improvement strategy.

Business Unit Managers' Induction

This component uses the outline provided to introduce the business unit managers to the project, its vision, expected benefits, impact on the participants' business units, and the potential benefits of the system on the business unit operations.

High-Level Systems Overview

This component presents a high-level overview of how the new systems can integrate and function together and how they impact the results of the individual business units.

Organizational Performance Improvement Strategy

This component assists the business unit managers to determine how their business unit's organizational performance can be improved by the capabilities of the new technology. This component includes a menu of items from which to select the organizational performance area and measurement items that will have the highest impact on the success of the business unit.

Change Leadership Guiding Principles and Roles

This component provides the guiding principles for effective change leadership and assists them to reach consensus on how they can best be applied at the business unit managers' level. This component helps business unit managers define and prepare for their role as change agents for their respective organization.

Appendices

This component includes Appendices A and B. Appendix A, Business Unit Managers' Interview Guide, provides an interview guide to gather insights from business unit managers. Appendix B, Table of Common Information Technology and Organizational Development Terms, shows a table of common information technology and organizational development terms.

Tools

Deliverable Template

Use the Business Unit Managers' Readiness Plan template to create the deliverable for this task.

Use the Session Planning Checklist template to prepare for the work sessions.

Use the Invitation Memorandum template to invite business unit managers to the planning sessions.

Use the High-Level Systems Overview Customization template to guide the preparation of the on-site systems overview.

Balanced Scorecard Software

Balanced scorecard software, such as Oracle Balanced Scorecard is a key enabler to help organizations monitor their strategic plans and measure whether operations are performing as projected. Balanced scorecard software provides a common platform for executives to identify performance measures as a result of the system technology implementation, and apply those measures across the organization.

AP.070 - Develop Project Readiness Roadmap (Core)

In this task, you generate key findings and recommendations toward the development of strategies that help the organization become ready to use and accept the new technology and work expectations at production cutover.



If your project includes an *organizational change management focus*, you should perform the additional task steps indicated below.

Deliverable

The deliverable for this task is the **Project Readiness Roadmap**. It captures key findings and recommendations that drive the development of a change readiness strategy, a communication strategy, a learning strategy for users, and a validation of previous organizational implementation decisions.

Prerequisites

You need the following input for this task:

☐ **Project Management Plan [initial complete] (PJM.CR.010)**

The Project Management Plan defines the scope, objectives, and approach for the project and refers to the detailed standards and procedures employed during the execution of the project.

☐ **High-Level Process Vision (BP.060)**

The High-Level Process Vision includes a high-level statement of the planned change described in terms of how the new processes operate, and the principles that regulate the operation of the new processes. If Develop High-Level Process Vision was not performed, this deliverable will not exist. (See the task description for BP.060 for more information on when this task should be performed.)

Executive Project Strategy (AP.010)

The Executive Project Strategy provides executive insights and the framework for the project and the related plans and expectations. It also identifies key stakeholder groups. If Define Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for more information on when this task should be performed.)

Oriented Project Team (AP.020)

The Oriented Project Team captures insights regarding the organization's readiness, communications, and user learning strategies required for the project.

Business Unit Managers' Readiness Plan (AP.060)

The Business Unit Managers' Readiness Plan captures insights regarding the organization's readiness, communications, and user learning strategies required for the project. If the Develop Business Unit Managers' Readiness Plan was not performed, this deliverable will not exist. (See the task description for AP.060 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Finish gathering insights from representative audiences that are not part of the project team (middle managers, first-line managers, and users) .	Appendices
2.	Consolidate insights from all sources and groups of respondents.	Appendices
Δ 3.	Tailor survey.	Appendices

No.	Task Step	Deliverable Component
Δ 4.	Administer the survey.	Appendices
Δ 5.	Compile and analyze results.	Appendices
Δ 6.	Document process.	Roadmap Process
Δ 7.	Document findings.	Change Readiness Strategy; Organizational Implementation Decisions; Communication Strategy; Learning Strategy for Users; Executive Summary
Δ 8.	Prepare readiness recommendations session.	Introduction
Δ 9.	Review background information.	Introduction
Δ 10.	Review key findings.	Change Readiness Strategy; Organizational Implementation Decisions; Communication Strategy; Learning Strategy for Users
Δ 11.	Co-develop recommendations for readiness, communication, and learning strategy, with stakeholders.	Change Readiness Strategy; Organizational Implementation Decisions; Communication Strategy; Learning Strategy for Users
12.	Finalize recommendations for communication strategy.	Communication Strategy
13.	Finalize recommendations for user learning strategy.	Learning Strategy for Users
Δ 14.	Review and augment implementation decisions.	Organizational Implementation Decisions

Table 10-19 Task Steps for Develop Project Readiness Roadmap

Approach and Techniques

The Project Readiness Roadmap results from an organizational assessment that evaluates an organization's ability to adapt to major technology-induced change. The information gathered through the assessment serves as a benchmark against which further measurements can be compared to track progress. The assessment itself is a mechanism for getting people involved in the change process, and can also be used as an early step in disseminating information.

The project readiness assessment activity begins by identifying how the users across the organization interact with the new system and the strengths/risks inherent in the culture and nature of the organization. Next, it identifies user "roles" which are driven by what users are expected to be able to do with the new system: which users interact with the new system within each function / process that the system supports? The analysis of users and stakeholders is important to formulating an overall assessment of how much change users are expected to absorb. This analysis drives the readiness, communication, and learning strategies necessary for the system to be successful.

Findings

Findings are the conclusions we draw based on analysis of the data gathered in the assessment. Organizations usually value assessments to discover strengths in their organizations which can be used to increase the likelihood of a successful implementation, and areas of weakness whose impact can be eliminated or lessened.

Recommendations

Involving organizations in the process of creating recommendations allows us to tap into their expansive knowledge of their own organization, and more importantly, to create acceptance. After validating the findings, we create a draft set of recommendations which can be used as a straw man in the facilitated Project Readiness Roadmap Recommendations work session.

Project Readiness Roadmap Recommendation Work Session

This Project Readiness Roadmap work session assembles key members of the steering committee, project team, and key stakeholder leadership. In the session, participants refine the strategies required to assist the

organization in readying itself for the technology-induced change. These strategies will drive the organization’s success in the acceptance and use of the new system. Participants convert the assessment findings into effective recommendations compatible with their culture and their budget.

Linkage to Other Tasks

The Project Readiness Roadmap is an input to the following tasks:

- AP.080 - Develop and Execute Communication Campaign
- AP.090 - Develop Managers’ Readiness Plan
- AP.130 - Conduct User Learning Needs Analysis
- AP.140 - Develop User Learning Plan
- AP.180 - Conduct Effectiveness Assessment
- PM.010 - Define Transition Strategy

Role Contribution

The percentage of total task time required for each role follows:

Core Task Steps

Role	%
Project Manager	100
Business Line Manager	*
Client Staff Member	*
Project Sponsor	*

Table 10-20 Role Contribution for Core Task Steps of Develop Project Readiness Roadmap

* Participation level not estimated.

Optional Task Steps

If your project is addressing the optional task steps for this task, then you need to allocate additional time and the following resources:

Role	%
Assessment Specialist	45
Communication Specialist	30
Adult Learning Specialist	20
Project Manager	5
Business Line Manager	*
Client Staff Member	*
Steering Committee Member	*

Table 10-21 Role Contribution for Optional Task Steps of Develop Project Readiness Roadmap

* Participation level not estimated.

Deliverable Guidelines

Use the Project Readiness Roadmap deliverable to capture key findings and recommendations toward the adoption of the new technology by stakeholders.

This deliverable should address the following:

- insights from all required stakeholders
- change readiness survey administration
- data analysis
- assessment finding positioning within the guiding principles framework for effective change leadership
- findings, formulation, and presentation to key stakeholders

- recommendation formulation
- change readiness strategy
- organizational implementation decisions
- communication strategy
- learning strategy for users

Deliverable Components

The Project Readiness Roadmap consists of the following components:

- Introduction
- Executive Summary
- Roadmap Process
- Change Readiness Strategy
- Organizational Implementation Decisions
- Communication Strategy
- Learning Strategy for Users
- Appendices

Introduction

This component outlines the activities to generate the readiness strategy, communication strategy, and learning strategy for users.

Executive Summary

This component captures the high-impact findings, and after the co-development work session, facilitates documenting the recommendations to ready the business for the implementation.

Roadmap Process

This component describes the approach to conduct the organizational assessment.

Change Readiness Strategy

This component highlights the key findings and recommendations related to each of the eight guiding principles for effective change

leadership. It presents these principles as high-level, high impact practices shown to have a profound effect on groups undergoing a large technology implementation. The practices identified form the basis of the change readiness strategy, in both the leadership approach recommended and the foundation for the topics which are explored in the assessment. It assists in positioning the findings and recommendations per stakeholder groups when appropriate.

Organizational Implementation Decisions

This component assists in reviewing the decisions made so far by the executives, the steering committee, the project team, and the business unit managers to verify congruence and relevance to the organizational readiness for the project. Up to this point, decisions have been made at multiple levels which are critical to the success of the implementation. This section should be used as a checkpoint where you revisit key decisions to make sure that nothing is neglected inadvertently and that these decisions are aligned with the decisions resulting from the Project Readiness Roadmap.

Communication Strategy

This component develops the approach for creating effective communications to all stakeholder groups throughout the life of the project. It specifies the communication structure, objectives, roles, and other components of the communication strategy on the basis of the relevant findings and recommendations. It positions the communication strategy as a primary component for implementation success where you migrate users from existing comfort zone to involvement and acceptance.

Learning Strategy for Users

This component develops a high-level plan for leveraging the current capabilities of the organization and conducting an analysis of the organization's specific learning needs, further in the life-cycle of the project. It focuses on the known requirements and outlines the approach to gather the information needed to make cost-effective decisions about who should participate in the learning events, when, where, how and by whom. It specifies the learning objectives, the types of users who are the target groups of learners, and organize findings and recommendations around learning options, content, reinforcement, measurement, and overall capabilities.

Appendices

This component includes appendices A, B, and C. Appendix A, List of Respondents by Group, lists respondents by group. Appendix B, Interview, Focus Groups and Survey Questions per Group of Respondents, documents the questions used in interviews, focus groups, and surveys per groups of respondents. Appendix C, Consolidated Insights per Group of Respondents, documents the consolidated insights per groups of respondents.

Tools

Deliverable Template

Use the Project Readiness Roadmap template to create the deliverable for this task.

Use the Session Planning Checklist template to prepare for the recommendations co-development session.

Use the Invitation Memorandum template to invite participants to the recommendations co-development session.

AP.080 - Develop and Execute Communication Campaign (Core)

In this task, you reach the stakeholder audiences with tailored messages at the optimal times and through the optimal media throughout the life of the project.



If your project includes an *organizational change management focus*, you should perform the additional task steps indicated below.

Deliverable

The deliverable for this task is the **Communication Campaign**. It describes the steps in creating a campaign to meet the communication requirements.

Prerequisites

You need the following input for this task:

☐ **Project Management Plan [initial complete] (PJM.CR.010)**

The Project Management Plan defines the scope, objectives, and approach for the project and refers to the detailed standards and procedures employed during the execution of the project.

☐ **Project Progress Report (PJM.CR.070)**

The Project Progress Report includes information on the change impact to be communicated.

☐ **Project Orientation Guide (PJM.RM.025)**

The Project Orientation Guide contains a compilation of the standards, policies, and procedures for the project team members.

☐ **Business and Process Strategy (BP.010)**

The Business and Process Strategy focuses on the organization’s strategy for improving its business processes and aligning them with the system to be implemented. If Define Business and Process Strategy was not performed, this deliverable will not exist. (See the task description for BP.010 for more information on when this task should be performed.)

☐ **Executive Project Strategy (AP.010)**

The Executive Project Strategy defines the highlights of the project, including the project vision, to be communicated in order to obtain acceptance. If Define Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for more information on when this task should be performed.)

☐ **Project Readiness Roadmap (AP.070)**

The Project Readiness Roadmap defines the communication strategy and roles that guide the development of the communication campaign.

Optional

You may need the following input for this task:

☐ **High-Level Process Designs (BP.070)**

The High-Level Process Designs illustrate how the new organization will operate after the new system is implemented.



Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Clarify objectives of the communications.	Communication Campaign Matrix

No.	Task Step	Deliverable Component
Δ 2.	Identify and assess target audiences (how they are impacted).	Target Group Analysis
Δ 3.	Determine audience key messages based on impact.	Communication Campaign Matrix
Δ 4.	Select appropriate communicators and media for each audience.	Communication Campaign Matrix
5.	Develop messages and events for each audience.	Communication Campaign Matrix
6.	Plan, develop, and roll out the campaign.	
7.	Measure the effectiveness.	Communication Campaign Matrix

Table 10-22 Task Steps for Develop and Execute Communication Campaign

Approach and Techniques

The Communication Campaign supports the change effort and provides the vehicles for generating the two-way communications around the key messages related to the project. The Communication Campaign also includes key measures for evaluating the effectiveness of the communications and reinforcement. Deliverables generated throughout the life of the project can be used as input sources for the Communication Campaign over time.

The Project Vision

Communicating the vision for the corporation, the project and the alignment between them is a high-leverage activity for the success of the implementation. If stakeholders lack a clear a picture of what they are striving for, creating buy in, commitment, and enthusiasm is almost impossible. Conversely, if effective communication makes it possible for them to visualize the business benefits and make the connection

between their own work and the achievement of those results, they can start to take ownership of the outcome of the project.

Project Kick-Off Information

Messages sent at the time of the project kick-off are particularly valuable because they let everyone know, “Now it really has begun.” Effective communication dispels the rumors which are inevitably circulating, gives stakeholders a heads up that things are really going to change, and helps to generate enthusiasm and excitement around the project.

Change Impact

It is important for all stakeholders to understand how they will be impacted by the change. The formal messages imbedded in the communication campaign on the change impact are intended to inform and manage the reactions to the changes, and foster acceptance. They should dispel uncertainty and unjustified fears around the consequences of the change. Make sure you also communicate the support provided to stakeholders throughout these changes to help them achieve the planned state. Target messages to specific audiences to best address how the audiences are being impacted.

Project Progress

To keep stakeholders involved, interested and supportive of the change, make sure you keep them abreast of the progress of the project and broadcast successes regularly. This is a key strategy to maintain the momentum of the project.

Production Cutover Information

Again, to confirm stakeholders’ support of the new environment, keep them abreast of all Production Cutover events, with emphasis on how they will be supported during the transition phase. Use communications to celebrate successes and results achieved.

Linkage to Other Tasks

The Communication Campaign is an input to the following task:

- AP.090 - Develop Managers’ Readiness Plan

Role Contribution

The percentage of total task time required for each role follows:

Core Task Steps

Role	%
Project Manager	100
Client Staff Member	*
Project Sponsor	*

Table 10-23 Role Contribution for Core Task Steps of Develop and Execute Communication Campaign

* Participation level not estimated.

Optional Task Steps

If your project is addressing the optional task steps for this task, then you need to allocate additional time and the following resources:

Role	%
Communication Specialist	100
Project Manager	*
Client Staff Member	*

Table 10-24 Role Contribution for Optional Task Steps of Develop and Execute Communication Campaign

* Participation level not estimated.

Deliverable Guidelines

Use the Communication Campaign deliverable to create the communication events around the key communication themes.

This deliverable should address the following:

- target audience reaction
- an approach describing message, objective, events, media, and timeline per audience
- communication agents' assignment definition
- required resources and logistics for deployment
- communication campaign plans

Deliverable Components

The Communication Campaign deliverable consists of the following components:

- Introduction
- Target Group Analysis
- Communication Campaign Matrix
- Appendix - Using Communications to Build Various Levels of Commitment

Introduction

This component contains the overview of the communication campaign.

Target Group Analysis

This component captures current key information related to the receptivity and communication needs of each target audience. It describes the target audience groups along with their expectations, concerns, issues, and impact. The better the understanding of the target audience groups, the more effective the communication campaign. It helps to make sure all stakeholders are represented in the target audience list.

Communication Campaign Matrix

This component highlights the elements of the communications for each AIM phase: communication events, audiences, objectives, key messages, media, timing, effectiveness, reinforcement, and so on. It identifies the communication agents and shows to which events they are assigned. It also captures the logistical information and resources required and describes the activities needed to produce and deploy the communication events, with integrated sequence of events, resources, start and end dates, overlaps, and so on.

Appendix - Using Communications to Build Various Levels of Commitment

This component describes how to use communications to build various levels of commitment.

Tools

Deliverable Template

Use the Communication Campaign template to create the deliverable for this task.

AP.090 - Develop Managers' Readiness Plan (Optional)

In this task, you conduct work sessions for middle and first-line managers, who are not on the project team, to assume their role in a successful people, process, and technology implementation.

- △ If your project includes an *organizational change management focus* or it *impacts a large user population*, you should perform this task.
- △ If your project includes *process change*, you should perform the additional task steps indicated below.

Deliverable

The deliverable for this task is the **Managers' Readiness Plan**. It describes a series of events which prepare the middle and first-line managers, who are not on the project team, to understand the goals for the project, the project strategy, and their roles in making the project successful.

Prerequisites

You need the following input for this task:

☐ **Business and Process Strategy (BP.010)**

The Business and Process Strategy includes key programs that will require changes to the organization. If Define Business and Process Strategy was not performed, this deliverable will not exist. (See the task description for BP.010 for more information on when this task should be performed.)

☐ **Future Process Model (BP.080)**

The Future Process Model guides the selection of team members who will be involved in the human performance support systems update work as appropriate.

Executive Project Strategy (AP.010)

The Executive Project Strategy provides executive insights and the framework for the project and the related plans and expectations. If Define Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for more information on when this task should be performed.)

Business Unit Managers' Readiness Plan (AP.060)

The Business Unit Managers' Readiness Plan includes the performance improvement strategy and transition strategy to be refined by middle managers. If the Establish Business Unit Managers' Readiness Plan was not performed, this deliverable will not exist. (See the task description for AP.060 for more information on when this task should be performed.)

Project Readiness Roadmap (AP.070)

The Project Readiness Roadmap includes the readiness strategy targeted for middle and first-line managers.

Communication Campaign (AP.080)

The Communication Campaign announces the project namely to middle and first-line managers.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Prepare for managers' readiness session.	Introduction
2.	Facilitate managers' readiness sessions.	Managers' Induction
3.	Conduct applications to process overview.	Applications to Process Overview

No.	Task Step	Deliverable Component
Δ 4.	Explain how to convert performance objectives to role levels.	Organizational Performance Improvement Strategy
Δ 5.	Set up human performance support systems task forces.	Organizational Performance Improvement Strategy
6.	Facilitate change leadership session.	Change Leadership Guiding Principles and Roles
7.	Define managers' communication roles.	Managers' Communication Guide

Table 10-25 Task Steps for Develop Managers' Readiness Plan

Approach and Techniques

The readiness materials for managers start with an orientation to the system project primarily focused on creating acceptance and commitment among the middle and first-line managers. The meeting should be led by a middle manager or a member of the organization's project change team.

After the agenda has been presented, the point should be made that the first-line managers have the responsibility of communicating information about the system implementation to their teams. First-line managers may want to devote a team meeting to explaining the business reasons for the change and the link with the objectives for the project, describing the impacts on the processes they use and the individuals on the team, and answering questions. This session is geared to prepare them to carry out that responsibility. This task targets two levels of managers: middle and first-line managers, through a cascading approach.

Middle Managers

Middle managers are the level of managers who report to business unit managers, such as vice presidents and directors who own the operational areas most heavily impacted by the technology change. Middle managers are the leaders who are closest to business processes

that cross organizational boundaries and who can bring the business unit measures down to role-specific expectations. Project team members frequently report to this level of management, and middle managers often lead the task forces associated with the change.

First-Line Managers

First-line Managers have direct supervisory responsibility over the roles impacted by the new technology. They are on the first-line of operations and have the largest number of users who report to them.

Cascade Approach

First, you assemble middle managers. After the agenda for the readiness session has been presented, the point should be made that middle managers will in turn be facilitating readiness meetings similar to the one in which they are participating, to their first-line managers. The content of the managers' readiness sessions is subject to change over time, as the project evolves through the Solution Design and Build phases. Through the Communication Campaign (AP.080), managers are kept apprised of project changes that need to be reflected in the readiness sessions.

The readiness meetings with first-line managers should be led by a middle manager or a member of the organization's project change team. First-line managers have the responsibility of communicating information about the implementation to their teams. First-line leaders may want to devote a team meeting to explaining the business reasons for the change and the link with the objectives for the project, describing the impacts on the processes they use and the individuals on the team, and answering questions. This session is geared to prepare them to carry out that responsibility.

Managers' Induction

The induction is a formal session facilitated by business unit managers (for middle managers) and middle managers (for first-line managers) to expose their direct reports to the highlights of the project and elicit their early involvement. It gives middle and in turn first-line managers a high-level view of the impact of the project on technology, process and people, including the need for new organizational performance measures tied to the new business unit and department objectives, and new human resource management tools. The session ends with an application to process overview session to review the features and

functions of the application as they relate to the relevant business processes, in terms of how the applications can improve the way departments operate.

Organizational Performance Improvement Strategy

The Organizational Performance Improvement Strategy session is intended for middle managers only. The session starts with a discussion of why the activity of translating the high-level business objectives to the department level is valuable. The middle managers have the responsibility of taking those organizational performance areas prioritized by the business unit managers and adding performance indicators, measurement actions, and measurement criteria. The middle managers complete a draft set of organizational measures, evaluate them for feasibility, risks, and benefits, and reach a consensus on a set of measures. This approach is consistent with the balanced scorecard management technique.



Reference: Kaplan, Robert S. and Norton, David P., *The Balanced Scorecard*, Harvard Business School Press, Boston, 1996. ISBN: 0-87584-651-3.

The middle managers then set up the human performance support systems team. The human performance support systems team includes representatives of the stakeholder groups, for example, process owners, and subject matter specialists in relevant areas such as: job performance, job evaluation and classification, organizational structure, workflows, or performance support practices. This team will work on the Human Performance Support Systems (AP.110).

Change Leadership Guiding Principles and Roles

In this session middle and first-line managers are introduced in turn to the fundamental leadership principles associated with managing change and the role managers play — at their respective level — as change leaders. For example, they make the connection between the organization's anticipated business results and the project objectives so their people understand the “why” behind the implementation. Similarly, managers become role models on a more immediate level than the executive team does. Their people watch to see if their day-to-day decisions are consistent with their stated positions.

Of all the levels of management, the first-line managers have the most direct experience with the resistance which may result from the

implementation of the new technology. This session describes the reasons for resistance, the symptoms to look for, and the strategies for reducing resistance and building acceptance in the workforce. The strategy is based on eight guiding principles of change, but in this case, the principles are tailored to the audience. For example, the first-line leader does not create the corporate or information technology vision; however, he or she has the responsibility to not only present a clear picture of the anticipated future to his people, but also to make the connections between the vision and the day-to-day work of his team. Each of the principles needs to be presented in view of the authority and accountability of a first-line manager.

The change leadership work session addresses these differences in the application of the principles by tailoring the discussion to the managers' role.

Managers' Communication Guide

Both middle and first-line managers have significant communication responsibilities in conjunction with the implementation. As a starting place for a discussion of that role, they need to describe the role they usually play in organizational communications and think through how that role applies in the context of the implementation project. The managers' communication guide provides guidelines for managers to fulfill their communication role.

Linkage to Other Tasks

The Managers' Readiness Plan is an input to the following tasks:

- AP.110 - Align Human Performance Support Systems
- AP.140 - Develop User Learning Plan

Role Contribution

The percentage of total task time required for each role follows:

Core Task Steps

Role	%
Human Performance Technologist	25
Communication Specialist	25
Application Specialist	20
Business Analyst	20
Trainer	10
Business Line Manager	*

Table 10-26 Role Contribution for Core Task Steps of Develop Managers’ Readiness Plan

* Participation level not estimated.

Optional Task Steps

If your project is addressing the optional task steps for this task, then you need to allocate additional time and the following resources:

Role	%
Human Performance Technologist	25
Communication Specialist	25

Role	%
Application Specialist	20
Business Analyst	20
Trainer	10
Business Line Manager	*

Table 10-27 Role Contribution for Optional Task Steps of Develop Managers' Readiness Plan

* Participation level not estimated.

Deliverable Guidelines

Use the Managers' Readiness Plan deliverable to prepare and involve the middle and first-line managers in a successful system implementation.

This Managers' Readiness Plan should address the following:

- highlights of the project
- application overview as linked to relevant business processes
- organizational risks and benefits associated with the change
- performance area for roles
- human performance support systems team established
- middle and first-line managers skilled as communication agents
- middle and first-line managers roles in sponsorship, communication, and change leadership
- reduction of resistance to change

Deliverable Components

The Managers' Readiness Plan deliverable consists of the following components:

- Introduction
- Managers' Induction
- Applications to Process Overview
- Organizational Performance Improvement Strategy
- Change Leadership Guiding Principles and Roles
- Managers' Communication Guide

Introduction

This component highlights the purpose and outline for the sessions.

Managers' Induction

This component contains a checklist to introduce the project to the middle and first-line managers. It highlights the project vision, benefits, expected results, impact on the middle and first-line managers' operations and overall involvement of target stakeholders. It explains how this information is to be rolled out to middle managers, to the first-line managers and to their people.

Applications to Process Overview

This component highlights the features of the applications as they relate to the processes associated with the business areas.

Organizational Performance Improvement Strategy

This component uses a menu structure to guide the middle managers in their input to the organizational performance improvement strategy, by converting the expected performance results and measurements defined at the business processes level to the role level. It assists the middle managers in setting up the required task forces for human performance support systems.

Change Leadership Guiding Principles and Roles

This component exposes the middle and first-line managers to the guiding principles for effective change leadership and instructs them on how to best manage reactions to change. It helps to reach consensus on

how middle and first-line managers can best fulfill their role as change leaders at their respective level.

Managers' Communication Guide

This component uses the communication guide to instruct middle and first-line managers in their role as communication agents. It helps to guide them in the best use of the communication infrastructure at their disposal and obtain their commitment to fulfill their role as communication agents.

Tools

Deliverable Template

Use the Managers' Readiness Plan template to create the deliverable for this task.

Use the Session Planning Checklist template to prepare for the planning sessions.

Use the Invitation Memorandum template to invite middle and first line managers to the planning sessions.

Use the Application to Process Overview Customization template to guide the preparation of the on-site application to process overview.

Balanced Scorecard Software

Balanced scorecard software, such as Oracle Balanced Scorecard is a key enabler to help organizations monitor their strategic plans and measure whether operations are performing as projected. Balanced scorecard software provides a common platform for executives to identify performance measures as a result of the system technology implementation, and apply those measures across the organization.

AP.100 - Identify Business Process Impact on Organization (Optional)

In this task, you document the extent and the ramifications of the organizational change brought about by the new or improved business processes.



If your project includes *process change*, you should perform this task.

Deliverable

The deliverable for this task is the **Business Process Organizational Impact**. It describes the impact to the work, people, organizational structure, organizational culture, and physical layout caused by business process change.

Prerequisites

You need the following input for this task:

 Change Catalog (BP.020)

The Change Catalog documents the cost and benefits involved in making changes to the business process and roles of the organization. If Catalog and Analyze Potential Changes was not performed, this deliverable will not exist. (See the task description for BP.020 for more information on when this task should be performed.)

 Current Process Model (BP.040)

The Current Process Model details the current environment for change impact assessment on the organization. If Develop Current Process Model was not performed, this deliverable will not exist. (See the task description for BP.040 for more information on when this task should be performed.)

 Future Process Model (BP.080)

The Future Process Model provides the details of the future processes for change impact assessment on the organization.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Compare future to current process models to identify business process change.	Organizational Model
2.	Identify the impact of process change on each organizational system.	Change Impact by Organizational System
3.	Prioritize the impacts.	Change Impact by Organizational System
4.	Develop a plan to manage impacts based on priorities.	Plan for Alignment and Acceptance

Table 10-28 Task Steps for Identify Business Process Impact on Organization

Approach and Techniques

Organizational Model

The model used to analyze the impact of new /improved business process on the organization is derived from the open systems theory of organizations. This model describes the inner-workings of the organization, consisting of five main systems: work, people, organizational structure, organizational culture, and physical layout. Optimally these five systems reinforce each other, creating a consistent message throughout the organization. This congruence, or lack thereof, is then transformed to outcomes on the organizational, group, and individual level.

On the organizational level we can measure this through Return On Investment (ROI), growth, revenue, and profit. On the group level this is reflected in the ability of the various groups to function as a unit. Feeding into the groups, yet with a separate dynamic are individuals,

who can experience both personal and job related growth, measured by behavior on the job. Using the five organizational systems as a framework, you document the changes caused by the business processes which were updated or improved.

Linkage to Other Tasks

The Business Process Organizational Impact is an input to the following task:

- AP.110 - Align Human Performance Support Systems

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Organizational Development Specialist	80
Business Analyst	20
Business Line Manager	*
Client Staff Member	*
Key User	*

Table 10-29 Role Contribution for Identify Business Process Impact on Organization

* Participation level not estimated.

Deliverable Guidelines

Use the Business Process Organizational Impact to analyze and document the changes to the five organizational areas caused by business processes which have been redefined or improved.

This Business Process Organizational Impact should address the following:

- relationship between the open systems model and the five organizational systems
- impact of altered business processes in all areas of the organization
- change disruption mitigation, bringing systems into alignment, and building acceptance among stakeholders

Deliverable Components

The Business Process Organizational Impact deliverable consists of the following components:

- Introduction
- Organizational Model
- Change Impact by Organizational System
- Plan for Alignment and Acceptance

Introduction

This component provides the overview for the business process organizational impact analysis.

Organizational Model

This component uses the open systems model and the five organizational areas as the framework to analyze the business process change impact.

Change Impact by Organizational System

This component uses the elements listed to conduct a thorough analysis of how the new business processes will impact the organization. For continuity and acceptance, it helps to involve the team who worked with the process and a representative group from the affected departments to identify the impacts.

Plan for Alignment and Acceptance

This component helps to develop a plan to help manage the disruption caused by the changes, bring systems into alignment, and build acceptance in the people who are stakeholders.

It involves the stakeholders themselves in the creation of the plan as an effective strategy for creating shared ownership.

Tools

Deliverable Template

Use the Business Process Organizational Impact template to create the deliverable for this task.

AP.110 - Align Human Performance Support Systems (Optional)

In this task, you define the human performance support systems requirements and execute the approach to meet the requirements to support the performance of users in their new roles.



If your project includes an *organizational change management focus* or *process change*, you should perform this task.

Deliverable

The deliverable for this task is the **Human Performance Support Systems**. It describes the approach to facilitate the changes on human resource and work management tools and processes. It includes the definition of new roles, performance metrics, competencies, human performance support systems processes, and protocols, as well as learning materials for managers to learn how to implement the Human Performance Support Systems.

Prerequisites

You need the following input for this task:

☐ **Business Procedure Documentation (BP.090)**

The Business Procedure Documentation includes the list of tasks and screens per role.

☐ **Business Volumes and Metrics (RD.040)**

The Business Volumes and Metrics include projected operational processing volumes, transaction patterns, and volume requirements.

☐ **Executive Project Strategy (AP.010)**

The Executive Project Strategy defines the change induced by the new technology. If Define Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for more information on when this task should be performed.)

❑ Managers’ Readiness Plan (AP.090)

The Managers’ Readiness Plan defines performance measures by activity and the charter of the Human Performance Support Systems task force. If Develop Managers’ Readiness was not performed, this deliverable will not exist. (See the task description for AP.090 for more information on when this task should be performed.)

❑ Business Process Organizational Impact (AP.100)

The Business Process Organizational Impact details the changes induced by the new or improved processes, including changes to roles. If Identify Business Process Impact on Organization was not performed, this deliverable will not exist. (See the task description for AP.100 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review new roles as identified in Business Procedure Documentation (BP.090).	
2.	Analyze current performance management context.	Current Performance Management Context
3.	Identify new performance support systems requirements.	New Performance Support Systems Requirements
4.	Co-develop human performance support systems recommendations with stakeholders.	Human Performance Support Systems Recommendations
5.	Define user support requirements.	User Support Requirements

No.	Task Step	Deliverable Component
6.	Develop deliverables for workflows, job structure, and organizational structure.	Workflows, Job Structure, and Organizational Structure
7.	Develop deliverables for job performance.	Job Performance
8.	Develop deliverables for job evaluation and classification.	Job Evaluation and Classification
9.	Develop deliverables for performance support practices, procedures, tools, and systems.	Performance Support Practices, Procedures, Tools, Systems
10.	Develop deliverables for job-person matching.	Job-Person Matching
11.	Develop deliverables for human performance support systems measurements.	Human Performance Support Systems Measurements
12.	Validate performance metrics from the full-time equivalent (FTE) level to the organizational level (bottom-up consolidation).	Job Performance
13.	Design the flow of roll-out events for the new Human Performance Support Systems.	Human Performance Support Systems Roll-Out Plan
14.	Create learning materials for middle managers to learn how to use the new Human Performance Support Systems.	Human Performance Support Systems Learning Materials

No.	Task Step	Deliverable Component
15.	Create learning materials for first-line managers to learn how to use the new Human Performance Support Systems.	Human Performance Support Systems Learning Materials
16.	Pilot effectiveness of new Human Performance Support Systems.	

Table 10-30 Task Steps for Align Human Performance Support Systems

Approach and Techniques

The general goal of performance support documentation for each targeted job / role is to create materials that transform the performance support specifications and criteria into valid, user-friendly instruments and procedures that uniformly guide the performance support processes. This includes the development, validation, and production of the actual performance support materials, tailored to the targeted roles, and the development of the measurement materials to evaluate the effectiveness of the performance support materials to be deployed. The document also describes the process of rolling-out the new human performance support systems to the organization. It includes a sequence of events and the development of the materials intended for managers to learn how to use the new human performance support systems.

The Human Performance Support Systems cover the relevant areas for the five performance areas described below:

Performance Area	Description
Workflows, job structure, and organizational structure — to include job descriptions and possible reorganization of departments and business units.	Actual and optimal content and context of jobs and departments impacted by the new or changed work processes resulting from the new technology implementation.
Job performance — to include new competency models, productivity expectations and other performance metrics, human resource allocations.	For each job / role group, competencies and productivity required given the new or changed work processes, human resource allocation, gap between actual and optimal job performance.
Job evaluation and classification — to include grade levels, classifications, and compensation ranges.	Structure of content and optimal job classification systems, as well as compensation factors, given the new or changed work processes.
Performance support practices, procedures, tools, and systems — to interview, select, set performance goals, appraise performance, provide feedback, plan career development, reward, and recognize.	Content and scope of existing performance support practices, tools and systems, and requirements to support new performance expectations.
Job person matching — new skills profiles, learning per role, succession planning, and so on.	Competencies of current job incumbents and their supervisors, reskilling requirements, and succession plans.

The Human Performance Support Systems requirements and documents cover the relevant areas in job structure, job performance, job classification, performance support practices, and job-person matching. The new performance requirements and recommendations

are converted into materials, including materials for the rollout of the new human performance support systems from the time of implementation and thereafter. The preparation includes components to prioritize the deployment, validate the materials including the measurements, and develop the learning materials and the updating. When launching the new human performance support systems, you focus on the smooth use of the new practices to manage the performance of the new roles.



Suggestion: Critical to the success of your project is the establishment of a robust and capable internal and vendor external support infrastructure. Oracle Support Services provides a unique service to assess your current support capabilities and assist with planning, updating and aligning these capabilities to Oracle's support infrastructure. Contact your local Oracle Support Services representative to schedule an Oracle Support Assessment (OSA). This should coincide with the activities planned for this AIM task.

Linkage to Other Tasks

The Human Performance Support Systems Documentation is an input to the following tasks:

- AP.120 - Align Information Technology Groups
- AP.130 - Conduct User Learning Needs Analysis
- AP.140 - Develop User Learning Plan
- AP.150 - Develop User Learningware
- AP.170 - Conduct User Learning Events

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Human Performance Technologist	75
Adult Learning Specialist	25
Business Line Manager	*
Client Staff Member	*
IS Manager	*
Key User	*

Table 10-31 Role Contribution for Align Human Performance Support Systems

* Participation level not estimated.

Deliverable Guidelines

Use the Human Performance Support Systems deliverable to identify the and execute on the requirements to support the performance of users in their new roles, as defined through the new procedures. This deliverable provides user-friendly performance tools, measurements, roll-out, and learning materials.

This deliverable should address the following:

- change impact on human performance systems brought about by changed processes
- performance area and roles to address in priority
- performance context assessment
- current performance systems analysis

- new performance systems projection
- user support requirements definition
- user-friendly performance tools based on user support requirements
- effectiveness measurement approach for the new performance tools
- new Human Performance Support Systems roll-out sequence for performers and their managers
- performance management learning materials for performers and their managers

Deliverable Components

The Human Performance Support Systems deliverable consists of the following components:

- Introduction
- Current Performance Management Context
- New Performance Support Systems Requirements
- Human Performance Support Systems Recommendations
- User Support Requirements
- Workflows, Job Structure, and Organizational Structure
- Job Performance
- Job Evaluation and Classification
- Performance Support Practices, Procedures, Tools and Systems
- Job-Person Matching
- Human Performance Support Systems Measurements
- Human Performance Support Systems Roll-Out Plan
- Human Performance Support Systems Learning Materials
- Appendices

Introduction

This component provides the overview of the human performance support systems.

Current Performance Management Context

This component provides a framework to guide the analysis of the performance support systems in which current roles are performed. It helps to assess strengths and limitations of the current systems. This will clarify the organization's thinking around the human resource and performance management direction and the implications associated with change. It uses these insights as the basis for deriving the performance support objectives for the technology-induced change effort. Refer to the User Learning Needs Analysis (AP.130) deliverable for background on the first line managers' practices to manage performance.

New Performance Support Systems Requirements

This component helps to identify and prioritize the human performance support systems requirements for the target roles and the approach to meet those requirements, based on the findings from the current human resource and performance management context and the assessment of its relevancy to the required performance environment. It helps to prioritize performance support needs for each target job / role by considering both the value they add to achieving the objectives of the technology-induced change effort, as well as the extent of the gap between existing and optimal states. It uses a checklist of questions as a guide.

Human Performance Support Systems Recommendations

This component lists the recommendations for the development of the new human performance support systems, the required documents, the revisions to the Human Resource Information System (HRIS) and the rollout. It focuses on defining job / role group specific approach to the performance support needs and objectives. Refer to the User Reference Manual (DO.060) to understand the functionality of each custom extension, in addition to the regular features, and how it impacts each role.

User Support Requirements

This component can be used as a complement to the human performance factors; it focuses on the technical support needs of the

users whose performance is now tied to the new applications. It includes a menu to guide the set up of the technical help. For the user support requirements, a spreadsheet relates to the help desk requirements. This component might include the involvement of an Oracle Support Specialist.

Workflows, Job Structure, and Organizational Structure

This component uses as a starting point the new procedures, for each role, to document the new workflows, job structure, and organizational structure, as defined in the new requirements.

Job Performance

This component uses as a starting point the new procedures and workflows, for each role, to document the performance metrics and resource allocation, as defined in the new requirements.

Job Evaluation and Classification

This component is used to identify the classification and compensation level for each role, as defined in the new requirements, based on the new job structure and performance.

Performance Support Practices, Procedures, Tools, and Systems

This component focuses on materials to guide managers in performance management practices, for example, interview, select, appraise performance, set performance goals, provide feedback, plan career development, reward and recognize, and terminate employees in the new roles.

Job-Person Matching

This component maps the potential of current employees to become job incumbents to the new roles, as defined in the new requirements. It can also specify learning path and succession plans per individual.

Human Performance Support Systems Measurements

This component converts the measurement recommendations into measurement tools compatible with the performance practices and culture of the organization. It creates a deliverable that effectively measure the performance support objectives, determine the effectiveness of the performance support materials and approaches, and allow for revisions and recommendations for ongoing support.

Human Performance Support Systems Roll-Out Plan

This component develops steps to deploy the new Human Performance Support Systems to the various levels of management and performers, for acceptance and use. It merges the events for all various targeted job / role groups into a single sequence of events covering all approaches. It includes learning, validation, communication, and measurement events.

Human Performance Support Systems Learning Materials

This component develops learning materials to make sure that the performers know the new system within which their performance will be managed, and that their managers are skilled at managing their performance to the new expectations. It helps to tailor the learning materials to the needs of the managers, based on the information gathered in the User Learning Needs Analysis (AP.130). Make sure to cover the new expectations per role, how each role fits into the new workflows, the new work management structure, and the tools that will now be used to manage performance within the new work context.

Appendices

This component contains Appendices A, B, and C. Appendix A, Location Information Spreadsheet, contains the location information spreadsheet. Appendix B, Sample Questions to Gather Human Performance Support Systems Data, provides sample questions to gather human performance support systems data. Appendix C, Organization's Policies and Procedures Impacted by Oracle Applications, contains organization's policies and procedures impacted by Oracle applications.

Tools

Deliverable Template

Use the Human Performance Support Systems and Appendix A - Location Information Spreadsheet templates to create the deliverable for this task.

Oracle Tutor

Oracle Tutor provides users with relevant process documentation and learning materials. These materials are process- or role-based, easy to update, cost-effective and developed in conjunction with the implementation. Oracle Tutor helps apply functionality specific to Oracle Applications within the context of the organization's unique set of policies and procedures. The product is comprised of three components: an authoring tool, a publishing tool, and a repository of Oracle Applications courseware and procedure document objects.

AP.120 - Align Information Technology Groups (Optional)

In this task, you help transition the information technology groups to their new / updated mandate and roles, given the implementation of the new systems.



If your project impacts a *large user population* and includes an *organizational change management focus*, you should perform this task.

Deliverable

The deliverable for this task is the **Aligned Information Technology Groups**. It represents the members of the information technology groups with their updated mission and skills to perform their new roles.

Prerequisites

You need the following input for this task:



Executive Project Strategy (AP.010)

The Executive Project Strategy defines the business expectations enabled by the new technology. If the Define Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for more information on when this task should be performed.)



Business Unit Managers' Readiness Plan (AP.060)

The Business Unit Managers' Readiness Plan defines performance requirements by business units. If Develop Business Unit Managers' Readiness Plan was not performed, this deliverable will not exist. (See the task description for AP.060 for more information on when this task should be performed.)



Human Performance Support Systems (AP.110)

The Human Performance Support Systems define the user support requirements. If Align Human Performance Support Systems was not

performed, this deliverable will not exist. (See the task description for AP.110 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Define new / updated information technology groups' mission.	Information Technology Groups' Mission
2.	Analyze current human performance support systems in information technology groups.	Current Information Technology Human Support Performance Systems
3.	Identify new / updated human performance support systems requirements for the information technology groups.	New Information Technology Human Performance Support Systems Requirements
4.	Make Human Performance Support Systems recommendations for the information technology groups.	Information Technology Human Performance Support Systems Recommendations
5.	Develop Human Performance Support Systems deliverables for the information technology groups.	Information Technology Human Performance Support Systems Materials
6.	Convert transition requirements to roll-out plan.	
7.	Develop learning plan for information technology personnel.	Information Technology Learning Plan

No.	Task Step	Deliverable Component
8.	Create roll-out communications.	
9.	Administer logistics.	

Table 10-32 Task Steps for Align Information Technology Groups

Approach and Techniques

The general goal of this task is to update the charter of the information technology groups within the new or updated business and technology contexts, so the information technology members can sustain and pursue the technical gains achieved. The implementation becomes a trigger to look at the mission and role of the groups given the corporate strategy, and align them as required. From the mission and role, you drill down to the new individual roles and human performance support systems, including learning requirements for each key information technology role. Finally, you put in place the events to migrate the groups from their current context to their new performance environment, including learning, communication, measurement, and contingency planning.



Suggestion: The establishment of a robust and capable internal and vendor external support infrastructure is critical to the success of your project. As you start discussions with the information technology groups, you should initiate the discussions on technical support infrastructure. Oracle Support Services provides a unique service to assess your current support capabilities and assist with planning, updating and aligning these capabilities to Oracle's support infrastructure. Contact your local Oracle Support Services representative to schedule an Oracle Support Assessment (OSA). This should coincide with the activities planned for this AIM task.

Information Technology Learning

Based on the skills required from the information technology personnel, determine the skills gaps between required and existing skills. Then develop the learning plan for the information technology personnel.



Web Site: For a list and description of the latest Oracle Education offerings, such as the professional development paths, course details (Reusable Content Object Matrix), and interactive seminars, refer to Oracle Education’s web site ([http:// education.oracle.com /](http://education.oracle.com/)).

Linkage to Other Tasks

The Aligned Information Technology Groups are an input to the following task:

- PM.020 - Design Production Support Infrastructure

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Human Performance Technologist	100
Client Staff Member	*
IS Manager	*

Table 10-33 Role Contribution for Align Information Technology Groups

* Participation level not estimated.

Deliverable Guidelines

The Aligned Information Technology Groups represent the members of those groups whose mission, roles, organizational structure, and human performance support systems have been updated to reflect the new business and technology context.

The Aligned Information Technology Groups should understand the following:

- information technology groups' mission
- new information technology human performance support systems to support the preferred new scenario
- information technology groups' infrastructure alignment plan for the new mandate
- communications describing the purpose behind the transition, the support provided to the group members to carry them through the transition, and the planned events
- effectiveness measurement of the new mandate, new human performance support systems, learning, and communications
- contingency development to provide for extra resource requirements during transition
- departmental transition logistic deployment

Deliverable Components

The Aligned Information Technology Groups template consists of the following components:

- Introduction
- Information Technology Groups' Mission
- Current Information Technology Human Performance Support Systems
- New Information Technology Human Performance Support Systems Requirements
- Information Technology Human Performance Support Systems Recommendations
- Information Technology Human Performance Support Systems Materials
- Information Technology Learning Plan
- Appendices

Introduction

This component provides the overview of the information technology groups alignment, key considerations and alignment approach.

Information Technology Groups' Mission

This component provides steps to help the information technology groups formulate or update each group's mission, to reflect its new capability and align with the corporate strategy. This component helps to form the new mission for the information technology groups.

Current Information Technology Human Performance Support Systems

This component uses a framework to assess the strengths and deficits of the current information technology human performance support systems for a full understanding of the current work and performance management context within the information technology groups. This approach provides the basis for deriving the performance support objectives that allow the information technology groups to sustain the technical gains.

New Information Technology Human Performance Support Systems Requirements

This component lists the requirements for the new information technology human performance support systems, based on the new mission and capabilities, and the approach to meet these requirements. It uses questions as a guide.

Information Technology Human Performance Support Systems Recommendations

This component develops specific approaches to the performance needs and objectives of the key information technology roles. It lists the recommendations for the development of the new human performance support systems, the required deliverables, the revisions to the Human Resource Information Systems and the rollout.

Information Technology Human Performance Support Systems Materials

This component develops the actual materials that were listed in the Information Technology Human Performance Support Systems Recommendations, to meet the Human Performance Support Systems

Requirements for the new / updated mission and roles. When looking into new information technology roles, refer to the Technical Reference Manual (DO.080) (as available) which describes the components and the design of the customizations and extensions; it is a reference document for application maintenance personnel. Also refer to the System Management Guide (DO.090) (as available) which contains the final set of procedures for operating the application system; it is a guide for system administrators.

Information Technology Learning Plan

This component contains the details of the learning paths for the information technology group members, to include learning objectives, content, approach, rewards and recognition, measurement, schedule and administration details.

Appendices

This component contains Appendices A and B. Appendix A, Common Information Technology Job Roles, provides a list of common information technology roles. Appendix B, Traits for Information Technology Competencies, lists sample skills profiling traits for information technology competencies.

Tools

Deliverable Template

Use the Information Technology Groups Alignment template to create the supporting document for this task.

Oracle Tutor

Oracle Tutor provides users with relevant process documentation and learning materials. These materials are process- or role-based, easy to update, cost-effective and developed in conjunction with the implementation. Oracle Tutor helps apply functionality specific to Oracle Applications within the context of the organization's unique set of policies and procedures. The product is comprised of three components: an authoring tool, a publishing tool, and a repository of Oracle Applications courseware and procedure document objects.

AP.130 - Conduct User Learning Needs Analysis (Optional)

In this task, you gather insights on the learning needs of all audiences of users.



If your project includes an *organizational change management focus* and impacts a *large user population*, you should perform this task.

Deliverable

The deliverable for this task is the **User Learning Needs Analysis**. It builds on the Learning Strategy for Users component in the Project Readiness Roadmap (AP.070) and describes the gaps in knowledge, skills, and aptitudes between the current and future states for all audiences of users.

Prerequisites

You need the following input for this task:

☐ **Business Procedure Documentation (BP.090)**

The Business Procedure Documentation describes the new procedures users will be following in their new roles.

☐ **Project Readiness Roadmap (AP.070)**

The Project Readiness Roadmap describes the Learning Strategy for Users.

☐ **Human Performance Support Systems (AP.110)**

The Human Performance Support Systems Documentation details the competency requirements for the targeted roles and the new performance management context for both performers and their immediate supervisors. If Align Human Performance Support Systems was not performed, this deliverable will not exist. (See the task description for AP.110 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Identify respondent groups.	Scope and Objectives
2.	Tailor interview guide for the learning needs analysis.	Appendix A - Interview Guide
3.	Conduct learning needs analysis interviews.	Scope and Objectives
4.	Consolidate interview data.	Consolidated Findings
5.	Map findings to competencies for each role found in the Human Performance Support Systems.	User Profiles by Role
6.	Determine gaps between current and desired skills.	User Profiles by Role
7.	Derive user learning recommendations.	User Learning Recommendations

Table 10-34 Task Steps for Conduct User Learning Needs Analysis

Approach and Techniques

During this task, gather information about the knowledge, skills, and aptitudes of all categories of users. You record current knowledge, skills, and aptitudes, compare them to the new competencies and identify the gaps between the present and the ideal. Those gaps mark the parameters of the needed learning. The profile is a snapshot in time which serves not only as a guide for creating learning paths, but also as a baseline against which gains can be measured. In addition, the User Learning Needs Analysis provides insights in these audiences’ readiness for the project.

For representation, you interview a cross-section of users who are selected based on their ability to impact the business results anticipated as a result of the implementation. Choose people who represent a large number of stakeholders, such as super-users.

Linkage to Other Tasks

The User Learning Needs Analysis is an input to the following task:

- AP.140 - Develop User Learning Plan

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Adult Learning Specialist	70
Human Performance Technologist	30
Business Line Manager	*
Client Staff Member	*
User	*
Project Sponsor	*

Table 10-35 Role Contribution for Conduct User Learning Needs Analysis

* Participation level not estimated.

Deliverable Guidelines

Use the User Learning Needs Analysis deliverable to assess the gaps in knowledge, skills, and aptitudes between the current and future competencies for all audiences.

This User Learning Needs Analysis should address the following:

- cross-section sampling of stakeholders among users
- profiles of users' knowledge, skills, and aptitudes
- gap assessment between required and existing skills (system literacy, procedural, and business skills)
- project acceptance assessment

Deliverable Components

This User Learning Needs Analysis consists of the following components:

- Introduction
- Scope and Objectives
- Consolidated Findings
- User Profiles by Role
- User Learning Recommendations
- Appendices

Introduction

This component provides the overview off the user learning needs analysis.

Scope and Objectives

This component describes the steps and people involved in the conduct of the learning needs analysis, for example, list of respondents and interviewers, sampling techniques, interview guides, and so on.

Consolidated Findings

This component captures the high-impact findings on the Skills, Knowledge and Aptitudes (SKA) gaps and readiness for the project. It highlights the gaps between existing and required SKAs, as per the findings derived from the input collected from the respondent groups. It organizes the findings per groups of performers / roles and highlights common learning needs.

User Profiles by Role

This component lists the current system literacy, procedural and business SKAs compared to the required SKAs per group of performers / roles, as defined in the Business Procedure Documentation (BP.090) and Human Performance Support Systems (AP.110) deliverables. It includes key findings and recommendations related to the target groups' receptivity and readiness for the project. It positions the findings around the eight guiding principles for effective change leadership.

User Learning Recommendations

This component captures the recommendations derived from the findings and the skills gap analysis. It includes recommendations for target groups of learners, timing, learning approach, and so on.

Appendices

This component includes Appendices A and B. Appendix A, Interview Guide, contains the interview guide for the User Learning Needs Analysis data gathering. Appendix B, List of Respondents by Group, lists respondents by group.

Tools

Deliverable Template

Use the User Learning Needs Analysis template to create the deliverable for this task.

AP.140 - Develop User Learning Plan (Core)

In this task, you create learning path approaches that allow users to become skilled in the new technologies, apply new / updated procedures, and fulfill their new roles.



If your project includes an *organizational change management focus*, you should perform the additional task steps indicated below.

Deliverable

The deliverable for this task is the **User Learning Plan**. It describes a customized approach for reskilling those employees whose knowledge, skills, and aptitudes need to change so the full benefit of the new technology can be realized.

Prerequisites

You need the following input for this task:



User Reference Manual (DO.060)

The User Reference Manual contains the final description of the detailed functionality of each custom module and extension, to be used by users. If the final version of the User Reference Manual is not available, a draft of this document is acceptable to use as input. If Publish User Reference Manual was not performed, this deliverable will not exist. (See the task description for DO.060 for more information on when this task should be performed.)



User Guide (DO.070)

The User Guide describes the final set of procedures for using the application system in response to business events and is a guide for new users. If the final version of the User Guide is not available, a draft of this document is acceptable to use as input. If Publish User Guide was not performed, this deliverable will not exist. (See the task description for DO.070 for more information on when this task should be performed.)

☐ **Project Readiness Roadmap (AP.070)**

The Project Readiness Roadmap details the learning strategy for users on which the User Learning Plan is built.

☐ **Managers' Readiness Plan (AP.090)**

The Managers' Readiness Plan details the involvement of the managers and end-users in the learning plan. If Develop Managers' Readiness Plan was not performed, this deliverable will not exist. (See the task description for AP.090 for more information on when this task should be performed.)

☐ **Human Performance Support Systems (AP.110)**

The Human Performance Support Systems documents the skills required for the new roles. If Align Human Performance Support Systems was not performed, this deliverable will not exist. (See the task description for AP.110 for more information on when this task should be performed.)

☐ **User Learning Needs Analysis (AP.130)**

The User Learning Needs Analysis details the skills gaps for the various groups of users. If Conduct User Learning Needs Analysis was not performed, this deliverable will not exist. (See the task description for AP.130 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review and select approved recommendations from the User Learning Needs Analysis (AP.130), if available.	

No.	Task Step	Deliverable Component
2	For each targeted role, select the role-based procedure knowledge from the Business Procedure Documentation (BP.090) (or Oracle Tutor, if applicable).	
3.	For each targeted role, select the role-based content related to product skills from the standard applications courseware (or Oracle Tutor, if applicable).	
Δ 4.	For each group of learners, create learning objectives by role.	Learning Objectives by Role
Δ 5.	Tailor learning content for each role.	Learner Profiles
Δ 6.	Select the learning approach and the delivery method for each group of learners.	Learning Paths by Role
Δ 7.	Finalize the measurement methods.	Learning Measurement Methods
Δ 8.	Describe the learning reinforcement methods and rewards and recognition process.	Learning Reinforcement, Rewards and Recognition Methods
Δ 9.	Describe the approach for creating needed learning materials.	Material Development Recommendations

No.	Task Step	Deliverable Component
10.	Determine resource requirements such as facilities, equipment, materials, and supplies, including learning environment.	Resource Requirements
11.	Describe the plan for learning logistics and administration.	Learning Administration Process

Table 10-36 Task Steps for Develop User Learning Plan

Approach and Techniques

The general goal of the User Learning Plan is to provide guidelines for the development of the curriculum and to create criteria for measuring the effectiveness of the learning approach. The starting point for creating a learning approach was the identification of target groups of learners and the gaps in their knowledge, skills, and aptitudes. The learning objectives define the ending point for example, where we want to be when the learning is complete. The Learner Profiles component forms the bridge between the two, by defining what it is that people need to know, or learn, or do in order to reach those objectives. Each Learner Profile covers the three areas of learning: system literacy (which includes computer skills), procedural skills, and business skills.

The learning path combines the content, the media, and the sequence to create a flow of learning events for each targeted learning group. In creating the learning path, you consider the optimal methods for transferring the needed skills to each group of learners, given their unique needs and the constraints or the opportunities provided by the organization environment. The goal is to focus on the full range of the learning experience for example, the preparation, the support offered during learning, and the follow up when the learners return to the job.

Linkage to Other Tasks

The User Learning Plan is an input to the following tasks:

- AP.150 - Develop User Learningware
- AP.160 - Prepare User Learning Environment
- AP.170 - Conduct User Learning Events
- PM.030 - Develop Transition and Contingency Plan

Role Contribution

The percentage of total task time required for each role follows:

Core Task Steps

Role	%
Trainer	100
Client Staff Member	*
User	*

Table 10-37 Role Contribution for Core Tasks Steps of Develop User Learning Plan

* Participation level not estimated.

Optional Task Steps

If your project is addressing the optional task steps for this task, then you need to allocate additional time and the following resources:

Role	%
Adult Learning Specialist	100
Client Staff Member	*
User	*

Table 10-38 Role Contribution for Optional Task Steps of Develop User Learning Plan

* Participation level not estimated.

Deliverable Guidelines

The User Learning Plan is the blueprint of learning events covering what users need to know to fulfill their new roles and achieve their new performance expectations. The user groups represent all stakeholders who need to use the new technology, from executives who will read reports to the clerks who will enter data. This plan is based on the Learning Strategy for Users component defined in the Project Readiness Roadmap (AP.070) and reflects any new findings, issues, and requirements that were added or occurred since the roadmap was initially developed.

The User Learning Plan should address the following:

- learning objectives per role
- learning content per role
- reinforcement and learning paths per role that provide variety and foster effective performance
- required learningware materials development plan
- resource requirements, including staff, space, equipment, materials, and supplies

- learning effectiveness measurement plan

Deliverable Components

The User Learning Plan deliverable consists of the following components:

- Introduction
- Learning Objectives by Role
- Learner Profiles
- Learning Paths by Role
- Learning Reinforcement, Rewards and Recognition Methods
- Learning Measurement Methods
- Material Development Recommendations
- Resource Requirements
- Learning Administration Process
- Appendices

Introduction

This component provides the overview for the user learning plan.

Learning Objectives by Role

This component documents the learning objectives by role, such as enabling and learning objectives in each of the Skills, Knowledge, and Aptitudes (SKA) categories. It ties to the materials around learner verification.

Learner Profiles

This component describes the logical flow of the content to be learned by groups of performers. The content to be learned is outlined in a modular format or reusable content objects, for optimal leveraging of content across groups of learners. This content is articulated in a learning flow in the next component.

Learning Paths by Role

This component converts the logical flow of the content to be learned into a learning flow, describing how a learner is first be exposed to the

content, understand it, observe it, practice it, and so on. It builds the flow from more simple to more complex and includes interactive learning and reinforcement activities in the flow.

Learning Reinforcement, Rewards and Recognition Methods

This component describes the methods to reinforce the performers' learning of the new content, how they complement one another, who administers them, for example, self-administered, included in the learning paths (for example, learner verification), administered by the performer's learning agent, manager. It reflects the reinforcement activities in the learning paths per role. The reinforcement methods encourage users to transfer the skills acquired in the work place.

The learning reinforcement, rewards and recognition methods component provides the following benefits:

- It describes the methods to encourage, reward and recognize the performers' participation and involvement in the learning events.
- It specifies how they complement each other, who administers them, and how to prepare those administrators.
- It suggests guidelines for recognition for each target group of learners and the time frames in which they should be given.
- It integrates with the Human Performance Support Systems for the impacted roles.

Learning Measurement Methods

This component describes the methods to measure the effectiveness of the learning, as defined in the user learning strategy. The Learning Measurement Methods component provides the following benefits:

- It helps to select methods that are compatible to the culture of the organization and the selection of learning paths, from self-administered measurement, for example, self-assessment, learner verification, to more formal methods such as, use of a control group.
- It assists in looking for leveraging opportunities.

- It links the administration of measurement methods to the update of the learningware and the final evaluation of its effectiveness.
- It specifies who will administer the measurement methods and what preparation they will need.

The general goal of describing measurement methods is to specify how the success of the learning will be measured against the desired outcome, and compared to data collected during the User Learning Needs Analysis (AP.130).

Material Development Recommendations

This component provides recommendations required to support the learning flows by role. It includes material for all parties involved in the learning process, for example, the learner, the learning agent (for example, coach, instructor, and manager), and the administrator. Consider developing learningware on the model of “reusable content object” for leveraging material development across groups of learners. Wherever possible, available courseware should be used in order to minimize the effort required; the focus is on modifying existing courseware and on producing those materials needed to support self-guided learning.

Resource Requirements

This component details the requirements in terms of people, money, equipment, facilities, and so on, for the development and the delivery of the learningware. When appropriate, it specifies the skills required from the people, for example, instructional design, knowledge of authoring software, and so on. It provides provisions for learningware in multiple languages if appropriate.

Learning Administration Process

This component describes the processes to schedule learning, distribute materials, and communicate with learners. It specifies the tools required, the procedures, and people involved in the administration of the learning, for example, logging learning events, tracking participation in learning events, consolidating measurement data, offering catch-up learning to those who missed an offering of a learning event, and so on.

Appendices

This component contains Appendices A, B, and C. Appendix A, Medial Selection Flowchart, contains a decision flowchart to select learning media. Appendix B, Electronic Performance Support Systems (EPSS), describes electronic performance support systems (EPSS). Appendix C, Legal Implications for Learner Verification, describes legal implications for learner verification.

Tools

Deliverable Template

Use the User Learning Plan template to create the deliverable for this task.

Oracle Tutor

Oracle Tutor provides users with relevant process documentation and learning materials. These materials are process- or role-based, easy to update, cost-effective and developed in conjunction with the implementation. Oracle Tutor helps apply functionality specific to Oracle Applications within the context of the organization's unique set of policies and procedures. The product is comprised of three components: an authoring tool, a publishing tool, and a repository of Oracle Applications courseware and procedure document objects.

AP.150 - Develop User Learningware (Optional)

In this task, you develop the learningware defined in the User Learning Plan (AP.140) to support the reskilling of employees.



If your project includes an *organizational change management focus, customizations, interfaces or process changes*, you should perform this task.

Deliverable

The deliverable for this task is the **User Learningware**. It includes materials for preparing learning agents; for assisting user learning, reinforcement, and measurement; and for administering learning events.

Prerequisites

You need the following input for this task:



User Reference Manual (DO.060)

The User Reference Manual defines the functionality and use of applications extensions. If the final version of the User Reference Guide is not available, a draft of this document is acceptable to use as input. If Publish User Reference Manual was not performed, this deliverable will not exist. (See the task description for DO.060 for more information on when this task should be performed.)



User Guide (DO.070)

The User Guide represents the final user documentation that the users need when working with the system. If the final version of the User Guide is not available, a draft of this document is acceptable to use as input. If Publish User Guide was not performed, this deliverable will not exist. (See the task description for DO.070 for more information on when this task should be performed.)

☐ **System Management Guide (DO.090)**

The System Management Guide represents the final system management documentation that the information technology resources need when working with the system. If the final version of the System Management Guide is not available, a draft of this document is acceptable to use as input. If Publish System Management Guide was not performed, this deliverable will not exist. (See the task description for DO.090 for more information on when this task should be performed.)

☐ **Human Performance Support Systems (AP.110)**

The Human Performance Support Systems Documentation documents the skills required for the new roles. If Align Human Performance Support Systems was not performed, this deliverable will not exist. (see the task description for AP.110 for more information on when this task should be performed.)

☐ **User Learning Plan (AP.140)**

The User Learning Plan details the materials required to support the learning project.

☐ **Production Support Infrastructure Design (PM.020)**

The Design Production Support Infrastructure Design describes how the desktop operates, how to log and get resolution to technical issues. Users need to know about the new desktop and related procedures.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Develop the Learning Materials per learning path by role.	Learning Materials
2.	Develop the Learning Reinforcement, Rewards And Recognition Materials.	Learning Reinforcement, Rewards and Recognition Materials

No.	Task Step	Deliverable Component
3.	Develop the Measurement Materials.	Measurement Materials
4.	Develop the Learning Administration Materials.	Learning Administration Materials
5.	Develop the Learning Agents Materials.	Learning Agents Materials

Table 10-39 Task Steps for Develop User Learningware

Approach and Techniques

In this task, you tailor learningware and other skills-change materials to meet the learning objectives developed in the User Learning Plan (AP.140). Tailor the existing learningware to reflect the new procedures derived from the business process changes and the custom extensions if appropriate. Organize the content around the new roles if the role-based approach is selected in the User Learning Plan. The general goal of this task is to support the reskilling of those employees whose knowledge, skills and aptitudes need to change, to achieve optimal benefit from the new technology.

Use templates (as provided by Oracle Tutor or other authoring software) for ease of development and consistency across deliverables. Focus on developing materials that are user-friendly, attractive, and meet good communication standards.

Linkage to Other Tasks

The User Learningware is an input to the following tasks:

- AP.160 - Prepare User Learning Environment
- AP.170 - Conduct User Learning Events

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Adult Learning Specialist	90
Application Specialist	10
Client Staff Member	*
Key User	*

Table 10-40 Role Contribution for Develop User Learningware

* Participation level not estimated.

Deliverable Guidelines

Use the User Learningware for the planned learning events, as listed in the User Learning Plan (AP.140).

This User Learningware should address the following:

- user-friendly, attractive, interactive documents for learning as per the planned path, by role
- custom extensions to the applications
- updated roles and procedures
- system literacy and business skills
- materials to reinforce learning, measuring the success of the learning, administering the learning events, and preparing the learning agents

Deliverable Components

The User Learningware deliverable consists of the following components:

- Introduction
- Learning Materials
- Learning Reinforcement, Rewards and Recognition Materials
- Measurement Materials
- Learning Administration Materials
- Learning Agents Materials
- Appendices

Introduction

This component contains the overview of the User Learningware.

Learning Materials

This component helps to develop the learning materials specified in the User Learning Plan (AP.140), as per the learning paths by role. It guides piloting materials before full deployment.

Learning Reinforcement, Rewards and Recognition Materials

This component develops materials to reinforce, reward, and recognize learning as specified in the User Learning Plan (AP.140).

Measurement Materials

This component develops the measurement tools as specified in the User Learning Plan (AP.140).

Learning Administration Materials

This component develops the tools to help the administration of learning events, as specified in the User Learning Plan (AP.140). It includes materials for announcing and logging learning events, tracking participation, and so on.

Learning Agents Materials

This component develops the materials to skill the learning agents in their role related to the success of the learning events, for example, coaching skills/ plans for super-users, and so on.

Appendices

This component contains Appendices A, B, and C. Appendix A, Development Checklist, is a checklist to develop user learning materials. Appendix B, Communications Template, is a template to communicate about the learning events. Appendix C, Learning Event Participation Event, is used to list participants who partake in the learning events.

Tools

Deliverable Template

Use the User Learningware template to create the deliverable for this task.

AP.160 - Prepare User Learning Environment (Core)

In this task, you establish the technical and physical infrastructure required for the actual user learning, including preparing an environment that reflects the production applications environment.

Deliverable

The deliverable for this task is the **User Learning Environment**. It includes the computer systems and networks for web-based learning, production of paper-based support materials, and setup of learning facilities. In addition, it covers the application servers, configuration, and setups containing a controlled set of data that is synchronized with the learningware.

Prerequisites

You need the following input for this task:

☐ **Prepared Infrastructure (PJM.RM.050)**

The Prepared Infrastructure helps plan the approach to learning environments early in the project. Follow this plan to maintain consistency with expected resource allocations and schedules.

☐ **Application Setup Documents (BR.100)**

The Application Setup Documents include description of setup documents captured during mapping activities that can be used in learning.

☐ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy helps define the requirements for Project Team Learning Environment.

❑ Integration-Tested System (TE.120)

The Integration-Tested System contains an environment that has had the new applications, custom extensions, and interfaces to remaining legacy applications tested in its entirety.

❑ User Learning Plan (AP.140)

The User Learning Plan lists the requirements for user learning.

❑ User Learningware (AP.150)

The User Learningware dictates the amount and type of data that must be available for learning. If Develop User Learningware was not performed, this deliverable will not exist. (See the task description for AP.150 for more information on when this task should be performed.)



Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review Prepared Infrastructure (PJM.RM.050), Application Setup Documents (BR.100), and User Learning Plan (AP.140).	
2.	Review and update checklists.	Preparation Checklists
3.	Install User Learning Environment	Systems Environment
4.	Set up applications.	
5.	Set up support infrastructure	
6.	Convert or add necessary sample data.	

Table 10-41 Task Steps for Prepare User Learning Environment

Approach and Techniques

The environment needs to be configured so it fully supports the user learningware and learning plan. Use the checklists provided to set up the technical and physical infrastructure, such as learning centers and web-based learning.

Preparation of Technical Environment

As part of preparing the user technical learning environment, make sure you test the client desktop devices and other hardware, such as printers.

User Learning Environment

Existing environments may be appropriate for user learning. Use the Prepared Infrastructure (PJM.RM.050) to guide you in the creation of learning environments.

Separate Learning Environments

A separate environment is the preferred choice for learning because it can contain clean setups and actual data and remain unaffected by concurrent testing and mapping activities. Create an export of the target live environment with additional data entered manually or converted from the legacy system. This scenario assumes that conversion programs are complete and tested at this point.

A refreshed separate environment is important if many groups learn on the same modules and the learning database includes required examples or exercises for specific setups. All required separate installations must follow the optimal flexible architecture (OFA) standard.

If you cannot create a separate user learning environment, the best choice is one that closely resembles production and offers the most stability. The alternatives include the following:

- Demonstration Database
- Requirements Mapping Environment
- System Test Environment

Demonstration Database

Project team learning often uses the demonstration database shipped with the standard applications. This environment is most appropriate if you plan to use the standard learning materials for user learning, because all examples in the standard notes reference this database. However, it is not ideal because the data and scenarios will not be familiar to users.

Requirements Mapping Environment

The environment that the project team used for mapping business requirements should contain the appropriate setups and adequate sample data, but it also includes many invalid setup codes and data. Install customizations in this environment. If you do not have enough room for a separate environment, this requirements mapping environment might be your best choice.

System Test Environment

The system test environment should contain sample production data and include all custom modules and extensions. By using this environment, you are more likely to include all current modules in the learning.

Linkage to Other Tasks

The User Learning Environment is an input to the following task:

- AP.170 - Conduct User Learning Events

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Administrator	35
Database Administrator	25
Application Specialist	20
Trainer	20
Client Staff Member	*
User	*

Table 10-42 Role Contribution for Prepare User Learning Environment

* Participation level not estimated.

Deliverable Guidelines

The User Learning Environment includes existing or newly installed standard applications, physical and technical setups, and data required to support user learning.

This User Learning Environment should address the following:

- application setups
- demonstration data
- learning facilities reservation and preparation
- desktop client for testing and other hardware such as printers
- usage plan for shared concurrent learners' environment

Deliverable Components

The User Learning Environment deliverable consists of the following components:

- Preparation Checklists
- Systems Environment

Preparation Checklists

This component contains checklists to help guide the preparation of the physical and technical learning environment.

Systems Environment

This component describes the application installations, application server and database server instances needed to support the learning environment.

Tools

Deliverable Template

Use the User Learning Environment template to create the deliverable for this task.

AP.170 - Conduct User Learning Events (Core)

In this task, you conduct the skills-change events developed in the User Learning Plan (AP.140), as well as track and document their progress.



If your project includes an *organizational change management focus*, you should perform the additional task steps indicated below.

Deliverable

The deliverable for this task is **Skilled Users**. Skilled Users have learned what they need to succeed in their new roles. This knowledge covers system literacy, procedural, and business skills.

Prerequisites

You need the following input for this task:



User Reference Manual (DO.060)

The User Reference Manual should be used as a complement to the learningware during the learning events. If Publish User Reference Manual was not performed, this deliverable will not exist. (See the task description for DO.060 for more information or when this task should be performed.)



User Guide (DO.070)

The User Guide should be used as a complement to the learningware during the learning events. If Publish User Guide was not performed, this deliverable will not exist. (See the task description for DO.070 for more information or when this task should be performed.)



System Management Guide (DO.090)

The System Management Guide should be used as a complement to the learningware during the learning events. If Publish System Management Guide was not performed, this deliverable will not exist.

(See the task description for DO.090 for more information or when this task should be performed.)

☐ **Human Performance Support Systems (AP.110)**

The Human Performance Support Systems includes learning materials to assist managers in learning the new human performance support systems for the roles they manage. If Align Human Performance Support Systems was not performed, this deliverable will not exist. (See the task description for AP.110 for more information on when this task should be performed.)

☐ **User Learning Plan (AP.140)**

The User Learning Plan details the events planned to support the learning project.

☐ **User Learningware (AP.150)**

The User Learningware includes the materials for the learning project. If Develop User Learningware was not performed, this deliverable will not exist. (See the task description for AP.150 for more information on when this task should be performed.)

☐ **User Learning Environment (AP.160)**

The User Learning Environment provides the physical and technical infrastructure required to support the learning events.

☐ **Return On Investment (ROI) Analysis**

A Return On Investment (ROI) analysis, such as Oracle Solution Value Assessment (SVA), can document the business case and the return that the executive management team can realistically expect from its investment in technology. If an ROI, such as Oracle Solution Value Assessment (SVA) has been performed, you will need it as an input.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
Δ 1.	Create human performance support communications.	
2.	Create user learning communications.	
Δ 3.	Conduct human performance support orientation and learning events.	Human Performance Support Orientation and Learning Administration Materials
4.	Hold user learning events.	User Learning Event Log
Δ 5.	Monitor human performance support deployment progress.	Human Performance Support Orientation and Learning Administration Materials
Δ 6.	Monitor skills-change progress.	Skills-Change Measurements

Table 10-43 Task Steps for Conduct User Learning Events

Approach and Techniques

The general goal of the user learning events is to conduct and track the skills-change events designed to provide the groups of learners with the skills they need to meet the performance objectives of their new roles. You monitor the pulse and progress of the user learning events as they unfold, to make sure that the momentum and quality are maintained.

Δ New/Updated Human Performance Support Systems Learning Events

For learning events around the new / updated human performance support systems, you provide communication, orientation, and learning to help the managers learn how to use the new tools to manage the performance of the new roles reporting to them. Target any individual

who will be using the human performance support materials. First-line managers in turn will orient target job group incumbents to the new performance expectations. Human Resource personnel may also be included in the learning target groups.

Learning Events Communication

As a success factor for the learning events, you develop communications on the purpose, value, context, and overall logistics for the learning events. Tailor the messages to the various groups of learners. The communications set the proper tone for the learning, for example, important effort, tailored to the learning styles, providing variety, reinforcement, and so on. Position the learning events in the context of the whole project and the expected business benefits. Ideally, develop a highly interactive campaign to address the changes in roles and performance expectations in a positive and motivational manner. Link to ongoing communications dedicated to the project.

Linkage to Other Tasks

The Skilled Users is an input to the following tasks:

- TE.130 - Perform Acceptance Test
- PM.060 - Implement Production Support Infrastructure
- PM.070 - Verify Production Readiness

Role Contribution

The percentage of total task time required for each role follows:

Core Tasks Steps

Role	%
Trainer	100
Business Line Manager	*
Client Staff Member	*
User	*

Table 10-44 Role Contribution for Core Task Steps of Conduct User Learning Events

* Participation level not estimated.

Optional Tasks Steps

If your project is addressing the optional task steps for this task, then you need to allocate additional time and the following resources:

Role	%
Adult Learning Specialist	85
Human Performance Technologist	15
Client Staff Member	*

Table 10-45 Role Contribution for Optional Task Steps of Conduct User Learning Events

* Participation level not estimated.

Deliverable Guidelines

Skilled Users represent users from across the organization who, by going through the learning paths associated to their roles, are able to take advantage of the new technology and to use the system, and meet their new performance expectations.

Skilled Users should understand the following:

- learning paths organization for users and managers
- learning content (systems literacy, procedure and business skills) as associated to their new roles
- new human performance support systems and performance management tools for managers, job incumbents, and HR personnel

Deliverable Components

To support the skilling of users, create a document with the following components:

- Introduction
- Human Performance Support Orientation and Learning Administration Materials
- User Learning Event Log
- Skills-Change Measurements

Introduction

This component gives an overview of the user learning events administration.

Human Performance Support Orientation and Learning Administration Materials

This component helps to manage the orientation and learning events dedicated to the human performance support systems tools.

User Learning Event Log

This component helps to track the learning events, as per the administration procedures specified in the User Learning Plan (AP.140).

Skills-Change Measurements

This component helps to consolidate the data from the measurement deliverables and to determine whether corrective action is required, for example, adjusting the User Learning Plan (AP.140) and User Learningware (AP.150).

Tools

Deliverable Template

Use the User Learning Events Administration template to create the supporting document for this task.

AP.180 - Conduct Effectiveness Assessment (Core)

In this task, you determine how well the production system, business and organizational performance meet the business objectives set at the beginning of the project.



If your project includes an *organizational change management focus*, and impacts a *large user population*, you should perform the additional task steps indicated below.

Deliverable

The deliverable for this task is the **Effectiveness Assessment**. It captures key benchmarked findings and recommendations for continuous improvement in all organizational, business and technical systems (for example, system, business and organizational performance, and so on).

Prerequisites

You need the following input for this task:

Change Catalog (BP.020)

The Change Catalog documents the changes to the business process and roles of the organization which are being measured now in terms of effectiveness. If Catalog and Analyze Potential Changes was not performed, this deliverable will not exist. (See the task description for BP.020 for more information on when this task should be performed.)

High-Level Process Vision (BP.060)

The High-Level Process Vision documents the vision the organization formulated for the business changes. If Develop High-Level Process Vision was not performed, this deliverable will not exist. (See the task description for BP.060 for more information on when this task should be performed.)

☐ **Business Procedure Documentation (BP.090)**

The Business Procedure Documentation provides the baseline against which the actual business procedures being executed on the system are compared to determine whether the system is meeting business objectives.

☐ **Business Volumes and Metrics (RD.040)**

The Business Volumes and Metrics provide the indicators that help determine the business performance.

☐ **Project Readiness Roadmap (AP.070)**

The Project Readiness Roadmap contains the first sets of organizational measurements.

☐ **Production System (PM.080)**

An audit such as this assessment is performed on a post-production live system.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the Business Process Measures component in RD.040.	
2.	Set up the mechanism to measure business performance.	
3.	Measure business performance.	
4.	Analyze performance gain or loss.	

No.	Task Step	Deliverable Component
5.	Summarize current business performance.	Executive Summary
6.	Document current business metrics.	Business Performance Metrics
Δ 7.	Conduct organizational effectiveness assessment.	Appendices
8.	Compile and analyze results.	Appendices
9.	Document Findings.	Organizational Effectiveness Strategy; Performance Transition Decisions; Communication Strategy; Learning Strategy for Users; and Executive Summary
Δ 10.	Create post-learning skills profiles.	Learning Strategy for Users
Δ 11.	Document the assessment process.	Assessment Process
Δ 12.	Prepare organizational effectiveness recommendation session.	Introduction
Δ 13.	Co-develop recommendations for continuous organizational effectiveness, communication, and learning, with stakeholders.	Organizational Effectiveness Strategy; Performance Transition Decisions; Communication Strategy; Learning Strategy for Users; and Executive Summary
14.	Review and augment transition decisions.	Performance Transition Decisions
15.	Consolidate and integrate action items.	

Table 10-46 Task Steps for Conduct Effectiveness Assessment

Approach and Techniques

The Effectiveness Assessment results from a business performance and organizational audit that evaluates the organization's effectiveness with its new systems and compares it to the initial readiness measurements. From the findings, new strategies are defined to optimize the organizational effectiveness.



Warning: Consultants should not drive these types of audits. User management should take full ownership of assessing how well they are performing. The consultant's role in this task is to assist them in accessing the information. This task is a reminder of the importance of the reporting and analysis elements of performance auditing.



Suggestion: The Effectiveness Assessment provides a great opportunity to merge in an Oracle Support Assessment. As the organization is measuring its performance on the business, process and organization's effectiveness with the systems, if you have not done so, you can assess the organization's technical support infrastructure to determine the support package best suited to the organization. Consult the Oracle Support Specialist.

Measuring Business Performance

Tools

Some tools may be required to measure business performance. For example, some businesses want to measure their organization performance by looking at detailed gross margin figures for each product. This type of analysis may require some ad hoc query/report development or SQL scripts.

Standard Reports and Inquiries

There are various standard reports and inquiries provided in the applications that already measure common business metrics. These reports should be used to analyze performance before considering building scripts.

Using Key Performance Indicators

There are many ways to measure how well a business is running. There may be performance metrics in place based on current operations. These business metrics may have been documented in Business Volumes and Metrics (RD.040) and / or the Business Unit Managers' Readiness Plan (AP.060). Often organizations adopt certain metrics based on the information access provided in their current system.

Period-End Close Audit

A true test of whether an organization has successfully implemented a system is how efficiently and accurately an organization can go through a period-end close. A period-end close summarizes, analyzes, and reports the performance of the business across multiple criteria. It includes not only the traditional financial performance, but the operations performance which is often buried in representations of revenue and margin.

Although financial periods vary from organization, a period of one month typically characterizes a full business cycle. When you create a production system audit, attach the balanced financial and operation reports as supporting documentation.

Organizational Effectiveness Recommendation Work Session

The Organizational Effectiveness Recommendation work session assembles key leaders to help them refine strategies to favor the organization's performance with the new systems. They are gathered to convert the assessment findings into effective recommendations compatible with their culture, budget, and on-going performance expectations. The recommendations, focused on continuous organizational effectiveness, performance transition, communication, and learning, are accompanied by an integrated action plan. Recommendations address how to adjust and fill any gap in the expected performance.

Linkage to Other Tasks

The Effectiveness Assessment is an input to the following tasks:

- PM.130 - Propose Future Business Direction
- PM.140 - Propose Future Technical Direction

Role Contribution

The percentage of total task time required for each role follows:

Core Task Steps

Role	%
Organizational Development Specialist	40
Business Analyst	40
Project Manager	20
Business Line Manager	*
Client Executive	*
Client Project Manager	*
Client Staff Member	*
User	*

Table 10-47 Role Contribution for Core Task Steps of Conduct Effectiveness Assessment

* Participation level not estimated.

Optional Tasks Steps

If your project is addressing the optional task steps for this task, then you need to allocate additional time and the following resources:

Role	%
Assessment Specialist	40
Communication Specialist	30

Role	%
Adult Learning Specialist	30
Business Line Manager	*
Client Executive	*
Client Project Manager	*
Client Staff Member	*
User	*

Table 10-48 Role Contribution for Optional Task Steps of Conduct Effectiveness Assessment

* Participation level not estimated.

Deliverable Guidelines

Use the Effectiveness Assessment deliverable to summarize the measurements of key business aspects of the production system, drive the administration of this assessment, and generate a renewed and updated emphasis toward organizational effectiveness. This assessment represents the second time that you measure the organization — the Project Readiness Roadmap (AP.070) established the first benchmark of organizational readiness. Now that the organization readied itself for acceptance and use of the new system, the focus is on how the organization absorbed the change and can continue improving.

The Effectiveness Assessment should address the following:

- measurement for operational and departmental efficiency of processes, including margin results, cycle times, and the number of non-value added steps eliminated
- productivity impact assessment, ease of use, availability, and other such attributes
- organizational effectiveness survey
- data analysis and documented findings

- findings positioned within the framework of the guiding principles for organizational effectiveness
- key stakeholders findings and co-develop recommendations
- organizational effectiveness strategy
- communication strategy
- learning strategy for users
- performance transition decisions

All analyses should be supported by measurement against key business indicators (such as efficiency, response to customers, and accuracy) and system performance attributes (such as system mean time to respond by transaction type).



Suggestion: During this critical task external quality audit services are recommended.

Deliverable Components

The Effectiveness Assessment consists of the following components:

- Introduction
- Executive Summary
- Business Performance Metrics
- Assessment Process
- Organizational Effectiveness Strategy
- Performance Transition Decisions
- Communication Strategy
- Learning Strategy for Users
- Appendices

Introduction

This component describes the activities performed in this task.

Executive Summary

This component captures the high-impact findings and recommendations to optimize the organizational effectiveness for continuous improvement.

Business Performance Metrics

This component describes the key performance indicators and other expectations defined for the business and system performance.

Assessment Process

This component describes the approach followed to develop this second roadmap.

Organizational Effectiveness Strategy

This component highlights the key findings and recommendations related to the organizational effectiveness, as per the congruency model of organizational systems. It positions the findings and recommendations per stakeholder groups when appropriate.

Performance Transition Decisions

This component helps to review the decisions made in the transition of the project to the organization's operations to confirm congruence and relevance to the effectiveness of the organization.

Communication Strategy

This component develops the approach for creating effective communications to all stakeholder groups on an ongoing basis. It specifies the communication structure, objectives, roles and other components of the communication strategy on the basis of the relevant findings and recommendations.

Learning Strategy for Users

This component develops a high-level plan for leveraging the current capabilities of the organization and conducting an analysis of the organization's ongoing learning needs. It specifies the learning

objectives, the types of users who will be the target groups of learners, and organize findings and recommendations around learning options, content, reinforcement, measurement, and overall capabilities.

Appendices

This component includes appendices A, B, and C. Appendix A, List of Respondents by Group, lists respondents by group. Appendix B, Interview, Focus Groups and Survey Questions per Group of Respondents, documents the questions used in interviews, focus groups, and surveys per groups of respondents. Appendix C, Consolidated Insights per Group of Respondents, documents the consolidated insights per groups of respondents.

Tools

Deliverable Template

Use the Effectiveness Assessment template to create the deliverable for this task.

Use the Session Planning Checklist template to prepare for the recommendations co-development session.

Use the Invitation Memorandum template to invite participants to the recommendations co-development session.

11

Production Migration (PM)

This chapter describes the Production Migration process.

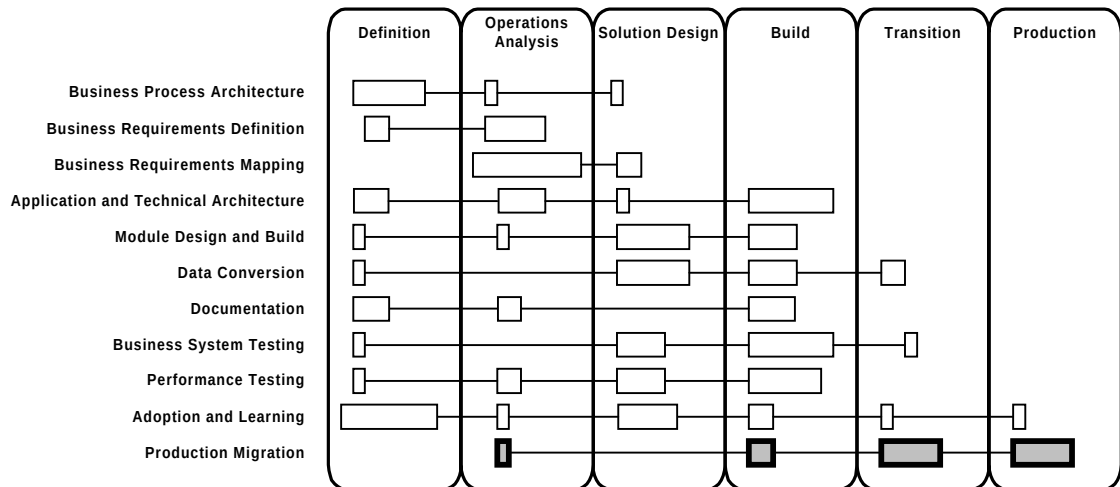


Figure 11-1 Production Migration Context

Process Flow

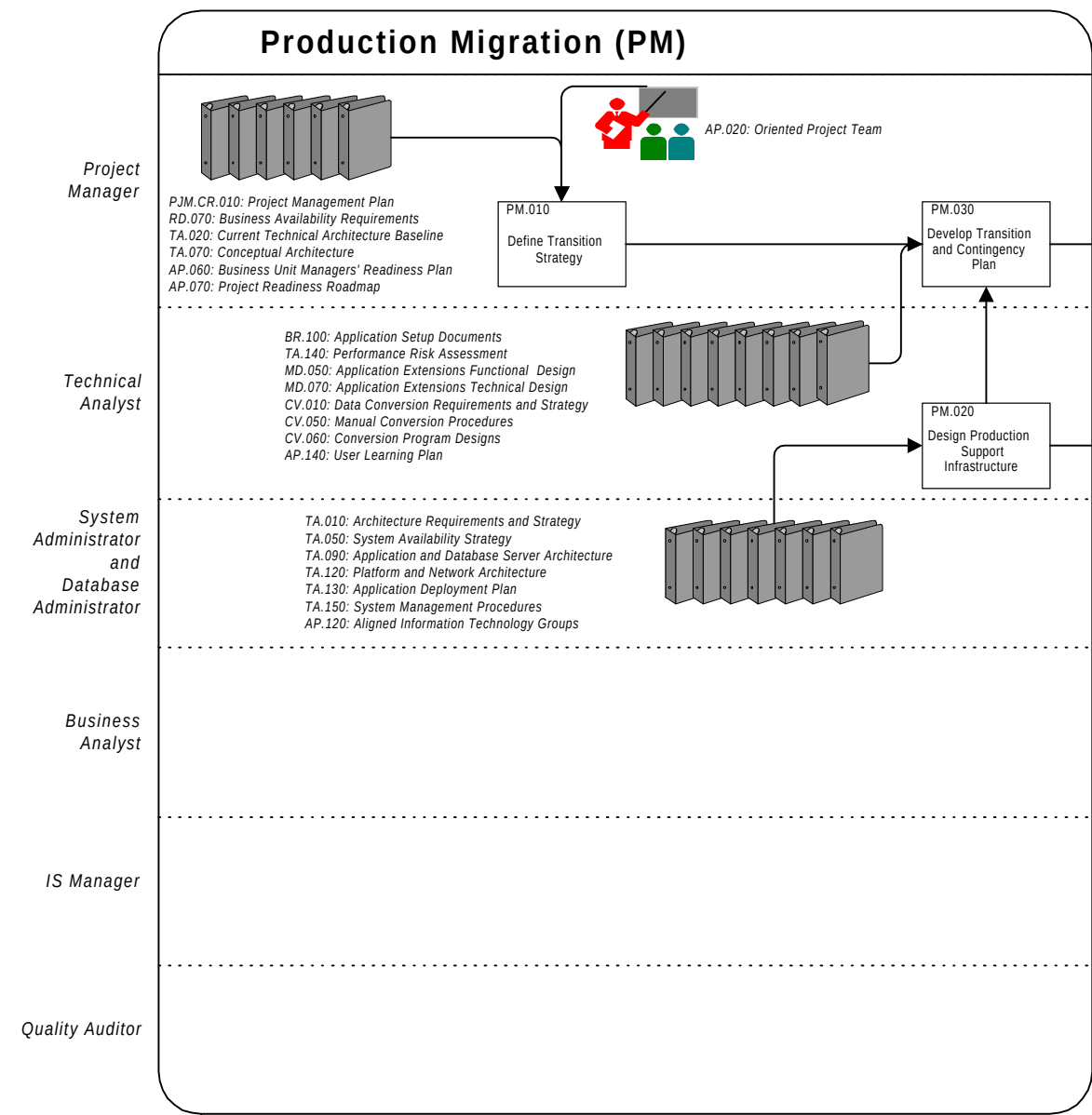


Figure 11-2 Production Migration Process Flow Diagram

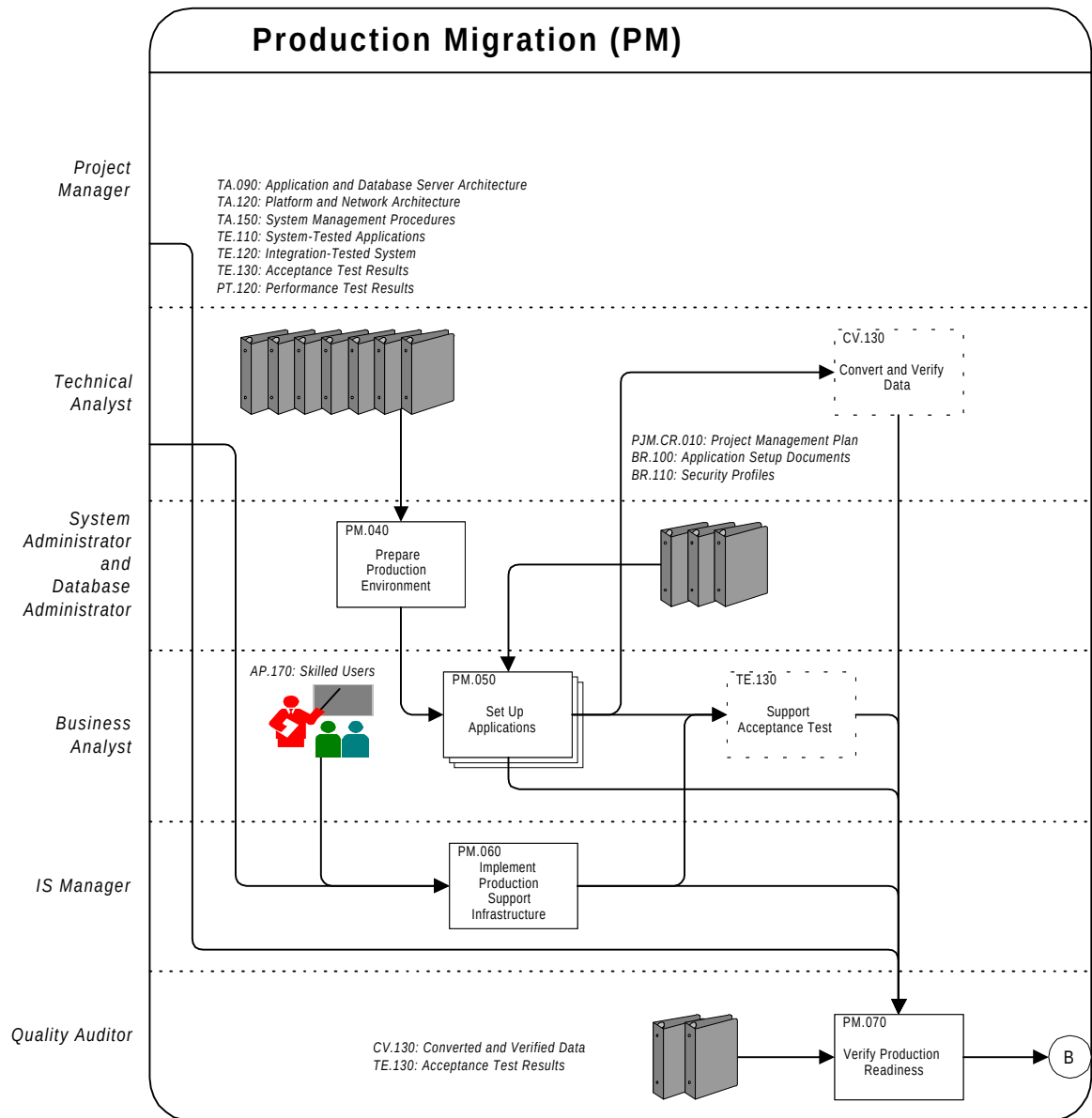


Figure 11-2 Production Migration Process Flow Diagram (cont.)

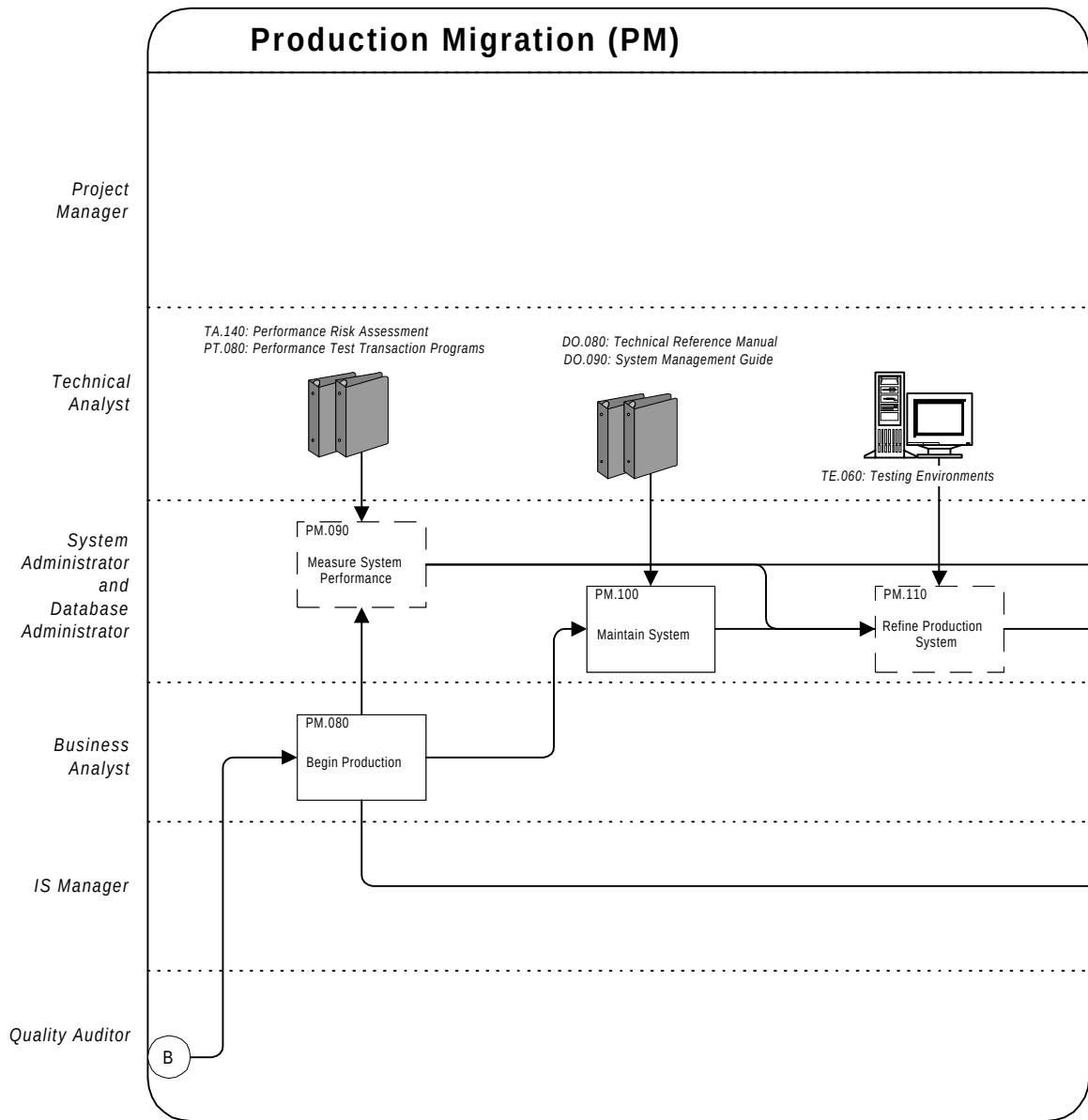


Figure 11-2 Production Migration Process Flow Diagram (cont.)

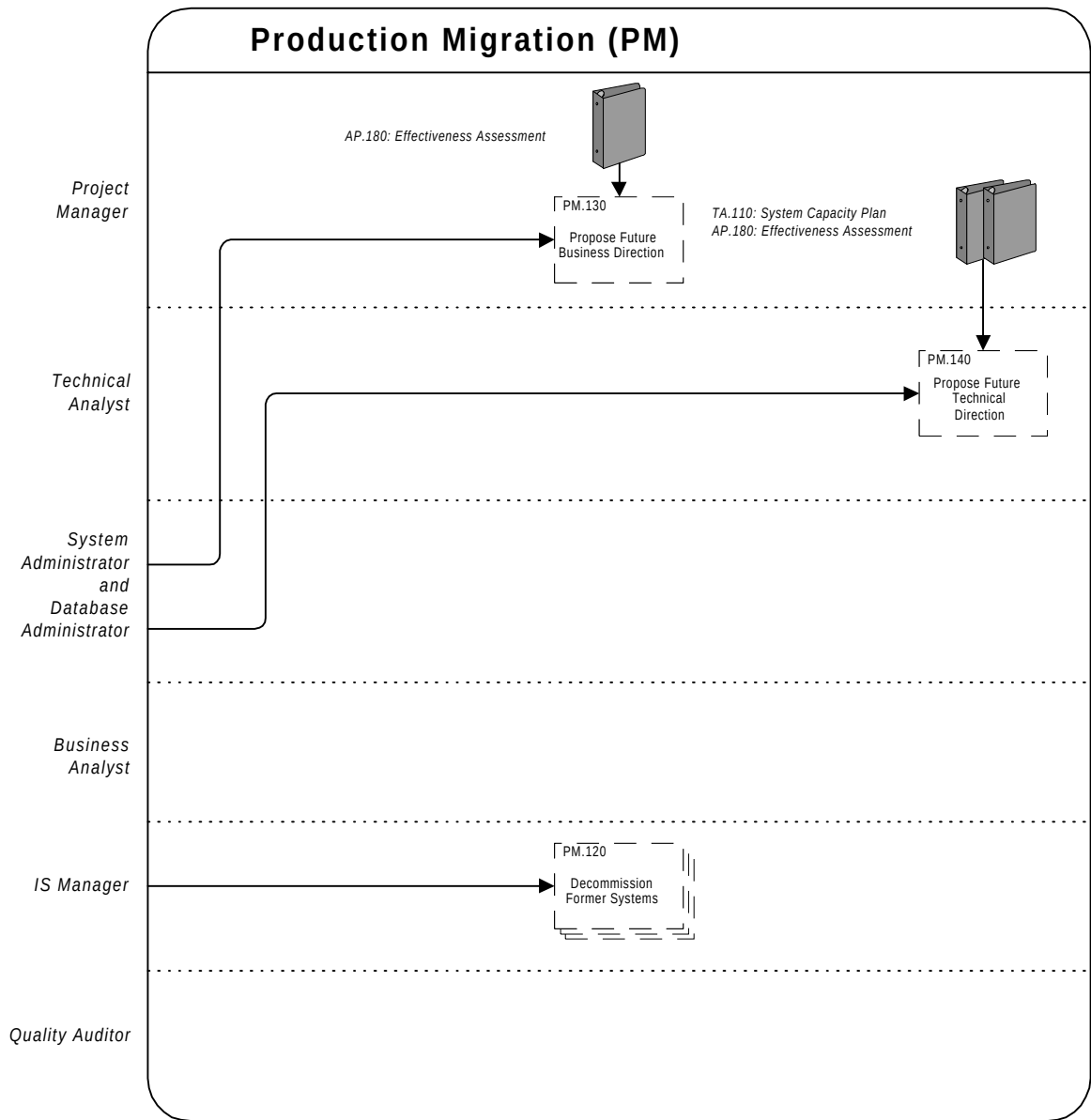


Figure 11-2 Production Migration Process Flow Diagram (cont.)

Approach

The objective of Production Migration is to migrate the organization, systems, and people to the new enterprise system. Following production cutover, additional objectives include monitoring and refining the production system and planning for the future.

Production Migration includes the following activities:

- assessing readiness for transition to production
- executing cutover to the new system
- conducting post-production support activities

Assessing Production Readiness

During Production Migration, the project team deploys the finished business system into the organization. This transition is dependent on Application and Technical Architecture (TA), Module Design and Build (MD), Data Conversion (CV), Documentation (DO), Business System Testing (TE), Performance Testing (PT), and Adoption and Learning (AP) for architecture designs, a fully tested business system, custom extensions, conversion programs, documentation, training (learning) materials, and organization communication services. Upon completion of transition, the current business data is converted and verified, and users start live production.

All elements of the implementation must come together in order to transition successfully to production status. The project team trains the users while the technical team configures the production environment and converts data from the legacy systems. In addition, the new roles and human performance enhancing systems are put into place. The transition aspect of Production Migration ends when the cutover to live production has been concluded. It is at this time that users must begin performing their work using new system functions and procedures.

Transition is a demanding experience for the project team, and in particular for the users who may have to maintain two systems until production readiness is achieved. As a result, managing change and buffering the organization from any negative impact — through contingency planning, for example — must be top priority. Careful

preparation and planning make the transition process easier and more risk free.

Production Cutover and Post-Production Support

Production status is obtained immediately after cutover to the new production system. Production cutover marks the end of the implementation life cycle and the beginning of the system support cycle. Your information systems personnel should work as quickly as possible to stabilize the system and begin a routine of normal maintenance and use.

Production Migration activities include all production and system support activities as well as some specific post-production tasks (such as refinement and measurement activities). After the production system has stabilized, you measure the business improvements against project objectives, the actual implementation metrics against planned metrics, and actual system performance against predicted system performance. System refinement must be controlled so that it minimizes the impact to users.

During Production Migration, you gather information about the organization's future business and technical direction and identify opportunities for continuous improvement.

Tasks and Deliverables

The tasks and deliverables of this process are as follows:

ID	Task Name	Deliverable Name	Required When	Type*
PM.010	Define Transition Strategy	Transition Strategy	Always	SI
PM.020	Design Production Support Infrastructure	Production Support Infrastructure Design	Always	SI
PM.030	Develop Transition and Contingency Plan	Transition and Contingency Plan	Always	SI
PM.040	Prepare Production Environment	Production Environment	Always	SI
PM.050	Set Up Applications	Configured Applications	Always	MI

ID	Task Name	Deliverable Name	Required When	Type*
PM.060	Implement Production Support Infrastructure	Production Support Infrastructure	Always	SI
PM.070	Verify Production Readiness	Production-Ready System	Always	SI
PM.080	Begin Production	Production System	Always	SI
PM.090	Measure System Performance	System Performance Assessment	Project includes <i>performance testing</i>	SI
PM.100	Maintain System	Maintained Production Environment	Always	SI
PM.110	Refine Production System	Refined Production Environment	Project includes <i>post-production system enhancement objectives</i>	SI
PM.120	Decommission Former Systems	Decommissioned Systems	Project includes <i>legacy system decommissioning objectives</i>	MI
PM.130	Propose Future Business Direction	Business Direction Recommendations	Project includes <i>post-production business process improvement or post-production system enhancement objectives</i>	SI
PM.140	Propose Future Technical Direction	Technical Direction Recommendations	Project includes <i>post-production system architecture enhancement or technology migration recommendation objectives</i>	SI

*Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Table 11-1 Production Migration Tasks and Deliverables

Objectives

The objectives of Production Migration are:

- Prepare the production environment according to the transition plan.
- Move to the production environment.
- Establish support for the production environment.
- Measure actual performance against expectations and plans.
- Refine and tune the system to reflect business process change.
- Determine future direction for business and technology opportunities.

Deliverables

The deliverables of this process are as follows:

Deliverable	Description
Transition Strategy	Documents the approach for migrating the people, organization, and system to production is outlined. It includes the transition plan, transition resources, cutover contingency plan, and transition support strategy.
Production Support Infrastructure Design	Documents the operational infrastructure for managing and maintaining the application environment, database, and network communications.
Transition and Contingency Plan	States the transition plan, implementation contingency alternatives, and former systems decommission plan.

Deliverable	Description
Production Environment	A working environment that is installed and prepared to receive final application setup and data conversion.
Configured Applications	Application setups that are applied to the Production Environment.
Production Support Infrastructure	The implemented operational infrastructure as documented in the Production Support Infrastructure Design (PM.020).
Production-Ready System	The system ready for production. Production readiness is verified using detailed checklists and optionally, a quality assurance review.
Production System	A fully operational system. Production status is declared following verification that all aspects of the system are operational.
System Performance Assessment	Documents that system transaction and reporting performance is quantified. Metrics are defined and applied to illustrate how well the system supports the technical needs of the business.
Maintained Production Environment	A working environment that is carefully maintained using procedures that focus on necessary and preventative actions, such as timing of maintenance procedures, reporting unusual occurrences, interpreting system indicators, logging completion of maintenance work, and adjusting the software and hardware for anticipated business changes.

Deliverable	Description
Refined Production Environment	A production system that has been refined through tuning, application setup changes, and other performance adjustments.
Decommissioned Systems	Discontinued former systems. These systems may be discontinued during or after cutover.
Business Direction Recommendations	Documents a plan for future business improvement opportunities.
Technical Direction Recommendations	Documents a plan for opportunities to improve or implement new technology.

Table 11-2 Production Migration Deliverables

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Business Analyst	Provide guidance in determining the future business direction of the organization.
Business Line Manager	Verify that users in each business area are trained and prepared to perform their jobs on the new system. Adjust the organizational structure (for example, roles and workflows) as needed.
Client Project Manager	Verify availability of user personnel, hardware, software, and facilities.

Role	Responsibility
Client Staff Member	Convert data, migrate custom extensions, and test the production database. Provide technical support during Production Migration activities. Evaluate existing support procedures, develop new procedures to reflect new support requirements, and incorporate new hardware and software suppliers' support offerings into the production support infrastructure.
Database Administrator	Install and configure the production database. Verify that all users have access to the database. Apply patches. Monitor and measure system performance. Provide support during Production Migration activities.
IS Manager	Participate in the planning, development, and execution of the infrastructure necessary to support the transition to and ongoing maintenance of the new system.
Network Administrator	Responsible for administering the network for the production environment, including configuration, maintenance, and performance monitoring.
Project Manager	Plan and estimate resources required to support Production Migration. Manage logistics and verify that the hardware and software are properly configured. Coordinate transition activities. Verify the availability of technical support, including system, applications, and database administrators.

Role	Responsibility
Quality Auditor	Participate in the overall review of the production readiness to determine the decision for going live with the new system.
System Administrator	Verify that application login IDs and application access responsibilities exist for all users. Create and maintain custom menus. Monitor and administer concurrent managers. Install and configure the hardware and operating system. Verify that operating system login IDs exist for all users. Provide support during Production Migration activities, including managing the user help desk and performing system maintenance.
System Architect	Provide the blueprint of the physical components of the database, hardware, and network for the production environment in support of the application architecture. Participate in the efforts to determine the future technical direction.
Technical Analyst	Provide direction on the use of new technologies to support the future technical direction of the organization.
Tester	Participate in the overall review of the production readiness to determine the decision for going live with the new system.

Role	Responsibility
Trainer	Participate in the development of Transition and Contingency Plan and review of the production readiness to determine the decision for going live with the new system.
User	Provide information on computer literacy level, roles performed, and availability. Perform job tasks on the new system. Report system problems to management and information systems personnel.

Table 11-3 Production Migration Key Responsibilities

Critical Success Factors

The critical success factors of Production Migration are as follows:

- organizational and process changes accepted and supported by all levels of the organization and senior-level managers
- demonstration by users that they can perform their jobs using the new documentation, tools, and systems
- data converted accurately and within the planned time frame
- installed and properly tested production hardware and software
- all application modules function as expected in support of business requirements
- agreement by all involved parties on the transition schedule
- proven measurable performance objectives for achieving high levels of service
- effective structure in place to provide timely response to user queries and requests
- availability of technical experts to tune the production environment

PM.010 - Define Transition Strategy (Core)

In this task, you outline the business transition approach for migrating the system, organization, and people to production status.

Deliverable

The deliverable for this task is the **Transition Strategy**. It addresses bringing transition resources and business systems online, contingency situations that must be anticipated and solved beforehand, the contingency plan and course of action, and support types and levels required during the production cutover.

Prerequisites

You need the following input for this task:

☐ **Project Management Plan [initial complete] (PJM.CR.010)**

A high-level strategy for production migration is introduced in the Project Management Plan.

☐ **Business Availability Requirements (RD.070)**

The Business Availability Requirements describe the method for supporting business contingency situations.

☐ **Current Technical Architecture Baseline (TA.020)**

The Current Technical Architecture Baseline provides an inventory of the systems, applications, hardware, and networks that constitute the existing technical infrastructure for the business. If Identify Current Technical Architecture Baseline was not performed, this deliverable will not exist. (See the task description for TA.020 for more information on when this task should be performed.)

☐ **Conceptual Architecture (TA.070)**

The complexity of the transition process is directly related to the complexity of the system architecture (number of modules, interfaces,

legacy interface programs, custom extensions, and so on). The Conceptual Architecture provides a high-level view of the future architecture to assist in preparing a strategy for transition. If Revise Conceptual Architecture was not performed, this deliverable will not exist. (See the task description for TA.070 for more information on when this task should be performed.)

 Oriented Project Team (AP.020)

Prior to defining the Transition Strategy, the project team members should attend the initial project team orientation.

 Business Unit Managers' Readiness Plan (AP.060)

The Business Unit Managers' Readiness Plan initiates discussions on the Transition Strategy that are picked up and finalized in this task. If Develop Business Unit Managers' Readiness Plan was not performed, this deliverable will not exist. (See the task description for AP.060 for more information on when this task should be performed.)

 Project Readiness Roadmap (AP.070)

The Project Readiness Roadmap defines organizational and implementation decisions that might impact the Transition Strategy.

Task Steps

The steps for this task are as follows:

<u>No.</u>	<u>Task Step</u>	<u>Deliverable Component</u>
1.	Review the proposed system architecture, locations, and resources within locations.	
2.	Determine the migration priority of applications and critical business functions and organizations.	Transition Plan

No.	Task Step	Deliverable Component
3.	Derive the resources and tools required in support of migrating business systems, software, and hardware.	Transition Resources
4.	Anticipate implementation contingency situations by reviewing similar implementation projects.	Cutover Contingency Plan
5.	Review the business contingency requirements component of the Business Availability Requirements (RD.070)	
6.	Define a Transition Support Strategy.	Transition Support Strategy
7.	Update the schedule to reflect the need-by dates for receiving and installing production hardware and software.	Workplan (PJM.WM.020)
8.	Review the draft deliverable with senior management and seek approval.	
9.	Identify any material changes to project scope and associated task estimates with the project manager.	

Table 11-4 Task Steps for Define Transition Strategy

Approach and Techniques

In order to complete the Transition Strategy, the following questions must be answered:

- What is the cutover approach?
- When should cutover occur?
- What is the time window for cutover?

What Is the Cutover Approach?

The two basic choices here are to either perform a clean cutover from existing legacy systems to the new systems, or to run the systems in parallel for a period of time, maintaining the existing system as a contingency, if needed. This question is not always so clear cut, since there may be a desire for a clean switch to the new systems, but the existing system may need to be kept online to maintain existing active data that will not be converted into the new system due to cost or time constraints. This would require users to enter transactions in the new system and the old system. There may also be issues in reporting or consolidating data across the systems.

When Should Cutover Occur?

The timing of the cutover to the new systems is usually determined early on in a project; cutover is one of the key milestones in any high-level project plan. The precise timing is determined by the competing factors of the desire to cutover as soon as possible, thereby minimizing disruption of business operations, and minimizing the need to perform cross-system transactions, reporting, and consolidation. The cutover milestone is usually at the end of financial quarter or year.

What Is the Time Window for Cutover?

The allowable time window for the production cutover is critical in determining what is or is not feasible and may influence the transition strategy profoundly. The time window usually needs to be kept to a minimum while considering the amount of risk that the business is willing to assume.

Other Organizations

Arrange site visits to other organizations that have recently moved to production status to get an external view of the transition process. It is preferable for these sites to be similar in size, hardware and software configuration, and industry. These visits set expectations regarding the importance of preparation and proper testing.

Linkage to Other Tasks

The Transition Strategy is an input to the following tasks:

- PM.030 - Develop Transition and Contingency Plan
- PM.120 - Decommission Former Systems

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Project Manager	90
System Architect	10
Client Project Manager	*

Table 11-5 Role Contribution for Define Transition Strategy

* Participation level not estimated.

Deliverable Guidelines

Use the Transition Strategy to consider all aspects of the project and organizations affected by transition to the new system.

This deliverable should address the following:

- project team resources
- human resources
- user community

- internal support
- external support
- hardware preparation
- network preparation
- software preparation
- system management
- learning
- business management
- logistics
- administration
- external customers
- suppliers
- partners

Deliverable Components

The Transition Strategy consists of the following components:

- Transition Plan
- Transition Resources
- Cutover Contingency Plan
- Transition Support Strategy

Transition Plan

This component identifies all of the needs of the project and operation prior to cutover, as well as a high-level plan containing the tasks required for transition of the system into production.

Transition Resources

This component lists the individuals (and their role types) that are needed during transition.

Cutover Contingency Plan

This component documents the contingency strategy, support staff procedures, and project team functions that need to be implemented if cutover has failed or been postponed.

Transition Support Strategy

This component lists the available support groups that should be available during transition.

Tools

Deliverable Template

Use the Transition Strategy template to create the deliverable for this task.

PM.020 - Design Production Support Infrastructure (Core)

In this task, you identify the operational infrastructure for managing and maintaining the application environment, servers, and network infrastructure in production.

Deliverable

The deliverable for this task is the **Production Support Infrastructure Design**. It addresses the operational infrastructure for managing and maintaining the application environment, servers, and network infrastructure.

Prerequisites

You need the following input for this task:

☐ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy addresses standards for hardware and networks, office automation, and system management.

☐ **System Availability Strategy (TA.050)**

The System Availability Strategy provides the level of system availability required by the business and defines approaches to handling system failures and scheduled downtimes.

☐ **Application and Database Server Architecture (TA.090)**

The Application and Database Server Architecture details the server architecture and configuration for the database, application and desktop client tiers. If Define Application and Database Server Architecture was not performed, this deliverable will not exist. (See the task description for TA.090 for more information on when this task should be performed.)

☐ **Platform and Network Architecture (TA.120)**

The complexity of the platform and network architecture determines the time and resources required to set up and maintain the production environment and must be taken into account when developing the Production Support Infrastructure Design. If Define Platform and Network Architecture was not performed, this deliverable will not exist. (See the task description for TA.120 for more information on when this task should be performed.)

☐ **Application Deployment Plan (TA.130)**

The Application Deployment Plan provides information on the basic framework of the new application system and details the deployment of applications across data centers. If Define Application Deployment Plan was not performed, this deliverable will not exist. (See the task description for TA.130 for more information on when this task should be performed.)

☐ **System Management Procedures (TA.150)**

The System Management Procedures provide information about the tools used to manage the new production systems.

☐ **Aligned Information Technology Groups (AP.120)**

Review the responsibilities of the system administrator and other information systems personnel in the new system as outlined in the Aligned Information Technology Groups. If Align Information Technology Groups was not performed, this deliverable will not exist. (See the task description for AP.120 for more information on when this task should be performed.)

☐ **Current Support Requirements (internal)**

Review the existing support procedures for the current systems environment.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Identify the support requirements for application, database, and network administration.	Production Support Infrastructure Plan
2.	Review software and hardware providers' support programs.	External Support Procedures
3.	Review internal support procedures.	Internal Support Procedures
4.	Develop user support levels, with references to manuals, system-provided support, and online support.	User Support Levels
5.	Update the Problem Report Log (PJM.CR.050).	Problem Report Log (PJM.CR.050)
6.	Identify support equipment and materials.	Production Support Infrastructure Plan
7.	Identify and categorize expected problems for internal and external support.	Problem Identification Guide
8.	Coordinate and plan additional equipment and support.	
9.	Plan production support learning events.	Production Support Infrastructure Plan

Table 11-6 Task Steps for Design Production Support Infrastructure

Approach and Techniques

The Production Support Infrastructure Design includes the human resources, facilities, and system reference materials needed to answer all user requests regarding the new system. Implement the support infrastructure in parallel with configuring the final production environment during production cutover.

Oracle Support Assessment (OSA)

The establishment of a robust and capable internal and vendor external support infrastructure is critical to the success of your project. Oracle Support provides a unique service to assess your current support capabilities and assist with planning, updating, and aligning these capabilities to Oracle's support infrastructure. This should coincide with the activities planned for this AIM task.



Suggestion: Contact your local Oracle Support representative to schedule an Oracle Support Assessment (OSA).

Support Process Diagrams

A useful technique to help clarify and communicate the support processes is to create diagrammatic flows of the processes in much the same way that the business requirements definition and mapping teams create process diagrams for the business and system processes implemented in the new system. These diagrams should show the owner (or agent) for each step in the process and should indicate the relationship between the various support organizations or individuals that may be called upon to help resolve an issue. The organizations may include:

- help desk / support staff
- client staff member
- hardware / network vendor support
- software vendor support

The support process diagrams need to show how issues are triaged and classified, and may have alternative resolution paths depending on severity. You also need escalation procedures.

Supplier Support

Hardware and software suppliers often have support lines available to organizations. Identify the level of support available based on the organization's support licenses and clearly understand the process that must be followed in order to use these services. Often support has some restrictions, such as limited calling hours or support for specific products only.

You should augment or complement these standard services by:

- extending support using independent consultants
- enlisting user personnel to support users outside of the standard support hours
- developing expertise within the using organization to provide more localized support

Self-Help Materials

As technology becomes increasingly deployed to the desktop, users can become more independent and take on additional responsibility for analyzing and correcting system problems. Custom documentation, web pages, portable documents (pdf), and online help facilities allow users to investigate the nature of the problems and also provide recommendations for problem resolution. If the standard suggestions have been followed and the problem still exists, users can turn to supplier or internal support channels. The key in using self-help materials is to effectively train the users early and have an easily accessible reference library.



Suggestion: Self-help materials should be written in a non-technical manner. The target audience is for the novice system user. Self-help instructions should include all details such as keystrokes, prompts, and other navigational paths.

Self-Help Web

You may want to create a web site to house materials frequently referenced by users. Putting frequently asked questions (FAQs), email support information, problem reporting forms, and other information useful to users on a web site ensures that those materials are readily available to all users.

Support Location and Access

Establish a central location for obtaining help. Publish when, where, and how to access support resources. The following information should be posted either in electronic mail or in the organization communication directory:

- support telephone numbers for suppliers and internal personnel
- hours of operation
- alternate resources
- instructions for online help
- information you need to provide support personnel
- standard classification of problems
- expected response time

Relationship to Information Systems Operations and System Management

This task is related to the tasks that describe the management of the new system by the information systems organization. The key difference is that the system management tasks relate to the back end of support processes (how the information systems organization manages the systems and resolves issues that are routed to them for assistance and resolution). The issues that are routed to the information systems organization should be technical in nature and may concern the real or perceived malfunctioning of some component of the new system.

The System Management Guide (DO.090) should help the client staff to resolve the technical issues and then communicate resolution back to the support help desk for closure. This task does not need to address the resolution of technical systems problems in detail — the resolution should be covered by the System Management Guide (DO.090). For more information, see Define System Management Procedures (TA.150) and Publish System Management Guide (DO.090).

Online Support Log

If the organization does not have an existing online support log, build or acquire one. The online support log assists in managing and tracking calls, and is especially useful in large organizations.

Linkage to Other Tasks

The Production Support Infrastructure Design is an input to the following tasks:

- DO.090 - Publish System Management Guide
- TE.100 - Prepare Key Users for Testing
- TE.110 - Perform System Test
- TE.120 - Perform Systems Integration Test
- AP.150 - Develop User Learningware
- PM.030 - Develop Transition and Contingency Plan
- PM.060 - Implement Production Support Infrastructure

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	100
Client Staff Member	*
IS Manager	*

Table 11-7 Role Contribution for Design Production Support Infrastructure

* Participation level not estimated.

Deliverable Guidelines

Use the Production Support Infrastructure Design to identify the need for additional, modified, or upgraded hardware platforms and production support software.

This deliverable should address the following:

- communication
- support lines
- video capability
- support platform
- test environment
- issue tracking system
- configuration management system

In addition, use the Production Support Infrastructure Design to document a help desk environment that begins with a key policy statement that addresses:

- types of products supported
- method for acquiring support (especially the technology that is to be employed)
- response time targets
- user self-help routes and methods
- window of support
- accessing reference materials

Be sure to get agreement at the steering committee level confirming the commitment of human and facilities resources to the system throughout all implementation activities and beyond.

Document the types of problems you expect, so that the proper support structure is defined. Problems may arise in the following areas:

- user access
- user profiles definitions
- application setup
- system error messages
- project documentation
- process flows

- user transaction error correction
- operating system errors
- printer controls

Deliverable Components

The Production Support Infrastructure Design consists of the following components:

- Production Support Infrastructure Plan
- External Support Procedures
- Internal Support Procedures
- User Support Levels
- Problem Identification Guide
- Production Cutover Notification to External Support

Production Support Infrastructure Plan

This component lists the production support requirements for the enterprise hardware and software, the production support staff, and the required equipment and materials needed to build and implement the production support infrastructure.

External Support Procedures

This component documents external support procedures. These procedures define the processes for interacting with all of the vendor and provider organizations that are associated with the software, platform hardware, network infrastructure and systems management for the new production system.

Internal Support Procedures

This component documents a step by step processes for obtaining internal support. Usually, users are directed to initiate their support activities with an internal or internal contract support organization, prior to escalation to external organizations and agencies.

User Support Levels

This component documents the various levels of user support that are available for the production support infrastructure.

Problem Identification Guide

This component lists the types of problems you may encounter and categorizes them (to help support personnel determine their root cause).

Production Cutover Notification to External Support

This component helps to facilitate a smooth production cutover and provide advance notification of the scheduled go-live date to Oracle Support.

Tools

Deliverable Template

Use the Production Support Infrastructure Design template to create the deliverable for this task.

Use the Problem Report Log (PJM.CR.050) template to log technical issues.

PM.030 - Develop Transition and Contingency Plan (Core)

In this task, you develop the detailed transition plan for moving onto the production system, as well as an implementation contingency plan.

Deliverable

The deliverable for this task is the **Transition and Contingency Plan**. It addresses the transition plan, implementation contingency alternatives, and former systems decommission plan.

Prerequisites

You need the following input for this task:

☐ **Project Management Plan [initial complete] (PJM.CR.010)**

A high-level strategy for overall production migration is introduced in the Project Management Plan.

☐ **Application Setup Documents (BR.100)**

The complexity of the application setups determines the time and resources required to configure the production applications and must be taken into account when developing the transition schedule.

☐ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the overall information systems policies and the standards for hardware and networks, office automation, and system management.

☐ **Conceptual Architecture (TA.070)**

The Conceptual Architecture provides a high-level view of the favored future application and technical architecture and indicates the approximate deployment of applications in this architecture model. If Revise Conceptual Architecture was not performed, this deliverable will not exist. (See the task description for TA.070 for more information on when this task should be performed.)

Application and Database Server Architecture (TA.090)

The Application and Database Server Architecture details the server architecture and configuration for the database, application and desktop client tiers. If Define Application and Database Server Architecture was not performed, this deliverable will not exist. (See the task description for TA.090 for more information on when this task should be performed.)

Platform and Network Architecture (TA.120)

The complexity of the platform and network architecture determines the time and resources required to setup and maintain the production environment and must be taken into account when developing the detailed transition schedule. If Define Platform and Network Architecture was not performed, this deliverable will not exist. (See the task description for TA.120 for more information on when this task should be performed.)

Application Deployment Plan (TA.130)

The Application Deployment Plan provides information on the basic framework of the new application system and details the deployment of applications across data centers. If Define Application Deployment Plan was not performed, this deliverable will not exist. (See the task description for TA.130 for more information on when this task should be performed.)

Performance Risk Assessment (TA.140)

The performance risk assessment results may determine the need to schedule performance tuning resources and allocate time in the transition schedule to accommodate tuning activities.

Application Extensions Functional Design (MD.050)

The Application Extensions Functional Design describes each customization and communicates all of the features provided by the customization in non-technical terms. If Create Application Extensions Functional Design was not performed, this deliverable will not exist. (See the task description for MD.050 for more information on when this task should be performed.)

Application Extensions Technical Design (MD.070)

The number and complexity of custom or modified application screens, programs, and interfaces must be taken into account when developing the transition schedule. The transition plan also includes the names and locations of all modification components, database extensions, and integration components. If Create Application Extensions Technical Design was not performed, this deliverable will not exist. (See the task description for MD.070 for more information on when this task should be performed.)

Data Conversion Requirements and Strategy (CV.010)

The Data Conversion Requirements and Strategy outlines the legacy business object programmatic conversions for inclusion in the transition plan. If Define Data Conversion Requirements and Strategy was not performed, this deliverable will not exist. (See the task description for CV.010 for more information on when this task should be performed.)

Manual Conversion Procedures (CV.050)

The manual conversion effort needs to be considered when developing the detailed transition activities. If Define Manual Conversion Procedures was not performed, this deliverable will not exist. (See the task description for CV.050 for more information on when this task should be performed.)

Conversion Program Designs (CV.060)

The number and complexity of conversion programs must be taken into account when developing the detailed transition schedule, so that the time it will take to execute the conversion programs can be incorporated into the Transition and Contingency Plan. The transition plan also includes the names and locations of all conversion programs and the order in which the conversion programs should be executed. If Design Conversion Programs was not performed, this deliverable will not exist. (See the task description for CV.060 for more information on when this task should be performed.)

User Learning Plan (AP.140)

A significant part of the transition process is conducting learning activities for users immediately prior to production cutover. The number of users to be skilled, conducting learning activities for

resources, and scheduling of learning events must be taken into account when planning transition.

☐ **Transition Strategy (PM.010)**

Transition and implementation contingency plans are discussed in the Transition Strategy.

☐ **Production Support Infrastructure Design (PM.020)**

Production Migration activities include the rollout of the support infrastructure to users.



Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the Transition Strategy (PM.010).	
2.	Review the Production Support Infrastructure Design (PM.020).	
3.	Update the transition plan to include name and location of modification components, database extensions, conversions programs, and integration components.	Detailed Transition Plan
4.	Review the schedule of learning events in the User Learning Plan (AP.140).	
5.	Identify any priorities and sequences for transition.	

No.	Task Step	Deliverable Component
6.	Review cutover contingency situations outlined in the cutover contingency plan component of the Transition Strategy (PM.010).	
7.	Define final implementation contingency procedures.	Cutover Contingency Plan
8.	Develop the initial schedule for production activities.	Production Schedule - First Week
9.	Review the draft deliverable with senior management and seek approval.	
10.	Identify any material changes to project scope and associated task estimates with the project manager.	

Table 11-8 Task Steps for Develop Transition and Contingency Plan

Approach and Techniques

Transition

The general sequence for transition is:

1. Configure the Production Environment.
2. Configure (set up) the applications.
3. Execute the Conversion Programs.
4. Register modifications.
5. Verify conversion data and application setups.

Plan the transition according to these guidelines:

- Set up the communications and computing hardware.
- Install the software applications, tools, and custom modules.
- Prepare the environments.
- Configure the application environments.
- Integrate the applications.
- Convert the data.
- Verify the data and application configuration.
- Run the reconciliation.

Support Infrastructure

Allocate time in the transition plan and schedule for system administrators or key project team members to implement the support infrastructure. This includes development and rollout of a help desk learning event for information systems support personnel, as well as users.

The establishment of a robust and capable internal and vendor external support infrastructure is critical to the success of your project. Oracle Support provides a unique service to assess your current support capabilities and assist with planning, updating, and aligning these capabilities to Oracle's support infrastructure. This should coincide with the activities planned for this AIM task.



Suggestion: Contact your local Oracle Support representative to schedule an Oracle Support Assessment (OSA).

Vendor Awareness of Cutover

The transition period may make extra demands on the support services of the hardware, network, and software vendors. By giving the vendors advance warning of the production cutover, you can alert them to possible extra demands on support services and make them aware of the critical nature of the business situation.

Execution Responsibilities

Process Teams

Separate functional and setup responsibilities by process team. Project team members that have responsibility for the design of new business processes will also be responsible for the execution and validation of the application configuration and converted data in the production system.

Process teams verify the setup for the applications and responsibilities, and the integration of custom modules, reports, and inquiry screens. If some data is to be manually converted, the process teams can either divide the data entry work or subcontract a temporary employee to enter the data. The latter case requires temporary staff learning events as well as an additional audit of the data by the process team members.

System Administration

The earlier that system administrators assume their new role in supporting the system, the better. You should prepare a checklist for all system administration responsibilities including form and program registration, custom menu definition, application user accounts, integration of non-Oracle applications, and descriptive and key flexfield setup. Create a system administration checklist for all the tasks and activities that involve execution of the production system.

Contingency Planning

Planning for contingencies in the migration process requires, as a first step, identification of the risks and failure points in the process. Having identified these areas, you can outline a high-level course of action for handling each eventuality. Because of the focus on risks and failure points, this task is tightly linked to Verify Production Readiness (PM.070). The detailed analysis that is used in Verify Production Readiness can also be used here, at a higher level, for identifying general risk areas. The identification of contingency measures may be unpopular because it has connotations of project failure and mishandling, but a more thorough contingency plan can potentially reduce the amount and degree of crisis management needed should the unthinkable happen. Do not confuse implementation contingency with system contingency failure.

Linkage to Other Tasks

The Transition and Contingency Plan is an input to the following tasks:

- CV.100 - Perform Conversion Business Object Tests
- CV.130 - Convert and Verify Data
- TE.130 - Perform Acceptance Test
- PM.070 - Verify Production Readiness
- PM.120 - Decommission Former Systems

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Project Manager	75
Trainer	15
Technical Analyst	10
Business Line Manager	*
Client Project Manager	*
Client Staff Member	*
IS Manager	*

Table 11-9 Role Contribution for Develop Transition and Contingency Plan

* Participation level not estimated.

Deliverable Guidelines

Use the Transition and Contingency Plan to efficiently orchestrate the migration of systems and people, and include all the worksteps necessary.

This deliverable should address the following:

- production support infrastructure service levels
- customization implementation status
- user learning status
- transition sequence dependencies
- contingency scenarios
- initial production schedule

Consider the following aspects of the implementation and the organization:

- **Environment** — host system, production, development, learning, and support
- **Software** — version, compatibility, integration of custom extensions
- **Users** — support, users, project team, management, and external staff
- **Other Hardware** — network, communications, personal computers, and peripherals
- **Organization** — new roles, responsibilities, and reporting structure
- **Learning Methods** — self, formal, and reinforcement
- **Data** — historical, transitional, and new setup
- **Forms** — preprinted external and internal forms (for example, payable checks)
- **Maintain** — documents and key deliverables

For each transition requirement, document all detailed procedures, their sequence, and the role responsible for execution of the task.

Consider possible process failures due to faulty designs and craft preventative measures and logical alternatives to avoid such situations. The contingency plan for the implementation must define what the organization will do in the event that the cutover is not successful. This task presents the following challenges:

- Will the current system still be available and fully operational?
- Will data be lost or damaged if the organization is forced to implement the contingency plan?
- What is the maximum amount of time the contingency plan can be used without jeopardizing the organization situation?
- What are the manual backup procedures?
- What elements of the system must be enabled to enforce the plan?
- Are users trained to recognize contingency situations and to activate corrective actions?

The standard AIM workplan already includes the transition-oriented Production Migration tasks, however, preparation of the Transition and Contingency Plan may highlight tasks that require additional time or resources. Update the project plan to reflect these changes.

Deliverable Components

The Transition and Contingency Plan consists of the following components:

- Detailed Transition Plan
- Cutover Contingency Plan
- Preparing for Production
- Production Schedule - First Week

Detailed Transition Plan

This component uses a series of tasks, or checklists, during transition of the new system into production.

Cutover Contingency Plan

This component documents the contingency strategy, support staff procedures, and project team functions that are needed to implement if cutover fails or is postponed.

Preparing for Production

This component provides the organization with a plan to schedule operations in a manner capable of support by the information technology and consulting organizations.

Production Schedule - First Week

This component provides a table for the first week production schedule. If no major problems arise, normal operational procedures can begin as early as the second day of production.

Tools

Deliverable Template

Use the Transition and Contingency Plan template to create the deliverable for this task.

PM.040 - Prepare Production Environment (Core)

In this task, you set up, configure, and install the database and application software for the Production Environment.

Deliverable

The deliverable for this task is the **Production Environment**. It is the installed production environment. The Production Environment is prepared to receive final application setup and data conversion.

Prerequisites

You need the following input for this task:

☐ **Application and Database Server Architecture (TA.090)**

The Application and Database Server Architecture details the server architecture and configuration for the database, application and desktop client tiers. If Define Application and Database Server Architecture was not performed, this deliverable will not exist. (See the task description for TA.090 for more information on when this task should be performed.)

☐ **Platform and Network Architecture (TA.120)**

The Platform and Network Architecture provides information about the hardware and network infrastructure of the new production systems. If Define Platform and Network Architecture was not performed, this deliverable will not exist. (See the task description for TA.120 for more information on when this task should be performed.)

☐ **System Management Procedures (TA.150)**

The System Management Procedures provide a list of the procedures and tools that client staff members use to manage and monitor the system.

☐ **System-Tested Applications (TE.110)**

The System-Tested Applications demonstrate the successful operation and integration of the business system processes.

☐ **Integration-Tested System (TE.120)**

If custom interfaces are included, the Integration-Tested System demonstrates the successful operation of interfaces between target applications and the legacy and third-party systems. If Perform Systems Integration Test was not performed, this deliverable will not exist. (See the task description for TE.120 for more information on when this task should be performed.)

☐ **Performance Test Results (PT.120)**

If the project includes performance testing, the performance test logs document the execution of the performance testing programs against the test database and measures the various performance metrics for individual transactions and system components. If Execute Performance Test was not performed, this deliverable will not exist. (See the task description for PT.120 for more information on when this task should be performed.)

☐ **Hardware and Software Delivered to Installation Site**

Confirm that all hardware and software (the latest certified versions of the applications and database) are on site and organized prior to the scheduled installation.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the architecture deliverables.	
2.	Prepare the production installation.	Oracle Installation QA Checklist

No.	Task Step	Deliverable Component
3.	Install the production servers.	Oracle Installation QA Checklist
4.	Install the operating system and networking software on the servers.	Oracle Installation QA Checklist
5.	Connect the servers to the network.	Oracle Installation QA Checklist
6.	Create user accounts on the servers.	Oracle Installation QA Checklist
7.	Configure the file system on the servers.	Oracle Installation QA Checklist
8.	Install the user machines and connect them to the network.	Oracle Installation QA Checklist
9.	Install the Oracle database on the database servers.	Oracle Installation QA Checklist
10.	Install the applications servers.	Oracle Installation QA Checklist
11.	Configure the standard applications database objects.	Oracle Installation QA Checklist
12.	Create the custom database objects.	Oracle Installation QA Checklist
13.	Install the custom modules.	Oracle Installation QA Checklist
14.	Start up the database and verify its operation.	Oracle Installation QA Checklist
15.	Install the system management tools.	Oracle Installation QA Checklist

No.	Task Step	Deliverable Component
16.	Install printers and other peripheral devices.	Oracle Installation QA Checklist
17.	Verify the production system installation.	Oracle Installation QA Checklist
18.	Update the Installation Plan and Record.	Installation Plan and Record (PJM.RM.050)
19.	Review the draft deliverable with senior management and seek approval.	

Table 11-10 Task Steps for Prepare Production Environment

Approach and Techniques

This task creates a Production Environment that is available to all potential system users, but cannot actually be used until production readiness is verified.

Technical Architecture

In projects where there are complex hardware and software requirements, designing and implementing the architecture begins during Application and Technical Architecture (TA). The implementation of the production environment is based on the assumption that the system architecture has been created and all necessary preparation for the installation of the applications and custom modules has been performed.

Timing Considerations in Implementing the Production Environment

Although Prepare Production Environment is part of Production Migration, the installation of the Production Environment may have taken place earlier in the implementation when other working environments were installed. This is common in cases where management selected, purchased, or sized the hardware prior to or during the database and application selection process. It is also often true for smaller organizations that can only provide a single machine to

support all project environments, including production. Assuming that the hardware configuration is adequate to support production requirements, this task then serves as a verification that the production environment is ready for production and that any custom modules have been installed. If the adequacy of the hardware configuration is in question, performance testing should be considered to identify performance risks and alternatives.

If performance testing has already been performed, portions of the Production Environment may have been installed and configured to support this test. This task then serves as an opportunity to apply any configuration changes necessary to reflect recommendations made in the Performance Test Report (PT.130).

This task represents the culmination of many tasks that contributed to the final application and technical architecture that must be implemented in the production system. The activities that comprise this process may span a wide time frame and be performed in parallel with tasks in other processes.

Custom Modules

When implementing custom modules, you will need to be meticulous about linking each of them into the production environment. The basic elements of installing custom modules follow:

- compatibility with application version
- submission into source control
- appropriate privileges (grants, synonyms, and so on)
- placement on standard or other custom menus
- registration of modules
- program controls

Linkage to Other Tasks

The Production Environment is an input to the following tasks:

- CV.120 - Install Conversion Programs
- CV.130 - Convert and Verify Data
- TE.130 - Perform Acceptance Test

- PM.050 - Set Up Applications
- PM.070 - Verify Production Readiness

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Administrator	50
Database Administrator	25
Network Administrator	15
System Architect	10
Client Staff Member	*

Table 11-11 Role Contribution for Prepare Production Environment

* Participation level not estimated.

Deliverable Guidelines

Use the Installation QA Checklist template to verify that a clean Production Environment is configured and available for subsequent applications setup, conversion, and other live data entry.

This deliverable should address the following

- menus, security, and database access
- custom application forms
- conversion forms and tables

You should also update the Installation Plan and Record (PJM.RM.050) located in the project infrastructure library. A complete and thorough log of the changes and updates to all project environments is important for maintaining the integrity and performance of the system.

Deliverable Components

The Installation QA Checklist template consists of the following component:

- Oracle Installation QA Checklist

Oracle Installation QA Checklist

This component is the starting point for validating the installation (the checklist includes the following sections: *Environment, Applications QA Checklist, Database QA Items, Tools and Standards Items, and Installation Audit and Control*).

Tools

Deliverable Template

Use the Installation QA Checklist template to create the supporting deliverable for this task.

PM.050 - Set Up Applications (Core)

In this task, you implement the required setups in all of the applications as part of your configuration.

Deliverable

The deliverable for this task is the **Configured Applications**. The Configured Applications are complete once the application setups are applied to the Production Environment (PM.040).

Prerequisites

You need the following input for this task:

☐ **Project Management Plan [initial complete] (PJM.CR.010)**

The Project Management Plan outlines the processes to be followed during verification of applications setups and seed data, including which project team members are responsible for quality assurance in each area.

☐ **Application Setup Documents (BR.100)**

The Application Setup Documents should reflect the most current, approved setup. Any approved changes to setups resulting from Business System Testing (TE) and Adoption and Learning (AP) activities should be incorporated into the document.

☐ **Security Profiles (BR.110)**

The Security Profiles identify the role and function information and define the user security required to support the new system in a controlled manner.

☐ **Production Environment (PM.040)**

The application code must be fully and correctly installed prior to beginning applications setups; this is part of configuring the Production Environment.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Set up the application users in the new system.	Application User Accounts
2.	Set up and compile the key flexfields.	
3.	Enter the application setups.	Configured Applications
4.	Load the initial setup data.	
5.	Load the setup data for application extensions.	
6.	Implement the security configuration.	
7.	Register the application objects.	
8.	Set up and compile the descriptive flexfields.	
9.	Enter or load custom seed data.	
10.	Verify the online operation of the applications.	
11.	Verify batch operation of the operations.	
12.	Back up the production system.	Production System Backups

Table 11-12 Task Steps for Set Up Applications

Approach and Techniques

Cutover and Count Down Steps

The term *cutover* refers to the final steps in transitioning from the former systems to the new system. The cutover execution steps include application setup and data conversion. The final step in cutover is the data and system validation.

Instance Synchronization

If you have maintained a separate *master* environment to capture the users' final, validated mapping decisions, you need to migrate the application setups from the controlled master environment into the Production Environment. The overriding concern is that the integrity of the setup data is preserved in the migration between the two environments. The mechanism for migrating the setup data should have been addressed when the project environment strategy was created, so at this point you should only need to put the mechanism into operation. The mechanism used may include:

- cold backup of the master setups database (the backup files are used to create the initial production database)
- extract of all relevant setup data from the master database and load into the production database
- use of automated test tools that trap keystrokes or graphical interface events (by replaying the scripts these tools generate, you can effectively automate the manual reentry of data)

In this situation, the applications setup task belongs to the database administrator or information systems operations and the responsibility for review of applications setup would fall to the analysts (depending on which role has been most involved in the definition of the applications setups throughout the life of the project). In the case where the environment that captures the final mapping eventually becomes the production environment, you should verify that this environment is appropriately sized and free of sample data.

Manual Application Setup

In some projects it may be feasible to manually enter the application setups into the production applications database. The Application Setup Documents (BR.100) produced during Business Requirements Mapping (BR) serve as a guide for entering setup data and define the correct setups for each application. Although you already have the application setup recorded, you should also capture the production setup after the data has been entered. This lets you compare the setup from expected to actual as a form of quality assurance. Use screen shots or standard reports as a means to make a record of production setup.

If you discover that some data should be altered in production, then check with the key process or module owners to determine whether a slight change in setup would impact other operations. In this situation, the applications setup task belongs to the business analysts who have been most involved in the definition of the applications setups throughout the life of the project.



Suggestion: Use Oracle's Application Implementation Wizard (AIW) as a guide to help you set up each application.

Setup and Seed Data Verification

The conversion data that follows relies heavily on the correct application setup data references in the system. Perform a thorough quality check on the application setups and seed data. Refer to the Project Management Plan (PJM.CR.010) for more detail on the quality assurance review process, procedures, and responsibilities.

Linkage to Other Tasks

The Configured Applications are an input to the following tasks:

- CV.130 - Convert and Verify Data
- PM.070 - Verify Production Readiness

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	70
System Administrator	25
Database Administrator	5
User	*

Table 11-13 Role Contribution for Set Up Applications

* Participation level not estimated.

Deliverable Guidelines

During this task, you enter the final application parameters and definitions resulting from completed design, mapping, testing, and adoption tasks. If prerequisite deliverables have been completed, this task is primarily one of data entry rather than design or analysis. Application setup loading is the movement of approved configuration decisions to the target production system.

In addition, enter static seed data (for example, validation tables and codes) and manually entered operating data (for example, master files) at this time. Application setup is a prerequisite to live system cutover. Follow the previously defined approach (in the Project Management Plan - PJM.CR.010) for auditing all data-entry sessions before considering this task complete.

When executing this task, consider the following (with regard to standard and custom setups):

- Enable parameters on the system, site, and application.
- Enter all additional definition data.
- Enable user security.

- Implement audit components.
- Enable concurrent manager.
- View entries when possible.
- Run standard reports.

This deliverable should address the following:

- final application setup parameters
- setup data entry (manual and automated)
- user security
- verification of application operation

Tools

Deliverable Template

A deliverable template is not provided for this task. Use the Application Setup Documents (BR.100) as source for manual setup entry.

Migration of Setup Data

The tools that you use for migrating the setup data depend on the strategy chosen for managing the project environments and controlling the setup data.

Creation of Production Database from Master Setup Instance

If the strategy calls for direct creation of the production database from the master setup database instance, you should use the standard cold backup mechanisms for disk file copying.



Reference: Velpuri, Rama, and Anand Adkoli. *Oracle8 Backup and Recovery Handbook*. Osborne McGraw-Hill, 1998. ISBN: 0078823897.

Setup Data Extract and Load

If the strategy calls for extracting the setup data from the master instance and loading it into the production instance, there are various

tools that you can use. When using these tools, you must be careful to export and import all of the tables that have foreign key relationships in order to preserve the integrity of the data you wish to use in the production database.

SQL and PL/SQL can be used to perform distributed transactions directly between database instances and can be used to select and insert between the databases.

Oracle Database Server has special import/export utilities for importing and exporting data out of tables in a database. These utilities can be useful for copying data between databases, rather than writing PL/SQL or SQL scripts specifically for this action, or manually reentering data.

Automated Test Tools

Automated test tools used for quality assurance testing of code often have the capability of trapping keystrokes or GUI events and then replaying them in another environment. This technique can be used to reenter setup data in the production instance in exactly the way that it was entered into the master setup instance. You should review the capabilities of the automated test tool with the tool vendor.

Application Implementation Wizard (AIW)

The Application Implementation Wizard (AIW) is Oracle's workflow-based setup tool. Use the wizard to guide you through the set up steps in the appropriate sequence.

PM.060 - Implement Production Support Infrastructure (Core)

In this task, you activate the support personnel and procedures for the new business system and review requirements for support-related services from software suppliers, contractors, help desks, and other support services.

Deliverable

The deliverable for this task is the **Production Support Infrastructure**. It involves implementing the operational infrastructure documented in the Production Support Infrastructure Design (PM.020).

Prerequisites

You need the following input for this task:

☐ **Skilled Users (AP.170)**

Skilled Users are involved in implementing the Production Support Infrastructure.

☐ **Production Support Infrastructure Design (PM.020)**

The Production Support Infrastructure Design outlines the operational infrastructure for supporting the application environment, database, and network communications in production.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Set up a library.	Library Index

No.	Task Step	Deliverable Component
2.	Introduce the help desk to the organization.	Help Desk Announcement
3.	Install communication equipment.	
4.	Set up and distribute reference materials.	
5.	Implement the online help text.	
6.	Implement the online issue tracking system.	
7.	Establish a change request and bug reporting procedure.	Change Request
8.	Establish online support.	Online Support Access
9.	Review procedures for getting help.	
10.	Distribute all support materials throughout the organization.	

Table 11-14 Task Steps for Implement Production Support Infrastructure

Approach and Techniques

Support Personnel Rehearsal

It is important to rehearse the support procedures. You can simulate various types of support calls and evaluate whether the expected response time, accuracy of the resolutions, and the general flow of the process are adequate to support the request volume. This is also a good opportunity to test default support mechanisms, after-business-hours support, and supplier support hotlines.

The establishment of a robust and capable internal and vendor external support infrastructure is critical to the success of your project. Oracle Support provides a unique service to assess your current support capabilities and assist with planning, updating, and aligning these capabilities to Oracle's support infrastructure. This should coincide with the activities planned for this AIM task.



Suggestion: Contact your local Oracle Support representative to schedule an Oracle Support Assessment (OSA).

Linkage to Other Tasks

The Production Support Infrastructure is an input to the following task:

- PM.070 - Verify Production Readiness

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Client Project Manager	*
Client Staff Member	*
IS Manager	*

Table 11-15 Role Contribution for Implement Production Support Infrastructure

* Participation level not estimated.

Deliverable Guidelines

Since this is an execution task, it should closely follow the Production Support Infrastructure Design (PM.020) which specifies its content and approach.

This deliverable should address the following:

- handling of inquiry calls
- call and response tracking
- system direction to all types of audiences
- links to external support mechanisms or sources
- understanding of performance metrics by support personnel (response, history retention, follow-up, accuracy, audit, and so on).
- backup resources and tools

Distribute support procedures to users, either during user learning events, or in a special meeting to introduce and review the support process and infrastructure. Documented support procedures should include the following:

- support telephone numbers for suppliers and internal personnel
- hours of operation
- alternate resources
- instructions for online help
- information you need to provide support personnel
- standard classification of problems
- expected response time

Tools

Deliverable Template

A deliverable template is not provided for this task.

PM.070 - Verify Production Readiness (Core)

In this task, you formally verify that the organization's systems, host facilities, local area networks (LANs), wide area networks (WANS), and people are prepared for production.

Deliverable

The deliverable for this task is the **Production-Ready System**. It consists of detailed checklists and optionally, a quality assurance review that allows you to verify production readiness.

Prerequisites

You need the following input for this task:

☐ **Converted and Verified Data (CV.130)**

Converted data should be validated prior to production cutover. If Convert and Verify Data was not performed, this deliverable will not exist. (See the task description for CV.130 for more information on when this task should be performed.)

☐ **Acceptance Test Results (TE.130)**

The Acceptance Test Results provide verification that the new application system meets the acceptance criteria specified in the Project Management Plan (PJM.CR.010).

☐ **Skilled Users (AP.170)**

Users should participate in the learning events and be confident that they can execute their job responsibilities at production cutover.

☐ **Transition and Contingency Plan (PM.030)**

The contingency alternatives and the transition schedule should be reviewed and understood by the entire organization prior to production cutover.

☐ **Production Environment (PM.040)**

The Production Environment should be set up, configured, and validated prior to production cutover.

☐ **Configured Applications (PM.050)**

The production environment applications should be configured with validated setups and seed data prior to production cutover.

☐ **Production Support Infrastructure (PM.060)**

The Production Support Infrastructure should be implemented prior to production cutover.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the production readiness verification checklist.	
2.	Conduct production readiness verification using the checklists provided in the Detailed Transition Plan component of the Transition and Contingency Plan (PM.030).	
3.	Confirm the production cutover.	
4.	Prepare the support team.	
5.	Obtain agreement for the initial production schedule.	
6.	Verify that users are trained.	

No.	Task Step	Deliverable Component
7.	Verify commitment and readiness of internal and external support personnel.	
8.	Verify the completion of production environment.	
9.	Confirm senior management commitment via the steering committee.	
10.	Distribute the initial production schedule.	
11.	Obtain approval for beginning production.	

Table 11-16 Task Steps for Verify Production Readiness

Approach and Techniques

Although the predecessor to this task is acceptance criteria, the true system certification occurs after Perform System Test (TE.110). Success with this task means that project leadership can request approval for transition activities. This is the final approval step before the cutover to the new production system.

User Readiness

For all business processes, users should demonstrate that they are qualified to fulfill the requirements of the process as well as each downstream process step. Presentations by users, also known as *walkthroughs*, can be used to demonstrate competency levels. Finally, users should confirm that they feel prepared and confident to move to the new system.

Notify Outside Agents and Customers

As part of the overall communication program outlined in the Communication Campaign (AP.080), contact partners and suppliers to let them know when production cutover will occur. Ask for additional support coverage for a set period of time. The organization management may also want to notify its key customers of the transition to a new system. In this way they can be prepared to deal with any issues, potential delays in service, or changes to procedures or business documents.

Production System Readiness

As another means to confirm that the system is stable, key users should go through the Production Environment setup and data to confirm that every form can be accessed and all reports print successfully. It is also a good opportunity to test user menus and verify that each user has the appropriate security access and responsibilities and that their user profiles are set up correctly. Verify that all business system test actions are closed.

Contingency Plan Verification

The Transition and Contingency Plan (PM.030) for implementing the production system was created prior to beginning the transition to production. It is important to review the Cutover Contingency Plan component with the information systems organization and, in addition, communicate the essence of the fallback plan to the users. In this case, if there is a production delay, an organization-wide announcement directing users to the Cutover Contingency Plan can be made without having to provide detailed explanations.

Initial Production Schedule

The initial production week should be carefully managed to control the organizational impact. You should provide additional control over the operational and support flow for all organizations during the initial week of production. Specific recommendations are as follows:

- Determine production schedule for financial, distribution, and manufacturing operations.
- Dedicate analysts, super users, or consultants to answer questions, fix problems, or provide real time learning events as needed.
- Provide a mechanism to get user feedback.
- Set support priorities to manage expectations.
- Update internal and external support channels of go-live date.

As outlined in the Communication Campaign (AP.080), distribute the initial production schedule to users and support personnel prior to production cutover. In the schedule, list the departments or processes that can begin transactions at a set point in time. This allows information systems personnel to provide full support to the specified departments or processes, thereby avoiding excessive support demands throughout the organization.

Linkage to Other Tasks

The Production-Ready System is an input to the following task:

- PM.080 - Begin Production

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Quality Auditor	45
Project Manager	20
Technical Analyst	15
Business Analyst	5
System Architect	5
Tester	5
Trainer	5
Client Project Manager	*
Client Staff Member	*
IS Manager	*
User	*

Table 11-17 Role Contribution for Verify Production Readiness

* Participation level not estimated.

Deliverable Guidelines

When you Verify Production Readiness, you confirm that the organization’s systems and people are prepared for production.

This deliverable should address the following:

- readiness verification
- quality audit or implementation review

Readiness Verification

The verification process uses the general and detailed task checklists from the Transition and Contingency Plan (PM.030). Readiness verification should be checked on two levels:

- business processes
- technical components

This is the final step before approving cutover. Verify that all business process owners have conveyed production status and plans to customers, suppliers, other business partners, and internal resources who will either use or be affected by the new system.

Business preparation should confirm the following:

- All system, reference, and manual procedures are distributed to all users.
- All system access, support infrastructure, and other necessary connections are available.
- All users are fully trained and prepared for the cutover.

Technical considerations include the following:

- database operating system
- software
- peripherals
- support personnel process
- reference materials
- contingency communication process

Quality Audit or Implementation Review

A quality audit or implementation review provides management with a final opportunity to prevent possible failure when critical issues are discovered during this task. The goal of the quality audit or implementation review is to identify risks and submit a candid and detailed summary of problems related to production cutover.

Implementation reviews can:

- Utilize a structured technical process to evaluate the application environment from a performance and application architecture perspective.
- Provide a functional evaluation of organization readiness and acceptance of the new system.
- Mitigate risk by identifying common transition problems before production cutover.
- Predict and plan for post-production support requirements.

It is important to realize that no project is risk free. Revealing risks to project management provides the project team with the opportunity to correct problems. During production, potential problems become real problems and business consequences become more apparent. Understanding the relative risk associated with each problem will help the organization's senior management make the ultimate decision whether or not to postpone production cutover.

Deliverable Components

Use the following components (from the Transition and Contingency Plan - PM.030) to support completion of the Production-Ready System:

- Detailed Transition Plan
- Preparing for Production
- Production Schedule - First Week

Detailed Transition Plan

This component uses a series of tasks, or checklists, during transition of the new system into production.

Preparing for Production

This component provides the organization with a plan to schedule operations in a manner capable of support by the information technology and consulting organizations.

Production Schedule - First Week

This component provides a table for the first week production schedule. If no major problems arise, normal operational procedures can begin as early as the second day of production.

Tools

Deliverable Template

A deliverable template is not provided for this task. Use the components from the Transition and Contingency Plan (PM.030) listed above to support task completion.

PM.080 - Begin Production (Core)

In this task, you confirm that organization-wide use of all aspects of the production system is in place.

Deliverable

The deliverable for this task is a fully operational **Production System**. It is a system in which you have verified that all aspects of the system are operational and production status is achieved.

Prerequisites

You need the following input for this task:

- ☐
- Production-Ready System (PM.070)**

Acceptance of the Production-Ready System allows you to proceed to production status.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Initiate using the production system.	
2.	Initiate incident / technical issue management procedures.	
3.	Initiate open issues and create a resolution list.	
4.	Initiate support.	

No.	Task Step	Deliverable Component
5.	Confirm that all components of the production system are operational.	
6.	Declare the new system live.	

Table 11-18 Task Steps for Begin Production

Approach and Techniques

This task covers a period of time during which the following conditions apply:

- The environment is carefully controlled.
- Transactions are brought online in a priority sequence (most critical first) to keep support requirements low and focused.
- Access to replaced systems is restricted.

The best way to begin using the new system is to control the initial transactions and entries. Bring up departments or sets of related transactions in sequence to permit focused attention on limited areas until they are stabilized. Use the Production Schedule - First Week component of the Transition and Contingency Plan (PM.030) to coordinate this support model. You must anticipate issues and quickly remedy them. Priorities may shift during the production state and therefore you must set expectations with users on what types of problems to resolve first.

Take advantage of the communications tools, defined in the Communication Campaign (AP.080), to make the entire organization aware that production status has been achieved.

Linkage to Other Tasks

The Production System is an input to the following tasks:

- AP.180 - Conduct Effectiveness Assessment
- PM.090 - Measure System Performance

- PM.100 - Maintain System
- PM.110 - Refine Production System
- PM.120 - Decommission Former Systems
- PM.130 - Propose Future Business Direction

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	60
Project Manager	40
Business Line Manager	*
Client Project Manager	*
Client Staff Member	*
User	*

Table 11-19 Role Contribution for Begin Production

* Participation level not estimated.

Deliverable Guidelines

Begin Production certifies that all aspects of the system are operational. Before declaring production status, you should consider the following:

- All reports can be generated, printed, and viewed online.
- All inquiries can be made from all established terminals.
- Users have access to and have used all respective functions and features of the business system.
- All automated backup and recovery programs operate according to schedule.

- All background programs perform as designed.
- No alternate support or business systems are required to run portions of the business.

This deliverable should address the following:

- production system operation
- technical issue management
- maintenance and contingency plan management

Tools

Deliverable Template

A deliverable template is not provided for this task. Use the Problem Report Log (PJM.CR.050) template to help you monitor resolution to any problems, issues, and action items resulting from initial production cutover.

PM.090 - Measure System Performance (Optional)

In this task, you measure the performance of the system in the Production Environment.



If your project includes *performance testing*, you should perform this task.

Deliverable

The deliverable for this task is the **System Performance Assessment**. It quantifies system transaction and reporting performance. In addition, metrics are defined and applied to illustrate how well the system supports the technical needs of the business.

Prerequisites

You need the following input for this task:

☐ **Project Management Plan [initial complete] (PJM.CR.010)**

The Project Management Plan may include key performance objectives of the implementation and the new system.

☐ **Performance Risk Assessment (TA.140)**

The Performance Risk Assessment highlights areas of potential performance risk based on transaction volumes, system resources, and staff technical skill levels.

☐ **Performance Test Transaction Programs (PT.080)**

If Performance Test Transaction Programs were created during Performance Testing (PT), it may be possible to reuse them to measure the performance of the production system in a separate environment, depending on how this task is to be conducted. If Create Performance Test Transaction Programs was not performed, this deliverable will not exist. (See the task description for PT.080 for more information on when this task should be performed.)

Production System (PM.080)

A system performance assessment is typically performed on a post-production live system.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Define the scope of production system performance measurements.	Scope
2.	Define the objectives of production system performance tests.	Objectives
3.	Identify performance measurements to be made on the production system.	Approach
4.	Identify new tools and scripts needed to gather performance metrics.	Approach
5.	Design and build new tools and scripts for making performance measurements.	
6.	Gather system performance statistics.	Results
7.	Analyze performance statistics and compare them to the original performance objectives.	Conclusions

No.	Task Step	Deliverable Component
8.	Summarize business and system performance results and recommendations.	Recommendations

Table 11-20 Task Steps for Measure System Performance

Approach and Techniques

Measuring the performance of the system in the production environment may include transactions processing performance, system response time, and system usage. The purpose of this task is twofold. First, by measuring the actual production system performance, closure can be achieved on the performance quality programs and methods employed during the implementation of the system. It is possible to assess the effectiveness of the programs and establish where continuing performance problems may have originated. Second, there may be continuing opportunities to refine and tune the performance of the system, which can only be identified by system metrics gathering.

Performance Testing

Performance Testing (PT) is also relevant to the measurement of the production system performance; the only difference is that during Measure System Performance you conduct the performance measurements and testing on the actual production system. When making performance measurements, there is a concern that the measuring process itself may disrupt the production operations; this may be true to some extent for continuous monitoring. Unfortunately, the Production Environment is the best place to make measurements on the real production system with all its complex, dynamic effects. The most valuable measurements are often those made with a fully loaded system — therefore, some compromise may be necessary.

If extra disk space and machine resources are available, you may be able to copy the production database and environment into a separate performance testing environment and conduct some of the testing where it does not impact production operations. However, this approach requires simulation of user transaction flows or actual user resources to perform transactions in the test environment. If an automated testing tool was used for the performance testing in the

implementation project, the test transaction programs may be of value to continue the testing of the production system, although they may require minor changes for the data set in the production database. The comparison of the performance testing results with the results from the Production Environment helps to establish the accuracy of the original performance test transaction models, and enable refinement of the models, if necessary.

Task Scope

There are many process and system performance metrics that you can measure in a complex application environment. A sample of areas to test include:

- individual batch program performance
- individual online form performance and response times
- individual report performance
- batch scheduling dynamics during various system scenarios
- end-to-end business process throughout
- system resource usage, including CPU, disk I/O, and memory
- identification of the ten most resource intensive SQL statements or programs
- network performance

Carefully consider the scope of this task so that the effort expended provides maximum performance benefit. If there were performance issues at the initial cutover to the new Production Environment, you may wish to focus on these areas during this task and attempt to fully resolve the issues.

Client staff members should be familiar with the new system; they may also be able to provide valuable input into the scope and objectives of this task.

This task may not be a discrete work effort, but instead could be the beginning of a production system performance management program. You can prioritize all of the performance areas to focus on and then address them gradually in a process on continuous system performance improvement. You can institute this program in concert with learning events for technical staff in the information systems organization who need to become system performance experts.

Linkage to Other Tasks

The System Performance Assessment is an input to the following tasks:

- PM.110 - Refine Production System
- PM.130 - Propose Future Business Direction

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Database Administrator	40
System Administrator	40
Technical Analyst	20

Table 11-21 Role Contribution for Measure System Performance

Deliverable Guidelines

Use the System Performance Assessment to quantify system transaction and reporting performance. In this task, you define and apply metrics to support how well the system supports the technical needs of the business.

This deliverable should address the following:

- objectives
- strategy for performance measurement
- participants required
- conditions for test
- findings
- conclusions
- gains accomplished
- improvement recommendations

Deliverable Components

The System Performance Assessment consists of the following components:

- Introduction
- Scope
- Objectives
- Approach
- Results
- Conclusions
- Recommendations

Introduction

This component documents the purpose and background of the System Performance Assessment and lists the system performance team members.

Scope

This component describes the scope of the task and the system performance measurements.

Objectives

This component identifies the performance objectives before the transition to production.

Approach

This component describes the approach taken to achieve the objectives of the system performance study.

Results

This component describes the performance results.

Conclusions

This component lists the conclusions suggested by the performance measurements and results.

Recommendations

This component documents the recommendations made by the system performance team, based on the results of the system performance and the interpretation outlined in the conclusions component.

Tools

Deliverable Template

Use the System Performance Assessment template to create the deliverable for this task.

Performance Statistics Gathering

System Activity Record (SAR)

This is a generic UNIX utility that collects low-level performance statistics over a period of time. This information can be used to identify disk and CPU bottlenecks and balance system load. For more information, consult your UNIX documentation. Similar activities exist in other operating systems as well.

Beginning Statistics, Ending Statistics, and Monitor

Beginning Statistics (BSTAT) and Ending Statistics (ESTAT) are standard Oracle utilities. BSTAT begins the performance monitoring process and ESTAT stops performance monitoring and stores the results in system tables. You can then run reports against the system tables that hold the performance data. Monitor (MON) allows you to view ongoing use of the database. For more information, consult your Oracle database documentation.

Trace

You can enable the Oracle trace facility for individual transactions that are performing poorly or for an entire database instance.

Trace captures performance statistics for individual SQL statements and can pinpoint bottlenecks related to specific tables and indexes.

Network Monitoring

A variety of tools are available for network monitoring. These tools measure network latency, capacity, and throughput (often yielding a succinct identification of network bottlenecks). Several examples are listed below:

- "snoop" utility for Solaris
- third-party tools like Expert Sniffer by Network Associates
- Microsoft Network Monitor for NT

Oracle Applications based on internet computing architecture work on low bandwidth networks but are sensitive to latency. Therefore an analysis of network traffic should be done between the client machine and the Forms server. In addition, network traffic should be monitored between the Forms server and the database server.



Suggestion: The Forms server and the database server should be linked by a Fiber Data Distributed Interface (FDDI) or a similar network scheme that supports a minimum of 100mbps.

The dual token-passing ring architecture of FDDI protects against downtime because the data always has two paths: a primary path, and a secondary path that acts as a backup for disaster recovery and fault tolerance. Since FDDI carries data traffic on dedicated circuits it provides you with superior data security and high performance.

PM.100 - Maintain System (Core)

In this task, you execute the system management procedures. This task is performed in perpetuity.

Deliverable

The deliverable for this task is the **Maintained Production Environment**. It addresses maintaining the Production Environment using procedures that focus on necessary and preventative actions, such as timing of maintenance procedures, reporting unusual occurrences, interpreting system indicators, logging completion of maintenance work, and adjusting the software and hardware for anticipated business changes.

Prerequisites

You need the following input for this task:

☐ **Technical Reference Manual (DO.080)**

The Technical Reference Manual explains the components and the design of the customizations and extensions. If Publish Technical Reference Manual was not performed, this deliverable will not exist. (See the task description for DO.080 for more information on when this task should be performed.)

☐ **System Management Guide (DO.090)**

The System Management Guide documents the procedures for maintaining the system. If Publish System Management Guide was not performed, this deliverable will not exist. (See the task description for DO.090 for more information on when this task should be performed.)

☐ **Production System (PM.080)**

System maintenance in this task is performed on an operational, post-production system.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Perform regular backups.	
2.	Periodically defragment table spaces.	
3.	Monitor and respond to database growth.	
4.	Purge report and system log files.	
5.	Apply software patches.	
6.	Perform regular version upgrades.	
7.	Implement user-driven changes.	
8.	Update the Installation Plan and Record for implemented changes.	Installation Plan and Record (PJM.RM.050)
9.	Update the system configuration document to reflect architectural changes.	

Table 11-22 Task Steps for Maintain System

Approach and Techniques

Create a system or process to effectively maintain the production system and respond to users' requests.

Routine Maintenance

Routine maintenance is performed by the information systems operations group. Routine maintenance includes:

- running batch reports
- running extracts
- logging file purges
- enlarging table space
- monitoring table space defragmentation
- running hot backups
- running cold backups
- performing security audits
- runaway process management
- performance tuning
- archiving and purging data

On Demand Maintenance

On demand maintenance includes unplanned or requested activities such as the following:

- setting up new users
- changing new user access
- correcting interface data
- correcting data due to user error

Upgrade Maintenance

Software and hardware upgrades are also considered maintenance. The amount of effort required depends on whether the upgrade approach is characterized as one of the following types:

- **Basic** — base software upgrade; retrofit misaligned customizations
- **Minimal** — base software upgrade; retrofit misaligned customizations; remove customizations eclipsed by new features
- **Comprehensive** — base software upgrade; retrofit misaligned customizations; remove customizations eclipsed by new features; reengineer business process to take advantage of new features

Oracle Application Upgrades

Use Oracle's EasiPath Migration Method (EMM) to upgrade to newer releases of the Oracle Applications. EMM has been specifically tailored to address the following challenges associated with upgrading Oracle's Applications from one release to another:

- justifying and planning upgrades
- reviewing business requirements
- updating mapping requirements
- migrating the application and technical architecture
- migrating custom extensions
- migrating data
- upgrading documentation
- testing the business system
- updating and conducting learning events
- migrating the production environment

Migration versus Implementation

EMM is specifically designed for application migrations (upgrades). Prerequisite tasks in the migration method depend on output deliverables from the initial implementation. Some tasks in the migration method are required (core) and some are optional. For

example, migrating the infrastructure to support a new release is mandatory; defining new business processes is optional.

The table below summarizes the major differences between implementations and migrations.

Project Characteristic	Implementation	Migration
Starting Point	Legacy or manual system	Existing production Oracle Application system
Expectation	New and substantially different system	Essentially the same system with improvements
Time	Usually 6 to 36 months	Usually 3 to 9 months
Cost	Variable, depending on scope	Usually 5 to 50 percent of implementations
Risk	Relatively high	Moderate, relative to implementations
Staffing	Substantial applications, business, and technical expertise required for a relatively long period of time	Fewer people and less advanced expertise unless migration involves implementing new application features and technology
Estimating	Driven by information requirements for business functions	Driven by changes from current to new Oracle Application release
Scope Determination	Based on business requirements	Wider range of needs from simple, such as Year 2000 compliance, to complex

Table 11-23 Differences Between Implementations and Migrations

Linkage to Other Tasks

The Maintained Production Environment is an input to the following task:

- PM.110 - Refine Production System

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Database Administrator	50
System Administrator	30
Network Administrator	20
Client Staff Member	*

Table 11-24 Role Contribution for Maintain System

* Participation level not estimated.

Deliverable Guidelines

The ongoing system maintenance begins during this task. Maintenance is typically not triggered by technical issues that the support or help desk address; this type of maintenance focuses on necessary and preventative actions.

This deliverable should address the following:

- timing of maintenance procedures
- reporting unusual occurrences
- interpreting system indicators
- logging completion of maintenance work
- adjusting the software and hardware for anticipated business changes

Verify that all normal, ongoing activities keep the system protected, purged, and operating as efficiently as possible. Make sure the Installation Plan and Record (PJM.RM.050) updated during Prepare Production Environment (PM.040) is available for recording all change events. Standard checklists can be incorporated for normal, standing

tasks that must be performed (like backups and defragmentation of table space). In addition, use the Installation Plan and Record (PJM.RM.050) as a record when new users are added to the system.

Tools

Deliverable Template

A deliverable template is not provided for this task. Maintain the Installation Plan and Record (PJM.RM.050) that you updated in Production Environment (PM.040) to reflect the changes resulting from production system maintenance.

EMM Advantage

EasiPath Migration Method (EMM Advantage) is available to help you structure and manage your upgrade projects. EMM is a tested framework for quality migrations. It provides you with a comprehensive toolkit which also includes Project Management Method (PJM), Oracle's standardized approach to project management. The EMM Advantage toolkit is available from the Oracle Direct Marketing group in your country (or you can contact your local Oracle sales representative).

PM.110 - Refine Production System (Optional)

In this task, you solicit user feedback and act upon any requests relative to the implementation, production system, or support.



If your project includes *post-production system enhancement objectives*, you should perform this task.

Deliverable

The deliverable for this task is the **Refined Production Environment**. It is the production system refined through tuning, application setup changes, and other performance adjustments.

Prerequisites

You need the following input for this task:



Testing Environments (TE.060)

Refresh the test environment periodically to help make sure that refinements to the system can be tested in a controlled and realistic environment.



Production System (PM.080)

System refinement is typically performed on an operational, post-production system.



System Performance Assessment (PM.090)

The System Performance Assessment can identify opportunities to refine and tune the performance of the system, which can only be identified by system metrics gathering. If Measure System Performance was not performed, this deliverable will not exist. (See the task description for PM.090 for more information on when this task should be performed.)

Maintained Production Environment (PM.100)

The Maintained Production Environment represents the most current process for maintaining the production system.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Conduct user feedback on the system.	User Feedback Questionnaire
2.	Interview key users in areas where issues have been raised.	
3.	Identify issues and their source.	
4.	Evaluate issues and develop alternatives.	
5.	Assign bug fixes, new process definition, and enhancements to the project team.	Post-Implementation Issues List
6.	Design and build customizations.	Custom Programs
7.	Test customizations.	Tested Customizations
8.	Obtain user approval.	Acceptance Certificate
9.	Implement production system changes.	Refined Production Environment
10.	Update appropriate maintenance logs.	Updated Maintenance Logs

No.	Task Step	Deliverable Component
11.	Update reference documentation.	User Reference Manual (DO.060)

Table 11-25 Task Steps for Refine Production System

Approach and Techniques

Test Environment Refresh

Always keep the test environment synchronized with the application setup in production. You should refresh the test environment periodically so that refinements to the system are tested in a virtual Production Environment.

Requests Types

Users will provide a myriad of system refinement requests. You should be able to discern what is a *must have* versus *nice to have* requirement. You should also evaluate whether the requirements can be addressed by providing additional learning events or adjusting the current process without making any program changes. The types of requests you should be prepared for are:

- additional application definitions
- modified application setup
- report tuning and enhancement
- form tuning and enhancement
- program tuning and enhancement
- profile option adjustments
- responsibility or menu security
- processing time adjustments
- procedural changes
- control changes
- interface error or interruption

- update report printing priority
- apply support patches
- implement support recommendations

Linkage to Other Tasks

The Refined Production Environment is an input to the following task:

- PM.140 - Propose Future Technical Direction

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Administrator	25
Business Analyst	20
Database Administrator	20
Technical Analyst	20
Network Administrator	15
Client Staff Member	*

Table 11-26 Role Contribution for Refine Production System

* Participation level not estimated.

Deliverable Guidelines

Solicit feedback from system users and analyze recommendations for improvement during this task. Distinguish system *bug* corrections from functionality or performance enhancements as you prioritize the work.

This deliverable should address the following:

- type of feedback
- urgency of request
- impact on current business
- recommendations to resolve the root problem
- costs and benefits of implementing the recommendations

Create a formal mechanism for recording lessons learned and compile a list of future actions. Categorize information gathered into these areas for evaluation:

- interface with system components
- business processes
- support

Initiate system evaluation sessions and document all feedback from users. You should try to translate subjective complaints and criticisms into action-oriented discussions that relate to business requirements. Ask questions that refer to time and volume as much as possible; for example:

- What is the average response time to gain resolution to reported problems?
- What is the volume of transactions or requests by type during a process cycle?
- How is information accessed (views, reports, and so on)?

Remember that a number of factors influence response to enhancement requests (including management priority, budget, technology limitations, skill set levels, and so on).

Tools

Deliverable Template

A deliverable template is not provided for this task.

PM.120 - Decommission Former Systems (Optional)

In this task, you involve the full analysis and execution of decommissioning former business systems.



If your project includes *legacy system decommissioning objectives*, you should perform this task.

Deliverable

The deliverable for this task is the **Decommissioned Systems**. This deliverable involves discontinuing the use of the former systems.

Prerequisites

You need the following input for this task:

☐ **Transition Strategy (PM.010)**

The Transition Strategy provides information about the strategy for the transition from legacy to new systems.

☐ **Transition and Contingency Plan (PM.030)**

The Transition and Contingency Plan provides detailed information about the transition plan including the plan for the decommissioning of existing systems.

☐ **Production System (PM.080)**

The new Production System must be operational and stable before the former systems are decommissioned.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Determine financial close write-off requirements.	
2.	Determine whether to void open balances.	
3.	Determine whether to carry over final material transactions.	
4.	Determine the need to change former systems security to read-only access.	
5.	Determine whether the former systems require concurrent existence with production.	
6.	Determine archiving requirements.	
7.	Determine the need for data warehousing.	
8.	Determine the need for historical data hard copy, microfiche, and so on.	
9.	Define a decommissioning action plan.	Decommission Former Systems Action Plan
10.	Execute the decommissioning action plan.	

No.	Task Step	Deliverable Component
11.	Introduce reference data procedures to users.	Data Retrieval Procedure
12.	Perform permanent removal of former systems hardware and software (if necessary).	
13.	Test the data reference and retrieval process.	

Table 11-27 Task Steps for Decommission Former Systems

Approach and Techniques

Decommissioning former systems can begin after the team has decided that the new production environment is stable, and that the Transition and Contingency Plan (PM.030) is no longer needed.

Decommission Strategy

Usually, the subject of decommissioning former systems is raised during Define Conversion Requirements and Strategy (CV.010). When preparing the Data Conversion Requirements and Strategy (CV.010), you should decide which legacy systems will be decommissioned and the timing of when each system will be decommissioned. Legacy systems that will be decommissioned at the time the new applications go into production will directly impact Production Migration and specifically the Transition and Contingency Plan (PM.030).

Legacy systems that will be decommissioned using a phased approach may increase the complexity of the conversion process due to additional interface points that need to be built and maintained to keep the legacy system and new applications synchronized. You also need to take into consideration master data identification and synchronization requirements. Even though the former systems may be decommissioned over more than one year, you should document a detailed plan for performing the decommissioning.

Decommissioning of Former Systems Timing

Data conversion usually defines a stop and end point for transactions. It is common to find that the production date does not correspond to a fiscal period. Legal requirements may require some data to exist in the former systems. Therefore, the timing of the decommission activity depends on when the need to report on data crossing over two systems (in this case, a fiscal period) no longer exists.

Transition Plan

Create an action plan to remove the former systems. The original transition plan may need to be revised after the production cutover. You should revisit and revise this action plan, expanding on the steps to remove the former systems, if necessary. You may want to perform this task in stages — although online transaction capabilities should be removed, history inquiries may still be useful until sufficient histories are built up in the new system.

Some considerations for decommissioning the former systems include:

- reliance on and access to historical data not converted over in the new system
- returning rented, contracted, or licensed hardware or software
- physically removing equipment and closing all connections to the former systems
- disposing of obsolete documentation
- cleaning out system space occupied by the former application
- transitioning key users supporting the old system

Data Archiving

Storing

If you do not want to maintain historical data in the former systems beyond the already converted data, use tape or other media to store historical data.

Retrieving Data Test

Before completing this task, you should perform a test of stored data retrieval. If third-party storage has been contracted, follow their

procedures for retrieving data and determine how efficient it is to obtain information whenever there is an unexpected need. This may help you decide whether to convert and maintain a longer period of history on the new system. You may find that long delays in data retrieval can cause serious operational problems downstream.

Linkage to Other Tasks

The Decommissioned Systems are not a prerequisite to any subsequent tasks.

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Client Staff Member	*
IS Manager	*

Table 11-28 Role Contribution for Decommission Former Systems

* Participation level not estimated.

Deliverable Guidelines

This task affects housekeeping more than business operations because the use of the former systems is often discontinued at cutover time. Execution of this task does, however, provide a good control point for acknowledging the success and stability of the new system.

This deliverable should address the following:

- financial closure for legacy system balances
- data archiving requirements
- decommission planning and execution
- reallocation of resources, such as personnel, hardware, and networks

Tools

Deliverable Template

A deliverable template is not provided for this task.

PM.130 - Propose Future Business Direction (Optional)

In this task, you identify opportunities and challenges for the organization and propose a future business direction for the new implementation.



If your project includes *post-production business process improvement* or *post-production system enhancement objectives*, you should perform this task.

Deliverable

The deliverable for this task is the **Business Direction Recommendations**. These recommendations identify a plan to address future business improvement opportunities.

Prerequisites

You need the following input for this task:

 Platform and Network Architecture (TA.120)

The Platform and Network Architecture deliverable describes the deployment of the key hardware and network components of the new system and their relationship to the application and database architecture. If Define Platform and Network Architecture was not performed, this deliverable will not exist. (See the task description for TA.120 for more information on when this task should be performed.)

 Performance Test Results (PT.120)

The Performance Test Results are documented in the performance test log, which records the test date and time, specific test or configuration parameters, as well as the individual measurements made. If Execute Performance Test was not performed, this deliverable will not exist. (See the task description for PT.120 for more information on when this task should be performed.)

Effectiveness Assessment (AP.180)

The Effectiveness Assessment provides evaluation business performance data that can influence the Business Direction Recommendations.

Production System (PM.080)

The Production System must be fully operational prior to proposing future business direction.

System Performance Assessment (PM.090)

The System Performance Assessment is used to quantify system transaction and reporting performance. If Measure System Performance was not performed, this deliverable will not exist . (See the task description for PM.090 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the open issues list.	
2.	Examine industry trends.	Future Business Direction
3.	Consider enhancements and change requests that were considered out of scope during Module Design and Build (MD).	Business Process Enhancements
4.	Propose the future business direction.	Future Business Direction
5.	Recommend outside resources support requirements for the future business direction.	Future Business Direction

No.	Task Step	Deliverable Component
6.	Summarize recommendations.	Executive Summary

Table 11-29 Task Steps for Propose Future Business Direction

Approach and Techniques

Future Enhancements Collection During the Implementation

The content for the Business Direction Recommendations can be collected in parallel with Technical Direction Recommendations (PM.140) throughout the implementation — especially during Business Requirements Definition (RD) and Module Design and Build (MD). Luxury or wish list features may take an extraordinary amount of project team time to produce. These are captured in a separate document to be considered for a subsequent implementation, and should be categorized so that management can determine the urgency of need, estimate the effort required, and justify the enhancements.

Future Applications Releases Upgrade

Looking forward, you should consider the new features planned for the applications. Consultants should be involved with this process not only for the new features, but as a regular application implementation. If you have insight with regard to what the next release of the applications provide, then you should recommend or highlight these as a justification to upgrade.

Oracle's upgrade method, EasiPath Migration Method (EMM), was developed expressly to assist organizations in moving from one Oracle Applications release to another. Based on the collective experience of Oracle consultants worldwide, EMM embodies leading practices specific to Oracle Applications migrations to help accelerate upgrade time frames and maintain high standards of quality. EMM helps you organize, perform, and control your upgrade project using a four-phase, step-by-step approach that guides you through the upgrade cycle.



Web Site: You can find additional information on Oracle Corporation's EasiPath Migration Method (EMM) at <http://www.oracle.com>

Industry Trends

You may be in an industry that has undergone some dramatic changes. If so, then it is likely that the future business direction must be in line with these changes or be able to absorb them. For example, if there is an outsourcing trend in forward distribution, then you might want to consider the benefits and costs of implementing such a program. Collecting industry trends can be based on obtaining general knowledge or a detailed analysis of books and periodicals.

Linkage to Other Tasks

The Business Direction Recommendations are not a prerequisite to any subsequent tasks.

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Project Manager	50
Business Analyst	25
Technical Analyst	25
Client Project Manger	*

Table 11-30 Role Contribution for Propose Future Business Direction

* Participation level not estimated.

Deliverable Guidelines

Record enhancement requests in the Business Direction Recommendations during the implementation. Concentrate primarily on future initiatives that have relevant system impact and focus on areas that have high perceived business impact. These requests, along with management's vision of future processes provide invaluable information for this deliverable.

This deliverable should address the following:

- audience
- priority
- timing
- cost and benefit
- applicability to current system
- industry trends
- ability to change in the future

Use this opportunity to influence future business systems and logistics initiatives. If possible, estimate the cost of each proposed enhancement, along with a statement of benefit.

Deliverable Components

The Business Direction Recommendations consist of the following components:

- Executive Summary
- Future Business Direction
- Business Process Enhancements

Executive Summary

This component documents recommendations gathered by the project team during the business process audit or improvement assessment.

Future Business Direction

This component documents the strategy and next steps of future improvement projects.

Business Process Enhancements

This component documents non-critical requirements expressed during the implementation.

Tools

Deliverable Template

Use the Business Direction Recommendations template to create the deliverable for this task.

PM.140 - Propose Future Technical Direction (Optional)

In this task, you identify innovations and advancements in software and technology that can enhance the efficiency, functionality, distributed nature, performance, and cost of maintaining the new production environment.



If your project includes *post-production system architecture enhancement or technology migration recommendation objectives*, you should perform this task.

Deliverable

The deliverable for this task is the **Technical Direction Recommendations**. These recommendations identify a plan to address opportunities to improve or implement new technology.

Prerequisites

You need the following input for this task:



Architecture Requirements and Strategy (TA.010)

The Architecture Requirements and Strategy provides information about the strategy used for architecting the new systems and may mention the enabling features or constraining influences of the technology selected. In addition, it may mention whether the current architecture is a step towards a technical or information systems vision.



System Capacity Plan (TA.110)

The System Capacity Plan provides information about the projected system capacity requirements for processing and data for some period into the future. If there are architecture upgrade implications because of a predicted shortfall in capacity at some point in the future, this document may address this situation and provide some recommendations for the future.

☐ **Effectiveness Assessment (AP.180)**

The Effectiveness Assessment provides evaluation technical performance data that can influence the Technical Direction Recommendations.

☐ **Refined Production Environment (PM.110)**

The production system should be stable and tuned to support current loads before you begin considering technical enhancements. If Refine Production System was not performed, this deliverable will not exist. (See the task description for PM.110 for more information on when this task should be performed.)

☐ **Existing Information Systems Strategy Documents**

If information systems architecture or strategy documents exist external to the current project, they may provide information about strategic business systems directions.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the detailed technical architecture document.	
2.	Examine industry trends and new technology.	
3.	Identify architectural challenges.	
4.	Identify tool, operating system, and hardware changes.	
5.	Determine version compatibility requirements.	

No.	Task Step	Deliverable Component
6.	Assemble custom upgrade considerations from mapping and detailed design documents.	Future Technology Direction
7.	Identify possible future enhancements.	Technology Enhancements
8.	Identify components of the current system that will eventually become obsolete.	Future Technology Direction
9.	Propose future technical direction.	Future Technology Direction
10.	Define supplier support involvement with technology.	Future Technology Direction
11.	Summarize recommendations.	Executive Summary

Table 11-31 Task Steps for Propose Future Technical Direction

Approach and Techniques

Assess the behavior of the new production systems and begin thinking about how the system can be improved. There may be an opportunity to implement architectural enhancements or limited-impact additional systems that bolt-on to the core systems just implemented. A single newly implemented system may be the first stage in a wider distributed implementation rollout and the implementation of additional national or international sites may bring new requirements for the technical components of the systems. Finally, new software releases from packaged application vendors may contain the enhancements that were designed and built as customizations in the current system and there may be a drive to implement the vendor-supported system as quickly as possible.

New Technology

Technology changes rapidly — by the end of the implementation you should be thinking about new products or newer versions of hardware or software that can impact the business.

Multiple Site Needs

If this is the first implementation in a series of multiple-site implementations, you should consider the technical requirements of bringing other remote sites live on the application system and the impact this may have on the architecture and technical requirements.

Linkage to Other Tasks

The Technical Direction Recommendations are not a prerequisite to any other tasks.

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	40
Technical Analyst	40
Project Manager	20
Client Project Manager	*
IS Manager	*

Table 11-32 Role Contribution for Propose Future Technical Direction

* Participation level not estimated.

Deliverable Guidelines

Use the Technical Direction Recommendations to document innovative technologies that may provide the organization with a competitive or transformational edge. Examples of these technologies include:

- EDI-facilitated partnerships with customers and suppliers
- web-based intranet and internet systems
- electronic commerce and storefronts
- data warehousing and OLAP systems
- cooperative distributed systems
- modular/virtual production environments based on quick access to shared information

This deliverable should address the following:

- audience
- priority
- timing
- cost and benefit
- stability
- support structure
- applicability to the current system

Deliverable Components

The Technical Direction Recommendations consist of the following components:

- Executive Summary
- Future Technology Direction
- Technology Enhancements

Executive Summary

This component documents recommendations gathered by the project team during the business process audit or improvement assessment.

Future Technology Direction

This component documents the strategy and next steps of future improvement projects.

Technology Enhancements

This component documents enhancements identified during the project's life-cycle.

Tools

Deliverable Template

Use the Technical Direction Recommendations template to create the deliverable for this task.

A

Task Cross-Reference

Task names and IDs are different in AIM version 2.0 than in the current version 3.0. This appendix provides an index which cross-references the AIM tasks in version 2.0 and version 3.0. Two tables list the cross-referenced tasks:

- AIM Version 2.0 Tasks Cross-Referenced with AIM Version 3.0 Tasks
- AIM Version 3.0 Tasks Cross-Referenced with AIM Version 2.0 Tasks

AIM Version 2.0 Tasks Cross-Referenced with AIM Version 3.0 Tasks

This table lists by process the AIM version 2.0 tasks with the corresponding AIM version 3.0 tasks.

V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase	V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase
BR.010	Prepare Mapping Environment	Operations Analysis	BR.020	Prepare Mapping Environment	Operations Analysis
BR.020	Map Business Requirements	Operation Analysis	BR.030	Map Business Requirements	Operations Analysis
BR.030	Map Business Data	Operations Analysis	BR.040	Map Business Data	Operations Analysis
BR.040	Conduct Integration Fit Analysis	Operations Analysis	BR.050	Conduct Integration Fit Analysis	Operations Analysis
BR.050	Develop Information Flow Model	Operations Analysis	BR.060	Create Information Model	Operations Analysis
BR.060	Develop Information Access Model	Operations Analysis	BR.060	Create Information Model	Operations Analysis
BR.070	Conduct Reporting Fit Analysis	Operations Analysis	BR.070	Conduct Reporting Fit Analysis	Operations Analysis
BR.080	Test Business Solutions	Operations Analysis	BR.080	Test Business Solutions	Operations Analysis
BR.090	Confirm Integrated Business Solutions	Operations Analysis	BR.090	Confirm Integrated Business Solutions	Operations Analysis
BR.100	Create Process Narratives	Solution Design	BP.090	Document Business Procedures	Solution Design
BR.110	Define Application Setups	Solution Design	BR.100	Define Application Setups	Solution Design
BR.120	Design Security Profiles	Solution Design	BR.110	Define Security Profiles	Solution Design
CM.010	Define Configuration Management Strategies, Standards, and Procedures	Definition	CM.010	Define Configuration Management Strategies, Standards, and Procedures	Definition
CM.060	Audit Key Deliverables	Definition Operations Analysis Solution Design Build Transition Production	CM.060	Audit Key Deliverables	Definition Operations Analysis Solution Design Build Transition Production
CR.010	Establish Scope, Objectives, and Approach	Definition	CR.010	Establish Scope, Objectives, and Approach	Definition
CR.020	Define Control and Reporting Strategies, Standards, and Procedures	Definition	CR.020	Define Control and Reporting Strategies, Standards, and Procedures	Definition
CR.030	Establish Management Plans	Definition	CR.030	Establish Management Plans	Definition

V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase	V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase
CR.080	Secure Client Phase Acceptance	Definition Operations Analysis Solution Design Build Transition Production	CR.080	Secure Client Phase Acceptance	Definition Operations Analysis Solution Design Build Transition Production
CV.010	Define Conversion Scope, Objectives, and Approach	Definition	CV.010	Define Data Conversion Requirements and Strategy	Definition
CV.020	Prepare Conversion Strategy	Definition	CV.010	Define Data Conversion Requirements and Strategy	Definition
CV.030	Prepare Conversion Standards	Operations Analysis	CV.020	Define Conversion Standards	Solution Design
CV.040	Prepare Conversion Environment	Solution Design	CV.030	Prepare Conversion Environment	Solution Design
CV.050	Perform Conversion Data Mapping	Solution Design	CV.040	Perform Conversion Data Mapping	Solution Design
CV.060	Design Manual Conversion Strategy	Solution Design	CV.050	Define Manual Conversion Procedures	Solution Design
CV.070	Design Conversion Programs	Solution Design	CV.060	Design Conversion Programs	Solution Design
CV.080	Prepare Conversion Test Plans	Solution Design	CV.070	Prepare Conversion Test Plans	Solution Design
CV.090	Develop Conversion Programs	Build	CV.080	Develop Conversion Programs	Build
CV.100	Perform Conversion Unit Test	Build	CV.090	Perform Conversion Unit Tests	Build
CV.110	Perform Conversion Business Object Test	Build	CV.100	Perform Conversion Business Objects Tests	Build
CV.120	Perform Conversion Validation Test	Build	CV.110	Perform Conversion Validation Tests	Build
CV.130	Install Conversion Software	Transition	CV.120	Install Conversion Programs	Transition
CV.140	Convert and Verify Data	Transition	CV.130	Convert and Verify Data	Transition
DO.010	Prepare Glossary	Definition	DO.030	Prepare Glossary	Definition
DO.020	Specify Documentation Requirements	Definition	DO.010	Define Documentation Requirements and Strategy	Definition
DO.030	Determine Documentation Standards and Procedures	Definition	DO.020	Define Documentation Standards and Procedures	Definition
DO.040	Prepare Documentation Environment	Operations Analysis	DO.040	Prepare Documentation Environment	Operations Analysis
DO.050	Produce Documentation Prototypes and Templates	Operations Analysis	DO.050	Produce Documentation Prototypes and Templates	Operations Analysis
DO.060	Produce Initial User Reference Manual	Solution Design	DO.060	Publish User Reference Manual	Build

V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase	V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase
DO.070	Produce Initial User Guide	Solution Design	DO.070	Publish User Guide	Build
DO.080	Produce Initial Technical Reference Manual	Solution Design	DO.080	Publish Technical Reference Manual	Build
DO.090	Produce Initial System Management Guide	Solution Design	DO.090	Publish System Management Guide	Build
DO.100	Complete User Reference Manual	Build	DO.060	Publish User Reference Manual	Build
DO.110	Complete User Guide	Build	DO.070	Publish User Guide	Build
DO.120	Complete Technical Reference Manual	Build	DO.080	Publish Technical Reference Manual	Build
DO.130	Complete System Management Guide	Build	DO.090	Publish System Management Guide	Build
DO.140	Provide Online Help Text	Build	This task is not included in Version 3.0.		
MD.010	Prepare Customization Strategy	Definition	MD.010	Define Application Extension Strategy	Definition
MD.020	Define and Estimate Custom Extensions	Operations Analysis	MD.020	Define and Estimate Application Extensions	Operations Analysis
MD.030	Define Design Standards	Solution Design	MD.030	Define Design Standards	Solution Design
MD.040	Define Build Standards	Solution Design	MD.040	Define Build Standards	Solution Design
MD.050	Design Database Extensions	Solution Design	MD.060	Design Database Extensions	Solution Design
MD.060	Produce Module Functional Design	Solutions Design	MD.050	Create Application Extensions Functional Design	Solution Design
MD.070	Produce Module Technical Design	Solution Design	MD.070	Create Application Extensions Technical Design	Solution Design
MD.080	Review Detailed Designs	Solution Design	MD.080	Review Functional and Technical Designs	Solution Design
MD.090	Prepare Development Environment	Build	MD.090	Prepare Development Environment	Build
MD.100	Implement Database Extensions	Build	MD.100	Create Database Extensions	Build
MD.110	Create Custom Modules	Build	MD.110	Create Application Extension Modules	Build
MD.120	Create Installation Routines	Build	MD.120	Create Installation Routines	Build
PM.010	Prepare Transition Strategy	Operations Analysis	PM.010	Define Transition Strategy	Operations Analysis
PM.020	Design Production Support Infrastructure	Solution Design	PM.020	Design Production Support Infrastructure	Build
PM.030	Develop Detailed Transition and Contingency Plan	Solution Design	PM.030	Develop Transition and Contingency Plan	Build
PM.040	Prepare Production Environment	Build	PM.040	Prepare Production Environment	Transition
PM.050	Set Up Applications	Transition	PM.050	Set Up Applications	Transition

V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase	V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase
PM.060	Implement Support Infrastructure	Transition	PM.060	Implement Production Support Infrastructure	Transition
PM.070	Verify Production Readiness	Transition	PM.070	Verify Production Readiness	Transition
PM.080	Commence Production	Transition	PM.080	Begin Production	Transition
PM.090	Audit Production System	Production	AP.180	Conduct Effectiveness Assessment	Production
PM.100	Measure System Performance	Production	PM.090	Measure System Performance	Production
PM.110	Maintain System	Production	PM.100	Maintain System	Production
PM.120	Refine Production System	Production	PM.110	Refine Production System	Production
PM.130	Decommission Former System	Production	PM.120	Decommission Former System	Production
PM.140	Propose Future Business Direction	Production	PM.130	Propose Future Business Direction	Production
PM.150	Propose Future Technical Direction	Production	PM.140	Propose Future Technical Direction	Production
PT.010	Define Performance Testing Scope, Objectives, and Approach	Definition	PT.010	Define Performance Testing Strategy	Definition
PT.020	Prepare Performance Testing Strategy	Definition	PT.010	Define Performance Testing Strategy	Definition
PT.030	Identify Performance Test Scenarios	Operations Analysis	PT.020	Identify Performance Test Scenarios	Operations Analysis
PT.040	Identify Performance Test Transaction Models	Operations Analysis	PT.030	Identify Performance Test Transaction Models	Operations Analysis
PT.050	Create Performance Test Scripts	Solution Design	PT.040	Create Performance Test Scripts	Solution Design
PT.060	Design Performance Test Transaction Programs	Solution Design	PT.050	Design Performance Test Transaction Programs	Solution Design
PT.070	Develop Performance Test Transaction Programs	Build	PT.080	Create Performance Test Transaction Programs	Build
PT.080	Design Performance Test Data	Solution Design	PT.060	Design Performance Test Data	Solution Design
PT.090	Design Test Database Load Programs	Solution Design	PT.070	Design Test Database Load Programs	Solution Design
PT.100	Develop Test Database Load Programs	Build	PT.090	Create Test Database Load Programs	Build
PT.110	Construct Performance Test Database	Build	PT.100	Construct Performance Test Database	Build
PT.120	Prepare Performance Test Environment	Build	PT.110	Prepare Performance Test Environment	Build
PT.130	Execute Performance Test	Build	PT.120	Execute Performance Test	Build
PT.140	Create Performance Test Report	Build	PT.130	Create Performance Test Report	Build

V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase	V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase
QM.010	Define Quality Management Strategies, Standards, and Procedures	Definition	QM.010	Define Quality Management Strategies, Standards, and Procedures	Definition
QM.050	Perform Quality Assessment	Definition Operations Analysis Solution Design Build Transition Production	QM.050	Perform Quality Assessment	Definition Operations Analysis Solution Design Build Transition Production
RD.010	Identify Financial and Operating Structure	Definition	RD.010	Identify Current Financial and Operating Structure	Definition
RD.020	Conduct Current Business Baseline	Definition	RD.020	Conduct Current Business Baseline	Definition
			BP.040	Develop Current Process Model	Definition
RD.030	Develop Future Process Model	Definition	BP.080	Develop Future Process Model	Operations Analysis
RD.040	Develop Future Business Function Model	Definition	BP.080	Develop Future Process Model	Operations Analysis
RD.050	Establish Process and Mapping Summary	Definition	RD.030	Establish Process and Mapping Summary	Operations Analysis
RD.060	Gather Business Volumes	Definition	RD.040	Gather Business Volumes and Metrics	Operations Analysis
RD.070	Create Business Requirements Scenarios	Operations Analysis	RD.050	Gather Business Requirements	Operations Analysis
RD.080	Determine Audit and Control Requirements	Operations Analysis	RD.060	Determine Audit and Control Requirements	Operations Analysis
RD.090	Identify Business Availability Requirements	Operations Analysis	RD.070	Identify Business Availability Requirements	Operations Analysis
RD.100	Develop Reporting Requirements	Operations Analysis	RD.080	Identify Reporting and Information Access Requirements	Operations Analysis
RM.010	Define Resource management Strategies, Standards, and Procedures	Definition	RM.010	Define Resource management Strategies, Standards, and Procedures	Definition
RM.020	Establish Staffing and Organization Plan	Definition	RM.020	Establish Staffing and Organization Plan	Definition
RM.030	Implement Organization	Definition	RM.030	Implement Organization	Definition
RM.040	Establish Physical Resource Plan	Definition	RM.040	Establish Physical Resource Plan	Definition
RM.050	Establish Infrastructure	Definition	RM.050	Establish Infrastructure	Definition

V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase	V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase
RM.080	Release Staff	Definition Operations Analysis Solution Design Build Transition Production	RM.080	Release Staff	Definition Operations Analysis Solution Design Build Transition Production
RM.090	Release Physical Resources	Definition Operations Analysis Solution Design Build Transition Production	RM.090	Release Physical Resources	Definition Operations Analysis Solution Design Build Transition Production
TA.010	Define Architecture Scope, Objectives, and Approach	Definition	TA.010	Define Architecture Requirements and Strategy	Definition
TA.020	Prepare Architecture Strategy	Definition	TA.010	Define Architecture Requirements and Strategy	Definition
TA.030	Establish Architecture Requirements	Definition	TA.010	Define Architecture Requirements and Strategy	Definition
TA.040	Develop Conceptual Architecture	Definition	TA.030	Develop Preliminary Conceptual Architecture	Definition
TA.050	Conduct Technical Architecture Baseline	Definition	TA.020	Identify Current Technical Architecture	Definition
TA.060	Develop System Availability Strategy	Operations Analysis	TA.050	Develop System Availability Strategy	Operations Analysis
TA.070	Define Future Application Deployment	Operations Analysis	TA.130	Define Application Deployment Plan	Build
TA.080	Develop Reporting Strategy	Operations Analysis	TA.060	Define Reporting and Information Access Strategy	Operations Analysis
TA.090	Revise Conceptual Architecture	Operations Analysis	TA.070	Revise Conceptual Architecture	Operations Analysis
TA.100	Define Architecture Subsystems	Operations Analysis	TA.100	Define and Propose Architecture Subsystems	Build
TA.110	Propose Architecture Subprojects	Operations Analysis	TA.100	Define and Propose Architecture Subsystems	Build
TA.120	Design Application Security Architecture	Solution Design	TA.080	Define Application Security Architecture	Solution Design
TA.130	Design Application Functional Architecture	Solution Design	TA.040	Define Application Architecture	Operations Analysis
TA.140	Design Logical Application and Database Architecture	Solution Design	TA.090	Define Application and Database Server Architecture	Build
TA.150	Design Physical Database Architecture	Solution Design	TA.090	Define Application and Database Server Architecture	Build

V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase	V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase
TA.160	Design Hardware and Network Architecture	Solution Design	TA.120	Define Platform and Network Architecture	Build
TA.170	Develop System Capacity Plan	Solution Design	TA.110	Define System Capacity Plan	Build
TA.180	Assess Performance Risks	Solution Design	TA.140	Assess Performance Risks	Build
TA.190	Design System Management Procedures	Solution Design	TA.150	Define System Management Procedures	Build
TE.010	Develop Testing Strategy	Solution Design	TE.010	Define Testing Requirements and Strategy	Definition
TE.020	Develop Unit Test Scripts	Solution Design	TE.020	Develop Unit Test Script	Solution Design
TE.030	Develop Link Test Scripts	Solution Design	TE.030	Develop Link Test Script	Solution Design
TE.040	Develop System Test Scripts	Solution Design	TE.040	Develop System Test Script	Solution Design
TE.050	Develop Systems Integration Test Scripts	Solution Design	TE.050	Develop Systems Integration Test Script	Solution Design
TE.060	Prepare Testing Environments	Build	TE.060	Prepare Testing Environments	Build
TE.070	Perform Unit Testing	Build	TE.070	Perform Unit Test	Build
TE.080	Perform Link Testing	Build	TE.080	Perform Link Test	Build
TE.090	Perform Installation Test	Build	TE.090	Perform Installation Test	Build
TE.100	Prepare Key Users for Testing	Build	TE.100	Prepare Key Users for Testing	Build
TE.110	Perform System Testing	Build	TE.110	Perform System Test	Build
TE.120	Perform Regression Testing	Build	TE.110	Perform System Test	Build
TE.130	Perform Systems Integration Testing	Build	TE.120	Perform Systems Integration Test	Build
TE.140	Support Acceptance Test	Transition	TE.130	Perform Acceptance Test	Transition
TR.010	Prepare Training Strategy	Definition	AP.070	Develop Project Readiness Roadmap	Definition
TR.020	Prepare Project Team Training Environment	Operations Analysis	AP.040	Prepare Project Team Learning Environment	Definition
TR.030	Conduct General Education Classes	Definition	AP.050	Conduct Project Team Learning Events	Definition
TR.040	Conduct High-level Overview of Application Features	Definition	AP.060	Develop Business Unit Mangers' Readiness Plan	Definition
TR.050	Prepare Project Team Training Environment	Operations Analysis	AP.040	Prepare Project Team Learning Environment	Definition
TR.060	Train Project Team	Operations Analysis	AP.050	Conduct Project Team Learning Events	Definition
TR.070	Develop User Training Materials	Solution Design	AP.150	Develop User Learningware	Build

V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase	V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase
TR.080	Prepare User Training Environment	Transition	AP.160	Prepare User Learning Environment	Build
TR.090	Train Users	Transition	AP.170	Conduct User Learning Events	Transition
WM.010	Define Work Management Strategies, Standards, and Procedures	Definition	WM.010	Define Work Management Strategies, Standards, and Procedures	Definition
WM.020	Establish Workplan	Definition	WM.020	Establish Workplan	Definition
WM.030	Establish Finance Plan	Definition	WM.030	Establish Finance Plan	Definition

AIM Version 3.0 Tasks Cross-Referenced with AIM Version 2.0 Tasks

This table lists by process AIM version 3.0 tasks with the corresponding version 2.0 tasks.

V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase	V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase
AP.010	Define Executive Project Strategy	Definition	This task is new in version 3.0.		
AP.020	Conduct Initial Project Team Orientation	Definition	This task is new in version 3.0.		
AP.030	Develop Project Team Learning Plan	Definition	This task is new in version 3.0.		
AP.040	Prepare Project Team Learning Environment	Definition	TR.020	Prepare Project Team Training Environment	Operations Analysis
AP.050	Conduct Project Team Learning Events	Definition	TR.030	Conduct General Education Classes	Definition
AP.060	Develop Business Unit Managers' Readiness Plan	Definition	TR.040	Conduct High-Level Overview of Application Features	Definition
AP.070	Develop Project Readiness Roadmap	Definition	TR.010	Prepare Training Strategy	Definition
AP.080	Develop and Execute Communication Campaign	Definition	This task is new in version 3.0.		
AP.090	Develop Managers, Readiness Plan	Operations Analysis	TR.040	Conduct High-Level overview of Application Features	Definition
AP.100	Identify Business Process Impact on Organization	Solution Design	This task is new in version 3.0.		
AP.110	Align Human Performance Support Systems	Solution Design	This task is new in version 3.0.		
AP.120	Align Information Technology Groups	Solution Design	This task is new in version 3.0.		
AP.130	Conduct User Learning Needs Analysis	Solution Design	This task is new in version 3.0.		
AP.140	Develop User Learning Plan	Solution Design	This task is new in version 3.0.		
AP.150	Develop User Learningware	Build	TR.070	Develop User Training Materials	Solution Design
AP.160	Prepare User Learning Environment	Build	TR.080	Prepare User Training Environment	Transition
AP.170	Conduct User Learning Events	Transition	TR.090	Train Users	Transition
AP.180	Conduct Effectiveness Assessment	Production	PM.090	Audit Production System	Production
BP.010	Define Business and Process Strategy	Definition	This task is new in version 3.0.		
BP.020	Catalog and Analyze Potential Change	Definition	This task is new in version 3.0.		
BP.030	Determine Data Gathering Requirements	Definition	This task is new in version 3.0.		

V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase	V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase
BP.040	Develop Current Process Model	Definition	RD.020	Conduct Current Business Baseline	Definition
BP.050	Review Leading Practices	Definition	This task is new in version 3.0.		
BP.060	Develop High-Level Process Vision	Definition	This task is new in version 3.0.		
BP.070	Develop High-Level Process Designs	Definition	This task is new in version 3.0.		
BP.080	Develop Future Process Model	Operations Analysis	RD.030	Develop Future Business Model	Definition
BP.090	Document Business Procedures	Solution Design	BR.100	Create Process Narratives	Solution Design
BR.010	Analyze High-Level Gaps	Operations Analysis	This task is new in version 3.0.		
BR.020	Prepare Mapping Environment	Operations Analysis	BR.010	Prepare Mapping Environment	Operations Analysis
BR.030	Map Business Requirements	Operations Analysis	BR.020	Map Business Requirements	Operations Analysis
BR.040	Map Business Data	Operations Analysis	BR.030	Map Business Data	Operations Analysis
BR.050	Conduct Integration Fit Analysis	Operations Analysis	BR.040	Conduct Integration Fit Analysis	Operations Analysis
BR.060	Create Information Model	Operations Analysis	BR.050	Develop Information Flow Model	Operations Analysis
BR.070	Conduct Reporting Fit Analysis	Operations Analysis	BR.070	Conduct Reporting Fit Analysis	Operations Analysis
BR.080	Test Business Solutions	Operations Analysis	BR.080	Test Business Solutions	Operations Analysis
BR.090	Confirm Integrated Business Solutions	Operations Analysis	BR.090	Confirm Integrated Business Solutions	Operations Analysis
BR.100	Define Application Setups	Solution Design	BR.110	Define Application Setups	Solution Design
BR.110	Design Security Profiles	Solution Design	BR.120	Design Security Profiles	Solution Design
CM.010	Define Configuration Management Strategies	Definition	CM.010	Define Configuration Management Strategies	Definition
CM.060	Audit Key Deliverables	Definition Operations Analysis Solution Design Build Transition Production	CM.060	Audit Key Deliverables	Definition Operations Analysis Solution Design Build Transition Production
CM.070	Conclude Configuration Management	Production	This task is new in version 3.0.		
CR.010	Establish Scope, Objectives, and Approach	Definition	CR.010	Establish Scope, Objectives, and Approach	Definition
CR.020	Define Control and Reporting Strategies	Definition	CR.020	Define Control and Reporting Strategies	Definition

V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase	V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase
CR.030	Establish Management Plans	Definition	CR.030	Establish Management Plans	Definition
CR.080	Secure Client Phase Acceptance	Definition Operations Analysis Solution Design Build Transition Production	CR.080	Secure Client Phase Acceptance	Definition Operations Analysis Solution Design Build Transition Production
CV.010	Define Data Conversion Requirements and Strategy	Definition	CV.010	Define Conversion Scope, Objectives, and Approach	Definition
CV.020	Define Conversion Standards	Solution Design	CV.030	Prepare Conversion Standards	Operations Analysis
CV.030	Prepare Conversion Environment	Solution Design	CV.040	Prepare Conversion Environment	Solution Design
CV.040	Perform Conversion Data Mapping	Solution Design	CV.050	Perform Conversion Data Mapping	Solution Design
CV.050	Define Manual Conversion Procedures	Solution Design	CV.060	Design Manual Conversion Strategy	Solution Design
CV.060	Design Conversion Programs	Solution Design	CV.070	Design Conversion Programs	Solution Design
CV.070	Prepare Conversion Test Plans	Solution Design	CV.080	Prepare Conversion Test Plans	Solution Design
CV.080	Develop Conversion Programs	Build	CV.090	Develop Conversion Programs	Build
CV.090	Perform Conversion Unit Tests	Build	CV.100	Perform Conversion Unit Test	Build
CV.100	Perform Conversion Business Object Tests	Build	CV.110	Perform Conversion Business Object Tests	Build
CV.110	Perform Conversion Validation Tests	Build	CV.120	Perform Conversion Validation Test	Build
CV.120	Install Conversion Programs	Transition	CV.130	Install Conversion Software	Transition
CV.130	Convert and Verify Data	Transition	CV.140	Convert and Verify Data	Transition
DO.010	Define Documentation Requirements and Strategy	Definition	DO.020	Specify Documentation Requirements	Definition
DO.020	Define Documentation Standards and Procedures	Definition	DO.030	Determine Documentation Standards and Procedures	Definition
DO.030	Prepare Glossary	Definition	DO.010	Prepare Glossary	Definition
DO.040	Prepare Documentation Environment	Operations Analysis	DO.040	Prepare Documentation Environment	Operations Analysis
DO.050	Produce Documentation Prototypes and Templates	Operations Analysis	DO.050	Produce Documentation Prototypes and Templates	Operations Analysis
DO.060	Publish User Reference Manual	Build	DO.100	Complete User Reference Manual	Build

V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase	V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase
DO.070	Publish User Guide	Build	DO.110	Complete User Guide	Build
DO.080	Publish Technical Reference Manual	Build	DO.120	Complete Technical Reference Manual	Build
DO.090	Publish System Management Guide	Build	DO.130	Complete System Management Guide	Build
MD.010	Define Application Extension Strategy	Definition	MD.010	Prepare Customization Strategy	Definition
MD.020	Define and Estimate Application Extensions	Operations Analysis	MD.020	Define and Estimate Custom Extensions	Operations Analysis
MD.030	Define Design Standards	Solution Design	MD.030	Define Design Standards	Solution Design
MD.040	Define Build Standards	Solution Design	MD.040	Define Build Standards	Solution Design
MD.050	Create Application Extensions Functional Design	Solution Design	MD.060	Produce Module Functional Design	Solution Design
MD.060	Design Database Extensions	Solution Design	MD.050	Design Database Extensions	Solution Design
MD.070	Create Application Extensions Technical Design	Solution Design	MD.070	Produce Module Technical Design	Solution Design
MD.080	Review Functional and Technical Designs	Solution Design	MD.080	Review Detailed Designs	Solution Design
MD.090	Prepare Development Environment	Build	MD.090	Prepare Development Environment	Build
MD.100	Create Database Extensions	Build	MD.100	Implement Database Extensions	Build
MD.110	Create Application Extension Modules	Build	MD.110	Create Custom Modules	Build
MD.120	Create Installation Routines	Build	MD.120	Create Installation Routines	Build
PM.010	Define Transition Strategy	Operations Analysis	PM.010	Prepare Transition Strategy	Operations Analysis
PM.020	Design Production Support Infrastructure	Build	PM.020	Design Production Support Infrastructure	Solution Design
PM.030	Develop Transition and Contingency Plan	Build	PM.030	Develop Detailed Transition and Contingency Plan	Solution Design
PM.040	Prepare Production Environment	Transition	PM.040	Prepare Production Environment	Build
PM.050	Set Up Applications	Transition	PM.050	Set Up Applications	Transition
PM.060	Implement Production Support Infrastructure	Transition	PM.060	Implement Support Infrastructure	Transition
PM.070	Verify Production Readiness	Transition	PM.070	Verify Production Readiness	Transition
PM.080	Begin Production	Transition	PM.080	Commence Production	Transition
PM.090	Measure System Performance	Production	PM.100	Measure System Performance	Production
PM.100	Maintain System	Production	PM.110	Maintain System	Production

V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase	V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase
PM.110	Refine Production System	Production	PM.120	Refine Production System	Production
PM.120	Decommission Former Systems	Production	PM.130	Decommission Former System	Production
PM.130	Propose Future Business Direction	Production	PM.140	Propose Future Business Direction	Production
PM.140	Propose Future Technical Direction	Production	PM.150	Propose Future Technical Direction	Production
PT.010	Define Performance Testing Strategy	Definition	PT.010	Define Performance Testing Scope, Objectives, and Approach	Definition
PT.020	Identify Performance Test Scenarios	Operations Analysis	PT.030	Identify Performance Test Scenarios	Operations Analysis
PT.030	Identify Performance Test Transaction Models	Operations Analysis	PT.040	Identify Performance Test Transaction Models	Operations Analysis
PT.040	Create Performance Test Scripts	Solution Design	PT.050	Create Performance Test Scripts	Solution Design
PT.050	Design Performance Test Transaction Programs	Solution Design	PT.060	Design Performance Test Transaction Programs	Solution Design
PT.060	Design Performance Test Data	Solution Design	PT.080	Design Performance Test Data	Solution Design
PT.070	Design Test Database Load Programs	Solution Design	PT.090	Design Test Database Load Programs	Solution Design
PT.080	Create Performance Test Transaction Programs	Build	PT.070	Develop Performance Test Transaction Programs	Build
PT.090	Create Test Database Load Programs	Build	PT.100	Develop Test Database Load Programs	Build
PT.100	Construct Performance Test Database	Build	PT.110	Construct Performance Test Database	Build
PT.110	Prepare Performance Test Environment	Build	PT.120	Prepare Performance Test Environment	Build
PT.120	Execute Performance Test	Build	PT.130	Execute Performance Test	Build
PT.130	Create Performance Test Report	Build	PT.140	Create Performance Test Report	Build
QM.010	Define Quality Management Strategies	Definition	QM.010	Define Quality Management Strategies	Definition
QM.050	Perform Quality Assessment	Definition Operations Analysis Solution Design Build Transition Production	QM.050	Perform Quality Assessment	Definition Operations Analysis Solution Design Build Transition Production
RD.010	Identify Current Financial and Operating Structure	Definition	RD.010	Identify Financial and Operating Structure	Definition
RD.020	Conduct Current Business Baseline	Definition	RD.020	Conduct Current Business Baseline	Definition

V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase	V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase
RD.030	Establish Process and Mapping Summary	Operations Analysis	RD.050	Establish Process and Mapping Summary	Definition
RD.040	Gather Business Volumes and Metrics	Operations Analysis	RD.060	Gather Business Volumes	Definition
RD.050	Gather Business Requirements	Operations Analysis	RD.070	Create Business Requirements Scenarios	Operations Analysis
RD.060	Determine Audit and Control Requirements	Operations Analysis	RD.080	Determine Audit and Control Requirements	Operations Analysis
RD.070	Identify Business Availability Requirements	Operations Analysis	RD.090	Identify Business Availability Requirements	Operations Analysis
RD.080	Identify Reporting and Information Access Requirements	Operations Analysis	RD.100	Develop Reporting Requirements	Operations Analysis
RM.010	Define Resource Management Strategy	Definition	RM.010	Define Resource Management Strategy	Definition
RM.020	Establish Staffing and Organization Plan	Definition	RM.020	Establish Staffing and Organization Plan	Definition
RM.025	Create Project Orientation Guide	Definition	This task is new in version 3.0.		
RM.030	Implement Organization	Definition	RM.030	Implement Organization	Definition
RM.040	Establish Physical Resource Plan	Definition	RM.040	Establish Physical Resource Plan	Definition
RM.050	Establish Infrastructure	Definition	RM.050	Establish Infrastructure	Definition
RM.080	Release Staff	Definition Operations Analysis Solution Design Build Transition Production	RM.080	Release Staff	Definition Operations Analysis Solution Design Build Transition Production
RM.090	Release Physical Resources	Definition Operations Analysis Solution Design Build Transition Production	RM.090	Release Physical Resources	Definition Operations Analysis Solution Design Build Transition Production
TA.010	Define Architecture Requirements and Strategy	Definition	TA.010	Define Architecture Scope, Objectives, and Approach	Definition
TA.020	Identify Current Technical Architecture	Definition	TA.050	Conduct Technical Architecture Baseline	Definition
TA.030	Develop Preliminary Conceptual Architecture	Definition	TA.040	Develop Conceptual Architecture	Definition
TA.040	Define Application Architecture	Operations Analysis	TA.130	Design Application Functional Architecture	Solution Design
TA.050	Define System Availability Strategy	Operations Analysis	TA.060	Develop System Availability Strategy	Operations Analysis

V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase	V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase
TA.060	Define Reporting and Information Access Strategy	Operations Analysis	TA.080	Develop Reporting Strategy	Operations Analysis
TA.070	Revise Conceptual Architecture	Operations Analysis	TA.090	Revise Conceptual Architecture	Operations Analysis
TA.080	Define Application Security Architecture	Solution Design	TA.120	Design Application Security Architecture	Solution Design
TA.090	Define Application and Database Server Architecture	Build	TA.140	Design Logical Application and Database Architecture	Solution Design
TA.100	Define and Propose Architecture Subsystems	Build	TA.110	Propose Architecture Subprojects	Operations Analysis
TA.120	Define Platform and Network Architecture	Build	TA.160	Design Hardware and Network Architecture	Solution Design
TA.110	Define System Capacity Plan	Build	TA.170	Develop System Capacity Plan	Solution Design
TA.130	Define Application Deployment Plan	Build	TA.070	Define Future Application Deployment	Operations Analysis
TA.140	Assess Performance Risks	Build	TA.180	Assess Performance Risks	Solution Design
TA.150	Define System Management Procedures	Build	TA.190	Design System Management Procedures	Solution Design
TE.010	Define Testing Requirements and Strategy	Definition	TE.010	Develop Testing Strategy	Solution Design
TE.020	Develop Unit Test Script	Solution Design	TE.020	Develop Unit Test Scripts	Solution Design
TE.030	Develop Link Test Script	Solution Design	TE.030	Develop Link Test Scripts	Solution Design
TE.040	Develop System Test Script	Solution Design	TE.040	Develop System Test Scripts	Solution Design
TE.050	Develop Systems Integration Test Script	Solution Design	TE.050	Develop Systems Integration Test Scripts	Solution Design
TE.060	Prepare Testing Environments	Build	TE.060	Prepare Testing Environments	Build
TE.070	Perform Unit Test	Build	TE.070	Perform Unit Test	Build
TE.080	Perform Link Test	Build	TE.080	Perform Unit Tests	Build
TE.090	Perform Installation Test	Build	TE.090	Perform Installation Test	Build
TE.100	Prepare Key Users for Testing	Build	TE.100	Prepare Key Users for Testing	Build
TE.110	Perform System Test	Build	TE.110	Perform System Testing	Build
TE.120	Perform Systems Integration Test	Build	TE.130	Perform System Integration Testing	Build
TE.130	Perform Acceptance Test	Transition	TE.140	Support Acceptance Test	Transition

V.3.0 ID	V.3.0 Task Name	V.3.0 Typical Phase	V.2.0 ID	V.2.0 Task Name	V.2.0 Typical Phase
WM.010	Define Work Management Strategies	Definition	WM.010	Define Work Management Strategies	Definition
WM.020	Establish Workplan	Definition	WM.020	Establish Workplan	Definition
WM.030	Establish Finance Plan	Definition	WM.030	Establish Finance Plan	Definition

APPENDIX

B

AIM Roles

This appendix gives a brief description of each role, highlighting the main responsibilities of the role.

Role Descriptions

Adult Learning Specialist

The adult learning specialist is an individual with expertise in the education of adult learners. This project role assists in custom training engagements by developing client-specific curriculum plans and instructional designs.

Application Specialist

The application specialist provides knowledge and guidance regarding application functionality. This project role also supports and provides interpretation for tools, templates and methods. FastForward projects include an Oracle application specialist in each application included in the engagement.

Assessment Specialist

The assessment specialist provides the client with expertise in developing and administering assessments. The assessment specialist assists the client in generating an assessment that is best suited to the unique needs of the client's organization. This project role provides expertise in the use of the Oracle Survey Tool. The assessment specialist has an understanding of sampling techniques, survey development and administration, and data analysis and reporting.

Bid Manager

The bid manager prepares the bid, negotiation, and award. This project role assists in the hand over of materials and information accumulated during the bid to the project manager at the start of the project.

Business Analyst

The business analyst should be familiar with the business area that the system covers and the terminology and practices.

The business analyst performs many activities that define the scope of the project. They examine the client's business and define what the system should do. They obtain information from existing documentation, when available. The business analyst identifies interviewees who might be representative client staff, management, and

technical support staff. The business analyst obtains information by conducting interviews, working sessions, and site visits. These analysis activities determine the technical, interfacing, and operational requirements and constraints.

The business analyst must understand the business objectives and requirements in order to document the analysis. The business analyst creates the business models and business requirement scenarios, gathers and translates business functionality into object models, and models the requirements at the subsystem level, creating the most appropriate analysis model level.

The business analyst conducts reviews of the findings with client management and the staff.

Business Line Manager

The business line manager participates in interviews and working sessions providing information about business objectives and ways in which the units operate and respond to events in order to achieve those objectives. The business line manager hosts site visits with staff in order to collect information. Additionally, this project role is responsible for allocating staff time to provide detailed information about the day-to-day business.

Also, this project role describes problems the business unit faces and requirements for the computer system.

The business line manager should review the content of the analysis documentation to make sure it accurately describes the business and requirements.

The business line manager role should be filled by someone who will manage one of the business units that uses the system.

Client Executive

The client executive participates in the strategic decisions regarding implementation and project strategy and defines business performance expectations and metrics. The client executive also appoints steering committee members and is involved in measurement of business results.

Client Project Manager

The client project manager is responsible for the daily management of the client's contractual commitments to the project. This project role must understand the client's business objectives for the project to form the basis for resolving problems, conflicts of interest, and making compromises.

The client project manager obtains physical resources such as space accommodation, office equipment, computer equipment, and materials. The client project manager assists in the availability of users and endeavors to gain user commitment.

Additionally, this project role assists in the allocation of client time to the project. The client project manager introduces the consulting staff to the other client staff. This project role monitors the project's performance, progress against milestones, appropriateness of work, quality of work, and seeks to resolve any problems with work or relationships between the development and business staff.

The client project manager assists in obtaining user review and signoff of deliverables. This role usually performs intermediate and phase-end acceptance.

Client Staff Member

The client staff member reports directly to either the project manager or client project manager.

This project role may be responsible for technical support of the client's systems. The client staff member may provide information about existing systems with which the new system is to interface or replace. This project role provides information about any IS standards with which the project must comply, supports the business' software systems, and takes over support of the system during production.

Finally, a client staff member may participate in training programs for the system initially as consumers and later, possibly as providers.

Communications Specialist

The communications specialist assists the client in developing the communication model and standards for the project team, and the communication campaign to involve, inform and generate buy-in from the stakeholders throughout the client organization.

This project role provides expertise in the selection of the communication channels that are most compatible with the organization's culture and the stakeholders' preferred sources of information. This project role's expertise also drives the design of the campaign in terms of key messages, timing, repetition, and feedback mechanisms.

Configuration Management Specialist

The configuration management specialist with direction from the project manager plans, establishes, and controls the Configuration Management process on the project, with these responsibilities:

- develops, documents, and implements Configuration Management plans and procedures
- establishes project baselines and determines the content of project releases
- makes sure that no unauthorized changes are made to a project baseline
- enforces Configuration Management procedures across all project processes
- establishes the Configuration Management Repository and assists in the adequate maintenance and protection against damage or loss
- recommends improvements, offers implementation advice and supports the changes.

Database Administrator

The database administrator installs and configures database software for the development environment; creates the various databases required during the development life-cycle (for example, the data dictionary, unit testing, system testing, training); and maintains database access controls. Additionally, this project role provides consultation on performance; monitors growth and fragmentation of the development database; and provides database backup and recovery.

Database Designer

The database designer produces the Logical and Physical Database Designs. This project role reviews the module designs to provide efficient access to the database.

The database designer must understand how to translate application logic into a System Data Model and have a thorough understanding of the System Data Model. The database designer is responsible for producing the System Data Model, the Logical Database Design, and the Physical Database Design.

The database designer reviews the application design to check database access efficiency. Additionally, an understanding of the technical architecture and functionality of the system is required so that trade-offs in the design can be made where different functions place conflicting requirements on the database. The database designer may make design suggestions in order to mitigate conflicts between the application design and the technical requirements.

Developer

The developer understands the requirements from the business analysis and how to meet these requirements using the Technical Architecture and Data Model.

The developer produces application and module designs and generation of modules. This project role interfaces closely with the lead system developer to make sure the database design meets the data requirements of the module functionality and module access data efficiently. The developer may create the object structure, the database object logic, and the test scripts for the database.

The developer also produces partition integration test plans and performs testing of partitions and system. During the various testing activities and the production phase, this project role diagnoses faults and determines corrections.

Developers produce the initial versions of online help text, user reference, and technical reference documents.

Facilitator

The facilitator manages the workshop process and acts as the catalyst for preparation and communication. The facilitator creates the context and not the content of the workshop. This project role has the specific responsibilities of:

- agreeing on the scope of workshop with project manager
- planning the workshop

- becoming familiar with the business area
- interviewing participants to verify their suitability and that any pre-work is complete
- facilitating the workshop to meet the objectives
- reviewing the workshop against the objectives

The facilitator runs mapping and process design sessions and keeps momentum going. Finally, this project role may be responsible for workshop activities, not just mapping and process design.

Human Performance Technologist

The human performance technologist understands the organizational and motivational factors that influence the performance of users. This project role brings expertise in the design of the optimal organization of work and project roles to maximize performance results of users.

This project role assists the client in the following activities:

- defining the new work flows and project roles that are derived from the new business processes and procedures
- identifying the new competency profiles
- assessing the new performance support requirements
- developing the performance support systems and human resource management protocol to meet the expected organizational performance

IS Manager

The IS manager directs the client information systems organization within a business. The IS manager acts as a business line manager for the staff in the IS organization. This project role is responsible for the technical infrastructure of a business; including decisions about purchases, in-house development, and operational maintenance and support. The following information systems staff report directly or indirectly to the IS manager:

- application and technical architect
- technical analyst
- designer

- technical (database, network, system) administrator
- operations staff
- support staff

The IS manager helps define the information systems strategy for a corporation and puts the strategy into practice through standards, policies, practices, and information systems selection processes.

The IS manager is responsible for operating the existing platforms within a business. The IS manager participates in the verification and approval of the final operating infrastructure management and architecture.

Key User

The key user participates in workshops and is empowered by the project sponsor to refine and prioritize requirements. This project role provides information about their business objectives and the ways in which their units operate and respond to events in order to achieve those objectives. The key user describes the problems they face and requirements for the computer system.

The key users write the initial user guide.

Network Administrator

The network administrator manages and administers the network. This includes making sure that the network is correctly configured, configuring and maintaining the network environment, and monitoring the performance of the network. The network administrator also acquires tools and establishes procedures to monitor network components to provide for sufficient capacity.

The network administrator provides network consultation and impact assessment to the project team. This project role provides information on data communication solutions deployed throughout the enterprise and makes decisions on network product acquisition and implementation. This project role works with the development team to determine and implement the network hardware requirements of the application. These requirements include designing and implementing components such as cabling, terminals, bridges, and routes. The network administrator makes decisions regarding network infrastructure deployment. In addition, this project role consults with

the development team to determine and implement any foreign host access requirements.

Organizational Development Specialist

The organizational development specialist provides the client with expertise in the human and organizational facets of change. The organizational development specialist develops the Organizational Change Management plan and facilitates an efficient, intelligent transition to the technology-induced change.

Process Modeler

The process modeler specializes in modeling the client's existing business processes and capturing relevant process data.

Project Manager

The project manager represents the project to both the business and IS management. This project role is ultimately held responsible for the success or failure of the project. The project manager must understand the project business objectives and have a clear vision of how to achieve those objectives. The project manager must resolve the conflict among the differing objectives of the various parties to the project.

The project manager primarily faces outwards from the project and handles political conflicts and issues and makes sure they do not impede the project.

The project manager agrees on the scope of the project with the client, makes sure the implementation remains within the agreed upon scope, and guards against scope creep. The project manager should review major deliverables — particularly those from the earlier phases of the project.

The project manager is responsible for comparing the project planning, resourcing, monitoring, and reporting progress against the plan. This project role obtains any physical resources required for the project, recruits staff, and, if necessary, dismisses staff. The project manager is responsible for ensuring that activities are performed in accordance with the Project Management Plan (Quality Plan).

Internal responsibilities of the project manager should be delegated to subordinate team leaders, as documented in the project organization plan.

The project role of implementation project manager is used to distinguish between the two types of projects on a program: focus area or program office and implementation. The responsibilities of the implementation manager are the same as any project manager; however, the implementation project manager is also responsible for the rollout of the baseline solution to the various implementation sites.

In contrast, the program office project manager manages projects tasked with creating a common baseline solution for the implementation project to roll out. Again, the responsibilities of this project role are the same as any project manager.

Project Sponsor

The project sponsor controls the budget and finances the project. This project role is usually a member of senior management. On large, cross-functional projects the project sponsor may be a board member. This project role must have a clear understanding of the project objectives, particularly concerning delivery of the business benefits. The project sponsor empowers the key users to refine and prioritize requirements. The project sponsor is the ultimate arbiter on conflicting business requirements and scope changes. The project sponsor makes sure the project is delivered on time and within budget.

The project sponsor is responsible for ensuring other members of the management share commitment to the project. This project role may provide the resources, particularly staff time, required to make the project a success. The project sponsor usually performs the final approval on the recommendation of the verification coordinator, internal auditor ,and data administrator.

Project Staff Member

The project staff member reports directly to either the project manager or client project manager.

This project role may be responsible for technical support of the client's systems. The project staff member may provide information about existing systems with which the new system is to interface or replace. This project role provides information about any IS standards with which the project must comply, supports the business' software systems, and takes over support of the system during production. Finally, project staff members may participate in training programs for the system initially as consumers and later, possibly as providers.

Project Support Specialist

The project support specialist assists the project manager in the daily management of the project by performing routine or repetitive tasks. On larger projects, this project role may be performed by a project office staff.

Specific duties are delegated by the project manager but will normally include some or all of the following:

- establishing and maintaining the Project Management Plan, project standards, and project procedures
- coordinating and executing effective phase management
- establishing and maintaining the Workplan and Finance Plan
- procuring staff or physical resources for the project
- monitoring and performing analysis of risks, issues, and problems for trends requiring project manager corrective action
- performing coordination and communication functions within the project organization by preparing or maintaining reports, records, logs and other written communications
- collecting progress data from project leaders; and distributing project calendars, meeting agendas
- recording and distributing minutes, decisions, and actions from management meetings
- organizing the Project Library, assigning documents into the library, and maintaining control of documents in the library
- orienting new project members to the project environment, policies and procedures
- coordinating with administrators in client and subcontractor organizations
- preparing project progress reports
- generating routine status information from project records
- maintaining information on project staff such as grade, qualifications, training, parent business unit or subcontractor, telephone and address, project assignment history, and other pertinent information

- establishing project baselines and determining the content of project releases
- establishing change control so that no unauthorized changes are made to a project baseline
- developing, documenting, and implementing Configuration Management plans and procedures, enforce Configuration Management procedures across all project processes
- establishing the Configuration Management Repository and assist in the maintenance and protection against damage or loss
- verifying that the standards and procedures which have been defined for the project in the Project Management Plan are implemented
- verifying and tracking that quality reviews and quality audits are conducted as required

For applications implementation projects, the project support specialist also performs deliverable and template version management, gathers and checks out deliverables, assigns document names and records new documents into the library. This project role also provides some advice regarding process integration, and provides deliverable status.

Quality Auditor

The quality auditor conducts quality audits of the project to include a review of the Project Management Plan. This project role should be filled by a project role independent of the project staff in the consulting organization. The quality auditor needs training in the audit process. This project role prepares for, conducts, and reports on the quality audit or audits undertaken, following up on any actions raised.

Quality Manager

The quality manager, with guidance from the project manager, plans and prescribes all matters affecting quality of a project. This project role makes sure that the specific quality requirements defined in the contract and Project Management Plan are achieved. The quality manager also makes sure the standards and procedures which have been defined for the project in the Project Management Plan are implemented and quality reviews and quality audits are conducted as required.

Steering Committee Member

The steering committee guides the overall project review, adopts the recommendations, and provides sponsorship for implementing the changes. The steering committee includes senior management and a consulting director. Regular meetings should be held to review progress and resolve outstanding issues.

The steering committee members are responsible for the program approach buy-in, funding, issue resolution, and signoff.

Subject Matter Specialist

The subject matter specialist provides detail about the scope of the increment. This project role works closely with the information analyst to define the information requirements and provide valuable input for the data access requirements and assists in the validation of the source system data. The subject matter specialist may provide user input during the design of the end-user layer and queries and reports. They are also involved in the tool evaluation and selection tasks. The subject matter specialist may support the training and user acceptance activities and should provide client support for other users.

The subject matter specialist may have a broad knowledge of the current state of the computer industry and emerging trends and standards. This project role may be responsible for creating a library of industry standards for a project.

System Administrator

The system administrator is responsible for administering a development system. This project role's responsibilities include ensuring hardware is correctly configured; installing, configuring, and maintaining operating and development software; and ensuring daily backups of the system are performed. The system administrator designs and maintains the system's security; for example, establishing system accounts.

The system administrator provides first-line support for development system problems and makes sure faults are quickly rectified. This project role may perform the set-up and initial maintenance of a production system or advise the client's operational staff on these tasks. The system administrator works with the project team to optimize system performance.

System Architect

The system architect defines the system and technical architectures including the major software components of the system, interfaces, and hardware configuration and software foundation.

The system architect is generally the senior or lead technical designer on the project. This project role must understand the business and technical requirements for the system.

The system architect establishes the application architecture of a newly implemented system. In order to accomplish this, the application architect translates the future vision of the business into an application and data deployment strategy. This strategy includes decisions about centralizing or decentralizing business data and processing; identification of interface points and specific requirements for data transfer and synchronization across the business; critical setup of applications to support the business process mapping; strategies to support the reporting needs of the business; and other less general requirements that may impact the architecture such as multilingual requirements. The systems architect provides input to more detailed technical design efforts such as interfaces and custom components, to promote compatibility with the overall applications architecture.

The system architect oversees the application architecture activities. This project role reviews and approves the data deployment strategy and decisions. This project role works very closely with the technical architect to make sure the physical layers of the architecture are consistent with, and fully support, the business and information systems vision for the corporation. This synergism may extend all the way from scoping and planning the project through to the final architecture deliverables and client review.

The system architect also produces and maintains the Capacity Plan and reviews all aspects of the design to make sure it performs within any capacity constraints. This project role also performs any benchmarking exercises required to measure the performance of hardware or software. Often, on smaller projects, this project role is undertaken by the staff performing the system architect project role. On projects developing large systems, this project role is likely to be filled by a separate project role who has the specialized skills required for systems sizing and capacity prediction.

The system architect is responsible for architecting the physical components of the database, hardware, and network in support of the application architecture strategic technical architecture and the

increment technical architecture. In order to accomplish this, the system architect needs to design the detailed database, hardware, and network architecture to support the application architecture and deployment strategy. This includes decisions about the physical distribution of processing across client and server machines, capacity planning of the technical infrastructure, detailed design of the layout of databases, and identification and advisement about performance risks. The system architect oversees detailed technical work on interface, system, data acquisition, data access, and metadata design to make sure the detailed work is consistent with the overall technical architecture.

Technical Analyst

The technical analyst poses solutions and technical assumptions and develops data profiles in support of testing.

Technical Writer

The technical writer becomes familiar with the business and technical requirements of the system and how the architecture, design, and modules achieve those objectives. The technical writer specifies, produces, and edits the user, technical, and operational documentation. This project role provides skills in language and presentation.

Tester

The tester develops and executes test script. Testers make sure test scripts are reviewed by the appropriate business analysts prior to test execution. This project role records test results during testing activities and documents test faults in the problem log. Testers update test scripts due to approved change requests or software faults that were not anticipated in the original development. When problems are resolved after re-testing, testers update the problem log.

Tool Specialist

The tool specialist provides knowledge and guidance regarding specific tool functionality. This project role also supports and provides interpretation for the tool capabilities and customization and design of the end-user layer.

Trainer

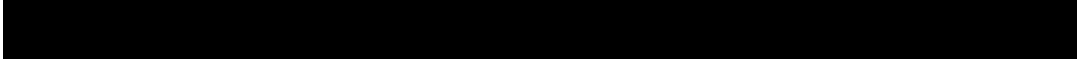
The trainer defines the training requirements, prepares a training plan, produces training material, and delivers courses.

The trainer is responsible for working with the client to develop and deliver the necessary project team training during a FastForward project. Within the scope of a FastForward project this resource is not responsible for developing or delivering the end-user training.

An external trainer may be needed to assist the key users in training end users.

User

The user is a member of the client's staff who actually uses the production system. This project role acts as a consumer of the training program and reports problems about the production system. The users are involved in testing in the later stages of development and assist the key users in performing the pre-production validation.



Glossary

24x7 A period or window of service availability that covers 24 hours day, seven days a week.

3GL see THIRD-GENERATION PROGRAMMING LANGUAGE.

4GL see FOURTH-GENERATION PROGRAMMING LANGUAGE.

A

ABT see APPLIED BUSINESS TECHNOLOGY.

Acceptance The approval, typically by a client or user, of a project deliverable.

Access Control The ability to manage which users or groups of users may have the privilege to retrieve, create, update, or delete data held in a repository, such as a relational database.

Activation A logical event corresponding to the enabling of one high level business function at one site for one business unit. Each activation represents a discrete unit of work that we can predict and measure.

Actuals Information gathered during a project concerning the actual amount of time, finances or resources expended on a task.

Administration Server The machine from which a user, typically an administrator, runs programs that maintain and update an Oracle Applications database. For example, AutoInstall and AutoPatch are both run on this machine to install, upgrade, or update the database, which may reside on another machine (the database server).

Advocate An individual who supports the project within the client environment, without being involved in the project's implementation. An advocate may be a formal or informal leader within the organization.

Agent A generic term for a party which is responsible for the execution and completion of a process step. An agent may be an organizational unit, a functional unit, a business system, an employee role, an external system, or an external party such as a customer or supplier.

Agent Channel A horizontal division on a process flow diagram that indicates which agent performs which particular functions within the process being modeled.

AIM see APPLICATION IMPLEMENTATION METHOD.

Adragogy The field of adult-centered learning.

Applet A Java program that is downloaded to a desktop client from a web server, and run within a Java-enabled web browser or appletviewer.

Appletviewer A program that runs a Java applet or application.

Application A collection of program modules that work together to support a set of related business functions; see also MODULE.

Application Extension A collective term referring to any of several custom program approaches to a business requirement that is not satisfied by Oracle Applications, including modification of base Oracle Application code, custom extensions or configurable extensions; see also CUSTOMIZATION.

Application Servers Machines that reside in a middle tier, between desktop clients and a high-end, or back-end, tier. Desktop clients send their requests to application servers, which process the request by sending it to another server, such as the database server. In this manner, the desktop clients never connect directly to the database server. The forms server and web server are types of application servers; see also TIER.

Application Environment (Oracle) A complete application installation along with other utilities used for business mapping, design, build, testing, or training. Usually, an application environment includes setups and test data to support business modeling, and procedures exist to recover to controlled starting points after sessions are complete.

Application Fit A recorded match between some aspect of a business requirement and the capability or features of the application system that satisfies the requirement.

Application Functional Configuration (Oracle) The definition of key architectural setup parameters in an application instance or product group to reflect the financial and operating environment of the business organizations that transact within that application instance or product group.

Application Gap A recorded variance between some aspect of a business requirement and the capability or features of the application system that are necessary to satisfy the requirement.

Application Implementation Method (AIM) A method that comprises a flexible approach for implementing Oracle Applications.

Application Instance A unique set of application tables and processes. Note that two instances of the same version of an application may share one set of program modules but they will not share the same set of application tables.

Application Interface A mechanism used to transfer data between applications. A data flow between applications is always implemented as a set of one or more application interfaces; see also NEAR REAL-TIME INTERFACE and REAL-TIME INTERFACE.

Application Process An indication of the sequential execution of a series of system functions, possibly including manual functions as well; see also MANUAL FUNCTION, SYSTEM FUNCTION, and SYSTEM PROCESS.

Application System An automated collection of business functions, entities, modules, technology platforms, and documentation that performs a specified set of business functions; see also BUSINESS SYSTEM, MODULE, PLANNED RESPONSE SYSTEM, and SYSTEM FUNCTION.

Applications Distributed Interface (Oracle) An interface between similar or dissimilar Oracle Applications products that have data stored in multiple separate databases. The databases may or may not be resident at multiple sites (data centers). The corollary of this is that applications distributed interfaces may exist at a single site.

Applications Interface (Oracle) An interface between similar or dissimilar Oracle Application products that have data stored in the same database. The interface may or may not be supported within the standard package products.

Applied Business Technology (ABT) A company that manufactures tools to profile, estimate, and plan projects. Oracle Services has a global licensing agreement with ABT. Oracle Method makes use of Project Workbench, Methods Architect, and Project Bridge Modeler as its worldwide methods and project management software standard; see also PROJECT BRIDGE MODELER and PROJECT WORKBENCH.

Approach A variation or subset of a method, packaged in order to efficiently support the delivery of a service; see also TECHNIQUE.

Arm's Length Transaction An inter-company transaction that is treated as a third party transaction.

Assessment The process of determining the value, extent, or significance of pre-determined topics for investigation. Methods may include surveys (questionnaires, interviews), observation, testing, and content analysis.

Attribute 1. Any detail that serves to qualify, identify, classify, quantify, or express the state of an entity; any description of a *thing of significance*. 2. A database object that links or relates the values of two dimensions. For example, you might define an attribute that relates the Sales District dimension to the Region dimension so that you can select data for sales districts according to region.

Audience A group of information recipients who possess similar communication and organizational attributes, e.g. content needs, communication styles, work emphases, etc. In a communication intervention, the audience is not passive, in that it shares the work of the project through two-way communication and direct involvement in the design and implementation of the project process.

AutoText Entry Microsoft Word allows you to store frequently used text, graphics, and other items and quickly insert them into documents. These stored items are referred to as AutoText entries. If you store text and graphics as AutoText entries, you can retrieve them by clicking a button, typing a few keystrokes, or choosing a command. Components of the Deliverable Templates are stored as AutoText entries in global templates used to store boilerplate text; see also **BOILERPLATE TEXT**.

B

Background Process A non-interactive process that runs in an operating system environment and performs a task.

Backup and Recovery Strategy A storage and recovery strategy that protects against business information loss resulting from hardware, software, or network faults.

Bandwidth The amount of data that can be sent through a network connection, measured in bits per second (BPS). The speed and capacity of a network depend on both bandwidth and latency; see also **LATENCY**.

Base Language The language used for seed data of products that do not support multiple languages. (Multiple language products include FND, AK, AS, AX, and AZ).

Baseline 1. A starting point or condition against which future changes are measured.
2. A named set of object versions which fixes a configuration at a particular point in time. A baseline normally represents a milestone or key deliverable of a project; see also **CURRENT BUSINESS BASELINE**.

Bid and Proposal Management (BPM)

Specifies the process, tasks, responsibilities, and deliverables regarding how business opportunities are qualified and responded to, eventually leading to the issue of an authorized bid to a client.

Billable Project Expenses The project expenses that are billable to a client; see also **PROJECT EXPENSES**.

Billable Utilization The utilization that is billable to a client; see also **UTILIZATION**.

Black Box Test A test of all or part of an application system based upon fulfilling business requirements or meeting functional specifications. A black box test does not require an understanding of the actual code under test. A system test is usually a black box test.

Boilerplate Text Prewritten collection of words, sentences, headings, and formatting in a template that you can modify to suit a specific need; see also **AUTOTEXT ENTRY**.

Bottom-Up Estimate A task-level estimate derived by calculating the estimating factors critical to completion of each task; see also **ESTIMATING FACTOR**.

BPM see **BID AND PROPOSAL MANAGEMENT**.

BPR see **BUSINESS PROCESS REENGINEERING**.

BRS see **BUSINESS REQUIREMENTS SCENARIO**.

Browser see **WEB BROWSER**.

Budget A plan for determining in advance the expenditure of time, money, etc.

Business An enterprise, commercial entity, or firm in either the private or public sector, concerned with providing products or services to satisfy customer requirements.

Business Aim A statement of business intent measured subjectively; for example, to move up market, or to develop a sustainable level of growth; usually strategic or tactical with a 3-5 year horizon.

Business Area The set of business processes within the scope of a project.

Business Constraint Any external, management, or other factor that restricts a business or system development in terms of resource availability, dependencies, timescales, or some other factor.

Business Data Mapping Verification that the underlying table structures and attributes will support business processes.

Business Function Something an enterprise does, or needs to do, in order to achieve its objectives; see also APPLICATION SYSTEM, BUSINESS PROCESS, MANUAL FUNCTION, and SYSTEM FUNCTION.

Business Function Model A representation of all the business functions within a defined scope. A wide range of techniques is available for modeling business functions; see also FUNCTION DECOMPOSITION and FUNCTION HIERARCHY.

Business Goal A statement of business intent.

Business Location A uniquely identifiable geographic location, site, or place from which one or more business units may wholly or partly operate.

Business Model A model or collection of models representing the definition of key components of a business. Components may include models of processes, objectives, functions, and information; see also BUSINESS PROCESS MODEL, ENTITY RELATIONSHIP DIAGRAM, and FUNCTION HIERARCHY.

Business Object A physical or logical object of significance to a business; for example, a sales order, department, assembly, item, balance, or invoice. A business object is analogous to a class in object-oriented terminology.

Business Objective Business conditions which, if met, will solve the business problem statement. Well-defined objectives are measurable and often relate directly to business processes and deliverables; see also PROBLEM STATEMENT.

Business Organization Any part of a business treated for any purpose as a separate unit within the parent business organization; for example, a department, division, or subsidiary; see also BUSINESS UNIT.

Business Organization Type A classification of a business organization into one of several functional categories. Each business organization type has a distinct set of business requirements. All the business organizations of a certain type will typically require similar applications and system capabilities. A given site may house one or more business organization types. Since business organizations may be related in a hierarchy, a high level business organization may be composed of several business organizations of different types. For the purposes of application architecture analysis and design, it is generally useful to decompose the hierarchy of business organizations until it is composed of atomic organization types; see also CUSTOMER SERVICE, DISTRIBUTION, FINANCE, HUMAN RESOURCES, INFORMATION SYSTEMS, MANUFACTURING, MARKETING, PLANNING, RESEARCH AND DEVELOPMENT, and SALES.

Business Priority A statement of the level or urgency of important business needs.

Business Process The complete response that a business makes to an event. A business process entails the execution of a sequence of one or more process steps. It has a clearly defined deliverable or outcome. A Business Process is defined by the business event that triggers the process, the inputs and outputs, all the operational steps required to produce the output, the sequential relationship between the process steps, the business decisions that are part of the event response, and the flow of material and/or information between process steps; see also BUSINESS FUNCTION, BUSINESS PROCESS MODEL, and EVENT RESPONSE.

Business Process Model The collection of process flow diagrams that comprise the complete set of business processes within the application scope; see also PROCESS FLOW DIAGRAM.

Business Process Reengineering (BPR) The activity by which an enterprise reexamines its goals and how it achieves them, followed by a disciplined approach of business process redesign. A method that supports this activity.

Business Requirement A formal statement that describes application features necessary to support a business process step.

Business Requirements Mapping An activity that describes the business requirements for a business process in business language, optionally compares the current solution for a business requirement to a proposed solution and specifies details for the type and nature of the solution in a descriptive format. The deliverable can also be used as a record of key decisions, or as a placeholder in anticipation of additional detailed design documentation.

Business Requirements Scenario (BRS) A formal statement of the detailed business requirements for a business process, the source of these requirements, how these requirements will be satisfied (either by the application, manual process steps, workarounds or other application solutions), and what prototyping steps must be taken to prove the designs.

Business Rule A rule under which an organization operates. A policy or decision that influences the process step.

Business Solution Testing A technique by which management agrees that business requirements will be satisfied if the application and other tools perform as specified by process designs.

Business System A combination of people and automated applications organized to meet a particular set of business objectives; see also APPLICATION SYSTEM and PLANNED RESPONSE SYSTEM.

Business Unit Part of an organization treated for any purpose as a separate entity within the parent organization. Examples include a department or distribution center; see also BUSINESS ORGANIZATION.

Business View 1. A frequently used subset of information, readily intelligible to users and defined in business terms, derived from definitions held in an entity model. It is based on one entity and can encompass renamed attributes from the base entity or any other entity. 2. A custom-tailored presentation of the data in one or more tables. A business view can be thought of as a "stored query."

C

Cartridge A software component that, when linked with other cartridges, forms a complete software solution. Cartridges are "wrappers" around a program. The program may be written in any programming language, such as C, C++, or Java. The wrapper enables the program to interface with other cartridges on the same machine or across a network. Cartridges may be used in many places: the Oracle ConText Cartridge is an example of a data cartridge (residing on the database tier), while the Oracle Forms Load Balancing Cartridge is an example of an application cartridge (residing on the application tier).

Cascade Method Model of communication that begins at the top by having managers inform their direct reports, who then inform their direct reports, etc. so that information cascades down the organizational hierarchy until all stakeholders have received the necessary information. Effective in building commitment and cooperation among stakeholders, as most individuals prefer to receive information from their supervisors.

CASE see COMPUTER-AIDED SYSTEMS ENGINEERING.

Case Method the presentation of real or fictional situations or problems to learners to analyze, to discuss, and to recommend actions to be taken. Used to develop critical thinking skills and to gain new perceptions of concepts and issues.

CASE Tools A set of integrated Computer-Aided Systems Engineering (CASE) and application development tools that assist in software development, for example, analyzing business requirements, designing applications, generating application code, etc.

Certificate File A certificate file contains the identity of a "trusted source" that the desktop client uses to guarantee the authenticity of a JAR file. Information contained within the certificate file allows the desktop client to decrypt the digital signature of a JAR file. If the identity can be confirmed, the desktop client assumes the JAR file is safe to download and execute; see also DIGITAL SIGNATURE.

CDM see CUSTOM DEVELOPMENT METHOD.

Change A deviation from a currently established baseline.

Change Effort Any activity consciously undertaken by an enterprise, business unit, or individual, that will result in critical change in one or more areas of the organization, for example, new technology implementation; see also ORGANIZATIONAL EFFORT.

Change Management 1. The complete set of processes employed on a project to make sure that changes are implemented in a visible, controlled and orderly fashion. 2. The activity, or set of activities, undertaken to govern systematically the effects of organizational change. 3. A Focus Area Group within Program Management that incorporates all necessary Oracle Method processes to develop an integrated approach to change as part of a Program's Baseline Solution. 4. The management and control of organizational change, and its effects on staff, during a project. It includes key issues such as the impact of change on staff tasks, staff capabilities, retraining, cultural change and communication between the project team and staff in the organization.

Change Management Repository A system for maintaining configuration items. It provides other services such as version control, access control and information storage and retrieval which support the configuration management process.

Change Readiness A measure of an organization's state of readiness to realize change successfully; involves a consideration of an organization's high-impact leverage points for change, as well as any change impediments.

Change Request 1. A request for a change to the required behavior of a system, usually from a user as a result of reviewing current behavior. 2. The mechanism by which a change is requested, investigated, resolved and approved; see also IMPACT ANALYSIS.

Character Set A set of encoded binary values that represents the letters, numerals, and punctuation marks of a language, or of a group of languages that use similar written symbols. For example, the WE8ISO8859P1 character set can be used by English and many other languages that use a Latin-based alphabet and Arabic numerals. Terminals and printers handle text data by converting these encoded values to characters. A character set may also be called a codeset.

Class 1. A class refers to the delivery training to students on a certain topic(s). Usually, training is provided regarding Oracle applications such as bill of materials, work in process, etc. Training may also be provided regarding job policy and procedures, help desk operations, etc. 2. A description of a set of objects that share the same attributes, operations, methods, relationships, and semantics. A class may use a set of interfaces to specify collections of operations it provides to its environment. (UML 1.1 Semantics)

Class Session A class session consists of delivering a class to a specific set of students at a particular place and time.

Client A general term for the computer in a client/server configuration that requests services, data, or processing. The computer supplying the services, data, or processing is the server.

Client/Server A type of technical architecture that links many personal computers or workstations (clients) to one or more large processors (servers). Clients generally manage the user interface, possibly with some local data. Servers usually manage multiple-access databases, including ensuring data integrity and other invariants.

Client/Server Configuration A configuration in which one or several servers perform database processing or other functions for applications that are run on clients.

Coaching A technique of apprenticeship whereby the coach demonstrates and observes learners as they try to complete tasks and provides hints, help, and feedback as needed.

Codeset see CHARACTER SET.

Column A means of implementing an item of data within a table. It can be in character, date, or number format, and be optional or mandatory.

Common Business Function A business function that appears in more than one place in a hierarchy.

Communication Agent An individual who implements a communication activity, such as a communication event, who acts as a communication resource, information resource or channel.

Communication Campaign A series of activities undertaken to achieve the communication objectives for a particular audience.

Communication Event Those activities within the deployment of a communication campaign that are discrete and time-bound, such as a presentation, launch of a kiosk, or video conference.

Company A commercial business; see also BUSINESS.

Company Base Hardware Configuration The actual hardware configuration that supports the company base configuration; see also COMPANY BASELINE.

Company Baseline The Company Baseline is the supported configuration of hardware, software, bug patches, modifications, operating systems, etc. that is part of a common set of business systems for the entire enterprise. There may be lower level of Company Baselines such as a Latin America Baseline that is a subset of the Company Baseline.

Compensable Factors Aspects of a job that contribute to its level of compensation.

Competency Underlying and enduring characteristic of an individual that causes or predicts effective or superior performance in a job or activity.

Competency Model A set of underlying and enduring characteristics (such as skills and knowledge) of an individual that are required to perform a job effectively.

Completion Criteria Standards or rules which determine completion of a task to an acceptable level of quality.

Computer-Aided Systems Engineering (CASE) The combination of graphical, dictionary, generator, project management, and other software tools to assist computer development staff engineer and maintain high-quality systems, within the framework of a structured method.

Computer-Based Training Instructional technology with computer and software programs for individualized learning. The programs are designed to interface with the user in such a way as to simulate an encounter between a learning agent and a learner, for computer-assisted or -based learning; for example, the programs are configured to deliver instruction and evaluate learner progress; may include mechanisms that adapt the program to learner characteristics as the program progresses

Computer Network An interconnected group of computers.

Conceptual Architecture A high level model of an enterprise's business application that identifies the organizational and geographical deployment of the most critical application systems and the technology components required to support them. This model provides the direction for the detailed enterprise technical architecture analysis. It should reflect the vision of the client senior executive management for the future direction of the information systems in the enterprise.

Concurrent Manager A process manager that coordinates the processes generated by users' requests to run various data-intensive programs. An Oracle Applications product group can have several concurrent managers.

Concurrent Process A task run by a concurrent manager. A concurrent process runs simultaneously with interactive functions and other concurrent processes.

Concurrent Processing Server An Oracle Applications server that runs time-consuming, non-interactive tasks in the background. This server resides at the database tier of the internet computing architecture.

Concurrent Queue A list of concurrent requests awaiting completion. Each concurrent manager has its own queue of pending requests.

Concurrent Request A request issued to the concurrent processing server when you submit a non-interactive task, such as running a report.

Conference Room Pilot (CRP) A system test in an environment set up to simulate the future production environment.

Configuration A named set of configuration items. Configurations are used to hierarchically organize configuration items in order to facilitate their management; see also CONFIGURATION ITEM.

Configuration Change The implementation of one or more change requests which leaves the configuration in an internally consistent state; see also IMPACT ANALYSIS.

Configuration Item A deliverable or deliverable component which is placed under configuration management.

Configuration Management The process of managing hardware, software, data, and any other documentation needed during the development, testing, and implementation of information systems.

Congruence Model Organization model, based on open systems theory, that stresses the interdependence between systems and the ripple effect of inputs across systems. Views strategy as driven by organization context (inputs) -- company history, resources, the firm's environment, that require organizational transformation to maintain correspondence or balance between the interrelated systems.

Constraint see BUSINESS CONSTRAINT and BUSINESS RULE.

Consulting Grade Level A grade level assigned to Oracle Services consulting resources used to calculate the cost of a resource's labor; 1-Administrative Assistant to 10-Regional Vice President.

Context Diagram A high-level diagram, indicating the major functional components and major internal and external interdependencies of a system.

Contingency Work effort allotted in a workplan for all unforeseen but possible occurrences of additional work.

Contribution Margin see MARGIN AMOUNT and MARGIN PERCENTAGE.

Controlled Document A document which constitutes or represents a project deliverable for approval internally or by the client, and is subject to change control.

Controls Environmental information that affects what is possible; constraints or limits.

Conversion Execution An example of a Conversion Execution would be where open orders are converted from a legacy system to Oracle Order Entry in one region on a given date. Other regions may go live with Order Entry on a different date requiring other order entry Conversion Executions.

Conversion Fit Analysis A formal statement of subsystems and entities converted into the new applications database.

Core Tasks. The minimum set of tasks necessary to complete an Oracle Applications implementation or migration project.

Core Business Process A major, driver process that affects or influences business objectives. The set of business processes identified in association with the objectives represent the "core processes" of the business area.

Corporate Repository Location of a collection of documentation, customizations, modifications, or enhancements designed to alleviate the recreation of successfully completed work.

Corporation A group of businesses acting as part of a single legal entity.

Cost The amount allotted or spent to acquire, deliver or produce anything, for example, the cost of labor to deliver consulting services, the amount spent on incidental costs to deliver consulting services, the amount spent on hardware, software, etc.

CPM see CRITICAL PATH METHOD NETWORK.

CPU Support Identification (CSI) A number assigned to each Oracle customer. The number is required in order to use the Support hotline or RTSS. Associated with the CSI number is the client's name, RDBMS version, supported products, operating system, telephone, area code, and Technical contracts.

Critical Path Method (CPM) Network A network diagram that shows tasks, their dependency links, and their critical path.

Critical Success Factor (CSF) A business event, dependency, product, or other factor that, if not attained, would seriously impair the likelihood of achieving a business objective.

CRP see CONFERENCE ROOM PILOT.

CSF see CRITICAL SUCCESS FACTOR.

CSI see CPU SUPPORT IDENTIFICATION.

CT see CUSTOM TRAINING.

Current Business Baseline The set of business process and function models representing the current application system that supports the business area; see also BUSINESS AREA, BUSINESS PROCESS MODEL, and BUSINESS FUNCTION MODEL.

Custom Code Coding added to a packaged application or module generated by a CASE tool to implement functionality that the application or generator has not provided; see also CUSTOM EXTENSIONS.

Custom Development Method (CDM) A structured method for full life-cycle custom development projects; see also LINE OF BUSINESS.

Custom Extensions Custom modules that extend the functionality of packaged applications without modifying the base functionality; see also CUSTOM CODE.

Custom Support System A custom-built information system conceived, developed, and managed at the enterprise level. It provides a service to one or more sites or business units. Such systems are designed to be generic, flexible, and independent of the particular service sites. Examples of custom support systems are: enterprise data warehouses, data registry propagation systems, and distributed application data transport systems.

Custom Training (CT) An organizational change management area providing organizations with training programs designed to meet specific training needs identified through a preliminary training needs assessment. Custom training seeks to re-skill a work force to optimize performance with new technology.

Customer Service The business organization that 1. manages returns and repairs of products that may or may not have been purchased from the enterprise originally; 2. responds to customer complaints about service or products; see also BUSINESS ORGANIZATION TYPE.

Customization A change made to the standard Oracle software which implements a solution to a gap.

Cut-Over Transition to a new application system in a live, production mode of operation.

D

Data Cleansing The transformation of data in its current state to a pre-defined, standardized format using packaged software or program modules.

Data Conversion The movement of data from a legacy system or application to a replacement application or subsystem.

Data Definition The specification of a data element to be maintained. The specification includes datatype, size, and rules about processing: for example, derivation, and validation; see also BUSINESS RULE.

Data Definition Language (DDL) A subset of SQL used to create, alter, drop, and otherwise change definitions of tables, views, and other database objects.

Data Dictionary 1. A part of a database that holds definitions of data elements, such as tables, columns, and views. 2. A set of database tables and views that contains administrative information about users, data storage, and privileges. It is created and maintained automatically.

Data Integration The movement of data between two co-existing systems. The interfacing of this data may occur once every hour, once a day, etc.

Data Integrity Testing Verification that converted data is accurate and functions correctly within a single subsystem or application.

Data Migration The movement of data from one database to another database – but not necessarily to a working application or subsystem tables.

Data Model A representation of the specific information requirements of a business area; see also ENTITY RELATIONSHIP MODEL.

Data Partitioning A technique to improve application performance or security by splitting tables across multiple locations.

Data Profile A description of the business conditions that are needed to test the application system.

Data Registry The master copy of the data associated with a business object. Several databases may share access to a common data registry to provide consistency and eliminate redundant entries across multiple applications and databases. An example of a data registry would be a shared customer master. All updates and changes would be made to the customer master data registry and are then propagated to subscribing sites. All systems requiring customer information would interface with the customer data registry.

Data Registry Interface An interface that transfers data registry data between similar or dissimilar applications.

Data Replication The copying of data to and from sites to improve local service response times and availability; frequently employed as part of a backup and recovery strategy.

Data Transfer The physical movement of data between applications, perhaps across sites.

Data Transformation The process of redefining data based on some predefined rules. The values are redefined based on a specific formula or technique.

Data Translation The process of redefining data in a manner differing between its original representation and its final representation.

Database A collection of data, usually in the form of tables or files, under the control of a database management system. The term can also refer to the software used to create, store, and manage this data—for example, the Oracle9.5i Server.

Database Architecture The collective application and database instances that comprise the complete system.

Database Function A callable routine executed within a database server environment.

Database Index A mechanism to locate and access data within a database. An index may quote one or more columns and be a means of enforcing uniqueness on their values.

Database Instance One set of database management processes and an allocated area in memory for managing those processes. Typically, a database instance is associated with one database. Note that a database instance may process data for one or more applications. There is always a one-to-one correspondence between an ORACLE instance and a system global area (SGA).

Database Management System (DBMS) A software environment that structures and manipulates data, and provides data security, recovery, and integrity.

Database Object A logical entity created and stored in a database. Tables, views, synonyms, indexes, sequences, stored procedures, and triggers are all examples of database objects.

Database Package An Oracle database object comprised of PL/SQL code allowing execution of code at the server based on specific events or triggers.

Database Server A software server on which the database holding Oracle Applications data resides. The database server processes SQL requests from other machines, such as forms servers and concurrent processing servers. The database server executes on a hardware platform server.

Database Schema see ORACLE ID.

Database Space The amount of disk space used by a set of database objects.

Dataflow A named flow of information between business functions, datastores, and external entities represented as an arrow on a dataflow diagram; see also BUSINESS FUNCTION, DATASTORE, and EXTERNAL ENTITY.

Dataflow Diagram A diagram representing the use of data by business functions; see also DATAFLOW, DATASTORE, EXTERNAL ENTITY, and PROCESS.

Dataflow Diagramming A technique for expressing the significant dataflows of a business system.

Datastore A temporary or permanent storage concept for logical data items used by specified business functions and processes.

DB2 An SQL-based database management system.

DDL see DATA DEFINITION LANGUAGE.

Decision A point in a process flow where there is a choice of possible paths. The outcome of the decision determines which path the flow follows.

Decision Point The diagrammatic representation, on a process flow diagram, of a decision which results in the subsequent execution of one of two or more alternative sequences of process steps. The decision is actually made during the execution of the previous process step. The decision point is shown after that step for diagrammatic convenience.

Decision Support System (DSS) An application primarily used to consolidate, summarize, or transform transaction data to support analytical reporting and trend analysis.

Deliverable Something a project must produce in order to meet its objectives. A deliverable must be tangible and measurable.

Deliverable Component A part or section of a deliverable. A deliverable component may be the output of a task step.

Deliverable Guideline A detailed description of a deliverable that includes: detailed description, usage, audience, and distribution, format guidelines, control, template, and samples; see also DELIVERABLE.

Deliverable Management The process of managing the creation, review, modification, and distribution of deliverables provided to a client. Software may be included as one of the managed deliverables.

Deliverable Template A tool designed to aid in production of a deliverable; a template that gives the format and structure of a deliverable; see also DELIVERABLE.

Delivery Vehicle The mechanism for producing or implementing something. For example, SQL* Forms is a vehicle to produce computer programs.

Demonstration Product Group A product group that includes predefined transaction data for Oracle Applications products. It is used primarily for system testing and user training. See also PRODUCT GROUP.

Dependency 1. An indication that one task cannot begin until another task has ended, or progressed to a certain specified level of completion; see also PREDECESSOR and SUCCESSOR. 2. A relationship between two modeling elements, in which a change to one modeling element (the independent element) will affect the other modeling element. (UML 1.1 Semantics)

Dependent Product An Applications product that is not licensed, but whose files are shared in part by a fully installed Applications product. A dependent product is also known as a shared product.

Deployment The total set of activities associated with the production implementation of a set of applications in specific location(s) at a particular point in time.

Desktop Client A computer that sends user requests to the forms server and handles responses such as screen updates, pop-up lists, graphical widgets, and cursor movements.

Detailed Deliverable A deliverable whose structure originated from a high level deliverable, but that contains significantly more detail; see also DELIVERABLE and HIGH-LEVEL DELIVERABLE.

Development Life-Cycle A complete process of developing computer systems.

Digital Signature A means of guaranteeing the authenticity of a program or collection of data, such as a JAR file. It is typically an encrypted message that contains the identity of the code's author. See also certificate file.

Distributed Concurrent Processing See PARALLEL CONCURRENT PROCESSING.

Distributed Directory Structure Applications product files installed in more than one file system (for example, when there is insufficient disk space in a single file system for all Applications product files).

Distributed Database A database that is physically located on more than one computer processor. It is connected via some form of communications network. An essential feature of a true distributed database is that users or programs work as if they had access to the whole database locally.

Distributed Processing The ability to have several computers working together in a network, where each processor runs different activities for a user, as required.

Distribution A high-level business function responsible for configuring the firmware parts of inventory, housing finished goods inventory (FGI), and shipping finished goods to customers or to internal sites; often synonymous with a business organization; see also BUSINESS ORGANIZATION TYPE.

Distribution Management Distribution management is the process to manage the release and distribution of software in a controlled manner. This includes the kitting of software, documentation, patches and other components. It involves initial releases and subsequent patch tapes, upgrades and new releases. It also includes providing the organizational challenges of obtaining release approval from the Company Baseline Review Board (CBRB).

DSS see DECISION SUPPORT SYSTEM.

E

EBF see ELEMENTARY BUSINESS FUNCTION.

EF see ESTIMATING FACTOR.

Effort The amount of work, measured in person-hours, to perform a task.

Electronic Performance Support Systems An integrated electronic environment easily accessible by users and structured to provide immediate, individualized, on-line access to information, software, guidance, advice, assistance, data, images, tools, assessments, and monitoring systems to permit users to perform their jobs with a minimum of support and intervention by others.

Electronic Performance Support Tool Software designed to improve employee productivity by supplying immediate on-the-job access to integrated information, learning opportunities, and expert consultation with scope and sequence controlled by the user.

Element A thing of significance about which information is recorded; a component at the most useful, basic level.

Element Type Any element held in the repository is classified as a particular type. Examples of element type are entity, attribute, program module, process, table, diagram, and text. Occurrences or instances of these are called elements.

Elementary Business Function (EBF) A business function which if started, must either complete successfully or, if it cannot complete successfully, must undo any effects that it has had up to the point of failure. An elementary business function changes a business's data from one consistent state to another; see also FUNCTION HIERARCHY and COMPOSITE BUSINESS FUNCTION.

End User see USER.

Enterprise A group of departments, divisions, or companies which make up an entire business.

Enterprise Support Systems The set of all computer-based systems, documents, and procedures used in support of business enterprise operations.

Enterprise Technical Architecture (ETA) A series of rules, guidelines, and principles used by an organization to direct the process of acquiring, building, modifying, delivering, and integrating Information Technology resources throughout the enterprise. These resources can include equipment, software, business processors, protocols, standards, methodologies, IT organizational structures and more.

Entity A thing of significance, whether real or imagined, about which information needs to be known or held. It is implemented in a database as one or more tables.

Entity Integrity Rules The rules that specify valid values or combination of values for attributes of an entity. These may include unique identifiers, domains, and multi-attribute validation rules; see also BUSINESS RULE and REFERENTIAL INTEGRITY CONSTRAINT.

Entity Relationship Diagram (ERD) A diagram that pictorially represents entities, the relationships between them and the attributes used to describe them; see also ATTRIBUTE, ENTITY, and RELATIONSHIP.

Entity Relationship Model A type of data model. Part of the business model that consists of many Entity Relationship Diagrams; see also DATA MODEL.

Entity Test A detailed test of certain key data elements of a business entity that is being converted or interfaced from one system to another.

Environment File A command file that sets environment variables. Only servers running UNIX or Open VMS use these files. Windows NT servers use the Windows registry instead.

Environment Variable A variable maintained by the UNIX shell that can be referenced by any program running within the shell. Environment variables hold values used by many Oracle programs and utilities. On Windows NT, a string consisting of environment information, such as a drive, path, or filename, associated with a symbolic name. Use the System option in the Control Panel or the set command from the Windows NT command prompt to define environment variables.

ERD see ENTITY RELATIONSHIP DIAGRAM.

Ergonomics The use of good design techniques that emphasize ease-of-use.

Estimate A preliminary calculation of the time and cost of work to be undertaken. The *construct* option calculates estimates using bottom-up, percent adjustment, or top-down techniques in Project Bridge Modeler; see also BOTTOM-UP ESTIMATE, PERCENT ADJUSTMENT ESTIMATE, WORK ESTIMATE, and TOP-DOWN ESTIMATE.

Estimated Function Point Count 7 Function Points per System Entity + 5 Function Points per System Function.

Estimating Factor (EF) A metric that describes an important project characteristic, used to estimate either the amount of effort that project tasks will take, or project risk or complexity. The best EFs are those that represent counts (number of users, objects); see also BOTTOM-UP ESTIMATE.

Estimating Formula A formula that uses estimating factors to derive an estimate for a task.

Estimating Guideline Text which describes in detail how a task is estimated.

Estimating Model The combination of estimating factors and estimating formulas necessary to completely estimate a route.

ETA see ENTERPRISE TECHNICAL ARCHITECTURE.

Event 1. An occurrence in a business's environment to which that business must respond; see also BUSINESS SYSTEM and EVENT RESPONSE. 2. An occurrence or happening of significance to a task or program, such as the completion of an asynchronous input/output operation. 3. A transaction or other activity that affects the records in a file.

Event Mechanism A permissible way in which an event is recognized. For instance, a Customer Order may be entered into the system if received by fax, phone, mailed-in purchase order or internet-facilitated, but not by markings on a scratch pad.

Event Response An event and the business process which responds to that event; see also APPLICATION SYSTEM, BUSINESS SYSTEM, and EVENT.

Executive Information System (EIS) A reporting application targeted for use by executives. Usually such applications have extremely user-friendly, graphical interfaces with a small local datastore derived from connection to a data warehouse. It is often used synonymously with decision support system.

Expense The amount of money allotted or spent to cover incidental costs (for example, travel and living) or the cost of something (hardware, software, etc.) to deliver consulting services.

Export Utility An Oracle Database Server utility used to write database data to operating system files external to the database. These files can then be used with the Import utility to read the data back into the same database or into a different one.

Expense Reimbursement Expenses for reimbursement by the client either allotted or received.

Extension Enhancements to an Oracle Applications system made to fit the needs of a specific user community.

Extensibility (CASE Dictionary) It is often useful to add new elements, properties, and associations into the Dictionary. This is achieved by a facility known as user extensibility.

External Business Function A business function that is outside the scope of the application system, that acts as a source or recipient of dataflows.

External Entity An entity that is outside the scope of application system, that acts as a source or recipient of dataflows. An external entity might be a person, a business unit, another application system, or any other thing that might provide or receive information from a function within the application system; see also ENTITY.

External Equity Measure of the fairness of a job's compensation when compared to the labor market as a whole within the region, profession, or industry; see also INTERNAL EQUITY.

External Process Step A process step that is performed by an agent outside the business area; see also INTERNAL PROCESS STEP and PROCESS STEP.

F

Fee A charge, compensation, or payment for a service or product.

Feedback Response, including corrections, additions, and approval, elicited from users, stakeholders, sponsors, and others, to any deliverable or deliverable component.

Feedback Session A meeting organized to present work in progress in order to gain feedback; see also FEEDBACK.

FH see FUNCTION HIERARCHY.

Field A means of implementing an item of data within a file. It can be in character, date, number, or other format and be optional or mandatory.

File Transfer Protocol (FTP) The physical movement of data files between applications, often across sites.

Finance A high-level business function that handles accounting and financial functions of a company; for example, corporate, division, and subsidiary headquarters; often synonymous with a business organization; see also BUSINESS ORGANIZATION TYPE.

Financial Organization A business organization that performs one or more financial business functions. The data is segregated for organizational management or security. When mapped onto Oracle Applications, a financial organization is required to have a defined set of books. This concept is more general than the set of books. It is replacing the set of books as the key financial application functional configuration parameter in releases of Oracle Applications starting with 10.6.

Flexfield A user-definable field in an Oracle Application that is made up of segments. Each segment has a name you assign and a set of valid values.

Focus Area (Project) 1. A group of associated Oracle Method processes that produce program level deliverables during the Baseline Solution Development stage. 2. A scoped area of the client organization with common responsibilities and information requirements. The focus area provides the scope for incremental development efforts and may be referred to as the INCREMENT. The focus area can overlap business functions or may reside in a specific business function.

Focus Group A small group selected to provide opinions and responses to topics or issues presented in a group setting; an assessment technique.

Foreign Key One or more columns in a relational database table that implement a many-to-one relationship that the table in question has with another table or with itself.

Form 1. A program for viewing or entering information. Normally it refers to a rectangular area of the screen, which has the appearance of a paper form, through which you can view or manipulate information in a database. 2. A logical collection of fields, regions, and graphical components that appears on a single screen. Oracle Applications forms resemble paper forms used to run a business. Data is entered by typing information into the form.

Formal Build Construction of an information system by the well established steps of specify, code, test, and correct.

Format The type of data that an attribute or column may represent; for example, character, date, number, sound, or image.

Forms Client A Java applet that runs on a desktop client and provides the user interface and interaction with a forms server.

Forms Metrics Server A machine that manages and distributes requests from forms clients among multiple forms servers; see also LOAD BALANCING and METRICS SERVER.

Forms Server A type of application server that hosts the Oracle Forms Server engine. It mediates between the desktop client and the database, providing input screens to the desktop client and creating or changing database records based on user actions.

Fourth-Generation Programming Language (4GL) A language that manipulates high-level objects, such as screen items and database tables, by declaring what is to be done to them rather than procedurally describing how it is to be done, as in 3GLs.

FPE see FUNCTION POINT ESTIMATE.

FTP see FILE TRANSFER PROTOCOL.

Function see BUSINESS FUNCTION, ELEMENTARY BUSINESS FUNCTION, and SYSTEM FUNCTION.

Function Decomposition A technique for modeling business functions by decomposing a single business function into a number of lower level business functions, and then progressively decomposing these until the appropriate level of detail is reached. Function decomposition gives rise to functions arranged in groups or hierarchies known as a business function hierarchy; see also ELEMENTARY BUSINESS FUNCTION, FUNCTION HIERARCHY AND COMPOSITE BUSINESS FUNCTION.

Function Dependency The dependency of one function's commencement upon the completion of another function.

Function Dependency Diagram A visual means of recording dependencies between business functions.

Function Hierarchy A grouping of elementary business functions into one or more hierarchical levels. Typically the highest level corresponds to the company organization, and the middle level corresponds to a grouping of available application functions; see also BUSINESS FUNCTION and FUNCTION DECOMPOSITION.

Function Label A unique ID, within an application system, used for a business function.

Function Name A short, succinct sentence, starting with a verb, describing a business function; see also FUNCTION LABEL.

Function Point Analysis (FPA) A technique used to estimate system size. Using Function Point Analysis, you can calculate system size based on the number of functional element types in the system you are building, as adjusted by its general system characteristics and your technology productivity factor.

Function Point Estimate (FPE) An estimate of work effort produced using Function Point Analysis or an equivalent method.

Functional Currency The principle currency used to record most business transactions and maintain accounting data while working within a particular Oracle Application set of books.

G

Gantt Chart A scheduling tool used to display the status of a project's tasks. A Gantt chart shows each task's duration as a horizontal line. The ends of the lines correspond to the task's start and end dates.

Gap A gap is a relationship between a requirement and an application function where the standard application function will not meet the requirement.

Gap Analysis 1. The process of determining, documenting, and approving the variance between business requirements and system capabilities in terms of packaged application features and technical architecture. 2. The process of determining and evaluating the variance or distance between two items properties being compared. 3. The process of determining and documenting the variance between a business vision and the existing processes and organization.

General Education Course A course that educates the project team or users in fundamental business concepts. It exposes attendees to new business approaches practiced in industry and provides them with a common understanding of relevant business issues.

Generator A mechanism for transforming the specification of a module into executable program code, also known as a code generator.

Generator Template A skeleton or outline program from which a generator can reuse common elements; for example, boilerplate information, window sizes, *OK* and *Quit* buttons.

Gigabyte (GB) A unit of memory or disk space equal to 1,073,741,824 bytes. One Gigabyte is equal to 1,024 Megabytes. Often rounded to 1,000,000,000 bytes.

Goal see BUSINESS GOAL.

Grade Level Relative ranking of jobs based on their difficulty or sphere of responsibility, rather than their job content; see also JOB CLASSIFICATION.

Graphical User Interface (GUI) An interface used with personal computers and workstations that allows the user to access fields and regions of the screen with a pointing device, typically a mouse. The acronym is pronounced "goo-ee".

Group Interview Any session where users, stakeholders, or sponsors collectively discuss the requirements, priorities, design, or implementation of a business solution system; see also FEEDBACK and WORKSHOP.

Guideline Text that provides instructions and advice for performing a task and suggests possible approaches.

GUI See GRAPHICAL USER INTERFACE.

H

Hardware Node A computer on a network; for example, clients and servers.

Hardware Platform The actual physical hardware that supports one or more software server processes.

Help Desk A support system designed to assist end users with technical and functional questions and problems.

Help text see METHOD HELPTXT.

High-Level Deliverable A deliverable that specifies a framework into which further details can be added; see also DELIVERABLE, DETAILED DELIVERABLE, and INITIAL DELIVERABLE.

HRIS see HUMAN RESOURCE INFORMATION SYSTEM.

HTML See HYPERTEXT MARKUP LANGUAGE.

HTTP See HYPERTEXT TRANSFER PROTOCOL.

HTTP Listener A program on a web server that accepts and processes incoming HTTP requests from web browsers.

Human Performance Support Any process, tool and system that reinforces or encourages the execution of a particular behavior or set of behaviors to meet performance goals.

Human Performance Support Systems an integrated collection of all performance support components (infrastructure, processes, protocols, procedures, tools and practices) for a given group or organization, designed to reinforce or encourage the execution of prescribed behaviors within the group.

Human Performance Technology The field of expertise within organizational development/behavior to set up performance management systems in support of meeting performance expectations.

Human Resources The high-level business function involving the management of human resources and payroll functions of a company; often synonymous with a business organization; see also BUSINESS ORGANIZATION TYPE.

Human Resource Allocation Determining the number and type of job functions and staffing levels appropriate to organizational goals.

Human Resource Planning Forecasting the labor force needs (hiring, termination) for the organization.

Hypertext Markup Language (HTML) A simple language used to format documents, predominantly for viewing with a web browser. Portions of text or images, called hypertext, can be associated with other documents.

Hypertext Transfer Protocol (HTTP) The TCP/IP-based network protocol used to transmit requests and documents between a web server and a web browser.

I

Impact Analysis The process of understanding the complete effect of a particular change; see also CHANGE REQUEST.

Implementation Questionnaire A tool you use to collect business and system information during a business baseline interview. It consists of a pre-built set of questions organized by business function that are to be supplemented by the analyst with relevant company terms and other characteristics before use in driving the interview.

Import Utility An Oracle Database Server utility used to read operating system files written by the Export utility. You use it to restore data into a database.

Index see DATABASE INDEX.

Information Access Model A model that depicts access to key process and organization information for reporting or security purposes.

Information Flow Model A model that visually depicts information flows in the business between business functions, business organizations and applications.

Information Model A model of the information needed to support a business or other activity. Requirements are typically defined as lists of detailed items as well as summarized data from business transactions, such as orders and purchases, and master records, such as customers and vendors. How frequently this information must be made available is also part of the requirement. Information requirements (the what and when) are turned into functional specifications (the how) of an information system by systems analysts. The information is defined as a collection of data elements that are obtained by running query and report programs against a particular database or group of databases. The data and information that is stored in the databases in the first place is also derived from the information requirements; see also DATA MODEL.

Information Systems (IS) A system for managing and processing information, usually computer-based. Also, a functional group within a business that manages the development and operations of the business' information systems.

Information Services Strategy Planning

Method (ISSPM) A method that aligns information technology priorities with business strategies and defines the approach to take to achieve those goals.

Information Warehouse see DATA WAREHOUSE.

Initial Deliverable A deliverable is *initial* if it is intended to be updated later. An initial deliverable is usually preliminary and its content changeable by a later task when more information is known.

Initialization Parameters Parameters defined in an initialization file that configure an Oracle Database Server. The parameters affect how the database performs.

Input Process-Output (IPO) A technique is a generic modeling tool that was designed for framing complete process specifications by identifying: inputs, rules, process step descriptions, tools, and outputs.

Installation The loading of an instance of an application system that is complete, tested, operational, and ready. An installation includes all necessary software, hardware (including terminals, networks, etc.) and documentation, and includes all required data; see also APPLICATION SYSTEM.

Instance 1. An entity to which a set of operations can be applied and which has a state that stores the effects of the operations (UML 1.1 Semantics); see also DATABASE INSTANCE.

Integration Fit Analysis A statement of fit and gaps for integration points between unique applications and installations of the same application.

Integration Test A sequence of steps or set of procedures to verify the inter-operability of various system components; see also SYSTEMS INTEGRATION TEST.

Inter-Company Transaction A transaction between two legal entities that share common ownership, for example, under the same holding company. The two companies usually have an account relationship (for example, an inter-company suspense or clearing account) that facilitates inter-company accounting.

Internal Equity Measure of the fairness of a job's compensation when compared to similar jobs within the same organization.

Internal Concurrent Manager A concurrent manager process that monitors, controls, and dispenses requests to all other concurrent manager processes.

Interface 1. A linkage between systems which can be either automated (via software programs) or procedural (manual). 2. A declaration of a collection of operations that may be used for defining a service offered by an instance. (UML 1.1 Semantics).

Interface Programs A set of programs that systematically link two or more systems to each other.

Internal Process Step A process step that is performed within the business area; see also EXTERNAL PROCESS STEP and PROCESS STEP.

Internet Computing Architecture An Internet-based framework that provides communication in a multi-tier architecture. Also known as the *Internet Computing Model*.

Intranet A network of computers that are internal to a company. A company's intranet is basically a local extension to the globally-distributed Internet, which is a global inter-connected network of computers and smaller computer networks.

Inventory Organization A fundamental Oracle manufacturing and distribution application functional configuration parameter. The inventory organizations are derived by mapping inventory or manufacturing business organizations onto Oracle Applications. Typically they will be manufacturing plants, warehouses, divisions, or departments.

IPO see INPUT-PROCESS-OUTPUT.

IS see INFORMATION SYSTEMS.

ISSPM see INFORMATION SERVICES STRATEGY PLANNING METHOD.

Issue A situation or concern which requires a resolution. Some issues, if not addressed, could adversely impact the success of a project.

IT see ITERATED TASK.

Item Master The master list of all items available for transaction within a product group containing Oracle manufacturing applications. It is also a key application functional configuration parameter.

Iterated Task (IT) A task that is repeated once for each iteration in order to increase the quality of the deliverable to a desired level or to add more detail to the deliverable. Iterated tasks are shown as discrete in the workplan; see also TASK, ITERATION, and ONGOING TASK.

Iteration Indicates the number of times or degree to which a task or task group should be repeated, in order to either increase the quality of the task/group deliverables to a desired level, to add sufficient level of detail, or to refine and expand them on the basis of user feedback. A task or group may be “singly iterated” (the expectation is that it will be performed without repetition in the project), or “multiply iterated” (it will be performed successively, multiple times), resulting in a single deliverable for each task (note this is not true for multiply instantiated tasks). For multiple iterations, it is important for planning purposes to give notes on the number of times or the degree to which a task or group should be iterated, based on experience; see also ITERATIVE BUILD and ITERATIVE DEVELOPMENT.

Iterative Build Iterative construction of an information system by means of a cycle of code (or generate), test, review, starting from a prioritized (MoSCoW) list of requirements and guided by user feedback; see also MOSCOW LIST.

Iterative Development The application of a repeating cycle of the same or similar activities performed on the same piece of functionality that improves or grows into completion through the iterations.

J

JAR (Java Archive) File A collection of Java classes compressed into a single file for faster download to a desktop client.

Java A computing language used, among other things, to produce programs that can be downloaded and run on a desktop client using a web browser. It is also used to produce platform-independent programs that run on a server, either interactively or when invoked through a request from a web browser; see also **APPLET** and **SERVLET**.

Java Class Components of a Java program that define objects and operations performed on objects. In the Architecture and Installation manuals, *Java class* also identifies an operating system file that contains a program or part of a program written in Java.

Java Database Connectivity (JDBC) A Java programming interface that enables Java programs to execute SQL statements and access databases, such as the Oracle Database Server.

Java Virtual Machine An interpreter that translates a compiled Java program, called bytecode, to machine code. JVM makes Java portable, because each operating system's JVM translates bytecode to instructions that the microprocessor can execute. An appletviewer or Java-enabled web browser has an internal JVM that allows it to execute applets or applications written in Java.

JDBC see **JAVA DATABASE CONNECTIVITY**.

Jintitator A browser plug-in that runs a Java applet or application.

Job Analysis Process of obtaining information about jobs; may involve content analysis, interviews, observation, questionnaires, diaries.

Job Classification Process of using job descriptions to organize jobs within a given organization into classes on the basis of duties and responsibilities. Persons within the same job classification are thought to be equally qualified for all other positions within that class; see also **GRADE LEVEL**.

Job Content Duties and responsibilities of a particular position; see also **JOB STRUCTURE**.

Job Description Written description of a job; typically includes job title, summary of major responsibilities, authority and accountability (including reporting relationships), traits or attributes important to the job, and description of tasks.

Job Design/Redesign Creating or adjusting job content to meet new requirements (dictated by new organizational objectives and / or technology-induced change effort); some jobs may be absorbed by other departments or disappear.

Job Enlargement Adding additional but similar duties to a job; also called *Horizontal Loading*.

Job Enrichment Adding more responsibility or authority to a job; also called *Vertical Loading*.

Job Evaluation Analyzing the content of a job to determine level or grade and commensurate compensation.

Job Incumbent Person currently performing the job.

Job Inventory List of tasks presented to job incumbents to verify whether they perform these tasks and to answer qualifying questions about them.

Job Performance Demonstration of skills or competence in a job; sometimes measured by productivity.

Job Structure Duties and responsibilities of a particular position; see also JOB CONTENT.

Job-Person Matching Analyzing skills profiles and worker specifications to determine the best fit between workers and jobs.

JVM see JAVA VIRTUAL MACHINE.

K

Key A way of accessing something. Any set of columns used for retrieval of rows from a table; see also COLUMN and UNIQUE IDENTIFIER.

Key Deliverable A key deliverable represents the culmination, end result, or major milestone of activities performed during a Phase. A major deliverable that is usually reviewed with the client, signed off, and placed under change control. Key deliverables should not be confused with deliverables produced by core tasks. Core tasks produce the minimum set of deliverables needed for every project; key deliverables represent the end result and define the end of each phase. However, not every key deliverable is created for every project. For example, an integration-tested system will only exist on projects where interfaces are constructed; see also DELIVERABLE.

Key Performance Indicator (KPI) A significant measure used on its own, or in combination with other key performance indicators, to monitor how well a business is achieving its quantifiable objectives.

Key Resource A person with a wide range of skills or experiences who can be effective in many types of tasks, or is critical to the completion of a specific task.

KPI see KEY PERFORMANCE INDICATOR.

KSAO Knowledge, skills, aptitudes, and orientation (attitudes, values, etc.).

L

Labor Cost see COST.

Labor Cost Rate The rate (internal cost) for each consulting grade level for delivering services to a client.

Labor Fee see FEE.

Labor Fee Rate The rate (price) for each consulting grade level charged for delivering services to a client.

LAN See LOCAL AREA NETWORK.

Latency In networking, the amount of time it takes a packet of data to travel from a source to its destination. Latency is imposed on the network by devices that delay the forwarding of network packets. The speed and capacity of a network depend on both bandwidth and latency; see also BANDWIDTH.

Leaf Function An function that is not decomposed at the bottom of a function hierarchy. It may be a leaf function because definition is incomplete; see also ATOMIC FUNCTION.

Learner Any individual engaged in acquiring new skills, attitudes, or knowledge, whether with a specified sequence of instruction (intentional) or a random assortment of stimuli (incidental).

Learning Styles psychological traits that determine how individual learners and cultural groups of learners may perceive, interact with, and respond to learning environments.

Learningware Instructional material created to facilitate learning, including paper-based, technology-based and instructor-led media; in OCM, instructional material generated from a custom training curriculum plan, designed to re-skill individuals within the IS department, end-user, and management communities during a new technology implementation.

Legacy Application Interface A heterogeneous interface between Oracle Applications and a pre-existing and preserved legacy system.

Legacy System An existing system repository of information and processes.

Legal Entity A high level business organization that operates within the bounds of a specific set of legal requirements (including currency), typically pertaining to tax regulations and reporting requirements

Line of Business (LOB) Each service within Oracle Services is a line of business. For example, Custom Development, Application Implementation, and Business Process Reengineering are all lines of business.

Link Test A test to identify errors in linked modules of an application system. Link testing is an extension of module testing carried out on a number of levels of detail. Examples include, linked modules of a program, linked programs of a functional area or subsystem, and linked subsystems of the complete application system. The link test is usually a white box test; see also MODULE INTEGRATION TEST.

Live Implementation process has ended and the solution is put into production.

Load Balancing Distributing tasks to the server that is least busy when several servers are handling the same workload. For example, the Oracle Forms Load Balancing Cartridge has a single coordinator (the Metrics Server) that monitors load information from other forms servers and determines which server currently has the lightest load. Upon receiving a request from a forms client, the Forms Metrics Server can balance the processing load by instructing the forms client to connect to the forms server that has the lightest load. This server is called the *least loaded host*.

LOB see LINE OF BUSINESS.

LOCAL Under Windows NT, an environment setting that identifies the network alias of an ORACLE instance running on the local machine or on another networked machine. This variable overrides any setting for ORACLE_SID and causes the Net8 software to manage the connection request; see also ORACLE_SID and TWO_TASK.

Local Area Network A limited-distance, high-speed, data communications network that allows various data processing resources to be connected and shared. A LAN is a network contained within a single physical site (one or more buildings), as opposed to a WAN; see also WIDE AREA NETWORK.

Local Currency The denomination of currency used for transactions in a particular local finance business location.

Localization 1. An enhancement or modification necessary to support specific site requirements not addressed by the base configuration hardware and software. It is developed by and purchased from Oracle. These site specific requirements generally satisfy a government or regulatory agency requirement, although localizations are not limited to this purpose. 2. Modules that extend Oracle Applications to meet the unique business requirements of one or more geographic regions.

Localization Approval The agreement to modify the company standard software and hardware configuration relative to the site requirements.

Localization Tape A tape containing localization programs to be installed to the applications software.

Location see BUSINESS LOCATION.

Logical Application Architecture A complete map of the application instances required to support the applications architecture.

Logical System Design The task of designing a system to support business needs without making final decisions regarding the physical implementation. The same logical design should be appropriate for many physical implementations using; for instance, different versions of a database management system.

Look and Feel The appearance and behavior of a system facility as perceived by the end user. This includes the data, layout, and user interaction through menus, buttons, text editing, and other devices.

M

Management The process of planning, controlling, and completing the execution of an undertaking.

Manual Function A business function that is not system-assisted; see also BUSINESS FUNCTION.

Manufacturing A high-level business function responsible for manufacturing or assembling products; often synonymous with a business organization; see also BUSINESS ORGANIZATION TYPE.

Mapping A technique for establishing application fit to business requirements, identifying gaps and proposing initial solutions. Also a technique to define the relationship between objects; see also APPLICATION FIT and MATRIX DIAGRAM.

Mapping Scenario A plan for and record of business solution testing including: business processes involved in the test, the business conditions that are needed to test the application system, definition of test script execution, support tools required during execution of the test, and a record of test actions; see also BUSINESS SOLUTION TESTING.

Mapping Team A group of people responsible for the modeling, design or mapping for a particular business process.

Margin Amount The difference between the costs (labor, expenses, and overhead) and revenue plus expense reimbursements expressed as a currency amount.

Margin Percentage The ratio between the margin amount and costs, expressed as a percentage.

Marketing A high-level business function responsible for marketing functions within a company; often synonymous with a business organization; see also BUSINESS ORGANIZATION TYPE.

Matrix Diagram A spreadsheet diagram where the axes represent two associated types of elements of interest to information systems developers. A matrix diagram used to express a mapping.

MB see MEGABYTE.

Mechanism 1. A particular technique or technology for delivering a function. Examples might be a telephone, a computer, or an electronic mail service. 2. Resources that enable or facilitate the step/sequence in a test scenario.

Megabyte (MB) A unit of memory or disk space equal to 1,048,576 bytes (1024 x 1024). Often rounded to one million bytes.

Method 1. A system of doing things and handling ideas. A method establishes a network of common tasks, a vocabulary, a set of common processes, estimates, and guidelines for delivering services. 2. The implementation of an operation. It specifies the algorithm or procedure that effects the results of an operation (UML 1.1 Semantics).

Method HelpText The automated form of Oracle Method's handbooks included with every route. This may be accessed directly from the Windows™ desktop or from within Project Bridge Modeler or Project Workbench; also called online help.

Metrics Server A software server that manages and distributes requests from forms clients among multiple forms servers; see also LOAD BALANCING.

MI see MULTIPLY INSTANTIATED TASK.

MIPS (Millions of Instructions Per Second) a measure of computer processing capacity.

MLS see MULTIPLE LANGUAGE SUPPORT.

MO see MULTIPLY OCCURRING TASK.

Module A logical program unit. Examples include: forms, reports, user exits, C programs, PL/SQL procedures, and database triggers; see also SUPPORTING MODULE.

Module Integration Test A test of related modules in an application system using scenario test specifications.

Module Network A technical diagram of modules in an application system that expresses the possible execution paths of business transactions.

Module Process Test Model A detailed testing model for the development and execution of testing. It is identical to the System Process Test Model, with the addition of module references.

Module Test A procedure or sequence of steps that determines whether a module functions properly in isolation from other system components, and conforms to project standards.

MoSCoW List Must have, Should have, Could have, Won't have — a way of classifying and prioritizing requirements or facilities for inclusion in an information system. This mechanism is used in timeboxed development where the scope may need to be redefined according to the rate of progress. The Must have is vital and hence must be developed or included; the Should have is important, but not vital; the Could have is nice to have, but not really important; and finally the Won't have is not developed or included; see also TIMEBOX.

MRC see MULTIPLE REPORTING CURRENCY.

MSOB see MULTIPLE SETS OF BOOKS ARCHITECTURE.

MTEWA Machines, tools, equipment, and work aids used to perform a task.

Multiple Language Support (MLS) Oracle Applications MLS allows the use of multiple simultaneous character sets, and provides translated messages, help files, and seed data for languages other than American English.

Multiple Organization Architecture (Multi-Org) A single installation of any Oracle Applications product to support any number of organizations or different sets of books. The data contained in product schemas is for all organizations and is partitioned by the ORG_ID column in tables.

Multiple Reporting Currency (MRC) An Oracle Applications feature that allows you to create, maintain, and report on accounting records at the transaction level in more than one functional currency.

Multiple Sets Of Books Architecture (MSOB) An older Oracle Applications mechanism for implementing an organization or group of organizations that shares a common Accounting Flexfield structure, calendar, and functional currency. With MSOB architecture, you must define at least one set of books for each business location. This architecture is no longer supported by Oracle Corporation. MSOB architecture is superseded by the newer Multiple Organization Architecture; see also MULTIPLE ORGANIZATION ARCHITECTURE.

Multiply Instantiated Task (MI) A task that may be performed at various times during a project, such as status reports or healthchecks, or a task that is partitioned on a project plan, for example, by multiple teams or functional areas.

Multiply Occurring Task (MO) A task that is repeated at specific, known times during a project. A multiply occurring task is similar to a multiply instantiated task, except the specific occurrences of the task can be planned at the method level. Each occurrence has specific task dependencies which determine when it occurs.

Multi-Tier Architecture A computing model in which various functions are distributed among multiple levels, or tiers, of machines. Typically there are at least three such tiers: desktop clients issuing user requests, back-end servers used for database processing and other intensive tasks, and a middle tier that mediates connections between the two; see also INTERNET COMPUTING ARCHITECTURE.

N

National Language Support (NLS) Oracle Applications NLS allows the use of various character sets and provides translated messages, help files, and seed data for languages other than American English.

Near Real-Time Interface An application interface that supports asynchronous transfer of data between applications. It transfers data with a sufficiently small time delay so as to leave the interfaced applications in states that are very close to synchronized.

Net8 The Oracle product that enables network connectivity between a client machine and the Oracle8 Database Server. Net8 manages communication sessions between these machines by opening and closing sessions and by packaging and sending SQL statements and data responses. Prior to the Oracle8 Database Server release, Net8 was called SQL*Net.

NLS see NATIONAL LANGUAGE SUPPORT.

Node A single computer, group of computers, or mechanism for handling some communication traffic through a particular point on a computer network.

Non-Billable Project Expenses The project expenses that are not billable to a client; see also PROJECT EXPENSES.

Non-Billable Utilization The utilization that is not billable to a client; see also UTILIZATION.

Non-Revenue Entity A legal entity that performs services on behalf of another entity. Such services are usually performed in exchange for an inter-company service fee.

Normalization A step-by-step process of analyzing the data to be represented in order to eliminate data redundancy. It produces either entity or table definitions that have:

- no multi-value attributes or columns
- the same kind of values assigned to all occurrences of any given attribute or column
- a unique name
- uniquely identifiable rows

Null The state of a data item indicating no value. Null is not equivalent to zero.

O

O see ONGOING TASK.

Object Orientation (OO) The perspective that systems should be constructed from objects, which themselves may be aggregations of smaller objects.

Object Technology (OT) The technology, programming languages, development tools, databases, and so on, introduced to support object-oriented systems development.

Objective A statement of business intent that may be measured quantifiably.

OC see ORGANIZATIONAL COMMUNICATIONS.

OCM see ORGANIZATIONAL CHANGE MANAGEMENT.

ODS see OPERATIONAL DATASTORE.

OLAP see ON-LINE ANALYTICAL PROCESSING.

OM see ORACLE METHOD.

Ongoing Task (O) A task that occurs continuously throughout a project, rather than at a specific known time. Ongoing task are shown as continuous on the workplan; see also TASK, PHASE and ITERATED TASK.

On-Line Analytical Processing (OLAP) On-line retrieval and analysis of data to reveal business trends and statistics not directly visible in the data directly retrieved from a data warehouse. Also known as multi-dimensional analysis.

OO see OBJECT ORIENTATION.

Open Systems Organizational systems that are influenced by and interact with their environment.

Operating System The computer software that performs basic tasks such as allocating memory and allowing computer components to communicate.

Operational Datastore (ODS) A data warehouse that is a repository for near real-time operational data rather than long term trend data.

Optional Tasks Additional tasks that may need to be completed during an implementation or migration due to specific project circumstances.

ORACLE_HOME An environment setting that specifies the top directory for Oracle Database Server program files.

Oracle ID An account on a database, comprised of a database username and password.

Oracle Method (OM) Oracle Services' integrated service methodology which consists of workplans, handbooks, and templates used to provide enterprise business system solutions.

ORACLE Schema see SCHEMA.

Oracle Server The database management system used by Oracle Applications. The term refers in general to the product files or the ORACLE instances created using these files.

Oracle Server Manager An Oracle Database Server utility that allows DBAs to start, stop, monitor, and maintain databases.

Oracle Services An Oracle Corporation business organization that provides professional services.

ORACLE_SID An environment setting that identifies an ORACLE instance running on the current machine. See also TWO_TASK and LOCAL.

Oracle Survey Tool A customizable database of questions used by Organizational Change Management to generate surveys, data analyses, and graphical reports for assessments.

ORACLE User ID A username and password used to access an ORACLE instance.

Organization see BUSINESS ORGANIZATION.

Organization Type see BUSINESS ORGANIZATION TYPE.

Organization Unit An element that represents part of the structure of a business. An organizational unit can represent an entire business, a group or department within the business, a person within a group or department or a role; see also AGENT, BUSINESS UNIT, and AGENT CHANNEL.

Organizational Change Management (OCM) An Oracle Services line of business providing change management expertise to organizations seeking to manage the human and organizational factors involved with implementing new technology. Organizational change management includes five service areas: organizational effectiveness assessments, organizational communications, human performance and development, leadership development, and custom training; see also ORGANIZATIONAL EFFECTIVENESS ASSESSMENTS, ORGANIZATIONAL COMMUNICATIONS, and CUSTOM TRAINING.

Organizational Communications (OC)

A Organizational Change Management area that assists organizations in developing and implementing communication and feedback programs that seek to optimize information and buy-in while minimizing resistance to change and promoting a unified vision and open communication among levels of the organization.

Organizational Culture The shared constructs (concepts, modes of thinking, views of itself and others) of an organization.

Organizational Effectiveness The qualitative state at which the organization is functioning to meet its goals. Includes quality of organizational communication, relevance of the performance support system, ability of executives to manage to objectives, core competencies and skills, etc.

Organizational Effort Any activity or set of activities implemented systematically by an organization to achieve a particular goal.

Organizational Readiness The state of an organization's potential to manage change successfully. Involves a consideration of an organization's high-impact leverage points for change, as well as any change impediments.

Organizational Structure The arrangement of jobs into groups, departments, or business units within a given organization.

OT see OBJECT TECHNOLOGY.

Outsourcing The practice of appointing an external organization to provide any or all of the services of the IS department or any other internal service department.

Overhead The operating expenses associated with delivering services, such as rent, light, heat, taxes and non-billable utilization.

Overhead Factor A rate multiplier associated with a category of overhead, e.g., Corporate, Division, or Practice; see also OVERHEAD.

P

Parallel Concurrent Processing In a UNIX environment, distribution of concurrent processes among multiple concurrent processing servers. Also called DISTRIBUTED CONCURRENT PROCESSING.

Parameter A value passed to a command. A parameter can be a file specification, a symbol, or word understood by the program.

Partition A discrete part of the total functionality that is developed in a project. A partition may be a single module, but more often it will be an entire sub-system. A partition may be related to a business area. Partitioning a system also implies a partitioning of the project that develops the system. A separate team develops each partition; see also MODULE, and BUSINESS AREA.

Password An identification word associated with a username that the user must supply to access an ORACLE instance or an Oracle Applications system.

PAT see PERFORMANCE ASSURANCE TEST.

Payment Milestone A significant project event at which time a payment is due. Payment milestones can be progress points, dates, the completion of a task or the production of a deliverable.

Payment Terms The terms and conditions upon which payments will be received from a client.

PBM see PROJECT BRIDGE MODELER.

PCM see PRACTICE MANAGEMENT.

PDF see PORTABLE DOCUMENT FORMAT.

Percentage Adjustment Estimate Total estimate adjusted for organizational factors.

Performance Assurance Test (PAT) A project concerned with simulating an implemented business system for the purposes of predicting its likely performance.

Performance Management Establishing and monitoring performance criteria / standards that contribute to organizational objectives; typically involves goal-setting, measurement, feedback, rating / appraisal, rewards and recognition, and career development.

Performance Support Practice An established or customary set of actions that reinforces or encourages the execution of a particular behavior or set of behaviors, for example, performance review, to meet performance goals.

Performance Support System An integrated collection of all performance support components (infrastructure, processes, protocols, procedures, tools and practices) for a given group or organization, designed to reinforce or encourage the execution of prescribed behaviors within the group.

Performance Support Tool An instrument, such as a performance review form or formula for the calculation of bonuses, that is designed to meet the objectives of a given performance support practice or system.

Performance Technology The field of expertise within organizational development / behavior to set up performance management systems in support of meeting performance expectations.

PGM see PROGRAM MANAGEMENT METHOD.

Phase A chronological grouping of tasks in an approach. Services are delivered by phase in order to reduce project risk. Each phase allows a checkpoint against project goals, and measurement against quality criteria to be made.

Phase Completion The project management tasks which conclude and secure client signoff of a phase.

Phase Control The project management tasks which execute concurrently with phase execution, and perform project monitoring, directing, and reporting functions during a phase.

Phase Execution The method execution tasks performed during a project phase.

Phase Management The project management tasks required to plan, control and complete the execution of a project phase.

Phase Planning The project management tasks which update project plans and procedures for a phase and secure additional resources necessary to execute that phase.

Physical Application Architecture A complete map of the database instances, their sites, and the application instances and Oracle Ids that they support. This will incorporate aspects of the logical architecture, and the high level designs for security and interfaces.

Pilot An initial project which will serve as a model or template for future projects.

PJM see PROJECT MANAGEMENT METHOD.

Plan A scheme, method or design for the attainment of some objective or to achieve something.

Planned Response System The entire set of business processes in a business area that respond in a predetermined way to a known set of events; see also APPLICATION SYSTEM and EVENT.

Planning (M&D) A high-level business function responsible for manufacturing and distribution planning for one or more manufacturing and distribution business units; often synonymous with a business organization; see also BUSINESS ORGANIZATION TYPE.

Platform Specific computer hardware, as in the phrase “platform-independent.” It may also refer to a specific combination of hardware and operating system and any additional software required by the platform to execute applications code, connect to other platforms across a network, and provide services to other platforms (for example, printing, processing, data storing, etc.). It is also used to refer to support software for a particular activity. Sun SPARC Solaris and DEC Alpha Windows NT are examples of platforms. Some Oracle Applications functionality is platform-specific, meaning its behavior may differ on other platforms.

PL/SQL A procedural extension of SQL that provides programming constructs such as blocks, conditionals, and functions.

Policy A guiding principle, typically established by senior management, which is adopted by an organization or project to influence and determine decisions.

Portable Document Format (PDF) A file format developed by Adobe Systems. PDF files are *snapshots* of formatted document files and may contain formatted text, graphics, color, and images that are viewable, navigable, and printable on multiple platforms. Use the free program Adobe Acrobat Reader to view a PDF file.

Position Ranking Method of comparing whole jobs to one another to rank them in a hierarchy.

Practice Management (PCM) Specifies the process, tasks, and responsibilities regarding the management of a consulting practice. Specifically, this includes client management, project portfolio management, and staff training, recruitment, and growth.

Pre-Sales Cycle The series of activities that occur before the application was selected.

Predecessor A task that precedes another task and is related to it by a task dependency; see also SUCCESSOR.

Prerequisite Something needed by a task, which is produced by a previous task or an external source; see also DELIVERABLE.

Primary Functional Currency The currency you use to record transactions and maintain your accounting data within Oracle Applications.

Primary Set of Books A financial reporting entity in which you conduct business.

Problem A perceived variance between the expected and observed ability of an item to fulfill its defined purpose.

Problem Report The mechanism by which a problem is recorded, investigated, resolved, and verified.

Problem Statement A concise phrase, motto, or goal-oriented explanation of the motivation behind buying a new application system.

Pro*C An Oracle precompiler product that allows developers of C programs to embed standard database calls to an ORACLE database in C programs.

Procedure A written set of steps that specifies how to carry out a business function. If the business function is system-assisted, its corresponding procedure will indicate how the application system carries out that business function; see also APPLICATION SYSTEM and BUSINESS FUNCTION.

Process 1. The sequential execution of functions triggered by one or more events. 2. A discipline or sub-project that defines a set of tasks related by subject matter, required skills and common dependencies. A process usually spans several phases in an approach. Examples are: Data Conversion, Testing, Documentation; see also BUSINESS PROCESS and SYSTEM PROCESS.

Process Analysis A component of a business requirements scenario that facilitates quantitative analysis and measurement of each process step and compares the current to the proposed process in order to sell change to management and to key business people and systems users.

Process Characteristics Attributes that describe a process and how it works including descriptive narrative for its steps, tools required, skill levels required, controls, and performance measures.

Process Flow The passing of execution of a process from one process step to the next. It may include the passing of information or materials from the first step to the second.

Process Flow Diagram A diagram which shows the triggering event(s), sequential flow of process steps, decision points, and deliverable or outcome of a single process.

Process Label A unique reference, within an application system, for a process.

Process Modeling A structured approach used to identify, define, and document the activity performed by a business to produce business deliverables.

Process Narrative A reduction of a new business process design down to a job-level description, thereby defining how the work gets done and laying a foundation for development of a user guide, role-based user training and user certification or other types of readiness testing. One process narrative should be written for each business process; see also ROLE-BASED TRAINING.

Process Owner The agent with overall responsibility for a complete business process; could be the customer of the process, or the supplier (person or organization charged with fulfilling the request).

Process Research A technique used in testing the feasibility of a process model and gathering facts. The approach involves answering implementation questionnaires, reviewing current process documentation, gathering statistics regarding volumes or frequencies, understanding policy statements and measures of performance, interviewing key users to ascertain critical factors for success.

Process Step An instance of the execution of a function as a step in performing a process. In a fully-analyzed business process model, all process steps are instances of elementary business functions. A business process step is normally composed of a series of procedure steps, where tools such as application screens, reports and inquiries are used. Once a process steps begins, all of its procedure steps must be completed in order to achieve an accurate and quality output. Business requirements are defined at the process steps level, while job definitions are at the procedure level.

Product Group Each separate application instance of Oracle Application Object Library in an Oracle database. A product group may contain any number of Oracle Applications products in addition to the single instance of Oracle Application Object Library.

Production Environment The database, equipment, documentation, and procedures used in support of live business operations.

Productivity Measured comparison of the quantity and/or quality of outputs and the quantity of resources required to produce them.

Productivity Metrics Measure by which acceptable performance will be evaluated; may include a time limit, accuracy tolerance, proportion of correct responses required, and/or qualitative standards.

Profile Option An Oracle Applications control switch that a user can set to govern some aspect of system processing or user interface.

Program 1. A set of coded instructions that a computer executes or interprets to perform an automated task. 2. A interrelated group of projects that are either being run concurrently or sequentially and that share a system goal. Individual projects may have different goals, however the combined set of projects will have a program goal.

Program Library The physical location, typically a networked server, used as a shared repository for all program information produced during the life of a program. Contents include issue logs, meeting minutes, status reports, deliverable documents, working papers, project plans, standards, tools, and configured software deliverables. The library is designed to facilitate access to all archived program and project deliverables by all members of a program.

Program Management Method (PGM) Part of Oracle Method designed to provide value-added management, infrastructure, and control to successfully manage complex enterprise-wide implementations of Oracle technology. PGM emphasizes commonality and reusability for management and control processes, execution processes, and hard and soft deliverables.

Program Office A project executed under the sponsorship of program management that establishes commonality and reuse across multiple projects within a program.

Project A set of work processes, tasks with associated deliverables and resources executed over a specific period of time with predetermined budget and business objectives.

Project Bridge Modeler (PBM) Applied Business Technology tool used to build the project planning and estimating system. PBM allows project managers to select, edit, combine, and create project workplans or routes, that best fit the needs of the client.

Project Completion The third and final part of the PJM project life-cycle. The satisfactory conclusion of the project and settlement of all outstanding issues prior to hand over of the project deliverables to the client.

Project Earned Value A measure of the value of completed tasks in a project. There are various ways of measuring the value of a task. These include percentage on commencement, percentage on completion, and amount at milestone.

Project Execution The second part of the PJM project life-cycle. The carrying out of project plans determined during planning for a method approach. Project execution also encompasses elements of control which analyze project performance and take corrective action as needed.

Project Expenses Funds allocated or spent to cover incidental or non-labor costs of a project.

Project Infrastructure The framework for storing, maintaining, and referencing all implementation deliverables and supporting materials including office space, software tools, and standards.

Project Library 1. A system for storing, organizing and controlling all documentation produced or used by the project. 2. The physical location of all deliverables for a single project, plus administrative and support materials. An administrative office to which all members of a team have access.

Project Life-Cycle The organization of a project according to its three major parts: planning, execution, and completion.

Project Management Method (PJM) A method which defines how a project is managed when executed according to the requirements of Oracle Method.

Project Milestone A significant project event.

Project Objectives The set of criteria for measuring a project's success.

Project Office The management of the project library for a specific project.

Project Partition Part of a project usually representing a coherent set of facilities to be developed by a single developer, commonly organized and managed as a sub-project.

Project Planning The first part of the PJM project life-cycle. The definition of a project with respect to scope, quality, time and cost. Project planning also determines the appropriate organization of resources and responsibilities to execute a project.

Project Schedule A list of tasks to be carried out presented against a timetable for their completion.

Project Timeline A specification of work to be carried out together with the number of resources needed to achieve a target duration; see also PROJECT SCHEDULE.

Project Workbench (PMW) Applied Business Technology tool used to schedule, track, and analyze your project. PMW provides work breakdown structures based on the routes selected in Project Bridge Modeler and provides the capability to manage these plans.

Project Workplan A specification of the work to be performed for a project, expressed as a set of interdependent tasks with project resources allocated over time.

Property Any detail that serves to qualify, identify, classify, quantify, or express the state of an element in a repository.

Prototype A facsimile of an end product used to demonstrate a concept rapidly, check feasibility, or gain acceptance.

Prototyping The construction of a partial system to demonstrate some aspect or aspects of the intended system behavior in order to gain user acceptance or to establish technical feasibility.

Publish and Subscribe A data communication paradigm for implementing the data flow between two or more applications. An event in a source application causes the application to 'publish' a data object or message to the potentially interested applications who individually opt to 'subscribe' to the data object or message. The subscription of the interested applications to the data object or message causes a state change in the applications. Contrast Request-Reply.

PMW see PROJECT WORKBENCH.

Q

Quality Audit An audit used to assess the adherence of the project team to plans, procedures, and standards.

Quality Management The means of implementing quality policy. On a particular project this is achieved through quality planning; quality assurance; quality control; and quality improvement.

Quality Review A review used to assess the quality of a deliverable in terms of fitness for purpose and adherence to defined standards and conventions.

Questionnaire A written or electronic survey instrument comprised of a series of questions, designed to measure a specific item or set of items.

R

RBT see ROLE-BASED TRAINING.

RDBMS see RELATIONAL DATABASE MANAGEMENT SYSTEM.

Real-Time Interface An application interface that supports synchronous transfer of data between applications.

Real-Time Support System (RTSS) An electronic information system accessible by Oracle and the client.

Real-Time System A system in which events control actual mechanisms. Real-time systems often control machinery (for example, a control system for an aircraft) and are often time- or safety-critical; see also EVENT.

Record In a non-relational database system, a record is an entry in a file, consisting of individual elements of information, which together provide full details about an aspect of the information needed by the system. Individual elements are held in fields and all records are held in files. An example of a record might be an employee. Every detail of the employee for example, date of birth, department code, or full names will be found in a number of fields. In a relational system record is an alternate word for row; see also ROW.

Record Type A predetermined set of fields within a file.

Readiness State of a person's, group's or organization's potential to manage change successfully. Involves a consideration of high-impact leverage points for change, as well as any change impediments.

Reference Materials Documents that describe key aspects or samples for the current business. These are normally compiled before the project starts and may have been used during the pre-sales cycle.

Registry An embedded database that holds configuration information for the Windows NT operating system. During installation, Oracle Applications writes data to the registry. You can also edit the registry directly with the Registry Editor (regedt32.exe).

Registry Key A folder that appears in the left pane of the Registry Editor window. A key can contain subkeys and value entries. For example: Environment is a key of HKEY_CURRENT_USER; see also REGISTRY SUBKEY.

Registry Subkey A key within a key. Subkeys are analogous to subdirectories in the registry hierarchy. Keys and subkeys are similar to the section heading in .ini files. However, subkeys can carry out functions. Oracle Applications stores important information about a product group in a registry subkey; see also REGISTRY KEY and REGISTRY.

Regression Test Specific assurance testing on an application system release, after defects have been corrected or enhancements have been completed. Regression testing is required to re-validate an application system, confirming that prior validations have not regressed.

Relational Database Management System

(RDBMS) A database management system in which data can be viewed and manipulated in tabular form. Data can be sorted in any order and tables of information are easily related or joined to each other.

Relationship 1. What one entity has to do with another. 2. Any significant way in which two things of the same or different type may be associated.

Release A baseline issued from the CM Repository for delivery to an a destination. The destination may be internal to the project environment, such as for testing, or external, such as to the client.

Remote Procedure Call (RPC) A protocol that allows a client to execute a program on a server. The client sends a message to the server with appropriate arguments, and the server returns a message containing the program's results.

Report An organized display of Oracle Applications information. A report can be viewed online or printed. The content of a report can range from a summary to a complete listing of values.

Reporting Database A database used by reporting applications. Reporting databases are often duplicates of transaction databases used to off-load report processing from transaction databases.

Reporting Functional Currency A currency, other than your primary functional currency, for which you need to generate reports.

Report Review Agent A tool used by Oracle Applications to view concurrent processing files online.

Reporting Set of Books A financial reporting entity that is associated with a primary set of books. It has the same chart of accounts and accounting calendar, but usually a different functional currency.

Repository A mechanism for storing any information about the definition of a system at any point in its life-cycle. Repository services would typically be provided for extensibility, recovery, integrity, naming standards, and a wide variety of other management functions.

Representative Sample A scientifically selected group of individuals, who provide information that allows mathematical inferences to be made about the larger constituency they are chosen to represent.

Request for Proposal The formal mechanism by which a company conveys its business requirements during the search for a new application system. Known as the RFP, this document drives the pre-sales cycle and provides valuable information into the business requirements definition process of the implementation; see also PRE-SALES CYCLE.

Requirement A requirement is a specific, detailed business need which is to be compared with application functionality to determine if the need can be met with the standard Oracle software.

Requirements Scenario A requirements scenario is a formal statement of detailed business requirements for a process, the source of the requirements, how these requirements will be satisfied (either the application, manual procedure, workarounds, or other application solutions) and what prototyping steps must be taken to prove the design.

Requirements Workshop A workshop, usually attended by project sponsor, stakeholders, and developers, to provide a sufficiently detailed definition for a developer to commence build. The definition will be further defined and refined during development, particularly by user reviews; see also USER REVIEW and WORKSHOP.

Research and Development These business organizations are responsible for the research and product development functions within a company; see also BUSINESS ORGANIZATION TYPE.

Resource Any persons, equipment, or material needed to perform a task(s).

Resource Category see CONSULTING GRADE LEVEL.

Resource Database A record of the resources available, primarily human resources, including information about the skills and experiences of the resources.

Respondent In OCM, any individual who provides information in response to a survey, such as a questionnaire or interview.

Responsibility 1. A contract or obligation of a type or class. (UML 1.1 Semantics) 2. A collection of functions within an Oracle Application. Each Applications user is assigned one or more responsibilities to allow them access to Applications forms.

Revenue Income from services labor fees, training fees, licensed product sales and support fees.

Revision The authorized modification to a configuration item.

Rewards and Recognition Means by which individuals are reinforced or compensated for certain behaviors; may be tangible/symbolic, verbal/social, cash/non-cash, immediate or long-term. EX: bonuses, team celebrations, profit-sharing, outstanding contributor awards.

Risk 1. The potential of an adverse condition occurring on a project which will cause the project to not meet expectations. A risk requires management assessment and a strategy for its mitigation. 2. The logical product of the impact of the risk and the likelihood of it occurring.

Role A classification of staff member used on a project. Examples are: analyst, application developer, system architect; see also RESOURCE.

Role-Based Training (RBT) A user training approach that focuses on developing curriculum based on Oracle Application responsibilities (i.e., roles) that have been defined and assigned to users. Application responsibilities tie a user to a menu of functions that can be performed. An RBT curriculum trains a group of users on these specific functions, using client data and highlighting client procedures.

Rollback Segment A set of entries used to undo changes in the database in the event of transaction rollback, crash, or media recovery.

Rollback Tablespace A tablespace created for rollback segments.

Route A variation of a method containing all tasks required in order to deliver a service; a dependency network.

Row An entry in a table that typically corresponds to an instance of some real thing, consisting of a set of values for all mandatory columns and relevant optional columns. A row is often an implementation of an instance of an entity; see also COLUMN and TABLE.

RPC see REMOTE PROCEDURE CALL.

RTSS see REAL-TIME SUPPORT SYSTEM.

Rule of 3-2-1 A rule of thumb used during process research that stresses the importance of “walking the floor” and other practical investigation over conference room interviews. It means that roughly 3 hours of research are normally required for every 2 hours of process design and 1 hour of formal deliverable creation (like creating a BRS using a template or other tool); see also PROCESS RESEARCH and BUSINESS REQUIREMENTS SCENARIO.

S

Sales Business organizations that handle typical sales functions, including generating quotes and sales orders; see also BUSINESS ORGANIZATION TYPE.

Sample A statistically-significant subset selected and analyzed to estimate the characteristics of a larger group or population; a set of individuals within an organization assessed to provide information on the preferences, opinions, attitudes, and practices of the group they represent.

Schema 1. An information model implemented in a database. A schema may be a logical schema, which will define, for example, tables, columns, and constraints, but which may not include any optimization. It may be a physical schema that includes optimization, for example, table clustering. 2. An ORACLE account or ORACLE ID.

Scenario A discrete instance of a system process; see also **MAPPING SCENARIO**.

Scenario Test Specification A component of a test script which defines the test execution – it is comprised of scenario information, system process information, a series of test steps, and their associated data profiles.

Scope The boundaries of a project expressed in some combination of geography, organization, applications or business functions.

Scope Change A change to project scope. A scope change requires an adjustment to the project workplan, and nearly always impacts project cost, schedule or quality.

Scope Creep The common phenomenon where additional requirements are added after a project has started without reconsidering the resourcing or timescale of the project. Scope creep arises from the misapprehension that such small additions will not affect the project schedule.

Scoping Workshop A workshop, usually attended by the project sponsor and developers, with the objective of defining the boundaries of the scope for an intended project prioritizing requirements within the scope; see also **WORKSHOP**.

Script 1. A sequence of coded instructions executed or interpreted by computer programs.
2. A prescribed set of steps to follow when testing.

Security Profile A list of role-based security specifications.

Sequence A database object created such as a table used to generate unique keys (sequence numbers).

Server A program or machine that provides resources, such as files, processing, or other information, to other programs or machines, which are called clients. For example, the Oracle Forms Server process runs on a forms server (machine), and the Oracle Database Server (database process) runs on a database server. Server processes may communicate with other programs, such as desktop client processes or other servers. Do not use the term *server* in an unqualified sense. Use *Hardware Server*, *Applications Server*.

Service 1. Work performed (or offered) by a server. This may mean simply serving simple requests for data to be sent or stored (as with file servers, gopher or http servers, e-mail servers, finger servers, SQL servers, etc.) or a more complex work, such as that of irc servers, print servers, X Windows servers, or process servers. 2. A Windows NT internal software routine that provides support for other applications. The Oracle Database Server, Net8, and Oracle Applications concurrent managers all make use of services.

Servlet A Java program executed on a web server rather than downloaded to a desktop client; see also **APPLET**.

Set of Books A company or group of companies within Oracle Applications that share a common chart of accounts, accounting calendar, and functional currency. This concept is being superseded by the concept of a financial organization in releases of Oracle Applications starting with 10.6.

SGA see **SYSTEM GLOBAL AREA**.

Shared Localization A customization to an application product required by more than one country.

Shared Product see **DEPENDENT PRODUCT**.

Short Name An abbreviation for an Oracle Applications product (such as GL for Oracle General Ledger).

Shut Down The process of stopping an active process, such as an ORACLE instance or a concurrent manager, to make it unavailable.

SI see SINGLY INSTANTIATED TASK.

Signoff Agreement with a client of the successful completion of a project, project phase, or deliverable.

Singly Instantiated Task (SI) A task which occurs once, at a specific time, during a project.

Site A uniquely identifiable geographic location or place from which one or more business organizations may be wholly or partly operating.

Site Based Configuration Specific modifications or enhancements made to the company base configuration to support site requirements. The site base configuration must remain integrated with the company base configuration to maintain integrity of the company base configuration; see also COMPANY BASE HARDWARE CONFIGURATION.

Site Configuration The definition and management of the site base configuration.

Skills Analysis The collection of data from groups and individuals targeted in a specific training situation, to determine the existence and nature of any performance gap(s); used in a skills assessment to determine training needs and develop training goals and objectives to be translated into learningware recommendations.

Skills Database see RESOURCE DATABASE.

Skills Profile Description of the skills held by a particular job incumbent or group of incumbents at a given point in time.

Skills Set Collection of physical or mental capabilities inherent in and available to a given individual.

SmartClient An implementation used with Release 10 to provide client/server access to Oracle Applications. Use of SmartClient is no longer recommended by Oracle Corporation for new implementations.

Software Management (SWM) A program management focus area project that defines and establishes a common software infrastructure for a multi-site implementation. Included items are development environment; software distribution; version control; and software configuration management methodology.

Solution A solution is the resolution of the discrepancy between one or more gaps and one or more standard application functions. A solution may take the form of a procedure, policy or customization.

Source Module A physical program unit. An application system's repository of source code is controlled at the source module level.

SQL see STRUCTURED QUERY LANGUAGE.

SQL*Loader An Oracle Database Server tool used to load unstructured data from operating system files into Oracle Database Server database tables.

SQL*Plus An Oracle program you can use to enter and run SQL commands and PL/SQL blocks that store data in and retrieve data from an ORACLE database.

SQL Script A file containing SQL statements that you run, with a tool such as SQL*Plus, to query or update ORACLE data.

Stakeholder A person, group, or business unit that has a share or an interest in a particular activity or set of activities.

Standard A set of rules for measuring quality. Usually, standards are defined for products deliverables or deliverable components and processes.

Standard Request Submission A standard interface in Oracle Applications that lets you run and monitor concurrent requests.

State A recognizable or definable condition that a system or an object can be in at some point in its life-cycle.

State Transition A valid change of a system or an object from one state to another, modeled on a state transition diagram; see also STATE.

Store A collection of information or materials used in a process.

Storyboard A technique, borrowed from the film industry, for describing screen dialogues. A storyboard consists of an ordered series of pictures illustrating stages of the dialogue. The pictures are annotated with notes about logic and user input.

Structured Query Language (SQL) The ANSI internationally accepted standard for relational database systems, covering not only query but also data definition, manipulation, security, and some aspects of referential and entity integrity.

Subdirectory A directory that is contained within another directory.

Sub-Function A business function that has a parent function.

Sub-Process A process performed entirely within another process.

Successor A task that follows another task and is related to it by a dependency link; see also PREDECESSOR.

Support Profile A section of a test scenario that identifies support tools required during execution of the test.

Supporting Module A module that is not itself an entry point in an application system. Supporting modules are generally shared modules that provide functionality used by multiple primary modules. They are usually not referenced independently in user-oriented documentation. All PL/SQL packages, procedures, and database triggers are examples of supporting modules; see also MODULE.

Swim Lane see AGENT CHANNEL.

Synonym 1. A name assigned to a table or view that may then be used more conveniently for reference. 2. An alternate name for an entity.

Syntax The orderly system by which commands, qualifiers, and parameters are arranged together to form valid command strings.

SYS Username A database user which is automatically created in Oracle Database Servers upon installation and initialization. This user owns the tables and database objects that make up the Oracle data dictionary.

System A named, defined, and interacting collection of procedures and processes, along with the organized deployment of people, machines, various mechanisms, and other resources that carry out those procedures and processes; see also APPLICATION SYSTEM.

System Architecture A representation of the structure of an application system usually confined to essentials.

System Facility A part of an information system that supports an identifiable set of business functions. A facility may be a single module or it may be a whole sub-system.

System Function Something a computer system does in order to support one or more business functions; see also APPLICATION SYSTEM and BUSINESS FUNCTION.

System Global Area (SGA). A reserved section of main memory that provides communication between all database users and the ORACLE background processes.

System Interface The mechanism used to create connectivity between two systems.

System Operations Function A system function that serves to support the continuing operations of the application system. Examples are backup, recovery, audit; see also APPLICATION SYSTEM, SYSTEM FUNCTION, and SYSTEM OPERATIONS PROCEDURE.

System Operations Procedure A step-by-step indication of the way in which an application system executes a system operations function. The system operations guide is a CDM deliverable that is a compilation of all relevant system operations procedures; see also APPLICATION SYSTEM.

System Process The response which an application system makes to an event. It comprises the sequential execution of both manually executed process steps which are instances of manual functions and process steps automatically executed by a computer system which are instances of system functions.

System Process Test Model The overall basis for testing the functional requirements of a system. The system process test model contains scenarios, data profiles, and test specifications, but no module references.

SYSTEM Tablespace Holds data dictionary tables owned by the SYS account. It is created when you install the database.

System Test A project activity that tests an application system over its complete life-cycle, using scripts and associating scenario test specifications into chronological sequences.

System Test Flow A set of steps or a transaction sequence used in system or business process testing.

SYSTEM Username One of two standard usernames automatically created with each database (the other is SYS). This user owns the views created on the data dictionary tables. The SYSTEM username is the preferred username to use when performing database maintenance. See also SYS USERNAME.

Systems Integration Test A project activity that consists of testing related application systems using sequences or scripts from two or more systems which interface.

T

Table The basic unit of storage in a relational database management system, defined by one or more columns of data and a primary key. A table represents entities and relationships.

Table Constraint A set of rules constraining values in a combination of one or more columns of a database table.

Tablespace A logical portion of an ORACLE database used to allocate storage for data and to group related logical structures. For example, one tablespace may contain all of one Oracle Applications product's database tables and indexes.

TAR see TECHNICAL ASSISTANCE REQUEST.

Target Application The primary set of new application modules that are within project scope.

Task A unit of work that done in delivering a service. A task is the smallest trackable item on a project plan, and forms the basis for a work breakdown structure. The minimum elapsed time for a task iteration or instantiation should be one day. The maximum should be two weeks; see also WORK BREAKDOWN STRUCTURE.

Task Dependency The relationship between two tasks where the start or end date of the successor task is constrained by the start or end date of the predecessor task.

Task Dependency Network A network of tasks and task dependencies where each node is a task and each link is a task dependency; see also CRITICAL PATH METHOD (CPM) NETWORK.

Task Group A set of tasks that should be treated similarly. For example, they should be performed at the same time, should be iterated as a group, should be multiply instantiated as a group. A task group is represented on a process flow diagram by enclosing the tasks in a rectangle.

Task Step A discreet step to be done in executing a task.

TCP/IP See TRANSMISSION CONTROL PROTOCOL / INTERNET PROTOCOL.

Technical Architecture The physical hardware, network configuration, and software tools that support the system architecture.

Technical Assistance Request (TAR) Oracle's name for recorded problems. A TAR number is a unique number assigned by WWSUP to track the problem.

Technical Platform Architecture Setup rules, guidelines, and principles that define the structure, functions, and relationships among the hardware, systems software, communications, and network facilities that act as the technical foundation for the business process, information, and application architectures. The technical platform architecture specifies the common or shared facilities and function of the facilities needed to support the applications and data to meet the needs of the business. It supports such established principles as data, devices, and location independence and formal interfaces.

Technique A specific approach to performing a task. A methodical means of handling and communicating complex details.

Temporary Tablespace A tablespace used when a SQL statement requires the creation of temporary segments (for example, the creation of an index).

Test Data Profile Specific test data values for performing a test.

Test Environment A combination of software and possibly hardware that provides a stable environment in which to test newly developed software without elaborate preparation. A test environment might provide test data; data structures; test scripts; automatic test execution; test results recording; and test results analysis.

Test Scenario An instance of an event. A point-in-time sample of the business conditions and processing environment to be tested. Each scenario may include users working in multiple applications, performing online or batch processing being performed.

Test Script A document consisting of a test specification, a test data profile and instructions for performing a test.

Test Specification A set of steps or a transaction sequence used in module; module integration; system; systems integration; or entity testing.

Thin Client 1. A client is considered thin if the personal database and Express Server are installed on the server and all processing occurs on the server. 2. A low-maintenance desktop client, usually without a hard drive, intended for optimal use in a multi-tier architecture environment. In a multi-tier environment, server hardware platforms perform data processing actions instead of the thin client. 3. The client software that is running on the server hardware platform. See also TIER.

Third-Generation Programming Language (3GL) A programming language that uses procedural definitions to carry out tasks, and typically uses record-by-record processing of data. Procedural 3GL language structures include *if....then....else, do....while* statements and others.

Third-Party Application Interface A heterogeneous interface between Oracle Applications and the application product of another vendor or an application developed 'in-house'.

Tier A set of hardware platforms that perform similar tasks. For example, client/server is a two-tier architecture, with hardware platforms on the client tier connecting to machines on the server tier. A multi-tier architecture consists of three or more tiers. In Release 11, hardware platforms on the desktop client tier communicate with hardware platforms on the application tier, which in turn, communicate with each other and with the hardware platforms on the database tier.

Timebox A project management technique that fixes the duration and resources of a task, or a set of tasks, and forces the scope of the project to be adjusted based on the time available to complete the task(s). The contingency for under-estimation of the work is provided by a prioritized list of features left out if necessary. The contingency for over-estimation is provided by a prioritized list of features that should be added in if time allows.

Tool Software applications, deliverable templates, or any other utility suggested to facilitate the completion of a particular task.

Top-Down Estimate A high-level work effort estimate. This type of estimate is derived by taking a total project estimate and dividing it among the project's phases, activities, and tasks.

Topical Essay A high-level illustration of how to satisfy a business need. A section of a design document that specifies business needs, major features, terms, and processing overview.

Traceability The ability to trace an application system component to its business requirement.

Train-the-Trainer An approach toward training whereby inside resources take ownership of delivering training to their peers and associates.

Transaction Database A database used primarily by transaction oriented applications.

Transaction Interface An interface that transfers transaction data between similar or dissimilar applications.

Transmission Control Protocol / Internet Protocol (TCP/IP) A popular industry-standard networking protocol used for communication between computers.

Trigger 1. A database object that executes PL/SQL code based on a change to a database table. 2. An Oracle Forms function that is executed based on an event in the forms environment.

TWO_TASK With a UNIX operating system, an environment setting that identifies the network alias of an ORACLE instance running on the local machine or on another networked machine. This variable overrides any setting for ORACLE_SID and causes the Net8 software to manage the connection request. See also ORACLE_SID and LOCAL.

U

Uncontrolled Document A document which is produced once for information only, and is not subject to formal approval or change control.

Unique Identifier Any combination of attributes or relationships that serves, in all cases, to uniquely identify an occurrence of an entity.

Unit Test see MODULE TEST.

Usability That quality of a system that makes it easy to learn, easy to use and encourages the user to regard the system as a positive help in getting the job done.

User A person who uses a system to perform a business function.

User Preferences In many circumstances in computer systems there may be alternate ways a user can influence the behavior of a utility, user interface, or other system process. Typically set by adjusting values in a set of user preferences; for example, in a program generator, preferences may be set for style, performance, user interface behavior, and code standards.

User Review A meeting at which some of the facilities of a system are demonstrated to and reviewed by user. The objective of a user review is to elicit feedback on which to base future development and improvement of the facilities being reviewed.

Utility 1. A program or system function that performs a job for the users, but does not provide or require user interaction (for example, the calculation of interest); see also GENERATOR and TRANSFORMER. 2. A program that performs a job for developers or for administrators (for example, Headstart utilities).

Utilization The amount of time a staff member books to a project accounting system; see also BILLABLE UTILIZATION and NON-BILLABLE UTILIZATION.

V

Vanilla 1. Standard or plain. 2. An ice cream flavor.

Variance The difference between a planned and an actual value; for example, budgeted hours vs. actual hours.

Version The rendering of a configuration item which incorporates all of its revisions starting from a given point.

Version Control A mechanism to manage multiple revisions of files, documents, programs, applications, or other items that undergo change.

View 1. A means of accessing a subset of data in a database. 2. A custom-tailored presentation of the data in one or more tables. A view can be thought of as a *stored query*.

W

Walkthrough Review of work in progress, usually taking the form of a presentation by a developer to an audience of stakeholders or fellow developers who are encouraged to comment and ask questions. The objective is to make sure that work is proceeding in the right direction.

WAN see WIDE AREA NETWORK.

WBS see WORK BREAKDOWN STRUCTURE.

Web Browser A program running on a desktop client used for viewing documents formatted in HTML and running Java applets. A web browser sends requests to a web server using a special protocol (HTTP) to retrieve documents and Java applets.

Web Client see THIN CLIENT and TIER.

Web Server A type of application server that runs an HTTP listener. The term is also used by some in an informal manner to describe the software running the HTTP listener, as in "the Apache web server".

White Box Test A test of all or part of an application system that requires knowledge of the actual code being tested. Module tests are usually white box tests; see also BLACK BOX TEST.

Wide Area Network (WAN) A communications network that connects geographically separated areas; see also LOCAL AREA NETWORK.

Work Breakdown Structure (WBS) An organization of project tasks into a hierarchy for scheduling and reporting progress.

Work Effort see EFFORT.

Work Estimate see ESTIMATE.

Workflow 1. The flow of work, step-by-step, through a process, including inputs required, outputs produced, activities undertaken and decisions made. 2. The movement of documents around an organization for purposes including signoff, evaluation, performing activities in a process and co-writing. 3. The flow of inputs and outputs linking one job or one task to the next in a work process.

Work Schedule see PROJECT SCHEDULE.

Workshop 1. A meeting attended by users and developers to create a plan, specification or other documentation that can guide the developers in their development tasks. 2. A meeting designed to facilitate interaction and the exchange of information between individuals or groups; see also REQUIREMENTS WORKSHOP, SCOPING WORKSHOP, and USER REVIEW.

Workstep A step within a task that may produce a deliverable component.

Worldwide Support (WWSUP) Oracle's support organization.

World Wide Web (WWW) A hypertext-based system for browsing multimedia documents that uses the Internet as its transport mechanism. In a hypertext system, you navigate among documents by following hyperlinks, which take you to other documents. Often simply referred to as the *Web*. See also HTML and HTTP.

WWSUP see WORLDWIDE SUPPORT.

WWW see WORLD WIDE WEB.

